



Private Network Service Manager User's Guide

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A Microsoft Company

Private Network Service Manager User's Guide
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Preface

Affirmed Networks' Private Network Service (APNS) solution allows Enterprise customers to deploy, configure, and operate Private LTE networks on Microsoft's Azure platform. By deploying multiple VNFs and leveraging Azure Stack Edge technologies along with Affirmed Networks' powerful automation solutions, alarms, and stat reporting abilities, Private Enterprises can run their private LTE service platform on a secure virtualized infrastructure.

This guide discusses the Service Manager application used to configure and run a Private Network.

Prerequisite

Before you open PNSM, you must configure all Private Network deployments, including ASAP, Service Manager, and NetFoundry. Only after these have been configured in Azure is Service Manager functional. Please refer to the *Affirmed Private Network Service Deployment* guide for instructions on configuring these parameters.

Audience

This guide is intended for network engineers and operators who provision and maintain network services (sometimes referred to as Users or Operators in this guide).

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

This guide contains the following:

Section	Summary
Getting Started	Basic tasks such as navigating the system and accessing information about private network objects.
Configuring Enterprise Details	Advanced tasks such as service configuration and setting locations for the Private Network Services in your enterprise.
Creating Private Networks	Information on how add, edit, and delete private networks.
SIM Provisioning	Discusses process of SIM Provisioning, including how to add, edit, and delete parameters.
Monitoring Private Networks	Describes alarms and events to view private network issues, and the dashboards and reports available in the Private Network Service Manager to monitor key parameters such as Performance Indicators, Resource Utilization, Network Slicing, etc..
Network Backups and Upgrades	Tasks required to maintain backups for your private networks and perform upgrades to network software.
Security Settings	Indicates how to add users to a private network, and the privileges granted based on user definition.
Administrative Tasks	Describes tasks required to configure administrative features of the PNSM software, including configuring alarm color settings and external communications. Additionally, it describes how to set filters for viewing alarms and events for your private network service.
Additional Configuration Settings	Defines less frequently used configuration settings within the Service Manager.
End-to-End Security Overview for APNS	Discusses information regarding the security principles utilized in the Affirmed Private Network Service, and specifically defines how each component of the APNS solution is secured.
Northbound APIs	Provides the JSON syntax and Resources necessary for to run Northbound APIs in the Service Manager application.

Features In This Release

This release contains Security Patches for the APNS software.

Software Components

Release 9.3.1.0 consists of the following software components:

- PNSM version 1.4
- EMS version 12.2.1
- ASAP vTransactor version 7.3
- MCC version 13.0.0.5-2008
- MME version 15.0.20.15
- HPE HSS PNW-LTE version 04.03.02
- Azure Network Function Manager service
- Azure Stack Edge versions 2110 (**2.2.1758.4034**) , 2111 (**2.2.1777.4088**) and 2203 (**2.2.1902.4561**)

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menu	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management
Variable parameters for which you supply values	<courier bold italic>	configure_dr --host <host name>
User input	Courier bold font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular font	result success
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}

Description	Convention	Examples
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.		Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<host name>
Asterisk after a CLI object indicates a required object	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Private Network Service Manager User's Guide</i> . <i>Workflow</i> is an application running on the network element.
Images in online help		Many images can be enlarged by clicking the image with the left-mouse button. Alternately, they can be enlarged/reduced by pressing the Enter key when the image is highlighted (surrounded by a thick border) on the screen.

Note: Additional information that may apply to the subject text.

Tip: Helpful suggestions.

Caution: Proceed carefully to avoid possible equipment damage or data loss.

Customer Support

Affirmed Networks provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

[Email support](#)

[Open a support ticket](#)

Call support

Local: +1-978-268-0800

Toll-free: +1-888-283-2838

Getting Started

Overview

Affirmed Networks' Private Network Service Manager (PNSM) is the application that Operators use to deploy, monitor, and manage private mobile core networks on the Azure platform.

Using PNSM, you can configure private mobile networks as standalone, distributed, or all-in-the-cloud on the Azure platform. You can define the location for each of these private networks, and perform SIM provisioning. Then, you can use the graphics-based dashboards and reports to monitor your networks and ensure optimal performance.

This section provides a high-level overview of the PNSM interface and details functionality commonly used throughout the software, including icons and filters.

Logging Into Private Network Service Manager (PNSM)

To log into Service Manager, you need a User name and Password, which are provided by your system administrator.

To log into Service Manager:

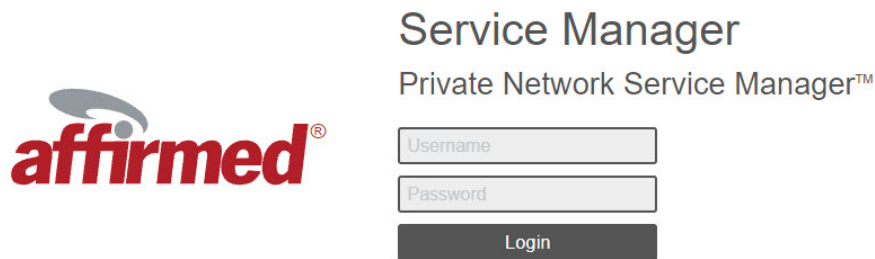
1. Start the Microsoft Edge (preferred), Google Chrome, or Mozilla Firefox web browser.
2. In the browser's address bar, enter the IP address of the machine where the Service Manager server is started, using either of the following URLs (the `http:` URL redirects you to the `https:` URL):

`http://<Service Manager server IP address>`

`https://<Service Manager server IP address>`

The browser warns you that Service Manager is an untrusted site.

3. To eliminate the warning screen from appearing each time you log into this Service Manager server, you can create a security exception in your browser's settings.

Figure 1-1 Service Manager Login Screen


The image shows the Service Manager login screen. On the left is the Affirmed logo, which consists of a stylized grey 'a' with a swoosh and the word 'affirmed' in red. To the right of the logo, the text 'Service Manager' is displayed in a large, dark grey font, followed by 'Private Network Service Manager™' in a smaller, dark grey font. Below the text are two input fields: 'Username' and 'Password', both with light grey borders. Below these fields is a dark grey button with the word 'Login' in white text.

4. At the Service Manager login screen, enter your Username and Password.
5. Select **Login**.

Note: If you attempt to log into Service Manager six consecutive times (by default), but fail (because you supplied the wrong password), Service Manager automatically locks your account. You cannot log into Service Manager until your administrator unlocks your account. For instructions, see [Unlocking Users](#).

Service Manager Password Policies

Service Manager enforces the following password policies. For any existing user, Service Manager enforces these password policies whenever the user or administrator changes the user account.

- Password must contain:
 - At least nine characters for newly created users.
 - Existing users with a password of at least six characters remain valid. When updating the password, existing users must create a new password at least nine characters long.
 - At least one upper case letter (A-Z)
 - At least one lower case letter (a-z)
 - At least one number (0-9)
 - At least one of the following special characters: ([!@#\$\$%^&*])
- Password *cannot* contain the user name with which it is associated
- Password *cannot* be reused until either of the following rules is met:
 - Minimum number of other passwords (default 4 with a range 1-12) to be used before allowing a reused password
 - or

— Minimum number of days (default 120 with a range of 0-365) to pass before allowing a reused password

- After consecutive failed logon attempts (default 6), Service Manager automatically disables the user account, which must be unlocked by the administrator.

The rules for password use and account lockout are configurable in the `securityconfiguration.properties` file. For instructions, see [securityconfiguration.properties](#).

securityconfiguration.properties

The `securityconfiguration.properties` file contains configurable attributes that control security settings:

The `securityconfiguration.properties` file is in the directory:

/opt/Affirmed/NMS/server/ems/conf

The configurable attributes are:

Maximum Failed Login Count

MAXIMUM_LOGIN_FAILED_COUNT= 6

Lockout time(in minutes) to enable automatically

LOCKOUT_TIME=30

Flag to enable/disable dormancy validation

If it is disabled DORMANCY_PERIOD configuration will not be used.

VALIDATE_DORMANCY=true

Dormancy Period is to be given in days

DORMANCY_PERIOD=30

Minimum number of other passwords to be used before allowing a reused password

PWD_REUSE_NUM_COUNT=4

Minimum number of days to pass before allowing a reused password

PWD_REUSE_DAY_COUNT=120

Minimum number of password characters

passwordConfig.minSize = 6

SSH_AUTO_ADD_KEY=true

User account lockout and password use:

- The maximum number of consecutive failed logins (when the user supplied the wrong password) before the user is locked out.
- Service Manager idle time (in minutes) before the user is locked out.
- Service Manager dormancy time (in days) before the user is locked out, including the following scenarios:
 - User attempts to login for first time, but the number of days between the account creation and the attempted login is greater than the dormancy period.
 - User attempts to login, but the number of days between the last successful login and the current login is greater than system's dormancy period.
- Password reuse rules. A password *cannot* be reused until either of the following rules is met:
 - Minimum number of other passwords (default 4 with a range 1-12) to be used before allowing a reused password.
 - Minimum number of days (default 120 with a range of 0-365) to pass before allowing a reused password.

A locked out user cannot log into Service Manager until the administrator unlocks the account. For instructions, see [Unlocking Users](#).

SSH key use:

- Specifies whether to automatically retrieve an SSH key when connecting to an SSH server:
 - When `SSH_AUTO_ADD_KEY` is set to true, Service Manager automatically retrieves the SSH host key during first SSH connection with each of the existing NE or file servers and uses the key for verifying the host in subsequent connections.
 - When `SSH_AUTO_ADD_KEY` is set to false, Service Manager *does not* retrieve an SSH key when connecting to an SSH server. In this case, the SSH connection fails.

Note: Irrespective of `SSH_AUTO_ADD_KEY` setting, Service Manager automatically includes the key of SSH hosts in its list of trusted SSH servers when using VNF Manager to instantiate MCC/MME clusters.

Any reconfiguration requires restart of the Service Manager server to take effect.

Navigating the System

PNSM has an easy-to-use graphical user interface with a tree-based representation of the private network in the left pane and tabs displayed in the right pane.

Tree Display

The tree display allows you to view devices deployed in different locations.

To open the tree display, select the **Toggle Resource Selection View**  icon (next to the **Settings** icon) to expand the left pane.

The tree hierarchy includes:

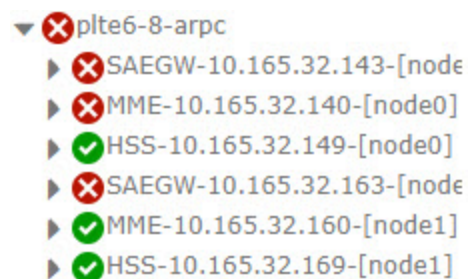
- Global Enterprise – A global view of all the Private Networks deployed. It depicts the topology of the elements installed at the Enterprise.
- Private Networks – A logical group composed of few network elements/sub-groups. It depicts the topology of the sub-groups/elements deployed in the selected private network.
- Server Deployments - A logical group composed of few network elements/sub-groups. It depicts the topology of the sub-groups/elements deployed on the selected server.
- Elements – Depicts the deployed network elements for each private network service.

In the tree display, the expansion icon indicates which objects contain other objects:

- If the expansion icon points to the right, the object is closed.



- If the expansion icon points down, the object is open, and the names of its contained objects are listed below it.



You navigate through objects in the tree much like you navigate through folders and files listed in a file management program. Select the expansion icons to open and close objects in the tree display.

Tabs

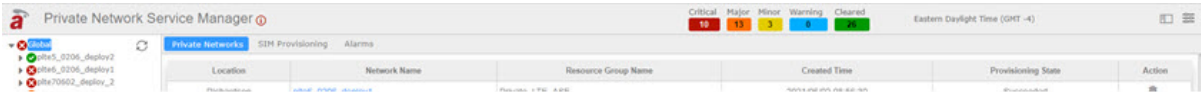
Tabs on the right panel apply to the Enterprise level, the Private Network level, or the Server level. The data displayed in each tab is filtered based on the elements in the selected group. Refer to each topic for additional information:

- [Enterprise Level Tabs](#)
- [Private Network Level Tabs](#)
- [Server Level Tabs](#)

Enterprise Level Tabs

These tabs show the functions available at the Enterprise level of the Service Manager application. The settings here show all private networks, SIM provisioning, and Alarms for monitoring your Private Network Services.

Figure 1-2 Enterprise Level Tabs



Private Networks

The Private Networks tab shows the location, network and resource group names, along with the time created and provisioning state of each network.

You can select a private network service to view network-specific parameters regarding deployment and service configuration status by selecting the private network name from the table. Additionally, you can delete the network by selecting the delete icon in the appropriate row.

Figure 1-3 Private Networks Tab

Private Networks					
SIM Provisioning					
Alarms					
Location	Network Name	Resource Group Name	Created Time	Provisioning State	Action
P1te11	plte11_0306_deploy01	Private_LTE_ASE	2021/06/03 07:11:44	Succeeded	[Delete Icon]
P1te12-13	plte13_12_0306_deploy02	Private_LTE_ASE	2021/06/03 07:13:34	Succeeded	[Delete Icon]

SIM Provisioning

The SIM Provisioning tab lists all SIM Configurations, and allows you to add, activate, deactivate, and delete SIMs from the Private Network. Please refer to [SIM Provisioning](#) for additional detail.

Alarms

Alarms are triggered based on events specific to the server parameters at this level. For additional information on alarms, please refer to [Alarms](#).

Private Network Level Tabs

Private Network Detail

At the Network level, the Private Network Detail tab displays all details of the private network selected, including the Name, Packet Core Version, Device, Resource Group, Region, Deployment Mode, and Provisioning Status. Additionally, it shows all VNF Deployment and Service Configuration parameters and status.

Figure 1-4 Private Network Detail Tab

The screenshot displays the 'Private Network Service Manager' interface. The top navigation bar includes tabs for 'Private Network Detail', 'Element Details', 'Alarms', 'Event Logs', 'PM Dashboard', 'Network Availability', 'Upgrades', 'RAN Info', and 'Subscriber Tracing'. The 'Private Network Detail' tab is active, showing a summary of the network 'plite2-3-0311-deploy1' with fields for Resource Group, Created Time, NTP Server, Time Zone, Packet Core Version, Region, Deployment Mode, NTP Server1, Location, Provisioning State, and Action.

Below the summary, the 'VNF Deployment' section is expanded, showing three columns of network function details:

MME		SAEGW		HSS	
Status	Done	Status	Done	Status	Done
MM-Management	10.165.32.100	Management	10.165.32.103	Audit Status	N/A
MM-Base	10.165.60.33	Peer Management	10.165.32.104	Audit Time	N/A
MM-Data	10.165.60.41	Logical Management	10.165.32.105	Management	10.165.32.109
MM-Management	10.165.32.101	Base	10.165.61.34	LAN	10.165.60.23
MM-Base	10.165.60.34	Peer Base	10.165.61.35		
MM-Data	10.165.60.42	Logical Base	10.165.61.36		
MM-North/South 1	10.165.60.10	East/West	10.165.61.42		
Management Logical	10.165.32.102	North/South 1	10.165.61.10		
SG IP	10.165.62.203	North/South 2	10.165.64.10		

The 'Service Config' section is also expanded, showing three columns of configuration details:

Service Attributes		GI Attributes		GN Attributes	
Network Name	Telecom.yyz	Interface	10.165.64.10	Interface	10.165.61.10
Home PLMN ID	99999	Interface Next hop	10.165.64.1	Interface Next hop	10.165.61.1
APN Name	plite	Local UE Pool Start	200.20.170.1	SGW S1 Loopback	10.165.61.49
DNS Primary	200.110.10.1	Local UE Pool End	200.20.170.255	SGW S588 Loopback	10.165.61.48
DNS Secondary		Loopback	10.165.65.200	PGW Loopback	10.165.61.50
Roaming PLMN ID	12345	Router ID	10.165.64.10	SGW S11 Loopback	10.165.61.51

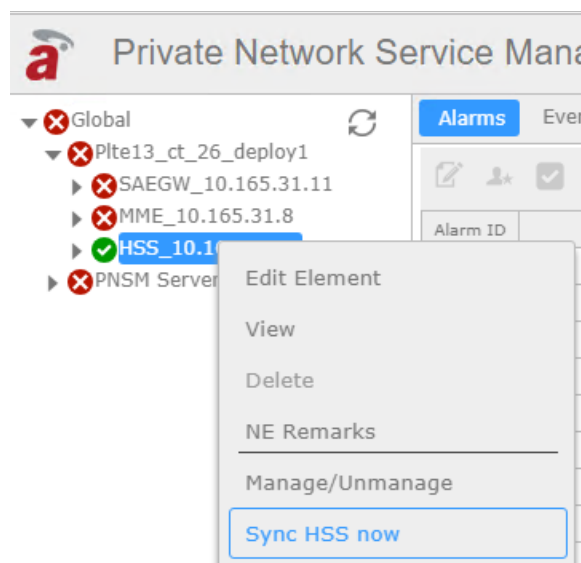
For information on adding a private network, please refer to one of the following topics:

- [Create Standalone Edge Network](#)
- [Create Distributed Edge Network](#)
- [Create All in Cloud Network](#)

Element Details

The Element Details tab shows each network function that is part of the selected private network. Elements can include HSS, MME, and SAEGW. You can edit, view or delete elements from this tab. Additionally, you can show SIM Instances associated with each network element, and view NE Remarks.

Note: While the HSS audit status and time are displayed based on a synchronization of SIMs between Service Manager and HSS occurring every 4 hours by default, you can also trigger a sync using the right-click menu of the network element.

Figure 1-5 Right-Click HSS on-demand Sync

Alarms

Alarms are triggered based on events specific to the server parameters at this level. For additional information on alarms, please refer to [Alarms](#).

Event Logs

Event logs can be viewed at each level of the Service Manager. For additional information, please refer to [Events](#).

PM Dashboard

The PM Dashboard displays the Key Performance Indicators (KPIs) collected during the previous 24 hours.

For additional information regarding dashboard configuration and reports, please refer to [Performance Reports](#).

Network Availability

The Network Availability tab displays a graphical representation of device metrics. For detailed information, please refer to [Network Availability](#).

Upgrades

The Upgrades tab allows you to view any network upgrade packages that are running or completed for PNSM, along with the status of the upgrade. For additional information on upgrading networks, please refer to [Network Software Upgrades](#).

Note: The information displayed on this screen is also available through the [Software Schedules](#) screen.

Server Level Tabs

Alarms

Alarms are triggered based on events specific to the server parameters at this level. For additional information on alarms, please refer to [Alarms](#).

Event Logs

Event logs can be viewed at each level of the Service Manager. For additional information, please refer to [Events](#).

Resource Utilization Dashboard

The Resource Utilization Dashboard provides a visual, graphical summary of the usage of server parameters over the past 24 hours. Information on this dashboard may include:

- Average CPU Usage (%)
- Average Memory Usage (%)
- Average Disk Usage (%)

For additional information regarding dashboard configuration and reports, please refer to [Performance Reports](#).

Element Level Tabs

Element level tabs display information specific to the highlighted network element.

Alarms

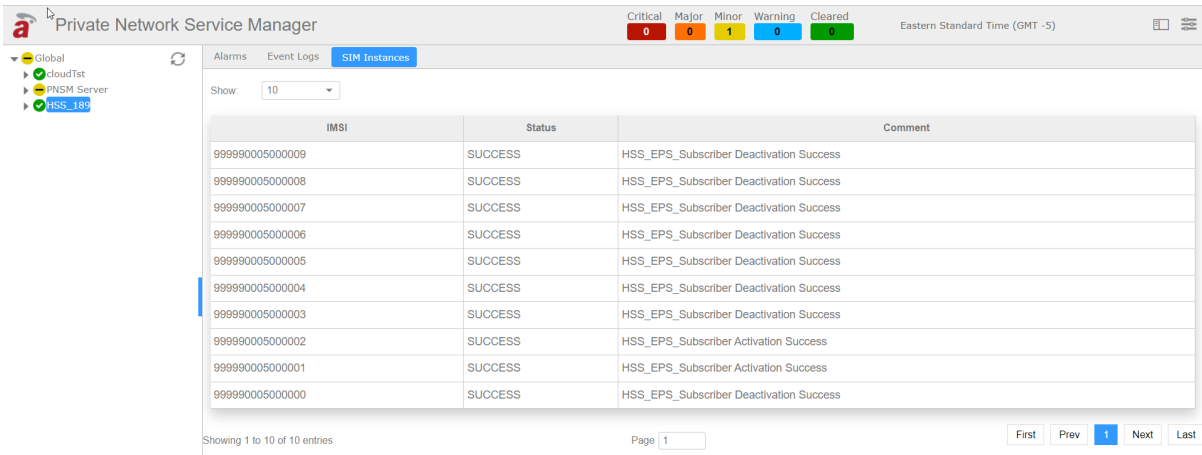
Alarms are triggered based on events specific to the server parameters at this level. For additional information on alarms, please refer to [Alarms](#).

Event Logs

Event logs can be viewed at each level of the Service Manager. For additional information, please refer to [Events](#).

SIM Instances

The SIM Instances tab displays only when the HSS device element is selected in navigation tree. It shows specific IMSI instances along with their status and a comment:



Using Icons

The following table defines PNSM icons.

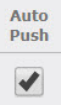
Table 1-1 PNSM Icons

Option	Icon	Use this option to...
General icons:		
Add, Create, New		Add a new record
Apply Page Size		Specify how many records you want displayed per page
Ascending Order		Sort devices in the Network View in ascending order
Calendar		Select the required date from the calendar
Change Theme		Change the GUI theme: ▪ Light Theme ▪ Dark Theme ▪ High Contrast
Close, Exit		Close a window
Copy		Copy an existing configuration
Descending Order		Sort devices in the Network View in descending order
Delete		Delete a record
Edit, Modify		Modify a record

Option	Icon	Use this option to...
Expand All/Collapse All		Expand All/Collapse All devices in the network view.
Export		Export records to a file
First Page		Navigate to first page of records
Import		Import a configuration template
Last Page		Navigate to last page of records
Manage/Unmanage		Manage/Unmanage a network element
Next Page		Navigate to next page of records
Obsolete		Set a software release obsolete so that it can no longer be used
Previous Page		Navigate to previous page of records
Print		Print the contents of the window or a specific record
Refresh		Refresh display
Save		Save settings
Search		Search for records that meet a condition
Settings		Present list of operations to perform
Stop		Stop the ongoing activity in the current context
View		Display the details of the current record
Alarm icons:		
Acknowledge Alarm		Acknowledge an alarm
Alarm Details		View alarm details
Alarm Forwarding		Forward an alarm to a user
Alarm Severity – Cleared		View Cleared alarms

Option	Icon	Use this option to...
Alarm Severity – Critical		View Critical alarms
Alarm Severity – Major		View Major alarms
Alarm Severity – Minor		View Minor alarms
Alarm Severity – Warning		View Warning alarms
Alarm Stats		View pie charts of the alarm statistics
Assign Owner		Assign a user to an alarm
Clear Alarm		Clear an alarm
Hide Cleared Alarms		Hide cleared alarms
Show Cleared Alarms		Show cleared alarms
Reports and Performance icons:		
Bar Chart		Display data as a Bar Chart
Clear Chart		Clear a plotted chart
Column Properties		Configure column properties
Export		Export Dashboard data as table PDF Export Dashboard data as table CSV Export Dashboard data as charts PDF Export Dashboard configuration as CSV
Export as CSV		Save data in the grid as a .csv file
Export as HTML		Save data in the grid as an .html file
Export as PDF		Save data in the grid as a .pdf file
Export as XLS		Save data in the grid as an .xls file

Option	Icon	Use this option to...
Export as XLSX		Save data in the grid as an .xlsx file
Export as XML		Save data in the grid as an .xml file
Import Dashboard Config as CSV		Import Dashboard configuration CSV file
Inspect Description		View description of an instance
Launch		Launch CSV statistics recovery schedule
Line Chart		Display data as a Line Chart
Metric Configuration		Specify metric to display in Dashboard
Outage Report		Display an Outage Report
Pie Chart		Display data as a Pie Chart
Purge Data		Purges data older than a specified number of days
Stacked Area Chart		Display data as a Stacked Area Chart
Stacked Bar Chart		Display data as a Stacked Bar Chart
Schedule icons:		
Disable		Disable a schedule
Enable		Enable a schedule
Quick Start		Implement a task immediately
Filter Management icons:		
Add Condition		Configure a filter based on specific parameters, values, and conditions
Apply to Table		Apply the filter to the Tabular view
Create AND Group		Create AND conditions
Create OR Group		Create OR conditions

Option	Icon	Use this option to...
Filter		Set up conditions to filter records
Recurrence Pattern		Specify the recurrence details of a schedule
Reset Filter		Restore the existing filter
Show Filter Manager		Display dropdown of configured filters on the toolbar
Show Quick Filter		Display the Quick Filter on the toolbar
Backup and Restore icons:		
Restore Database		Restore the PNSM database to an older backed up version
Scheduled Backup		Configure the PNSM server to backup the database based on a schedule
Scheduled Backup View		View a scheduled database backup
URL List Maintenance icons:		
Auto-Push		Enable Auto-Push to automatically push a clear-text URL file to all managed network elements every hour
Compare		Compare a clear-text URL file's current contents to its earlier file contents
Comparison File		View a comparison file listing URLs added or removed since the last time the file was marked as reviewed
Mark As Reviewed		Mark a clear-text URL file that is in the MODIFIED state as reviewed
Overwrite		Overwrite a clear-text URL file
Review and Edit		Review and edit URLs in a clear-text URL file
User Management icons:		
Active Users List		List active users
Audit Trail		List the activities for the user
Terminate User Sessions		Terminate active user sessions
Unlock User		Unlock a user whose access was blocked

Common Functionality

PNSM uses common functionality in various areas of the application. They are:

- Filters – Customize the details that PNSM displays based on specific parameters, values, and conditions.
- Column Properties – Specify which columns you want displayed and whether you want them to remain displayed so that you do not have to scroll to see them.
- Record Navigation – Specify how many records you want displayed per page. Navigate through multiple pages of records.
- Export Records
- Print Records
- Refresh the Screen

Filter Management

Creating Filters

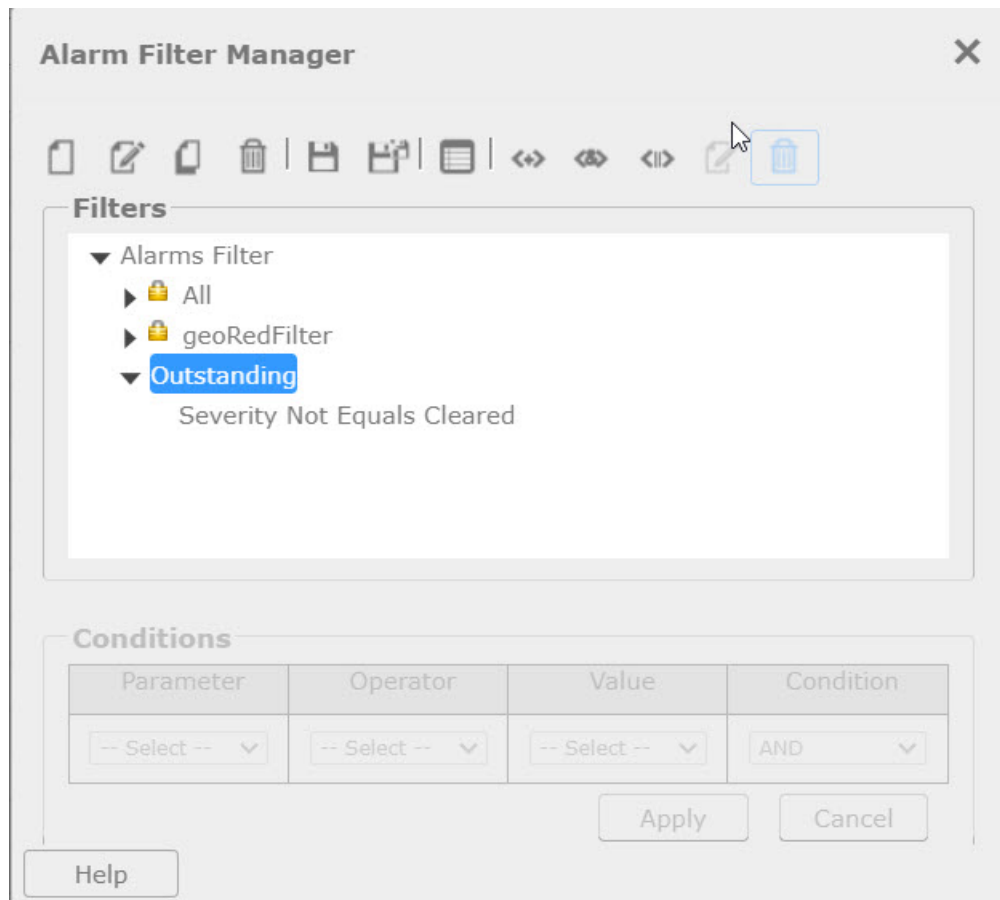
You can configure filters to customize the details that PNSM displays based on specific parameters, values, and conditions.

To create a filter:

1. From the tab or window, select the **Filter**  icon.
PNSM displays the **Filter Manager**.
2. Select the **New Filter**  icon.
3. In the **Add Filter** dialog, enter the Filter Name and select **Add**. A filter with that name is created.
4. In the **Filter Manager**, select the filter and select the **Add Condition**  icon:
 - a. Complete the Conditions fields by selecting appropriate values for **Parameter**, **Operator**, **Value**, and **Condition**.

The following example creates a filter of all outstanding alarms (filter name **Outstanding**). The Conditions are **Parameter** = Severity, **Operator** = Not Equals, and **Value** = Cleared.

Figure 1-6 Filter Manager



- b. Select **Apply**.
5. To add additional Conditions, repeat [step 4](#) . Specify **Condition** = AND or OR.
6. If the filter needs a combination of AND and OR conditions to achieve the desired result:
 - a. In the **Filter Manager**, select the filter and select the **Create AND group**  icon (or **Create OR Group**  icon).
 - b. In the **Filter Manager**, select the AND... (or OR...) group and select the **Add Condition**  icon and complete the Conditions for the group.
 - c. Select **Apply**.
7. To apply the filter to the table (to restrict the data in the table), select the **Apply to Table**  icon.
8. Select the **Save Filter**  icon.

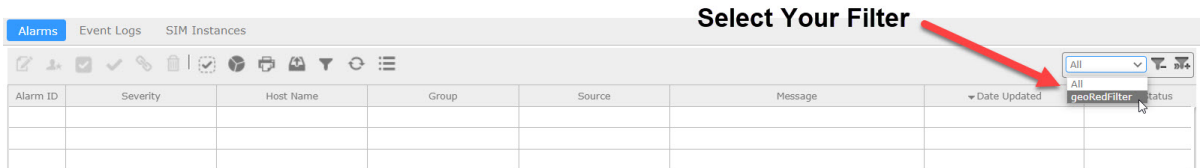
Applying a Filter

Once you create a filter, you apply it to filter the details that PNSM displays. From the tab or window, select your filter from the filter list to apply it.

If the filter list is hidden, select the **Show Filter Manager**  icon to display the dropdown list.

PNSM lists the details that meet the filter conditions. The following example shows an alarm filter that limits the list of alarms to all Outstanding alarms (alarms that have not cleared).

Figure 1-7 Select the Filter



Modifying Filters

To modify a filter:

1. From the tab or window, select the **Filter**  icon.
PNSM displays the **Filter Manager**.
2. To change the filter name:
 - a. Select the name and select the **Edit Filter Name**  icon.
 - b. Enter the new Filter Name and select **Update**.
3. To change a filter condition:
 - a. Select the condition and select the **Edit**  icon.
 - b. Complete the Conditions fields by selecting appropriate values for **Parameter**, **Operator**, **Value**, and **Condition**.
For example, to create a filter of all outstanding alarms, the Conditions could be **Parameter** = Severity, **Operator** = Not Equals, **Value** = Cleared.
 - c. Select **Apply**.
4. To delete a filter condition:
 - a. Select the condition and select the **Remove**  icon.
 - b. At the confirmation prompt, select **OK**.
5. To change an AND or OR group:
 - a. Select the AND or OR group and select the **Edit**  icon.
 - b. Complete the Conditions fields by selecting appropriate values for **Parameter**, **Operator**, **Value**, and **Condition**.
 - c. Select **Apply**.
6. To delete an AND or OR group:
 - a. Select the AND or OR group and select the **Remove**  icon.
 - b. At the confirmation prompt, select **OK**.

7. To apply the filter to the table (to restrict the data in the table), select the **Apply to Table**  icon.
8. Select the **Save Filter**  icon.

Deleting a Filter

To delete a filter:

1. From the tab or window, select the **Filter**  icon.
PNSM displays the **Filter Manager**.
2. Select the name, then select the **Delete Filter**  icon.
3. At the confirmation prompt, select **OK**.

Using Quick Filters

You can use the Quick Filter to customize the details that PNSM displays based on a specific parameter and value.

To use Quick Filter:

1. From the tab or window, select the **Show Quick Filter**  icon.
2. In the **Filter By** field, specify the filter conditions (parameter, operator, and value).
The following example creates a Quick Filter of alarms where the **Message** field contains **operationally up**.



3. Select the **Search**  icon to apply the Quick Filter.
PNSM lists the details that meet the filter condition.
To reset the Quick Filter to its original setting, select the **Reset Filter**  icon.

Setting Column Properties

In various screens, you can specify which columns you want displayed and whether you want them to remain displayed so that you do not have to scroll to see them.

To configure column properties:

1. From the tab or window, select the **Column Properties**  icon.
2. Mark the **Enabled** checkbox of the columns you want displayed. Unmark the checkbox of the columns you want hidden.
3. Mark the **Freeze** checkbox of the columns you want to remain displayed without requiring scrolling. Unmark the checkbox of the columns you want unfrozen.
4. Select **Apply**.

Sorting Column Data

To sort the data in a column in ascending alphanumeric order, select the column title. An up arrow in the column indicates ascending sort order.

To sort the data in a column in descending alphanumeric order, select the column title again. A down arrow in the column indicates descending sort order.

To return the sort order to the default setting, select the column title a third time.

Navigating through Records

In the case of a large number of records, PNSM displays them over multiple pages. The current page and record count are listed at the bottom of the window.

Figure 1-8 Page Navigation Bar



To specify how many records you want displayed per page, enter the value in the **Page Size** box and select the **Apply Page Size**  icon.

Use the icons to navigate over multiple pages:

- Previous Page 
- Next Page 
- First Page 
- Last Page 

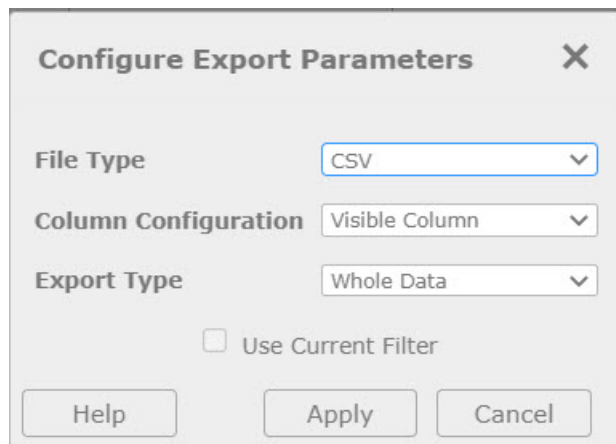
To navigate to a specific page, enter the page number in the **Page** box.

Exporting Records

To export records:

1. From the tab or window, if desired, use the Filter Manager or Quick Filter to filter the data displayed in the window.
2. Select the record(s) to be exported. To select multiple records, Shift-click or Control-click the records in the list. To export the Whole Data or Current View, do not select any records.
3. Select the **Export**  icon.

PNSM displays the **Configure Export Parameters** window.

Figure 1-9 Configure Export Parameters Window

4. Configure the parameters as follows:
 - **File Type:** Select the required file format (CSV, PDF, XLS, XML, HTML).
 - **Column Configuration:** Select the columns you want exported (Visible Column or All Columns)
 - **Export Type:** Select the export type (Whole Data, Current View, Selected Data)
 - **Use Current Filter:** Mark the checkbox to restrict data based on the existing filter conditions.
 5. Select **Apply**.
- PNSM displays the **Download Files** window.
6. Select **Download Exported File**.
 7. From the **Download Files** window, select **Close**.

Printing Records

To print records:

1. From the tab or window, select the **Print**  icon.
PNSM displays the records in the **Print Preview** screen.
2. Select **Print** to print.
3. Select the printer then select **Print**.

Refreshing the Screen

Select the **Refresh**  icon to refresh the screen with the latest data.

Logging Out of PNSM

To log out of PNSM:

1. Select **Logout** from the **Settings**  icon on the toolbar.
2. At the confirmation prompt, select **OK**.

Configuring Enterprise Details

This section discusses how to use the Service Manager application to configure your enterprise and define locations for Private Network Services.

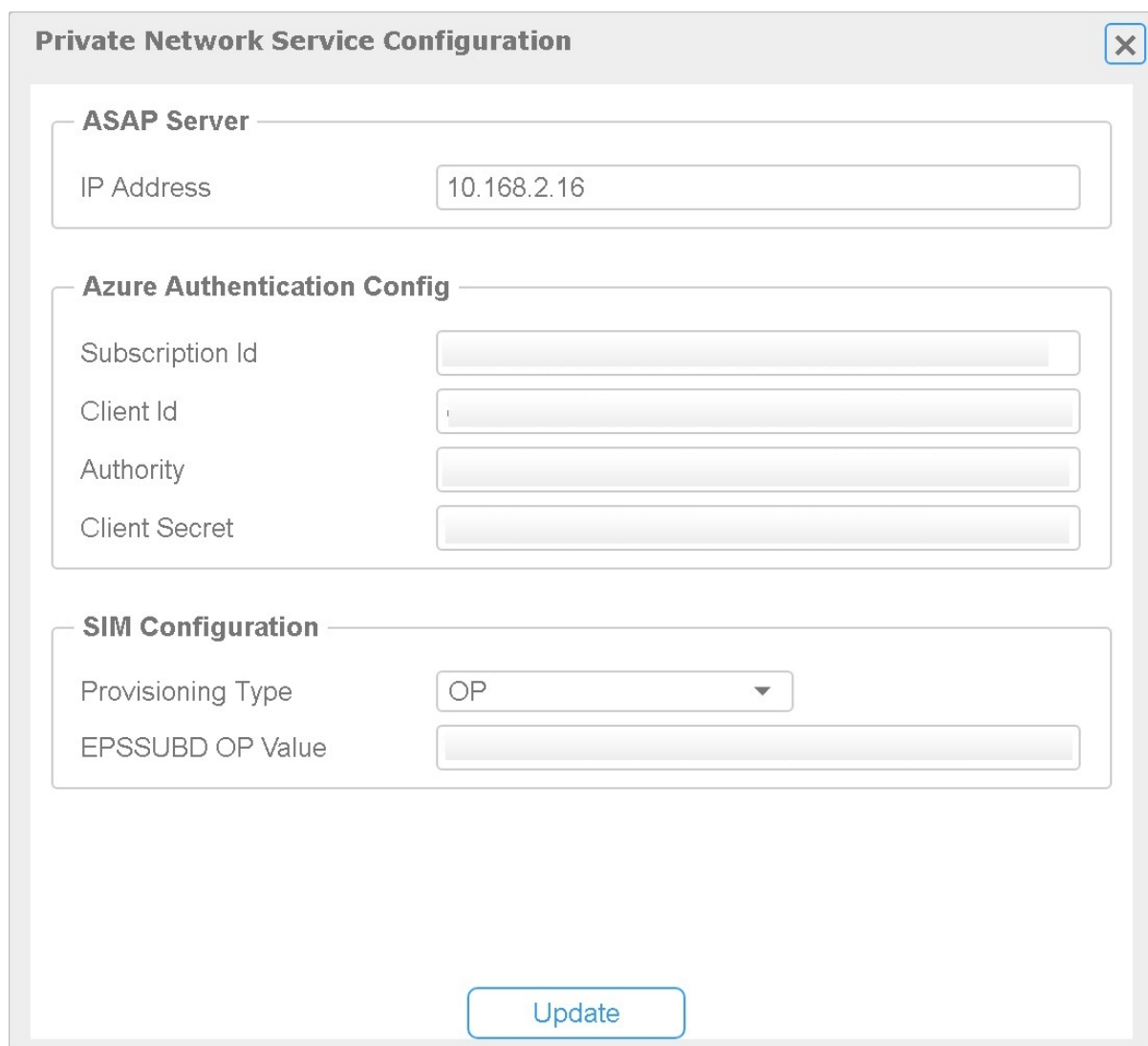
Service Configuration

In order for the Private Network Service Manager to communicate with Azure, you must configure the Service Manager subscription. This is done through the PNSM Configuration dialog, and must be completed before any Private Network is deployed.

***Note:** Follow the steps as described in the Private Network Service Manager Deployment Guide to ensure that the application is registered in the Azure Portal using both the subscription and client ID prior to completing this configuration process.*

To configure the Private Network Service Manager:

1. Select **Tools** then **PNSM Configuration** from the Settings () menu.

Figure 2-1 PNSM Configuration Dialog

The dialog box is titled "Private Network Service Configuration" and has a close button (X) in the top right corner. It contains three main sections:

- ASAP Server**: A section with a single text input field labeled "IP Address" containing the value "10.168.2.16".
- Azure Authentication Config**: A section with four text input fields labeled "Subscription Id", "Client Id", "Authority", and "Client Secret".
- SIM Configuration**: A section with a dropdown menu labeled "Provisioning Type" set to "OP" and a text input field labeled "EPSSUBD OP Value".

An "Update" button is located at the bottom center of the dialog box.

2. Enter the ASAP Server **IP Address**. This address is the private IP address of ASAP on Azure.
3. Complete the **Azure Authentication Configuration** fields. The parameters you enter here must be obtained from the Application Registration Overview page of the Azure Portal.
4. Select the **Provisioning Type** value. By default, this value is set to **OP**, with the default operator code displayed.

Note: If the Provisioning Type is set to OPC, then the EPSSUBD OP value is not shown. The Provisioning type is stored in the Service Manager database, and you must provide the OPC value for each IMSI when importing SIMs.

Private Network Service Configuration

ASAP Server

IP Address: 10.60.0.33

Azure Authentication Config

Subscription Id: 880949aa-c7ae-4cfe-b5cd-f90d8e81d776

Client Id: c1be777b-5b34-41a9-ba14-ecfef02b788f

Authority: https://login.microsoftonline.com/72f988bf-86f1-41af-

Client Secret: ~i97-0W-7q.MD1BF3EpcyCoG00nU07Fi.G

SIM Configuration

Provisioning Type: OPC

Update

5. Edit the **EPSSUBD OP Value** for the SIM Configuration as necessary. Note that this value must be configured prior to any SIMs can be imported and activated:
 - CSV Format for OP: *IMSI, KI_VALUE*
 - CSV Format for OPC: *IMSI, KI_VALUE, OPC*
6. Select **Update** to complete the configuration.

Creating a Location

Before you can create a private network, you must create a location. This is done through the Location Management option, available in the Topology submenu of the Settings Menu. You can define the number of locations based on your deployment needs.

To add a Location:

1. Select **Topology => Location Management** from the **Settings** menu.
2. Enter the **Location Name**. This is the physical location of the deployment.

3. Enter the location-specific subnet details for the **Management, LAN, and WAN Subnets**.
4. (Optional) Enter the **Gateway** fields for the Management, LAN and WAN subnets. If no gateway information is entered, the default Gateway will be created based on the defined Subnet.

Note: User-defined Gateway IP Addresses should be within the first five IP addresses of the corresponding subnet.

The screenshot shows the 'Location Management' dialog box. On the left, under 'Locations', there is a list with a '+' icon and a trash icon for each item: Richardson, CUPS_LOCATION, HA Setup Location, and plte11. On the right, under 'Location Details', the 'Location Name' is set to 'Richardson'. Below this are input fields for 'Management Subnet', 'Management Gateway', 'LAN Subnet', 'LAN Gateway', 'WAN Subnet', and 'WAN Gateway'. There are also buttons for 'Download Public Key', 'Download Private Key', and 'Next Available IP Addresses'. A 'Submit' button is located at the bottom right of the dialog.

5. Select **Submit** to create the location.

Once you have created a location, you can view its **Location Details** by selecting the Location on the left-side of the **Location Management** dialog. Along with the information you defined, the option to **Download a Public Key** and a **Private Key** displays.

These keys are generated automatically by the PNSM and stored in the database. These keys are used to communicate to Network Elements available in the Private Network Management System.

6. (Optional) Select **Next Available IP Addresses** to use the next set of Management IP Addresses for MME, MCC, HSS and NetFoundry before deploying the Network, and to provide the NetFoundry IP Addresses to deploy NetFoundry.
7. (For Non-Redundant Networks) Enter the **Resource Group** name, select the **Device** from the dropdown, and select **Populate** to view the available IP Addresses.

Next Available IP Addresses

Resource Group: **Private_LTE_ASE** Redundant: ☐

Select Device: **ase-plte-2** Populate

Service	IP Address
MME	
MRM-Management	10.165.32.8
MSM-Management	10.165.32.9
Management Logical	10.165.32.10
SAEGW	
Management	10.165.32.11
Peer Management	10.165.32.12
Logical Management	10.165.32.13
HSS	
Management	10.165.32.14
Net Foundry	
Management	10.165.32.7
LAN	10.165.60.24

8. (For Redundant Networks) Enter the **Resource Group** name and mark the **Redundant** option. Select the **Node0** and **Node1** Devices. Then select **Populate** to get the available IP Addresses for the Redundant Network.

Next Available IP Addresses

Resource Group: **Private_LTE_ASE** Redundant: ☒

Select Node0 Device: **ase-plte-9** Select Node1 Device: **ase-plte-10** Populate

Node 0 **Node 1**

Service	IP Address
MME	
MRM-Management	10.165.32.40
MSM-Management	10.165.32.41
Management Logical	10.165.32.42
SAEGW	
Management	10.165.32.43
Peer Management	10.165.32.44
Logical Management	10.165.32.45
HSS	
Management	10.165.32.46
Net Foundry	
..	10.165.32.22

Error message will appear for the following conditions:

- Location does not have enough IPs
- Invalid Resource Group entered
- For HA deployment, Node0 and Node1 are same

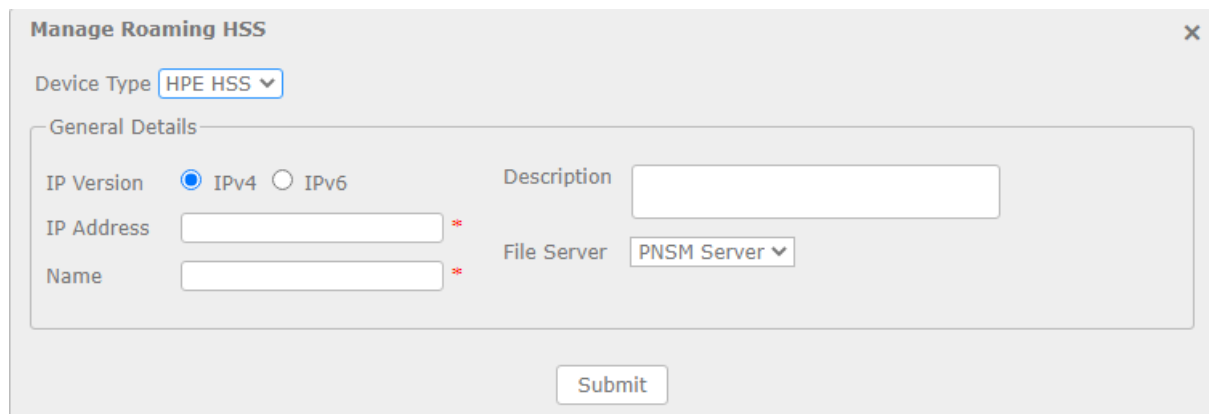
Managing the Roaming HSS

You can add Roaming HSS directly in the Service Manager application to provision the Private Network SIM and support roaming.

To configure the Roaming Gateway:

1. Right-click on the Global menu option and select **Manage Roaming HSS**.

The following dialog displays:

The image shows a 'Manage Roaming HSS' dialog box. At the top, 'Device Type' is set to 'HPE HSS'. Below this is a 'General Details' section. It contains fields for 'IP Version' (with radio buttons for IPv4 and IPv6, where IPv4 is selected), 'IP Address', 'Name', 'Description', and 'File Server' (set to 'PNSM Server'). There are red asterisks next to the IP Address and Name fields, indicating they are required. A 'Submit' button is at the bottom right.

2. Select the **IP Version** for the HPE HSS Device.
3. Enter the **IP Address**, **Name** and (optional) **Description** for the roaming private gateway.
4. Select the **File Server**. The options listed here are those configured in the **File Server** configuration window. Refer to [File Servers](#) for additional information.
5. Select **Submit**.

The HPE HSS is then added as a Managed Network Element. You can view the alarms, event logs, and SIM Instances by selecting the HSS from the tree on the left-side of the screen.

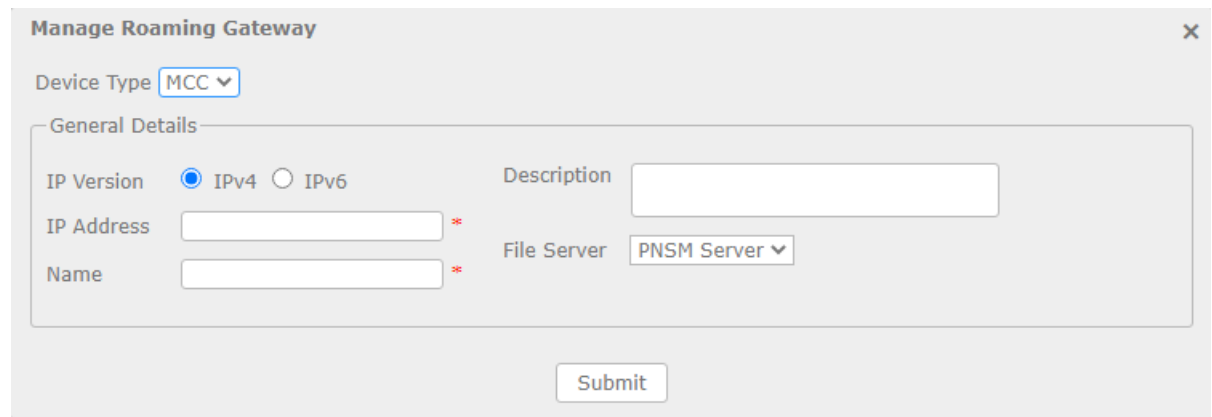
Managing Roaming Gateways

The Roaming Gateway option allows you to manage Private Network Gateways directly from the Service Manager application.

To configure the Roaming Gateway:

1. Right-click on the Global menu option and select **Manage Roaming Gateway**.

The following dialog displays:



The image shows a 'Manage Roaming Gateway' dialog box. At the top, 'Device Type' is set to 'MCC'. Below this is a 'General Details' section containing several fields: 'IP Version' with radio buttons for 'IPv4' (selected) and 'IPv6'; 'IP Address' and 'Name' text boxes, both marked with a red asterisk indicating they are required; 'Description' text box; and 'File Server' dropdown menu set to 'PNSM Server'. A 'Submit' button is located at the bottom right of the dialog.

2. Select the **IP Version** for the MCC Device.
3. Enter the **IP Address**, **Name** and **Description** for the roaming private gateway.
4. Select the **File Server**. The options listed here are those configured in the **File Server** configuration window. Refer to [File Servers](#) for additional information.
5. Select **Submit**.

The Private Gateway is then added as a Managed Network Element. You can view the details by selecting the private gateway from the tree on the left-side of the screen.

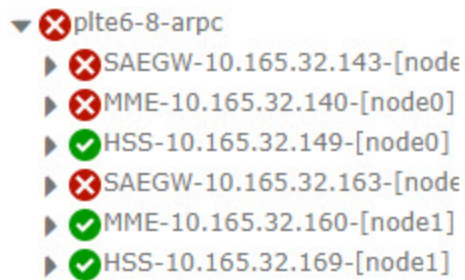
Creating Private Networks

Once you have completed the Private Network Service Configuration, and defined locations for your Private Network enterprise, you can create a network. PNSM allows you to create and configure three types of networks:

- [Standalone Edge](#) - The entire packet core including both Control Plane and User Plane is on-premises. This model is suited for deployments where the connectivity bandwidth between cloud and edge is either low or can be interrupted. eg. mining, remote oil rigs on offshore drilling, remote rural large agriculture farms etc.
- [Distributed Edge](#) - The Packet Core Control Plane is on Azure and User Plane is on-premises. This model is well suited for deployments in dense urban areas where there are many enterprise units well-connected with cloud but require lower latency with processing and breakout of data closer to edge. Some applications may be for tight industrial controls, robotics, AGVs inside an industrial plant, or inside factory premises.
- [All in Cloud](#) - In this model the entire packet core is on the cloud with only the RAN on the edge. This is well-suited for IoT sensor type of applications that are tolerant to higher latency and have low throughput requirements. This model allows customers to leverage the cloud to scale for millions of such devices and benefit from other Azure services such as IoT hub.

This section provides information on how add, edit, and delete private networks and how to monitor alarms and events occurring on your private networks.

Private networks are composed of Network Elements. These elements are configured when you are defining your private networks, and appear as submenus in the configuration tree. Note that for private networks with high-availability, each network element is listed, along with the node, so a network with one MME, SAEGW, and HSS network element with high availability would show six network elements, as shown here:



For information on configuring high availability, please refer to the appropriate topic as listed above.

Create Standalone Edge Network

Standalone Edge Network deploys the network as an onsite standalone network, where both the control and user plane is located onsite at the private enterprise.

To create a Standalone Edge Network:

1. Right-click on the Global Enterprise, select **Create Network** then **Standalone Edge** to open the **Create Private Network Instance on Standalone Edge** dialog:

***Note:** If no location has been created, the system prompts you to do so before creating a standalone edge private network.*

2. Enter a **Name** for the network. This should be a unique name specific for this private network.

3. Enter the **Resource Group**. This text should be identical to the Resource Group name used when configuring the Azure Network Function Manager as part of the subscription. Please refer to the *Private Network Management System Deployment Guide* for specific information regarding this setup.
4. Select the appropriate **Packet Core Version** to be used for this private network enterprise. This system software will be imported into the Service Manager.
5. Select the **Location** for the Standalone network. The Management, LAN, and WAN subnets default based on this selection.
6. Select the **Region**. The available options are based on those selected in the Azure Cloud Network Function Manager.
7. Enter the **MNC** (Mobile Network Code) and **MCC** (Mobile Country Code) based on the region of this private network deployment.
8. Select the **Redundant** checkbox to create a high availability deployment of the Private Network. Refer to [High Availability \(HA\)](#) for detailed information.
9. Select the **Device** or (**Node0 Device** and **Node1 Device** if you have selected an HA Deployment) to use for this Network Instance. The options listed are based on the customer subscription. These devices are specified in the Azure cloud.
10. Enter the **NTP Server** IP address. Select the **Add additional NTP** checkbox to add up to two optional NTP Servers.
11. Select the **Timezone** for this network instance.
12. To use Custom IP addresses, then activate the **Custom IP Addresses** option. This allows you to enter user-defined IP addresses for your private network.

Note: The NTP configuration entered here must match the server as configured in the Azure Stack Edge, and must be reachable via the MGMT network of the MME VM. Use of public time servers (such as the **default time.windows.com** on the Azure Stack Edge) is not supported.

Refer to the *Affirmed Private Network Service Deployment Guide* for additional details on this configuration.

13. (Optional) If you selected **Use Custom IPs**, enter the information for each device in the fields displayed.

Note: Each location must consist of only one IP type. You cannot create a network that has system-generated IPs and another with Custom IPs in one location.

14. Select **Submit**.

The system generates the IPs for the factory configuration, or uses the custom IP parameters you defined:

Create Private Network Instance on Standalone Edge

NamePrivate Network 1

Select LocationActon

Management10.165.32.0/22

Redundant☐

NTP Server

Add additional NTP☐

Resource Groupplte

Select RegionEast US

LAN10.165.60.0/22

Select Devicease-plte-4

Select TimezoneEurope/Andorra

Packet Core VersionAPNS 1.0

MNC099MCC999

WAN10.165.64.0/23

Custom IP Addresses?☐

MME

MRM-Management10.165.32.140

MRM-Base10.165.60.130

MRM-Data10.165.60.138

MSM-Management10.165.32.141

MSM-Base10.165.60.131

MSM-Data10.165.60.139

MSM-North/South 110.165.60.17

Management Logical10.165.32.142

SAEGW

Management10.165.32.143

Peer Management10.165.32.144

Logical Management10.165.32.145

Base10.165.61.130

Peer Base10.165.61.131

Logical Base10.165.61.132

East/West10.165.61.138

North/South 110.165.61.17

North/South 210.165.64.17

HSS

Management10.165.32.149

LAN10.165.60.237

Submit

Note: The default values are populated based on the subnets configured for this resource group. Each field can be updated manually as necessary. Select **Submit** to create the private network and push the factory configuration settings to the Azure Stack Edge device for this enterprise location.

After the network has been successfully created, you must configure the service attributes using the Network level [Private Networks](#) tab. Refer to [Service Configuration](#) for additional detail.

Service Configuration

Standalone Edge Networks

For Standalone Edge Networks, additional configuration is required after you have created the network. You must configure the private network's service attributes.

To configure the Standalone Edge Network's attributes:

1. Select the private network to configure from the tree, and view the **Private Network Detail** tab. This tab displays the network information previously defined, and shows the IP addresses for the VNF Deployment:

Figure 3-1 Private Network Detail Tab

The screenshot shows the 'Private Network Detail' tab in the Private Network Service Manager. The top navigation bar includes tabs for 'Private Network Detail', 'Element Details', 'Alarms', 'Event Logs', 'PM Dashboard', 'Network Availability', 'Upgrades', 'RAN Info', and 'Subscriber Tracing'. The 'Private Network Detail' tab is active, displaying a table with deployment information.

Resource Group	Private LTE_ASE	Packet Core Version	APNS 1.1	Location	Action
Created Time	2021/11/02 23:16:34	Region:	East US	Provisioning State	Provisioned
NTP Server	10.168.0.10	Deployment Mode	Standalone Edge Redundant	NTP Server2	
Time Zone	Europe/Andorra	NTP Server1	10.168.0.11		10.168.0.12

Below the table, there are two sections: 'VNF Deployment' and 'Service Config'. The 'VNF Deployment' section shows three columns: MME, SAEGW, and HSS, each with a list of components and their IP addresses. The 'Service Config' section shows a table with various attributes and their values.

Attribute	Value
Status	Configured
Service Attributes	
Network Name	Telecom-xyz
Home PLMN ID	99999
APN Name	plte
DNS Primary	200.110.10.1
DNS Secondary	
Roaming PLMN ID	12345
GI Attributes	
Interface	10.165.64.10
Interface Next hop	10.165.64.1
Local UE IP Pool Start	200.20.170.1
Local UE IP Pool End	200.20.170.255
Loopback	10.165.65.200
Router ID	10.165.64.10
GN Attributes	
Interface	10.165.61.10
Interface Next hop	10.165.61.1
SGW S1 Loopback	10.165.61.49
SGW S58 Loopback	10.165.61.48
PGW Loopback	10.165.61.50
SGW S11 Loopback	10.165.61.51

2. Scroll to the bottom of this screen and select the **Configure** button to perform Service Configuration.

Figure 3-2 Service Config

The screenshot shows the 'Service Config' dialog box. It has two tabs: 'ase-plte-9' and 'ase-plte-10'. The 'ase-plte-9' tab is active, displaying a form with three columns: 'Service Attributes', 'GI Attributes', and 'GN Attributes'. Each column contains a list of attributes with input fields for their values.

Service Attributes	GI Attributes	GN Attributes
Network Name	Interface	Interface
Home PLMN ID	Interface Next Hop	Interface Next Hop
APN Name	UE IP Pool Start	SGW S1 Loopback
DNS Primary	UE IP Pool End	SGW S58 Loopback
DNS Secondary	Loopback	PGW Loopback
Roaming PLMN ID	Router ID	SGW S11 Loopback
Data Fabric IP	Local As	Router ID
	Neighbor IP	Local As
	Remote As	Neighbor IP
	Fall Over BFD	Remote As

At the bottom of the dialog, there is a 'Submit' button.

3. Default IP addresses appear for each attribute type provided you selected the system defaults during the initial private network configuration. Otherwise, you must provide the IP addresses for each parameter to proceed with the configuration.
4. Enter the Primary and Secondary DNS IP Addresses.
5. Provide the respective values for **GI UE IP Pool Start** and **GI UE IP Pool End**.
6. Select **Submit**. A confirmation dialog appears, indicating the configuration has been completed. Select **OK** to continue.

7. The detailed **Service Configuration** appears on the **Private Network Service** tab as follows. Note that the configuration status for each attribute displays. In the following example, configuration has been completed successfully:

Figure 3-3 Detailed Service Configuration

Service Config

▼

Status

Service Attributes

Network Name

Home PLMN ID

APN Name

Roaming PLMN ID

Day 1 Configured

Telecom-xyz

99999

plte

12345

GI Attributes

Interface

Interface Next hop

UE IP Pool Start

UE IP Pool End

Loopback

Router ID

Local As

Neighbor IP

Remote As

Fall Over BFD

BFD Multi-Hop Peer

BGP GI LOOP

Remote Ip SubPool Range Start

Remote Ip SubPool Range End

V4ip SubPool Range Start

V4ip SubPool Range End

10.165.68.137

10.165.68.129

10.200.0.1

10.200.0.255

10.165.68.162

10.165.68.137

65320

10.165.68.129

64512

single-hop

10.165.68.129

10.165.68.178

10.201.0.1

10.201.0.255

10.202.0.1

10.202.0.255

GN Attributes

Interface

Interface Next hop

SGW S1 Loopback

SGW S5S8 Loopback

PGW Loopback

SGW S11 Loopback

Router ID

Local As

Neighbor IP

Remote As

Fall Over BFD

BFD Multi-Hop Peer

BGP GEO LOOP

BGP GN LOOP

10.165.68.10

10.165.68.1

10.165.68.75

10.165.68.72

10.165.68.73

10.165.68.74

10.165.68.10

65320

10.165.68.1

64512

multi-hop

10.165.68.1

10.165.68.114

10.165.68.115

MME Attributes

Status

Domain Name 1

Domain Name 2

REALM Peer

LB Instance IP1

GTP Profile

Node1 REALM Peer

Day 1 Configuration Completed

plte.apn.epc.mnc099.mcc999.3gppnetwork.org

tac-ib00.tac-hb01.tac.epc.mnc099.mcc999.3gppnetwork.org

plte21hss.affirmed.com

10.165.68.9/27

10.165.68.50

plte21hss.affirmed.com

Routing Next Hop GW Address

Diameter Client Instance

REALM Peer IP

S1 IF

Bfd

Node1 REALM Peer IP

Node1 External Ns1

10.165.68.1

10.165.68.49

10.165.68.11

10.165.68.51

enable

10.165.68.19

10.165.68.17/27

See [Optional Network Configurations](#) for further information about configuring your Standalone Edge Network.

Distributed Edge Networks

For Distributed Edge Networks, additional configuration is required after you have created the network. You must configure the private network's service attributes and enable communication between the Control and User Planes.

To configure the Standalone Edge Network's attributes:

1. Select the private network to configure from the tree, and view the **Private Network Detail** tab. This tab displays the network information previously defined, and shows the IP addresses for the VNF Deployment:

Figure 3-4 Private Network Detail Tab

The screenshot displays the 'Private Network Service Manager' interface. At the top, there's a status bar with counts for Critical (12), Major (20), Minor (2), Warning (9), and Cleared (14) alerts. Below this is a navigation menu with tabs: Private Network Detail (selected), Element Details, Alarms, Event Logs, PM Dashboard, Network Availability, Upgrades, RAN Info, and Subscriber Tracing. The main content area shows details for the deployment 'plite2-3-0311-deploy1'. It includes a table with resource group information and a section for VNF Deployment with three sub-sections: MME, SAE GW, and HSS. Each sub-section lists various management and data components with their respective IP addresses and status (e.g., 'Done'). Below the VNF Deployment section is the 'Service Config' section, which lists service attributes (Network Name, Home PLMN ID, APN Name, DNS Primary, DNS Secondary, Roaming PLMN ID) and their values, along with a 'Configure' button.

Resource Group	Private LTE ASE	Packet Core Version	APNS 1.1	Location	Action
Created Time:	2021/11/02 23:16:34	Region:	East US	Provisioning State:	Provisioned
NTP Server:	10.168.0.10	Deployment Mode:	Standalone Edge Redundant	NTP Server2:	10.168.0.12
Time Zone:	Europe/Andorra	NTP Server1:	10.168.0.11		

MME		SAEGW		HSS	
Status	Done	Status	Done	Status	Done
MRM-Management	10.165.32.100	Management	10.165.32.103	Audit Status	N/A
MRM-Base	10.165.60.33	Peer Management	10.165.32.104	Audit Time	10.165.32.109
MRM-Data	10.165.60.41	Logical Management	10.165.32.105	Management	10.165.60.23
MSM-Management	10.165.32.101	Base	10.165.61.34	LAN	
MSM-Base	10.165.60.34	Peer Base	10.165.61.35		
MSM-Data	10.165.60.42	Logical Base	10.165.61.36		
MSM-North/South 1	10.165.60.10	East/West	10.165.61.42		
Management Logical	10.165.32.102	North/South 1	10.165.61.10		
SS IP	10.165.62.203	North/South 2	10.165.64.10		

Service Config		GI Attributes		GN Attributes	
Status	Configured	Interface	10.165.64.10	Interface	10.165.61.10
Service Attributes		Interface Next hop	10.165.64.1	Interface Next hop	10.165.61.1
Network Name	Telecom-xyz	Local UE Pool Start	200.20.170.1	SGW S1 Loopback	10.165.61.49
Home PLMN ID	99999	Local UE Pool End	200.20.170.255	SGW S58 Loopback	10.165.61.48
APN Name	plite	Loopback	10.165.65.200	PGW Loopback	10.165.61.50
DNS Primary	200.110.10.1	Router ID	10.165.64.10	SGW S11 Loopback	10.165.61.51
DNS Secondary					
Roaming PLMN ID	12345				

2. Scroll to the bottom of this screen and select the **Configure** button to perform Service Configuration.

Figure 3-5 Service Config

Service Config

Control plane service configuration

Service Attributes		GI Attributes		GN Attributes	
Network Name	Telecom-xyz	GI Interface	10.71.5.9	Interface	10.71.4.9
Home PLMN ID	99999	Interface Next Hop	10.71.5.1	Interface Next Hop	10.71.4.1
APN Name	plte	Local UE Pool Start	200.220.170.1	SGW S1 Loopback	10.71.4.61
DNS Primary	1.1.1.1	Local UE Pool End	200.220.170.255	SGW S5S8 Loopback	10.71.4.62
DNS Secondary	8.8.8.8	Loopback	10.71.5.37	PGW Loopback	10.71.4.63
Roaming PLMN ID	12345			SGW S11 Loopback	10.71.4.64
Data Fabric IP	10.71.4.69			PFCP Peer	10.165.56.185

MME Attributes		Routing Next Hop GW	
Domain Name 1	plte.apn.epc.mnc099.mcc999.3gppnetwork.org	Address	10.71.4.1
Domain Name 2	tac-lb00.tac-hb01.tac.epc.mnc099.mcc999.3gppnetwork.org	Diameter Client Instance	10.71.4.38
REALM Peer	plte99hss.affirmed.com	REALM Peer IP	10.71.4.56
LB Instance IF1	10.71.4.4/27	S1 IF	10.71.4.37
GTP Profile	10.71.4.39	Bfd	disable

User plane service configuration

Location : Richardson

Service Attributes		GI Attributes		GN Attributes	
Network Name	Telecom-xyz	GI Interface	10.165.64.10	Interface	10.165.61.10
Home PLMN ID	99999	Interface Next Hop	10.165.64.1	Interface Next Hop	10.165.61.1
APN Name	plte	Local UE Pool Start	200.20.170.1	SGW S1 Loopback	10.165.61.49
DNS Primary	1.1.1.1	Local UE Pool End	200.20.170.255	SGW S5S8 Loopback	10.165.61.48
DNS Secondary	8.8.8.8	Loopback	10.165.64.33	PGW Loopback	10.165.61.50
Roaming PLMN ID	12345			SGW S11 Loopback	10.165.61.51
Data Fabric IP	10.165.61.37				

Submit

- Default IP addresses appear for each attribute type in the Control Plane section, provided you selected the system defaults during the initial private network configuration. Otherwise, you must provide the IP addresses for each parameter to proceed with the configuration. Only Gateway defaults are populated for the User Plane.
- Enter the Primary and Secondary DNS IP Addresses.
- Provide the respective values for **GI Local UE IP Pool Start** and **GI Local UE IP Pool End**.
- Enter the **PFCP** (Packet Forwarding Control Plane) **Peer**. This attribute must be populated to enable communication between the Control Plane and User Plane.
- Complete the fields in the User Plane section, as required.
- Select **Submit**. A confirmation dialog appears, indicating the configuration has been completed. Select **OK** to continue.
- The detailed **Service Configuration** appears on the **Private Network Service** tab.

See [Optional Network Configurations](#) for further information about configuring your Distributed Edge Network.

Optional Network Configurations

You can configure additional parameters for your private network after defining the standard attributes through the [Creating Private Networks](#) and [Service Configuration](#) screens. These parameters include the NAT configuration and EDR Status Configurations.

NAT Configuration

The NAT Config fields allow you to enter a NAT Pool Start and End address, and select to Provision. These settings define whether the configuration in the SAEGW on the private network is provisioned or unprovisioned.



The screenshot shows a form titled "NAT Config". It contains three input fields: "NAT Pool Start", "NAT Pool End", and a "Provision" toggle switch. The "Provision" toggle is currently turned off.

Create Distributed Edge Network

The Distributed Edge Network functions as a standard CUPS deployment, where the Control Plane operates in the Cloud and the User Plane is located on-site at the enterprise. Each Service Manager instance can support up to 5 CUPS deployments.

***Note:** A single control plane can have multiple user planes. Each user plane may exist in the same location, or in multiple locations.*

To create a Distributed Edge Network:

1. Right-click on the Global Network and select **Create Network**, then **Distributed Edge**.
The dialog opens:

2. Enter a **Name** for the network. This should be a unique name specific for this private network.
3. Enter the **Resource Group**. This text should be identical to the Resource Group name used when configuring the Azure Network Function Manager as part of the subscription. Please refer to the *Private Network Management System Deployment Guide* for specific information regarding this setup.
4. Select the appropriate **Packet Core Version** to be used for this private network enterprise. This system software will be imported into the Service Manager.
5. Edit the MNC and MCC fields as necessary.
6. Add an NTP IP address. Note that you can add additional NTPs as well.
7. To enable High Availability, mark the **Redundant** check box. For information regarding High Availability, please refer to the [High Availability \(HA\)](#) topic.

Caution: High Availability is not currently supported for Distributed Edge Networks . This feature will be added in a later release.

8. Select the appropriate TimeZone for this deployment.
9. Complete the Control Plane Configuration fields
 - a. Edit the **Resource Group** name, if necessary. The default name is identical to the parent Resource Group name above.
 - b. Enter the **Storage Account**
 - c. Enter the **Storage Container**
 - d. Enter the **VNET**
10. Complete the User Plane Configuration fields:

- a. Select the **Location** for the Standalone network.
- b. Select the **Region**. The available options are based on those selected in the Azure Cloud Network Function Manager.
- c. Select the **Device** to use for this Network Instance. The options listed are based on the customer subscription. These devices are specified in the Azure cloud.

***Note:** The Management, LAN and WAN default values are populated based on the subnets configured for this resource group. Each field can be updated manually as necessary.*

- d. (Optional) Activate the **Use Custom IPs** status and enter additional information as required.
11. Select **Submit** to create the private network and push the factory configuration settings to the Azure Stack Edge device for this enterprise location.

After the network has been successfully created, you can view the details using the [Private Networks](#) tab. Refer to that topic for additional detail.

Create All in Cloud Network

Using this private network configuration, all network functions exist solely in the cloud. This allows for a centralized location for both the control and user plane.

To create an All in Cloud Network:

1. Open the **Create Private Network Instance in Cloud** by right-clicking on the Global Menu option and select **Create Network**, then **All in Cloud**.

The dialog opens:

The screenshot shows a 'Create Private Network Instance In Cloud' dialog box. It has a title bar with a close button. The form is organized into two main sections. The top section contains fields for 'Name' (Cloud9), 'Resource Group' (cloud_resource_eastus), 'Packet Core Version' (APNS 1.1), 'MNC' (099), 'MCC' (999), 'NTP IP' (10.168.0.10), 'Select Timezone' (Europe/Andorra), 'Add additional NTP' (checkbox), and 'Redundant' (checkbox). The bottom section contains 'Storage Account' (cloudstorageeastus), 'Storage Container' (vhds), 'VNET' (cloud_test), and a 'Submit' button.

2. Enter a **Name** for the network. This should be a unique name specific for this private network.
3. Enter the **Resource Group**. This text should be identical to the Resource Group name used when configuring the Azure Network Function Manager as part of the subscription. Please refer to the *Private Network Management System Deployment Guide* for specific information regarding this setup.
4. Select the appropriate **Packet Core Version** to be used for this private network enterprise. This system software will be imported into the Service Manager.

Based on your selections, PNSM auto-populates the IP Addresses for MME, SAEGW, and HSS.

Note: The default values are populated based on the subnets configured for this resource group. Each field can be updated manually as necessary.

5. Select **Submit** to create the private network and push the factory configuration settings to the Azure Stack Edge device for this enterprise location.

After the network has been successfully created, you can view the details using the [Private Networks](#) tab. Refer to that topic for additional detail.

Retrying a Network

When a network fails, you can select to retry deployment. This removes any failed networks from the Azure portal, and deploys the functional resources.

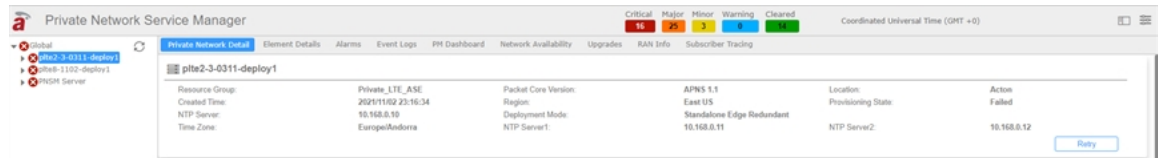
Prerequisites

A network can only be redeployed provided:

- the Location has not been changed.
- the ASE device has not been changed.
- any/all NFs are in a Failed state.

To Retry a Network Deployment:

1. Double click on the Failed Network Instance.
2. Select **Retry**.



Note: This option is available only for deployment-failed networks. Once a deployment has succeeded, this option is disabled. This feature is not supported for deleted-network failures.

Deleting a Network

To delete a Private Network:

1. Right-click on the Network you want to delete.
2. Select **Delete Network** from the menu.

Redeploying a Network

You can redeploy a Standalone Edge Network should issues with the ASE device persist, or following an upgrade. Redeploying a network requires that no configurations are changed, If User is facing any issues with ASE devices in long run or once after ASE Upgrades, User can redeploy the existing Networks in the corresponding ASEs with the same set of configurations with a single click option.

Note: Network redeploy is supported only for ONPREM and ONPREMHA Networks and is enabled by default for any network in the Provisioned state and the Network in Failed state on redeployment.

To redeploy a network:

1. Navigate to the Private Network's **View Network** screen.
2. Select **Redeploy**.

Private Network DetailElement DetailsAlarmsEvent LogsPM DashboardNetwork AvailabilityUpgradesRAN InfoSubscriber Tracing

plte2_3_0511_deploy1

Resource Group:Private_LTE_ASE
Created Time:2021/11/05 03:48:34
NTP Server:10.168.0.10
Time Zone:Europe/Andorra

Packet Core Version:Region:East US
Deployment Mode:Standalone Edge Redundant

Location:Provisioning State:Plte2-10
Provisioned

Redeploy

ase-plte-2ase-plte-3

VNF Deployment

MME

StatusDone

MRM-Management10.165.32.100
MRM-Base10.165.60.33
MRM-Data10.165.60.41
MSM-Management10.165.32.101
MSM-Base10.165.60.34
MSM-Data10.165.60.42
MSM-North/South 110.165.60.10
Management Logical10.165.32.102
S6 IP10.165.62.203

SAEGW

StatusDone

Management10.165.32.103
Peer Management10.165.32.104
Logical Management10.165.32.105
Base10.165.61.34
Peer Base10.165.61.35
Logical Base10.165.61.36
East/West10.165.61.42
North/South 110.165.61.10
North/South 210.165.64.10

HSS

StatusDone

Audit StatusN/A
Audit TimeN/A
Management10.165.32.109
LAN10.165.60.23

Service Config

StatusConfigured

Service Attributes
Network NameTelecom-xyz
Home PLMN ID9999
APN Nameplte
DNS Primary
DNS Secondary

GI Attributes
Interface10.165.64.10
Interface Next hop10.165.64.1
Local UE Pool Start10.200.0.1
Local UE Pool End10.200.0.255
Loopback10.165.65.200

GN Attributes
Interface10.165.61.10
Interface Next hop10.165.61.1
SGW S1 Loopback10.165.61.49
SGW S5/S8 Loopback10.165.61.48
PGW Loopback10.165.61.50

Configure

3. Confirm redeployment.

Network re-deploy message

Network re-deploy will delete the existing VNFs and re-deploy the VNFs with the same existing configuration. Are you sure, You want re-deploy plte2_3_0511_deploy1 network?

YesNo

A status message indicating a successful redeployment appears and the Provisioning State updates to In-Progress.

Private Network DetailElement DetailsAlarmsEvent LogsPM DashboardNetwork AvailabilityUpgradesRAN InfoSubscriber Tracing

plte2_3_0511_deploy1

Resource Group:Private_LTE_ASE
Created Time:2021/11/05 03:48:34
NTP Server:10.168.0.10
Time Zone:Europe/Andorra

Packet Core Version:Region:East US
Deployment Mode:Standalone Edge Redundant

Location:Provisioning State:Plte2-10
In-Progress

ase-plte-2ase-plte-3

VNF Deployment

MME

StatusDone

MRM-Management10.165.32.100
MRM-Base10.165.60.33
MRM-Data10.165.60.41
MSM-Management10.165.32.101
MSM-Base10.165.60.34
MSM-Data10.165.60.42
MSM-North/South 110.165.60.10
Management Logical10.165.32.102
S6 IP10.165.62.203

SAEGW

StatusDone

Management10.165.32.103
Peer Management10.165.32.104
Logical Management10.165.32.105
Base10.165.61.34
Peer Base10.165.61.35
Logical Base10.165.61.36
East/West10.165.61.42
North/South 110.165.61.10
North/South 210.165.64.10

HSS

StatusDone

Audit StatusN/A
Audit TimeN/A
Management10.165.32.109
LAN10.165.60.23

Service Config

StatusConfigured

Service Attributes
Network NameTelecom-xyz
Home PLMN ID9999
APN Nameplte
DNS Primary
DNS Secondary
Roaming PLMN ID12345
Data Evolved ID40.465.64.17

GI Attributes
Interface10.165.64.10
Interface Next hop10.165.64.1
Local UE Pool Start10.200.0.1
Local UE Pool End10.200.0.255
Loopback10.165.65.200
Router ID10.165.64.10

GN Attributes
Interface10.165.61.10
Interface Next hop10.165.61.1
SGW S1 Loopback10.165.61.49
SGW S5/S8 Loopback10.165.61.48
PGW Loopback10.165.61.50
SGW S11 Loopback10.165.61.51

Configure

3-14

Private Network Service Manager User's Guide, Release 1.4.0

Revision: A

High Availability (HA)

The high-availability (HA) feature provides failover of services from one cluster node to another in the event the active network becomes inoperative. The HA feature provides active-standby redundancy for the Service Manager server and its services as well as any underlying hardware.

High Availability in a Standalone Edge Network

When creating a Standalone Edge Network, you can ensure failover services by enabling the high availability option. The network creation process remains nearly identical, except you must select the devices to use in your high availability configuration.

To configure High Availability:

1. When creating a Standalone Edge Networks, select the **Redundant** option.
2. Select the **(Node0 Device** and **Node1 Device** to use for this Network Instance. **Node0** serves as the Primary Device, while **Node1** acts as the Standby Device.

***Note:** The options listed are based on the customer subscription. These devices are specified in the Azure cloud.*

3. Complete the remaining fields, and select **Submit**.

The devices you select for each node are then displayed as network elements, and can be viewed, edited, and monitored as standard network elements.

Figure 3-6 HA Configuration for Standalone Edge

The screenshot displays the configuration interface for a Standalone Edge Network, showing VNF Deployment and Service Config sections.

VNF Deployment

Component	Status	IP Address
MME	Done	
MME-Management		10.165.32.100
MME-Base		10.165.60.33
MME-Data		10.165.60.41
MSM-Management		10.165.32.101
MSM-Base		10.165.60.34
MSM-Data		10.165.60.42
MSM-North/South 1		10.165.60.10
Management Logical		10.165.32.102
S6 IP		10.165.60.51
SAEGW	Done	
SAEGW-Management		10.165.32.103
SAEGW-Peer Management		10.165.32.104
SAEGW-Logical Management		10.165.32.105
SAEGW-Base		10.165.61.34
SAEGW-Peer Base		10.165.61.35
SAEGW-Logical Base		10.165.61.36
SAEGW-East/West		10.165.61.42
SAEGW-North/South 1		10.165.61.10
SAEGW-North/South 2		10.165.64.10
HSS	Done	
HSS-Management		10.165.32.109
HSS-LAN		10.165.60.23

Service Config

Category	Attribute	Value
Service Attributes	Status	Configured
	Network Name	Telecom-xyz
	Home PLMN ID	99999
	APN Name	plte
Roaming PLMN ID	12345	
GI Attributes	Interface	10.165.64.10
	Interface Next hop	10.165.64.1
	UE IP Pool Start	200.20.1.1
	UE IP Pool End	200.20.200.255
	Loopback	10.165.65.200
	Router ID	10.165.64.10
	Local As	65320
	Neighbor IP	10.165.64.1
	Remote As	64512
	Fall Over BFD	single-hop
	BFD Multi-Hop Peer	10.165.64.1
	BGP GI LOOP	10.165.65.201
	Remote Ip SubPool Range Start	169.20.50.83
	GN Attributes	Interface
Interface Next hop		10.165.61.1
SGW S1 Loopback		10.165.63.204
SGW S58 Loopback		10.165.63.205
PGW Loopback		10.165.63.206
SGW S11 Loopback		10.165.63.207
Router ID		10.165.61.10
Local As		65320
Neighbor IP		10.165.61.1
Remote As		64512
Fall Over BFD	multi-hop	
BFD Multi-Hop Peer	10.165.61.1	
BGP GEO LOOP	10.165.63.201	

After the network has been successfully created, you must configure the service attributes using the Network level [Private Networks](#) tab. Refer to [Service Configuration](#) for additional detail.

Application-level Redundancy

The HA cluster provides application-level redundancy. For information regarding Service Manager and High Availability, please refer to the *Affirmed Private Network Service Deployment* guide.

SIM Provisioning

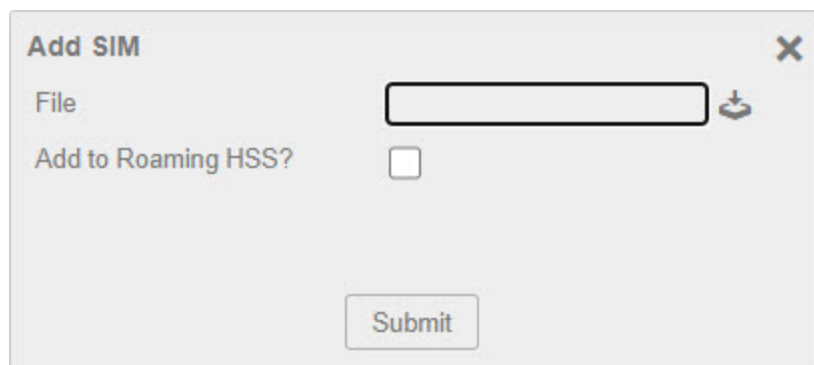
***Note:** Prior to SIM Provisioning, you must configure the EPSSUBD OP Value in the PNSM Configuration dialog. Refer to [Service Configuration](#) for additional details.*

Adding a SIM

To add a SIM

1. Open the **SIM Provisioning** tab at the Enterprise level.
2. Select **Add SIM** to open the Add SIM dialog:

Figure 4-1 Add SIM Dialog



3. Choose the CSV **File** to import. The IMSIs configured in this file are automatically added to the SIM.
4. Select if you want to include the SIM to the Roaming HSS.
5. Select Save to close the dialog. The SIMs are added to the **SIM Provisioning** tab.

Activating a SIM

Once you have added a SIM, you can choose to activate each SIM individually, or to perform a bulk activation by marking the row(s) to be activated.

To Activate a SIM or SIMS

1. Mark the checkbox in the first column for each SIM to be activated

Figure 4-2 Activating Multiple SIMS

☐	IMSI Number	User ID	Phone Number	Profile	Status	Action
☐	311180005000031	Not Assigned	Not Assigned	Default	Activated	  
☐	311180005000032	Not Assigned	Not Assigned	Default	Activated	  
☐	311180005000033	Not Assigned	Not Assigned	Default	Activated	  
☐	311180005000034	Not Assigned	Not Assigned	Default	Activated	  
☐		Not Assigned	Not Assigned	Default	Activated	  
☒		Not Assigned	Not Assigned	Default	Deactivated	  
☒		Not Assigned	Not Assigned	Default	Deactivated	  
☐	311180005000038	Not Assigned	Not Assigned	Default	Deactivated	  
☒	311180005000039	Not Assigned	Not Assigned	Default	Deactivated	  
☐	311180005000040	Not Assigned	Not Assigned	Default	Deactivated	  

All marked SIMS will be activated

2. Select the **Activate** button to activate all selected SIMs.
3. Enter the **User ID** and **Phone Number**, then select **Activate**.

To activate an individual SIM, you can follow the above steps, or select the Activate  icon in the **Action** column of the SIM's row.

Deactivating a SIM

Any activated SIM can be deactivated using SIM Provisioning tab. Deactivating a SIM removes a user's access, but the SIM remains in the Affirmed Private Network Service and can be re-activated at any time. You can choose to deactivate each SIM individually, or to perform a bulk activation by marking the row(s) to be activated.

To Deactivate a SIM or SIMS

1. Mark the checkbox in the first column for each SIM to be deactivated.
2. Select the Deactivate button.

To deactivate an individual SIM, you can follow the above steps, or select the Deactivate  icon in the **Action** column of the SIM's row.

Deleting a SIM

Deleting a SIM is a delayed process. When you choose to delete a SIM, it remains in a deactivated state for 24 hours, during which time you may choose to reactivate it. At the conclusion of the time, the SIM is permanently deleted from the private network.

To Delete a SIM or SIMS

1. Mark the checkbox in the first column for each SIM to be deleted.
2. Select the **Delete** button. The state of the SIM changes to **Deactivated for Delete**. Note that the **Status** column displays the time left before the SIM instance is permanently deleted.

Figure 4-3 Deleting a SIM

Private Networks SIM Provisioning Alarms						
☐	999990005000010	Not Assigned	Not Assigned	Default	Deactivated For Delete Will be deleted in 18 hours	  
☐	999990005000009	Not Assigned	Not Assigned	Default	Deactivated	  
☐	999990005000008	Not Assigned	Not Assigned	Default	Deactivated	  
☐	999990005000007	Not Assigned	Not Assigned	Default	Deactivated	  
☐	999990005000006	Not Assigned	Not Assigned	Default	Deactivated	  
☐	999990005000005	Not Assigned	Not Assigned	Default	Deactivated	  
☐	999990005000004	Not Assigned	Not Assigned	Default	Deactivated	  
☐	999990005000003	Not Assigned	Not Assigned	Default	Deactivated	  
☐	999990005000002	Not Assigned	Not Assigned	Default	Deactivated	  
☐	999990005000001	Not Assigned	Not Assigned	Default	Deactivated	  

Showing 1 to 10 of 11 entries Page 1 First Prev 1 2 Next Last

To deactivate an individual SIM, you can follow the above steps, or select the Delete  icon in the **Action** column of the SIM's row.

Showing Devices in a SIM

For each SIM, you can select to view the Devices by selecting the Show Devices  icon in the **Action** column of the SIM's row. This displays a dialog indicating the Configured Devices for the SIM.

Note: All actions performed during SIM provisioning are logged in the Audit Trail to verify the success of provisioning.

Monitoring Private Networks

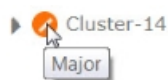
Once you have configured your Enterprise and its Private Networks, you can monitor the status to ensure optimal performance. This section discusses the alarms, events, dashboards and reports available in the Private Network Service Manager that work together to show private network issues and key parameters such as Performance Indicators, Resource Utilization, Network Slicing, etc..

Alarms

Identifying Network Elements with Alarms

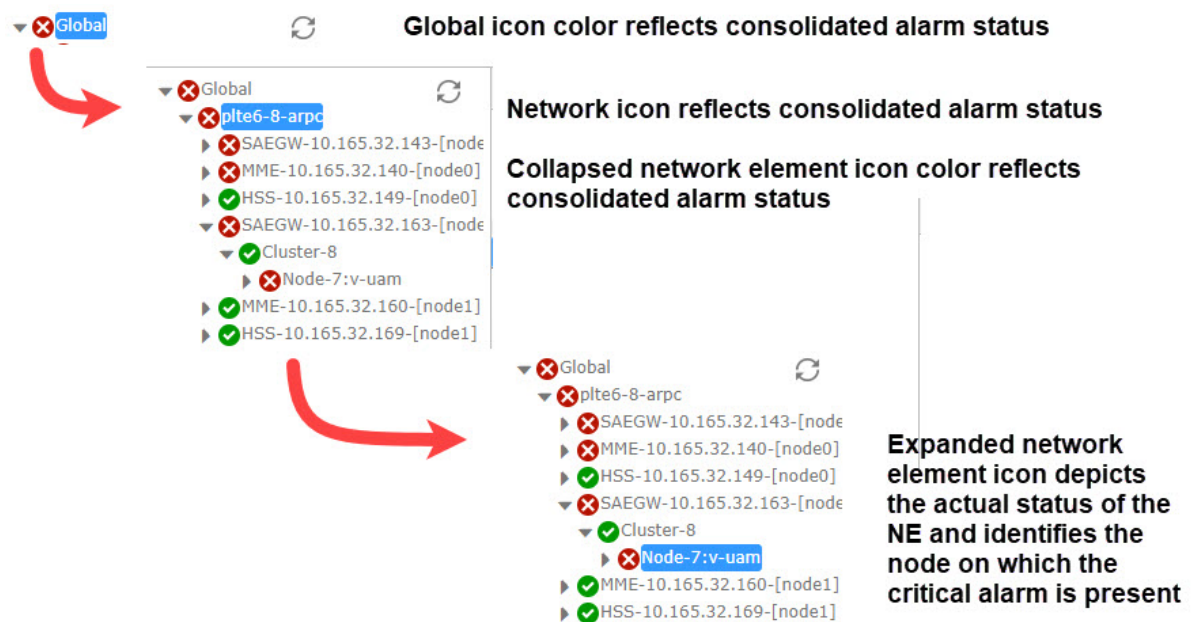
PNSM dynamically updates the tree display with current alarm information. As alarms are generated, the tree node icon changes color to represent the highest severity of alarms. Alarm color is propagated from the particular component for which the alarm was generated to the parent of the component. Specifically, when the parent component is collapsed in the tree, its icon color reflects the consolidated alarm status. When the parent component is expanded in the tree, its icon color depicts the actual status of the node.

To determine the alarm severity of the tree node icon, point the cursor at the icon. PNSM opens a pop-up window that lists the severity.



Note: Alarm severity color is configurable. For instructions, see [Configuring Alarm Severity Color Settings](#).

Alarm Propagation depicts how a user drills down to the actual problem after observing an alarm propagated to the Global level.

Figure 5-1 Alarm Propagation

Managing Network Element Fault Management State

When a network element is added to PNSM, its Managed Fault management state is set to Active so that PNSM automatically configures itself as a trap destination and can process traps from the network element.

There are scenarios where PNSM can resolve Fault Management problems in production systems. For example:

- If a managed network element is not properly configured or setup, it can send a high rate of traps to PNSM. If it overflows the PNSM traps staging buffer, many incoming traps to PNSM can be dropped. And, when encountering a trap storm, PNSM can consume a lot of resources to process the incoming traps and purge traps-related database tables, resulting in a significant negative impact on the PNSM performance and operations. In this scenario, the administrator can set the network element's Managed Fault management state to unmanaged so that PNSM removes itself as trap destination and no longer receives new traps from the network element. Once the issues are resolved, the administrator must set the network element's Managed Fault management state to managed.
- If, during the course of network element operation, the PNSM trap destination is inadvertently deleted from the network element, PNSM shows no notification or warning that the trap destination is deleted. As a result, PNSM no longer receives any traps from the network element. If this scenario arises, PNSM can automatically configure itself as a trap destination.

- Some network elements support a fixed number of trap destinations. If, when adding the element to PNSM for management, the maximum number is reached, PNSM cannot configure itself as a trap destination. If this scenario arises, PNSM can notify the user and periodically retry to configure itself as a trap destination.

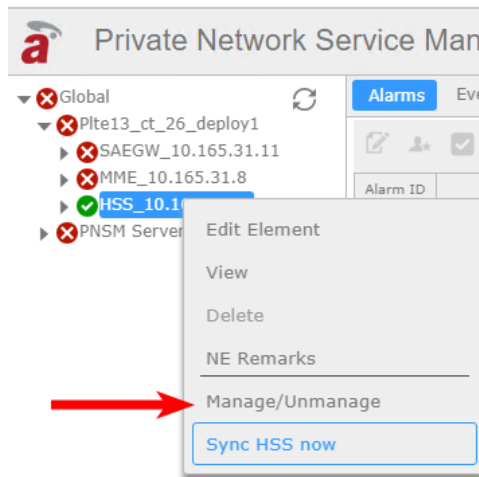
Changing Managed Fault management state to unmanaged is an infrequent operation performed only when PNSM is overwhelmed by a trap storm. The action required by the administrator is to change the Managed Fault management state to unmanaged and rectify the situation. Then, when the issues have been corrected, the administrator must change the Managed Fault management state back to managed.

To manage/unmanage the Managed Fault management state of a network element, the network element must be in the Managed state (is accessible by PNSM).

To manage/unmanage the Managed Fault management state of a network element:

1. Right-click a network element and select **FM Manage** or **FM Unmanage**.

Figure 5-2 FM Manage/Unmanage

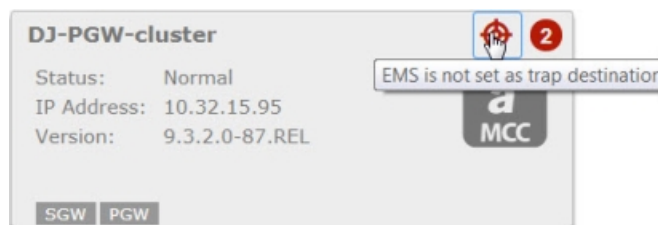


2. At the confirmation prompt, select **OK**.

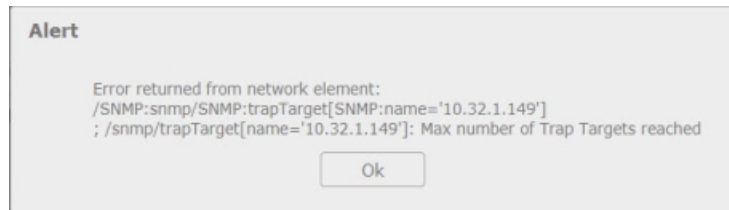
When the network element is in Managed state (is accessible by PNSM), the device icon in the Network View tab includes a status icon for Managed Fault management state and trap destination state indicating that administrator attention is required:

- When the Managed Fault management state is set to Managed but PNSM is not configured as a trap destination of the network element, the state  icon is red.

To determine the cause of the state icon, point the cursor at the icon. PNSM opens a pop-up window that lists the cause.



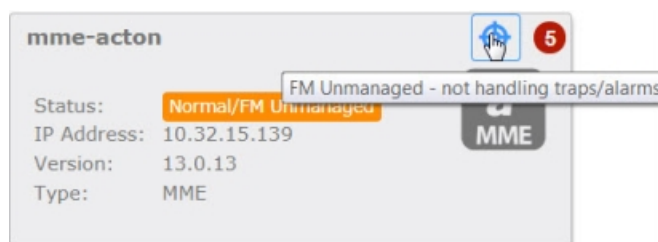
To determine the cause of the bad trap destination state, select the state icon. PNSM opens a pop-up Alert window that lists the cause.



If this scenario arises, PNSM periodically retries to configure itself as a trap destination.

- When the Managed Fault management state is set to Unmanaged, the device lists the Status as FM Unmanaged and the state  icon is gray.

To determine the cause of the state icon, point the cursor at the icon. PNSM opens a pop-up window that lists the cause.



If this scenario arises, the administrator can set the Managed Fault management state to Managed.

During normal operation, when the network element Managed Fault management state is set to Managed and PNSM is configured as a trap destination, no state icon is shown on the device icon because no user attention is required.

When the network element is in Managed state (is accessible by PNSM), its Managed Fault management state is set to Managed (Active) and PNSM automatically configures itself as a trap destination.

If the administrator unmanages the network element (is inaccessible by PNSM), PNSM sets the Managed Fault management state to unmanaged (Inactive) and removes the PNSM trap destination from the network element.

Configuring Trap OID Display Length

By default, PNSM displays the complete trap OID in the Trap OID field in the following areas of PNSM:

- Alarms and Event Logs tabs
- Host System Alarms tab
- Alarm Details window
- Event Logs View window
- Alarm Filter Conditions (under Parameter)

For example:

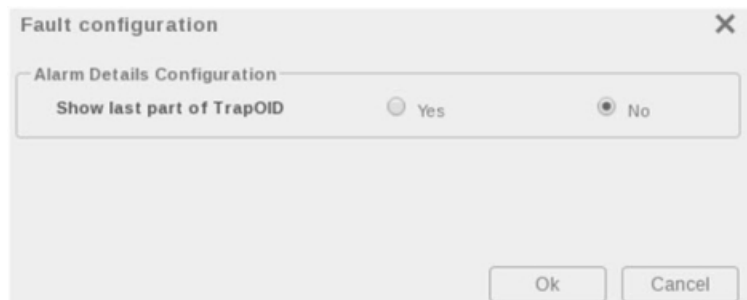
.1.3.6.1.4.1.116.6.41.1.1.1.1.1.65536002

You can configure PNSM to display only the last part of the full OID. For example:

65536002

To configure the Trap OID display length:

1. Select the **Settings**  icon on the toolbar and select **Fault => Configuration**.
2. In the **Show last part of TrapOID** field, mark the **Yes** box and select **OK**.



Viewing Alarms

The Alarm Browser provides a summary of the Alarms currently active for the selected group or network element in the tree. It displays the Alarm specific details in a tabular view indicating information on network element, affected component, reason for the alarm, severity of the alarm, and similar parameters.

To view alarms:

1. Select the object in the tree for which you want to view alarms:
 - To view alarms from all NEs, select **Global** in the tree.
 - To view alarms from NEs in a particular group, select the group in the tree.

- To view alarms for a particular NE, select the NE in the tree.

2. To view NE alarms, select the **Alarms** tab.

Figure 5-3 Alarms tab

Alarms Event Logs PM Dashboard										
Alarm ID	Severity	Host Name	Group	Source	Message	Date Updated	Ack. Status	Owner	Category	
130	CLEARED	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm/service=mm/unit=0	A critical NME application is missing. Missing: apvM	May-06-2021 03:27:742	N	admin	anMmeTrapAlarmMmeRm1303	F
129	CLEARED	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm/service=mm/unit=0	One or more VM types are not available. Unconfigure	May-06-2021 03:18:52:617	N	admin	anMmeTrapAlarmRm1300	F
140	CLEARED	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm0/service=mm0/unit=0	A configured VM is currently unlocked and disabled. I	May-06-2021 03:18:52:615	N	admin	anMmeTrapAlarmRm1301	F
156	CRITICAL	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm0/service=mm0	There are no peers on the Diameter Client that suppl	May-06-2021 03:18:16:000	N	admin	anMmeTrapAlarmMmeDcl302	F
155	MAJOR	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm0/hostid=pt66hss.affm	Diameter peer is unreachable. Diameter peer is dow	May-06-2021 03:18:16:000	N	admin	anMmeTrapAlarmMmeDcl300	F
154	MAJOR	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm0/hostid=pt66hss.affm	Diameter peer is unreachable. Diameter peer is dow	May-06-2021 03:18:16:000	N	admin	anMmeTrapAlarmMmeDcl300	F
153	CRITICAL	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm/vm-instance=pt66-mm	A network connectivity outage has been detected by	May-06-2021 03:18:07:000	N	admin	anMmeTrapAlarmPathMon1302	F
151	CRITICAL	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm0/vm-instance=pt66-mm	A network connectivity outage has been detected by	May-06-2021 03:18:07:000	N	admin	anMmeTrapAlarmPathMon1301	F
152	CRITICAL	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm0/service=mm0/unit=0	The virtual network interface connections do not me	May-06-2021 03:18:04:000	N	admin	anMmeTrapAlarmPathMon1304	F
141	MAJOR	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm0/service=mm0/unit=1	A configured VM is currently unlocked and disabled. I	May-06-2021 03:15:07:000	N	admin	anMmeTrapAlarmRm1301	F
139	MAJOR	NME-10.165.32.140-[node0]	pt66-8-arpc	/mm/service=mm/unit=1	A configured VM is currently unlocked and disabled. I	May-06-2021 03:15:07:000	N	admin	anMmeTrapAlarmRm1301	F

Alarms are listed with the following information:

- **Alarm ID** – Unique ID of the alarm
- **Severity** – Problem severity
- **Host Name** – Network element reporting the problem
- **Group** – Group having the fault
- **Source** – Network element component having the fault
- **Message** – Message/fault description obtained from the network element
- **Date Updated** – Date and time of last update of the alarm
- **Ack. Status** – Indicates whether or not the alarm is acknowledged
- **Owner** – Alarm owner
- **Category** – Alarm category, such as Temperature Threshold, CPU Threshold
- **Event Type** – (network element alarm/status poll/data collection)
- **Date Created** – Date and time when the alarm was raised
- **Age** – Numbers of days, hours, minutes, seconds, and milliseconds since the alarm was raised
- **Date Ack.** – Date and time the alarm was acknowledged, if applicable
- **Cleared Time** – Date and time the alarm was cleared, if applicable
- **Remedy** – Problem remedy suggested by the administrator
- **Additional Info** – Lists more information about the alarm
- **NE Alarm SeqNo** – NE Alarm Sequence Number
- **NE Alarm Idx** – NE Alarm Index

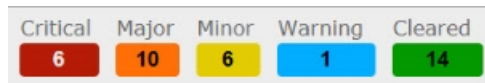
- **Trap OID** – Unique identifier for the device alarm type. Is listed as **N/A** for EMS Server alarms. To configure trap OID display length, see [Configuring Trap OID Display Length](#).

Filtering Alarms

Severity Filter

You can filter alarms based on alarm severity.

The alarm filter icons on the toolbar indicate the global alarm counts. They indicate the numbers and severity of all alarms on the network.



Select the alarm filter icon to view alarms of that severity.

PNSM lists **Alarms Details** for the severity you chose.

Figure 5-4 Filtered Alarms - Critical Severity

Alarm ID	Severity	Host Name	IP Address	Source	Message	Date Updated	Ack. Status	Owner	Alarm Name
156	CRITICAL	MME-10.165.32.140-[node0]	10.165.32.140	/mrm0/service+mrm0	There are no peers on the Diameter Client that supp	May-06-2021 03:18:16:000	N	admin	anMmeTrapAlarm
153	CRITICAL	MME-10.165.32.140-[node0]	10.165.32.140	/mrm0/vm-instance+pte6+mrm	A network connectivity outage has been detected by	May-06-2021 03:18:07:000	N	admin	anMmeTrapAlarm
151	CRITICAL	MME-10.165.32.140-[node0]	10.165.32.140	/mrm0/vm-instance+pte6+msi	A network connectivity outage has been detected by	May-06-2021 03:18:07:000	N	admin	anMmeTrapAlarm
152	CRITICAL	MME-10.165.32.140-[node0]	10.165.32.140	/mrm0/service+mrm0/unit=0	The virtual network interface connections do not mat	May-06-2021 03:18:04:000	N	admin	anMmeTrapAlarm
148	CRITICAL	SAEGW-10.165.32.143-[node0]	10.165.32.143	/geographicRedundancy/group	Geo-redundancy link is down.	May-05-2021 22:18:06:599	N	admin	anGeoRedundan
146	CRITICAL	SAEGW-10.165.32.163-[node0]	10.165.32.163	/cluster(8)/MCC	Approaching or crossed licensed service capacity en	May-05-2021 22:17:30:628	N	admin	anLicenseCapaci
137	CRITICAL	SAEGW-10.165.32.163-[node0]	10.165.32.163	/cluster(8)/node[7]	Communication over the data fabric is down.	May-05-2021 21:47:47:294	N	admin	anDataFabricOps
135	CRITICAL	SAEGW-10.165.32.163-[node0]	10.165.32.163	/cluster(8)/MCC	Approaching or crossed licensed service capacity en	May-05-2021 21:37:18:254	N	admin	anLicenseCapaci
127	CRITICAL	SAEGW-10.165.32.143-[node0]	10.165.32.143	/cluster(8)/node[7]	Communication over the data fabric is down.	May-05-2021 21:37:00:594	N	admin	anDataFabricOps
125	CRITICAL	SAEGW-10.165.32.143-[node0]	10.165.32.143	/cluster(8)/node[7]	Communication over the data fabric is down.	May-05-2021 21:37:00:594	N	admin	anDataFabricOps

Custom Filter

You can configure a specific Alarm Filter to view alarms based on filter conditions.

For instructions, see [Creating Filters](#).

Quick Filter

You can use the Quick Filter to view alarms based on a specific parameter and value.

For instructions, see [Using Quick Filters](#).

Viewing/Editing Alarm Details

You can view and edit an alarm's details. You can:

- Check the alarm's associated event logs for the detailed history of an alarm, such as how many times the same alarm has been raised.
- Change the owner for the alarm

- Acknowledge the alarm (if you are the alarm owner)
- Annotate the alarm by providing comments

PNSM saves these changes to maintain a history of changes.

To view/edit alarm details:

1. Select the **Alarms** tab.
2. Use any of the following methods to view/edit an alarm's details:

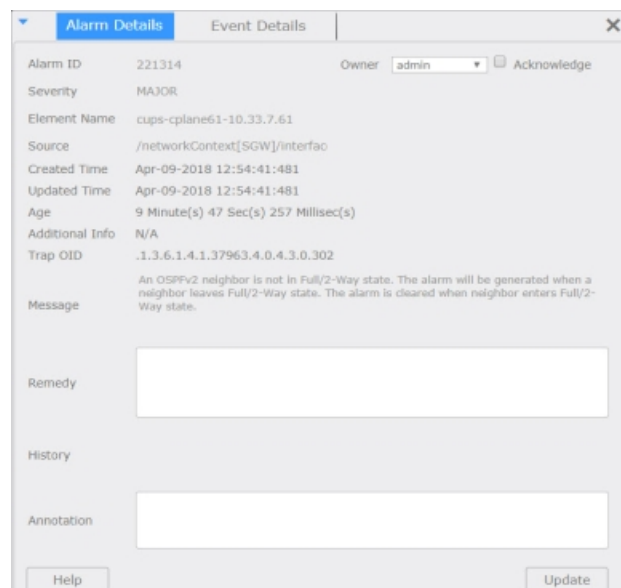
Toolbar: Select the alarm and select the **Alarm Details**  icon.

Pop-up menu: Right-click the alarm and select **Alarm Details**.

Or: Double-click the alarm.

PNSM displays the **Alarm Details** window.

Figure 5-5 Alarm Details Window



3. To edit the alarm, make the necessary changes and select **Update**.
 - **Owner** – You can change the field to assign the alarm to a new owner.
 - **Acknowledge** – If you are the alarm owner and the alarm has not been cleared, you can mark this checkbox to acknowledge the alarm. By acknowledging an alarm, you notify other Operators that someone has seen the alarm. Such information helps other Operators know what network issues are being addressed.
 - **Remedy** – You can enter a remedy
 - **Annotation** – You can enter an annotation.

PNSM updates the alarm information accordingly, including the **Owner**, **Ack. Status**, **Ack. Time**, **Remedy**, and/or **Annotation** fields.

4. Select the **Event Details** tab to view the Events of the alarm. By default, this tab lists event logs raised 15 minutes before and after the occurrence of each alarm log. It can help to debug the root cause of the alarm.

Assigning Alarm Owners

If you have sufficient privileges, you can assign an alarm. By assigning an alarm, you notify the Operator of the network issue to be addressed.

To assign an alarm:

1. Select the **Alarms** tab.
2. Use any of the following methods to assign an alarm:
 - Toolbar:** Select the alarm and select the **Assign**  icon.
 - Pop-up menu:** Right-click the alarm and select **Assign**.
3. In the **Assign To** field, select the required owner.
4. Select **Assign**.

PNSM updates the **Owner** field accordingly.

Acknowledging Alarms

If you are the alarm owner, you can acknowledge an alarm. By acknowledging an alarm, you notify other Operators that someone has seen the alarm. Such information helps other Operators know what network issues are being addressed.

***Note:** You cannot acknowledge an alarm that is cleared.*

To acknowledge an alarm:

1. Select the **Alarms** tab.
2. Use any of the following methods to acknowledge an alarm:
 - Toolbar:** Select the alarm then select the **Acknowledge**  icon.
 - Pop-up menu:** Right-click the alarm and select **Acknowledge**.

PNSM updates the alarm information accordingly, including the **Ack. Status** and **Ack. Time** fields.

Clearing Alarms

If you have sufficient privileges, you can clear alarms that have been acknowledged:

- When you clear an alarm generated by the PNSM server, it is automatically cleared.
- When you clear an NE alarm, PNSM forwards the clear request to the network element. If the alarm is cleared by network element, PNSM receives the clear trap that clears the alarm.

- Once an alarm is cleared, the clear event is added to the event log and the alarm is updated to show the cleared state.

To clear an alarm:

1. Select the **Alarms** tab.
2. Use any of the following methods to clear an alarm:

Toolbar: Select the alarm then the **Clear**  icon.

Pop-up menu: Right-click the alarm and select **Clear**.

3. At the confirmation prompt, select **OK**.

In the **Annotation** window, enter the reason for clearing the alarm (required). This annotation is sent to the MCC for inclusion in the actual clear trap.

4. Select **Save**.

PNSM updates the alarm **Severity** to Cleared and the severity color to green.

PNSM saves any annotation messages to maintain a history of changes. They are listed in the History section of the **Alarm Details** window.

Use any of the following methods to access the **Alarm Details** window:

Toolbar: Select the alarm then the **Alarm Details**  icon.

Pop-up menu: Right-click the alarm and select **Alarm Details**.

Or: Double-click the alarm.

Showing/Hiding Cleared Alarms

You can show or hide all the cleared alarms.

To hide cleared alarms:

1. Select the **Alarms** tab.
2. Select the **Hide Cleared Alarms**  icon.

PNSM updates the list of alarms (Cleared alarms are no longer shown).

To show cleared alarms:

1. Select the **Alarms** tab.
2. Select the **Show Cleared Alarms**  icon.

PNSM updates the list of alarms (Cleared alarms are shown).

Purging Alarms

PNSM supports an archive and purge mechanism whereby administrators can configure the Purge Interval for purging cleared alarms. PNSM uses purge settings to periodically purge the following alarms:

- Cleared alarms, regardless of whether the alarms have associated event logs
- Any event log mappings associated with the alarms

However, PNSM *does not* purge the associated event logs.

The purge action archives the alarms on the PNSM server in the following directory before removing them from database:

`/opt/Affirmed/NMS/server/ems/archive`

You can specify the following purge settings:

- **Maximum Number of Alarms to Retain:** When the number of cleared alarms increases beyond a specified limit (disabled by default)
- **Purge Alarms Older Than:** PNSM purges cleared alarms older than a specified number of days (default 30 days)

PNSM purges alarms every 30 minutes.

For details, see [Configuring Alarm Purge Settings](#).

In addition to these settings, you can manually trigger purging all cleared alarms. For details, see [Purging All Cleared Alarms](#).

Note: The maximum limit on purging alarms per run is 100k.

Configuring Alarm Purge Settings

To specify purge settings:

1. Select the **Settings**  icon on the toolbar and select **Fault => Alarms => Purge Settings**.
2. To purge alarms based on the maximum number to retain, mark the **Maximum no. of Alarms to retain** checkbox and specify the maximum number.
To purge alarms based on age, mark the **Purge alarms older than** checkbox and specify the number of days.
3. Select **OK**.

Purging All Cleared Alarms

You can purge all the cleared alarms regardless of alarm purge settings:

To purge cleared alarms:

1. Select the **Settings**  icon on the toolbar and select **Fault => Alarms => Purge All**.

2. At the confirmation prompt, select **OK**.

PNSM updates the list of alarms (Cleared alarms are no longer shown).

For instructions on purging event logs, see [Events](#).

Deleting Alarms

If you have sufficient privileges, you can delete alarms generated by the PNSM server that have been cleared. You cannot delete alarms generated by a network element until it has been removed from the network.

To delete an alarm:

1. Select the **Alarms** tab.
2. Use any of the following methods to delete an alarm:

Toolbar: Select the alarm then the **Delete**  icon.

Pop-up menu: Right-click the alarm and select **Delete**.

3. At the confirmation prompt, select **OK**.

PNSM updates the list of alarms (Deleted alarms are no longer shown).

Forwarding Alarms

You can forward alarms to a trap listener or to an e-mail listener.

For more information about the format of traps sent to northbound interfaces, such as to OSS systems, see [Interfacing with Northbound Interfaces](#).

To forward an alarm:

1. Select the **Alarms** tab.
2. Use any of the following methods to forward an alarm:

Toolbar: Select the alarm then select the **Alarm Forwarding**  icon.

Pop-up menu: Right-click the alarm and select **Alarm Forwarding**.

PNSM displays the **Alarm Forwarding** window.

3. To forward the alarm to a trap listener:
 - a. Select the **Trap** tab.
 - b. Select a trap listener from the list.

Note: To add a trap listener, select **Trap Config**. See [Configuring SNMP Alarm Forwarding for instructions](#).

To forward the alarm to an e-mail listener:

- a. Select the **Email** tab.
- b. Select an e-mail listener from the list.

***Note:** To add an e-mail listener, select **Email Config**. See [Configuring Alarm E-mail Notification](#) for instructions.*

4. Select **Send**.

Streaming Alarms to syslog

You can configure PNSM to stream alarms to a remote syslog server.

The syslog file is written in /var/log/affirmed.log.

To configure syslog on PNSM:

1. Shutdown PNSM server:

```
cd /opt/Affirmed/NMS/bin
./emsstop
```

2. Edit the following file:

```
/opt/Affirmed/NMS/server/ems/conf/ems-jboss-standalone.xml
```

3. Make the following changes.

- a. Add the following custom-handler under xml path:

```
"/server/profile/ <subsystem
xmlns="urn:jboss:domain:logging:2.0">" just before handler
for <size-rotating-file-handler name="SERVER_FILE">
<custom-handler name="SYSLOG"
class="org.apache.log4j.net.SyslogAppender"
module="org.apache.log4j">
    <level name="INFO"/>
    <formatter>
        <pattern-formatter pattern="%d %-5p [%c] %x -
%m%n"/>
    </formatter>
    <properties>
        <property name="Facility" value="LOCAL1"/>
        <property name="SyslogHost"
value="localhost:5140"/>
    </properties>
</custom-handler>
```

- b. Add the following logger:

```

"/server/profile/ <subsystem
xmlns="urn:jboss:domain:logging:2.0">" just after logger
using <handler name="SERVER_FILE"/> in <logger
category="org.jboss" use-parent-handlers="false">
    <logger category="syslogOutput" use-parent-
handlers="false">
        <level name="INFO"/>
        <handlers>
            <handler name="SYSLOG"/>
        </handlers>
    </logger>

```

- c. The syslogOutput includes only audit trail and alarms. If you need to stream any output in vm.log or other PNSM log files to syslog, add the SYSLOG handler to those loggers.
4. Save the file.
5. Start the PNSM server:

```

cd /opt/Affirmed/NMS/bin
./starttems.sh

```

Using rsyslog Server

This section describes how to install and use the rsyslog server.

To install and configure rsyslog on PNSM:

1. Obtain the rsyslog.repo file available at the following web site and place it in the /etc/yum.repos.d/ directory:

<http://rpms.adiscon.com>

2. Issue the following command to add the package:

```
yum update rsyslog
```

```
Loaded plugins: product-id, security, subscription-manager
```

```
This system is not registered to Red Hat Subscription
Management. You can use subscription-manager to register.
```

```
rsyslog_v5 | 2.5 kB 00:00
```

```
rsyslog_v5/primary_db | 24 kB 00:00
```

```
Setting up Update Process
```

```
Package(s) rsyslog available, but not installed.
```

```
No Packages marked for Update
```

3. Issue the following command to install the package:

```
yum install rsyslog
```

4. When the installation completes, edit the following file:

/etc/rsyslog.conf

5. Uncomment the following section to receive UDP syslog messages. Set the UDPServerRun parameter to the appropriate port. **Note:** The firewall must allow UDP on this port.

```
# Provides UDP syslog reception
$ModLoad imudp.so
$UDPServerRun 5140
# Provides UDP syslog reception
# for parameters see http://www.rsyslog.com/doc/imudp.html
module(load="imudp") # needs to be done just once
input(type="imudp" port="5140")
```

6. Add the following line to print the sender IP Address in the log. **Note:** Enter the entire string on a single line.

```
#### GLOBAL DIRECTIVES ####
$template EmsFileFormat,"%TIMESTAMP% [%FROMHOST%]
%syslogtag%%msg::: sp-if-no-1st-sp%%msg:::drop-last-1f%\n"
```

7. Add the following line to log into the desired log file using the template specified in [step 6](#) . You can specify any path and filename:

```
LOCAL1.* /var/log/affirmed.log;EmsFileFormat
```

8. Save the rsyslog.conf file.
9. Issue the following command to restart the rsyslog service:

```
service rsyslog restart
```

```
Shutting down system logger:      [ OK ]
Starting system logger:           [ OK ]
```

Viewing Alarm Statistics

PNSM provides pie charts of the alarm statistics for a specified time period. The charts show percentage of alarms by:

- Severity
- Acknowledged/Unacknowledged
- Owner
- Network Element

To view alarm statistics:

1. Select the **Alarms** tab.

2. Select the **Alarm Stats**  icon.
PNSM displays the pie charts of alarm statistics.
3. If desired, change the data time period.

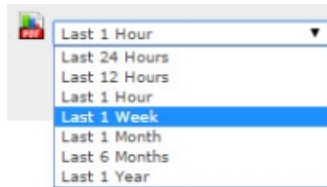
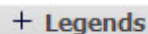
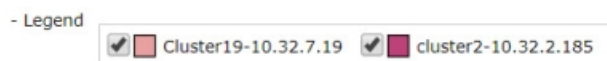


Figure 5-6 Alarm Stats

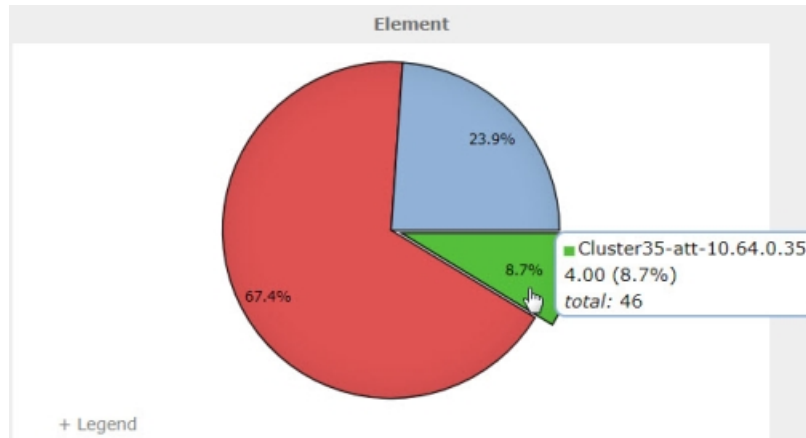


4. To view the chart legend, select the **Legends** icon.  If necessary, use the scrollbar to display the legend.

Mark/unmark the checkboxes in the legend to select/deselect instances to show/hide the respective data plots in the chart.



5. To view a single point in the graph, point the cursor at a point. PNSM opens a pop-up window that lists the data for that point.



6. To export the charts, select the **Export as PDF**  icon.
PNSM displays the **Download Files** window.
7. Right-click **Download PDF File** and select **Save Link As**.
8. Specify where you want the file saved and select **Save**.

To close the window, select the **Close**  icon.

Exporting Alarms

You can export alarms to a CSV/PDF/XLS/XML/HTML file.

To export alarms:

1. Select the **Alarms** tab.
2. If desired, use the Filter Manager or Quick Filter to filter the data displayed in the window.
3. Select the record(s) to be exported. To select multiple records, Shift-click or Control-click the records in the list. To export the Whole Data or Current View, do not select any records.
4. Select the **Export Alarms**  icon.
PNSM displays the Configure Export Parameters window.
5. Configure the parameters as follows:
 - **File Type:** Select the required file format (CSV, PDF, XLS, XML, HTML).
 - **Column Configuration:** Select the columns you want exported (Visible Column or All Columns)
 - **Export Type:** Select the export type (Whole Data, Current View, Selected Data)
 - **Use Current Filter:** Mark the checkbox to restrict data based on the existing filter conditions.
6. Select **Apply**.
PNSM displays the **Download Files** window.

7. Select **Download Exported File**.
8. From the **Download Files** window, select **Close**.

Printing Alarms

You can print alarms.

To print alarms:

1. Select the **Alarms** tab.
2. From the **Alarms** tab, select the **Print Alarms**  icon.
PNSM displays the **Print Preview** window.
3. Select **Print**.
4. Select the desired printer.
5. Select **Print**.

Generating an Alarm Report

To generate an alarm report:

1. Select the **Settings**  icon on the toolbar and select **Fault => Alarms => Generate Reports**.
PNSM displays the **Alarm Report Generator**.
2. To generate specific conditions for the report:
 - a. Select the **New Filter**  icon.
 - b. Enter the Filter Name and select **Add**.
 - c. Select the filter then select the **Add Condition**  icon.
 - d. Complete the Conditions fields by selecting appropriate values for **Parameter**, **Operator**, **Value**, and **Condition**.
 - e. Select **Apply**.
3. If the report needs a combination of AND and OR conditions to achieve the desired result:
 - a. Select the **Create AND group**  icon (or **Create OR Group**  icon).
 - b. Select the **Add Condition** icon  and complete the Conditions for the group.
 - c. Select **Apply**.
4. Select **Download Report** to download the report.
PNSM displays the **Download Alarm Report File** window.
5. Select **Download Exported File**.
6. From the **Download Alarm Report File** window, select **Close**.
7. From the **Alarm Report Generator** window, select **Cancel**.

Configuring SNMP Alarm Forwarding

PNSM supports forwarding alarms to other specific management systems as either SNMP TRAPs or as SNMP INFORMs.

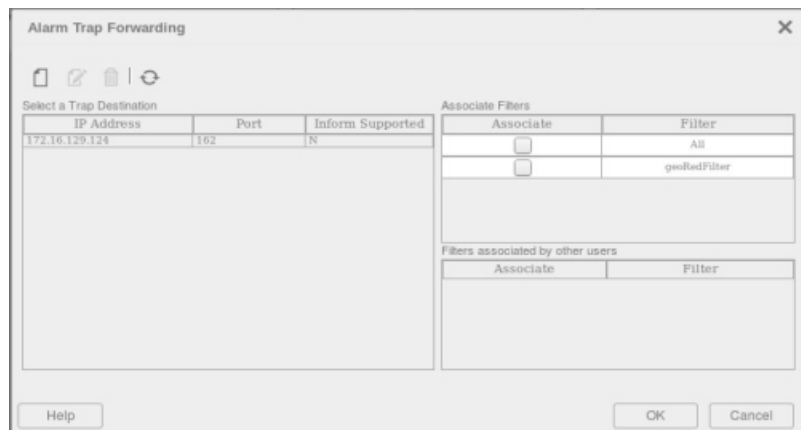
Note: To configure SNMP alarm forwarding, you must have the *Fault > Alarms > Manage > Manage Alarm Trap Forwarding* security group permission. For instructions, see *Specifying Permissions by Security Group*.

To configure SNMP alarm forwarding:

1. Select the **Settings**  icon on the toolbar and select **Fault => Manage => Trap Forwarding**.

PNSM displays the **Alarm Trap Forwarding** window. The left pane lists the destination addresses and their ports for which the alarm needs to be sent and lists whether the destination supports INFORMs. The right pane lists the alarm filters.

Figure 5-7 Alarm Trap Forwarding



Note: The *Inform Supported* column is shown only when the global flag *inform.enable=true* is set in the *NorthboundTrapConfig.properties* file. See [NorthboundTrapConfig.properties File](#).

2. To add a SNMP Listener, select the **Add**  icon.
PNSM displays the **Create New SNMP Listener** window.

Figure 5-8 Create New SNMP Listener

- a. Specify the **IP Version**, **IP Address**, and **Port**.
- b. To forward alarms as SNMP INFORMs, mark the **Inform Supported** checkbox.

Note: The *Inform Supported* field is shown only when the global flag `inform.enable=true` is set in the `NorthboundTrapConfig.properties` file. See [NorthboundTrapConfig.properties File](#).

- c. Specify the **SNMP Version** – SNMPv2 or SNMPv3.
If you specify SNMPv3, specify the following parameters:

- **User Name** – Security name
- **Security Level** – NOAUTH_NOPRIV, AUTH_NOPRIV, AUTH_PRIV

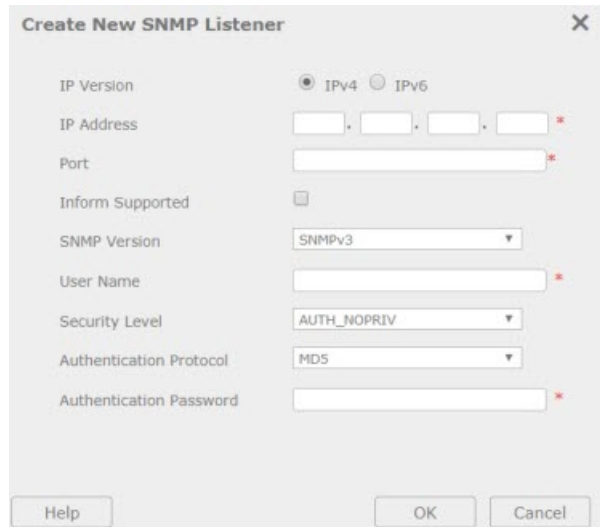
The **Security Level** value you specify determines what additional parameters to set:

Figure 5-9 NOAUTH_NOPRIV Settings

When you specify AUTH_NOPRIV, also specify:

- Authentication Protocol - MD5 or SHA
- Authentication Password - Secret Key

Figure 5-10 AUTH_NOPRIV Settings



The image shows a dialog box titled "Create New SNMP Listener" with a close button (X) in the top right corner. The dialog contains the following fields and controls:

- IP Version:** Radio buttons for IPv4 (selected) and IPv6.
- IP Address:** Four text input fields separated by dots, with a red asterisk to the right.
- Port:** A single text input field with a red asterisk to the right.
- Inform Supported:** A checkbox.
- SNMP Version:** A dropdown menu showing "SNMPv3".
- User Name:** A text input field with a red asterisk to the right.
- Security Level:** A dropdown menu showing "AUTH_NOPRIV".
- Authentication Protocol:** A dropdown menu showing "MD5".
- Authentication Password:** A text input field with a red asterisk to the right.

At the bottom of the dialog are three buttons: "Help", "OK", and "Cancel".

When you specify AUTH_PRIV, also specify:

- Authentication Protocol - MD5 or SHA
- Authentication Password - Secret Key
- Privacy Protocol - DES or AES
- Privacy Password - Privacy Key

Figure 5-11 AUTH_PRIV Settings



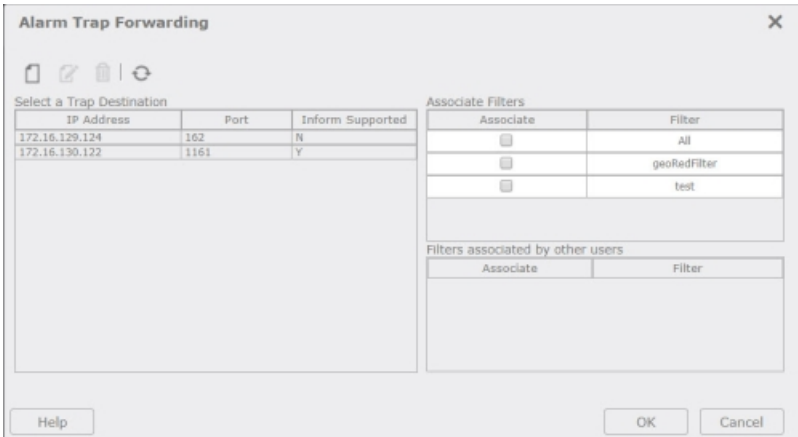
The 'Create New SNMP Listener' dialog box contains the following fields and controls:

- IP Version:** Radio buttons for IPv4 (selected) and IPv6.
- IP Address:** Four text input fields separated by dots, with a red asterisk on the right.
- Port:** A single text input field with a red asterisk on the right.
- Inform Supported:** A checkbox.
- SNMP Version:** A dropdown menu currently showing 'SNMPv3'.
- User Name:** A text input field with a red asterisk on the right.
- Security Level:** A dropdown menu currently showing 'AUTH_PRIV'.
- Authentication Protocol:** A dropdown menu currently showing 'MD5'.
- Authentication Password:** A text input field with a red asterisk on the right.
- Privacy Protocol:** A dropdown menu currently showing 'DES'.
- Privacy Password:** A text input field with a red asterisk on the right.
- Buttons:** 'Help', 'OK', and 'Cancel' at the bottom.

3. Select **OK** to commit the data. The newly created SNMP Listener is listed in the Alarm Trap Forwarding window.

A maximum of two Trap Destinations at a time can have the **Inform Supported** checkbox marked. The number of Trap Destinations to receive INFORMs is configurable in the `NorthboundTrapConfig.properties` file. See [NorthboundTrapConfig.properties File](#).

Figure 5-12 Alarm Trap Forwarding



The 'Alarm Trap Forwarding' dialog box is divided into two main panes:

- Select a Trap Destination:** A table with columns 'IP Address', 'Port', and 'Inform Supported'.

IP Address	Port	Inform Supported
172.16.129.124	162	N
172.16.130.122	1161	Y
- Associate Filters:** A table with columns 'Associate' and 'Filter'.

Associate	Filter
☐	All
☐	geofedFilter
☐	test
- Filters associated by other users:** A table with columns 'Associate' and 'Filter' (currently empty).
- Buttons:** 'Help', 'OK', and 'Cancel' at the bottom.

4. To associate the trap listener with alarm filters, select the Trap Destination in the left pane and mark the required check boxes in the right pane.
5. Select **OK** to associate the trap listener with alarm filters.

NorthboundTrapConfig.properties File

The NorthboundTrapConfig.properties file contains configurable attributes to control handling of SNMP INFORMs. If you change any property value, restart the PNSM server for the new value to take effect.

NorthboundTrapConfig.properties is in the directory:

/opt/Affirmed/NMS/server/ems/resources

The configurable attributes are as follows:

inform.enable=true

Global flag that enables INFORM support, including:

- Sending INFORM heartbeat to northbound systems
- Sending SNMP INFORMs
- Configuring INFORM for trap destination in PNSM

inform.timeout=10000

SNMP INFORM timeout. Default: 10 seconds.

inform.noOfRetries=3

SNMP INFORM number of retries. Default: 3.

inform.queueSize=500

SNMP INFORM queue size. The maximum limit of queue size is based on the JVM Heap Memory (any value configured greater than this value is overridden).

inform.noOfClientSize.max=2

SNMP INFORM number of northbound clients. Default: 2. Valid Range: 1 - 2.

If you configure a value out of the valid range, PNSM uses the default (2).

Decreasing the number of INFORM clients does not affect the previously configured INFORM clients. If two clients are already configured, they continue to work as INFORM supported. However, if one INFORM client is removed, PNSM does not allow configuration of a new second client.

heartBeat.isSupported=true

Global flag that enables heartbeat mechanism that checks the availability of both PNSM and the northbound system.

heartBeat.timeInterval=30000

Heartbeat polling time interval to update server cache used for INFORM queue processing
Default: 30 seconds

NMSSNMPAgent.properties File

The NorthboundTrapConfig.properties file contains configurable attributes to control SNMP Agent details.

NMSSNMPAgent.properties is in the directory:

/opt/Affirmed/NMS/server/ems/resources

The SNMP Agent port can be configured in the following field in the NMSSNMPAgent.properties file.

SNMP_AGENT_UDPPORT=1161

Default: 1161

Make sure that the port is open for external northbound access in the firewall configuration.

Configuring Alarm E-mail Notification

If an alarm requires generation of an e-mail alert, you can configure an e-mail server for alarm e-mail notification.

PNSM can be configured to either send all the alarms or alarms that match pre-defined match criteria. To be notified for only specific cases, first create an alarm filter with the match criteria and use it for configuring the e-mail notification. For example, create a filter with the match criteria of owner=<user ID> to forward all the alarms assigned to that user's e-mail. For instructions on creating an alarm filter, see [Filter Management](#).

The following alarm details are sent as part of the e-mail sent to the user:

- Alarm ID
- NE
- Source (shelf/slot/card/port)
- Severity
- Alarm Raised by (generated by NE/by PNSM)
- Created time
- Last updated time
- Owner assigned
- Acknowledgment status
- Acknowledged time (if acknowledged)

- Alarm description
- Remedy (If available)

To configure E-mail Notification:

1. Select the **Settings**  icon on the toolbar and select **Fault => Manage => Email Notification**.
2. If the mail server is not already configured:
 - a. At the confirmation prompt asking if you want to configure the Mail server, select **OK**.
 - b. Configure the Mail Server details:
 - **Primary SMTP Host Name** – Host name of the Simple Mail Transfer Protocol (SMTP) server.
 - **Secondary SMTP Host Name** – Another host name of the SMTP server, used in case of any failure in the Primary SMTP Host name.
 - **SMTP Port** – Port ID for the SMTP server:

If you specify port 25, PNSM creates an unauthenticated session with the mail server, so that no user or password should be configured when using that port.

If you need PNSM to create an authenticated session, specify a port other than port 25.

 - **SMTP User** – User name of the SMTP server.
 - **SMTP Password** – Password of the SMTP server.
 - **Confirm Password** – Confirmation of the entered password.
 - **Sender Name** – User name used for sending e-mail by the PNSM server.
 - **Sender Mail ID** – Mail ID used for sending e-mail by the PNSM server.
 - c. Select **Save**.

3. To add an E-mail Listener, select the **Add**  icon.
 To modify an existing E-mail Listener, select the E-mail Listener you want to modify then select the **Edit**  icon.
 PNSM displays the **Add (or Modify) Email Listener** window.
 - a. Specify the **Email address** and **Details**.
 - b. Select **OK** to commit the data. The newly created Email Recipient is listed in the Alarm Email Notification window.
4. To associate the Email Recipient with alarm filters, select the Email Recipient in the left pane.
5. To select the filters that need to be associated with the Email Recipient, select the required check boxes in the right pane.
6. Select **OK** to associate the Email Recipient with alarm filters.

Events

Viewing Event Logs

When a trap is generated as a consequence of a fault, by default an Event Log is generated, which is then consolidated to an alarm.

To view Event Logs:

1. Right-click on the Alarm for which you want to view the event log.
2. Select **Event Logs**.

Figure 5-13 Event Logs Details dialog

[illegible]

Events are listed with the following information:

- **Event Log ID** – Unique across all the network elements
- **Severity** - Informational for events and specific severity for alarms generated by the NE and PNSM
- **Network Element**
- **IP Address**
- **Source** – Source having the event
- **Message** – Message/fault description obtained from the network element
- **Event Time** – Date and time of the event
- **Age** – Numbers of days, hours, minutes, seconds, and milliseconds since the event occurred
- **Category** – Event category

- **Event Type** – Category of event, such as data collection based
- **Remedy** – Remedy for the event, if available

In the case of a large number of log entries, use the page navigational bar at the bottom of the window to page through the entries.

Troubleshooting Private Network Issues

The Private Network Service Manager application allows Admin users to access troubleshooting information directly from the **Tools** menu.

Figure 5-14 Troubleshooting Menu

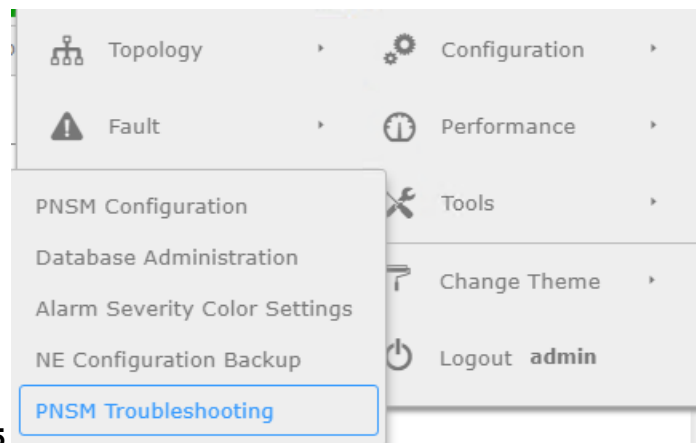
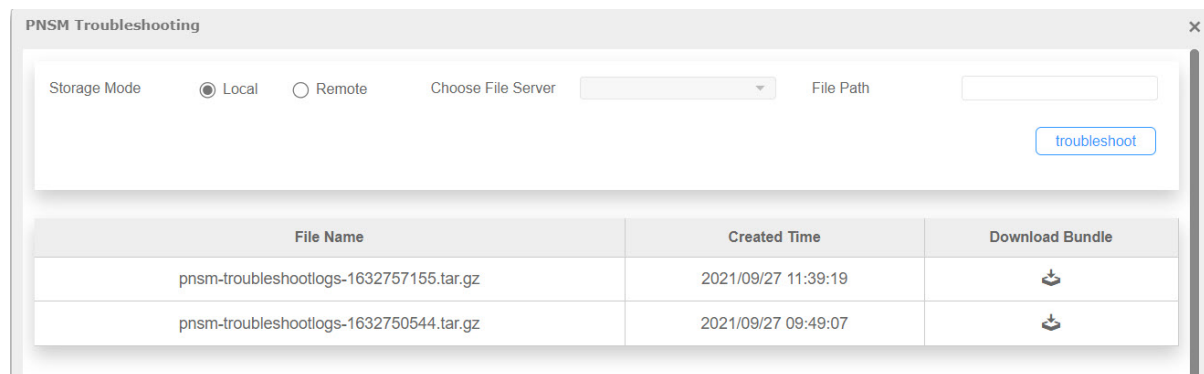


Figure 5-15

This information may be stored locally and downloaded directly in the application, or stored remotely and downloaded from the SFTP server.

Figure 5-16 PNSM Troubleshooting Window



To retrieve troubleshooting information:

1. Select **Tools** --> **PNSM Troubleshooting** from the **Settings** () menu.
2. Define the storage location of the troubleshooting file by selecting the Local or Remote radio button.

- If you select **Local**, the troubleshooting bundle will be stored on the local VM path.
 - If you select **Remote**, then select the **File Server**, and enter the **File Path** where the bundle will be stored.
3. Select the **Troubleshoot** button.
- Any troubleshooting bundles stored locally are listed, and are available for download.
 - Troubleshooting bundles stored remotely display the file name, stored location, and creation time. You must then navigate to the SFTP server to download these bundles.

***Note:** Service Manager stores the three most recent troubleshooting bundles; each is displayed and may be downloaded directly from the PNSM Troubleshooting screen.*

Enterprise Dashboard

The Enterprise Dashboard presents a graphical representation of statistics for defined subgroups over a specified period of time. By default, 24 hours of historical information for average task utilization is displayed. You can choose to view alternate time periods, and can define how this information is displayed. Additionally you can choose to export this information by creating a PDF or CSV file, or you can take a snapshot of the chart.

PM Dashboard

The PM Dashboard displays the Key Performance Indicators (KPIs) collected during the previous 24 hours. You can modify and configure this display using Performance Reports. The following display by default:

- CPU Usage (%) - Last 24 Hours
- Memory Usage (%) - Last 24 Hours
- Disk Usage (%) - Last 24 Hours

For a network element, PNSM lists preconfigured Dashboards in the toolbar of the Dashboard.

Configuring the PM Dashboard

You can configure the Dashboard to display multiple charts with different KPIs. The appropriate statistics to monitor depend on how the system is deployed. And, you can configure to periodically export the Dashboard as a report.

A single Dashboard report supports up to four graphs.

To configure the Dashboard:

1. Select the **PM Dashboard** tab.
2. To open the tree display, select the **Toggle Resource Selection View**  icon (next to the **Settings** icon) to expand the left pane.
3. Select the Global Group, Group, or device for which you want to display KPIs.
4. Add a new Dashboard or modify an existing Dashboard:
 - To add a new Dashboard, select the **New Dashboard**  icon on the tool bar of the Dashboard.
 - To modify an existing Dashboard, select the Dashboard you want to modify from the toolbar of the Dashboard. Then, select the **Edit Dashboard**  icon on the tool bar of the Dashboard.

Service Manager displays the **Dashboard Editor** window.

5. Configure the parameters as follows:

Table 5-1 Dashboard Parameters

Parameter	Description
Title	Enter a name for the Dashboard.
Statistics Type	Select the statistics type: ▪ Historical ▪ Realtime
Filter	Select the Filter  icon to set up conditions to filter records.
Set as default	Mark the checkbox to set this Dashboard as the default.
Layout Type	Select the screen layout for the data.
<i>Properties when Statistics Type is set to Realtime:</i>	

Parameter	Description
Metrics	Select the Metric Configuration icon.  to specify the metric to display. In the Metric Selection window: 1. To filter the tables and /or metrics displayed, enter a value in the Filter by Metric Name field and select the Filter Metric  icon. 2. In Metric Tables, select the data collection group name that contains the metrics you want to display. 3. In Metrics, select the metric. 4. Choose Select. 5. To clear the Filter by Metric Name field, select the Clear Filter  icon.
Chart	Select the chart type: ▪ Pie ▪ Bar ▪ Line
Refresh Interval	Select the time interval at which the chart is to be refreshed, in seconds.
<i>Properties when Statistics Type is set to Historical:</i>	
Metrics	Select the Metric Configuration  icon to specify the metric to display. In the Metric Selection window: 1. To filter the tables and /or metrics displayed, enter a value in the Filter by Metric Name field and select the Filter Metric  icon. 2. In Metric Tables, select the data collection group name that contains the metrics you want to display. 3. In Metrics, select the metric. 4. Choose Select. 5. To clear the Filter by Metric Name field, select the Clear Filter  icon.
Chart	Select the chart type: ▪ Pie ▪ Bar ▪ Line
Interval	Select the historical time interval you want for the chart.

Parameter	Description
Show	Specify how many devices you want to include: ▪ Show All – Shows statistics for all devices (by default, up to 50 instances) ▪ Show Top <i>x</i> – Shows statistics for the highest specified number of devices (up to 20 instances) ▪ Show Bottom <i>x</i> – Shows lowest statistics for the lowest specified number of devices (up to 20 instances)
Aggregate Type	Select the data aggregate type to control the aggregation function used to aggregate data over Resource or Resource Groups: ▪ Avg – Uses average function to aggregate for the resource or resource group ▪ Min – Uses minimum function to aggregate for the resource or resource group ▪ Max – Uses maximum function to aggregate for the resource or resource group ▪ Sum – Uses the sum function to aggregate for the resource or resource group
Filter By	Select the filter to filter the data displayed. ▪ All – Shows all statistics ▪ All Gateway (for QCI/ARP statistics only) – Show all gateway statistics ▪ All PGW (for QCI/ARP statistics only) – Shows statistics for gateway type PGW ▪ All SGW (for QCI/ARP statistics only) – Shows statistics for gateway type SGW

Parameter	Description
Group By	Select the grouping: ▪ Subgroup ▪ Resource ▪ Instance ▪ Category (use for the following metrics:) - DPI - % Protocol Network Usage - DPI - Protocol Network Usage (Mbps) ▪ Application (use for the following metrics:) - DPI - % Application Network Usage - DPI - Application Network Usage (Mbps) ▪ Protocol (use for the following metrics:) - DPI - % Protocol Network Usage - DPI - Protocol Network Usage (Mbps) ▪ Operating System (use for the following metrics:) - DPI - % OS Network Usage - DPI - OS Network Usage (Mbps) ▪ Gateway (use for the following metrics:) - QCI/ARP - Downlink Throughput(Mbps) - QCI/ARP - Uplink Throughput(Mbps) - QCI/ARP - Bearer Drop Rate - QCI/ARP - Handover Success Rate - QCI/ARP - Bearer Retainability Rate ▪ Apn (use for the following metrics:) - QCI/ARP APN - Downlink Throughput(Mbps) - QCI/ARP APN - Uplink Throughput(Mbps) - QCI/ARP APN - Bearer Drop Rate - QCI/ARP APN - Handover Success Rate - QCI/ARP APN - Bearer Retainability Rate ▪ QciArp (use for the following metrics:) - QCI/ARP - Downlink Throughput(Mbps) - QCI/ARP - Uplink Throughput(Mbps) - QCI/ARP - Bearer Drop Rate - QCI/ARP - Handover Success Rate - QCI/ARP - Bearer Retainability Rate - QCI/ARP APN - Downlink Throughput(Mbps) - QCI/ARP APN - Uplink Throughput(Mbps) - QCI/ARP APN - Bearer Drop Rate - QCI/ARP APN - Handover Success Rate - QCI/ARP APN - Bearer Retainability Rate

Parameter	Description
Enable Export Configuration	1. Mark the checkbox if you want to export the Dashboard as a daily report. 2. Select Next. 3. Enter a report title.
Apply this Export Configuration for	Specify to apply the Export configuration for NEs: ▪ To apply the export to all available NEs, select All. ▪ To apply the export for the NE for which the Dashboard is displayed, select Selected. ▪ To select a set of NE for which the export is applied, select Associate: 1. Select Next. 2. To select an NE, highlight it in the Available List and select the right-arrow to move it to the Assigned List. To remove a selected NE, highlight it in the Assigned List and select the left-arrow to move it to the Available List.

4. Select **Save**.

By default, Service Manager purges any file older than 30 days. To configure a different purge interval, see [Configuring Export Purge Interval](#).

If you marked the **Enable Export Configuration** checkbox to export the Dashboard, the export filename is `<Entity Name>_<Dashboard Title>_<Report Title>_<Date>.csv`. The file is exported to the Service Manager server, in the directory `/opt/Affirmed/NMS/server/ems/archive/<Date>/.`

Keep in mind that if the Dashboard you export has the **Show** setting set to Top x or Bottom x, the highest or lowest devices in the CSV file may differ from the devices shown when you run the Dashboard in real time. The difference results because the exported files contain data from the previous full day whereas the real time data uses data from the previous 24 hours. Depending on the actual time at which you view the real time Dashboard, the highest or lowest devices may differ between the export data and the real time Dashboard display.

Configuring Number of Data Points to Display in Dashboard Charts

You can configure Dashboard Reports to display between 50 and 150 data points in charts. To do so, modify the `PerformanceConfiguration.properties` in the directory:

`/opt/Affirmed/NMS/server/ems/resources`

The configurable attribute is:

```
# Randomize Window
```

```
ChartManager.instancecount.limit=50
```

In the case of PM Dashboard (Historical), the default value is 50. It can be set to any value from 50 to 150.

In the case of PM Dashboard (Realtime), the default value is 50. It can be set to any value from 50 to 150, but *only* for Historical charts with the Show All instance setting. For charts with the Show Top x and Show Bottom x instance settings, the value x can be set to a value up to 50.

Network Availability

The Network Availability dashboard provides graphical representations of the following parameters:

- Uptime % Meter: $\text{Uptime} / (\text{Uptime} + \text{Downtime}) * 100$
- Mean Time To Failure: Mean Uptime which equals (Sum of all Uptime/Number of Failures)
- Mean Time To Repair: Mean Downtime which equals (Sum of all Downtime/Number of Failures)
- Mean Time Between Failure: Mean Uptime + Mean Downtime which equals (Mean Time To Failure) + (Mean Time To Repair)

RAN (Radio Access Network) Information

The RAN (Radio Access Network) tab displays the RAN details pulled from the MME (Mobility Management Entity). This information is stored the Service Manager Database for each Private Network, and shows the current information for the MME Resources shown in the table. Select the **Refresh** button to re-synchronize the database to the MME.

Figure 5-17 RAN Info Tab

Private Network Detail	Element Details	Alarms	Event Logs	PM Dashboard	Network Availability	Upgrades	RAN Info	Subscriber Tracing
Overload Start Sent	Connection State	Cell Access Mode	Local IP Address1 Role	Local IP Address2 Role	Local Port	Connection Time	Network Name	
false	connected	open	*10.165.60.121	0.0.0.0	36412	0 days 0 hours 2 mins 34 secs	PLTES_2010_de	

***Note:** When Service Manager is configured with High Availability settings enabled, this table shows the RAN information for resources of the active HA node.*

Subscriber Tracing

The Subscriber Tracing feature is available for network elements, and provides the option to create, start, stop, download and delete subscriber tracing information for both MME and MCC.

Figure 5-18 Subscriber Tracing tab

Private Network Detail Element Details Alarms Event Logs PM Dashboard Network Availability Upgrades Subscriber Tracing					
					Create Delete
☐	Trace Name	State	Start Time	End Time	Action
☐	subTrace01_1631787826592	Running	2021/09/16 06:23:52		Refresh Details Delete
☐	sunTrace02_1631787888878	Stopped	2021/09/16 06:24:56	2021/09/16 06:26:32	Refresh Details Delete
☐	sunTrace02_1631788120090	Running	2021/09/16 06:28:47		Refresh Details Delete

Note: Subscriber Tracing Management is available only if all of the Network Resources are discovered in the Private Network Service. If any resource is not available, and error displays on the Subscriber Tracing Management tab. If both MME and Gateway Network elements are inactive or unreachable, the Subscriber Tracing Management window is disabled, allowing you to view only the existing traces.

Creating a Subscriber Tracing

To create a Subscriber Tracing:

1. On the Subscriber Tracing tab, select the **Create** button.

The Subscriber Tracing dialog appears:

Create Subscriber Tracing

Trace Name

Select Template

Save as Template

☐

MME

Target Type

imsi

Target Value

Session ID

Nms IP

0.0.0.0

Trace Depth

med

Trace Type

3gpp

Interfaces

MME Interfaces

all s10

Enb Interfaces

all uu

Sgw Interfaces

all gxc

Pgw Interfaces

all sgi

Sgsn Interfaces

all iu-ps

Ggsn Interfaces

all gn

Rnc Interfaces

all uu

Events

MME Events

all handover

Sgw Events

all bearer-act-mod-del

Pgw Events

all bearer-act-mod-del

Sgsn Events

all pdp-cbxt

Ggsn Events

pdp-cbxt

SAEGW

Create

2. Enter a unique **Trace Name**.

3. If you have created any templates in SM, you can choose **Select Template** to auto-populate the remaining fields.
4. If you manually enter the information, you can choose to save this configuration as a template for future use by selecting **Save as Template**. Then, provide a **Name** for this template.

The screenshot shows a dialog box titled "Create Subscriber Tracing". It has a close button (X) in the top right corner. Inside the dialog, there are four fields: "Trace Name" with the text "subTrace01", "Select Template" with a dropdown arrow, "Save as Template" with a checked checkbox, and "Template Name" with the text "subTrace01".

5. For MME tracing, provide a **Target Value** and **Session ID**.

Note: You can edit the values shown in the **Target Type**, **NMS IP**, **Trace Depth** and **Trace Type** fields as necessary.

6. Select the Interface(s) and Event(s) to include in the tracing, or select **ALL** to include all available Interfaces and Events.
7. For MCC tracing, provide a **Filter Value**.

Note: The **Filter** and **Sink** fields change based on the **Log Type** selected.

8. Create additional Filters and Sinks by selecting the + button in the Filter and Sink section of the dialog.

Note: You can create multiple filters with unique IMSI values to trace multiple IMSIs.

Create Subscriber Tracing

Pgw Interfaces: all, sgi | Sgsn Interfaces: all, iu-ps | Ggsn Interfaces: all, gn | Rnc Interfaces: all, uu

Events

MME Events: all, handover | Sgsn Events: all, pdp-cbxt | Sgw Events: all, bearer-act-mod-del | Ggsn Events: pdp-cbxt | Pgw Events: all, bearer-act-mod-del

SAEGW

Admin State: enabled | Log Type: debug

Filter

Filter Field: apn | Filter Value: | Exclude: ☐

Sink

Type: file | Display Field: timestamp | File Format: ascii | Display Type: default

Create

9. Select **Create**.

If the Network Element is inactive or unreachable:

- If both the MME and MCC Network Elements, then the corresponding section will not be rendered and a note displays.
- If MME Network Element, then you can create a Subscriber Tracing only for the MCC.
- If MCC Network Element, then you can create a Subscriber Tracing only for the MME.

If the tracing was successfully created:

- Defined configurations are applied to the Network Elements.
- The entry is added to the **Subscriber Tracing Management** table, with a status of **Running**.

Note: For HA Setups, only the Active/Primary MME status is considered. The MCC section will be rendered provided one MCC node is Active/Reachable.

Managing Subscriber Tracing

For each Subscriber Trace shown in the table, you can select to:

- **Start** - Displays only when subscriber trace is in a **Stopped** state. When you select **Start**, corresponding configurations are applied to the MME and MCC, and a new entry is added to the **Subscriber Tracing Management** table. This entry displays the **Trace Name** and **Start Time** with the corresponding time stamps for each.
- **Stop** - Displays only if subscriber trace is in a **Running** state. When you select **Stop**, the configurations are deleted from both MME and MCC, and the **Subscriber Tracing** entry shows the **End Time**.
- **View Details** - displays a dialog showing all subscriber trace data you provided.

Trace Name	State	Start Time
subTrace01_1631787826592		
sunTrace02_1631787888878		
sunTrace02_1631788120090		

Subscriber Tracing Data

```
{
  "traceName": "sunTrace02_1631787888878",
  "sessionTraceKey": "imsi#999990005000601",
  "traceTemplateID": 0,
  "saveAsTemplate": false,
  "mmeActiveAndReachable": true,
  "mccActiveAndReachable": true,
  "mmeEpcSessionTrace": {
    "targetType": "imsi",
    "targetValue": "999990005000601",
    "sessionID": "90271.000146",
    "nmsIP": "0.0.0.0",
    "traceDepth": "med",
    "traceType": "3gpp",
    "mmeInterfaces": [
      "all"
    ],
    "enbInterfaces": [
      "all"
    ]
  },
  "mccEventLogPolicy": {
    "policyName": "sunTrace02_1631787888878",
    "adminState": "enabled",
    "logType": "debug",
    "eventLogFilters": [
      {
        "filterName": "imsi",
        "filterValue": "999990005000601",
        "exclude": false
      }
    ]
  },
  "eventLogSinks": [
    {
      "sinkName": "sunTrace02_sink_0",
      "sinkType": "file",
      "fileName": "sunTrace02_1631787888878",
      "fileFormat": "ascii",
      "displayField": "time_stamp"
    }
  ]
}
```

- **Download** - Downloads the trace files of any trace in a **Running** or **Stopped** state to a compressed file containing both MME and MCC subscriber tracing logs.

- **View Status** - Displays dialog with the **Subscriber Trace** Action status.

Trace Name	State	Start Time	End Time
subTrace01_1631787826592	Running	2021/09/16 06:23:52	
sunTrace02_1631787888878	Stopped	2021/09/16 06:24:56	2021/09/16 06:26:32
subTrace02_1631788120090	Running	2021/09/16 06:28:47	

Subscriber Tracing Status			
Element	Action	Status	
MME_10.165.31.8-[node0]	Create / Start	Session Trace created/started successfully	
SAEGW_10.165.31.11-[node0]	Create / Start	Event Log Policy created/started successfully	
SAEGW_10.165.31.26-[node1]	Create / Start	Event Log Policy created/started successfully	
MME_10.165.31.8-[node0]	Delete / Stop	Session Trace deleted/stopped successfully	
SAEGW_10.165.31.11-[node0]	Delete / Stop	Event Log Policy deleted/stopped successfully	
SAEGW_10.165.31.26-[node1]	Delete / Stop	Event Log Policy deleted/stopped successfully	

Deleting Subscriber Tracing

The **Delete** option appears when you select one or more subscriber tracings.

***Note:** Only traces in a **Running** or **Stopped** state may be deleted.*

When you Delete a trace, all corresponding configurations are deleted from the MME and MCC Network Element, and all related entries are deleted from the Affirmed Private Network Service application.

Virtual Slicing Profile

This feature is currently in development.

Performance Reports

Service Manager provides reports on:

- **Historical** – Report on the data collected for a specified period. It is a view of the historical data for the performance statistic attributes configured in the data collection tasks for devices.
- **Real Time** – Live data collected report. It is a graphical view of the current data for the performance statistic attributes configured as a part of the data collection task.
- **Outage Report** – Report on the outage of network elements.

Configuring Data Type and Unit Conversion at the Global Level

The Administrator can configure the statistic data type at the global level. Service Manager applies the configured data type to Historical Reports and KPIs (displayed in Dashboard charts):

- **Delta Value** — Statistic values are plotted based on the following formula:

Delta value = current counter value - previous counter value

- Rate Value — Statistic values are plotted based on the following formula:

Rate value = Delta value / collection time interval in seconds

By default, the Rate Value is applied.

The Administrator can configure the unit conversion at the global level. Service Manager applies the configured unit conversion when formulating derived metrics KPIs (displayed in Dashboard charts):

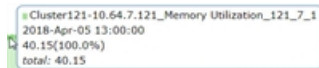
- Base-2
- Base-10

Understanding Delta Value and Rate Value Effects on Charts

The decision to display charts as Delta Value or Rate Value depends on whether you are interested in *summation* values or *average* values in a collection period:

- Delta value for Global Statistic sets each data point to represent the summation of the collected values in that collection period.
- Rate value for Global Statistic sets each data point to represent the average of the collected values in that collection period.

In the chart, collection time interval plays a major role in the charted results. To determine the collection time interval for the chart, point the cursor at a particular data point. Service Manager opens a pop-up window that lists the node name, timestamp, and value for that data point.



Point the cursor at an adjacent data point to see its timestamp.

The collection time interval is the ending timestamp minus the beginning timestamp.

The following examples demonstrate the difference between Delta and Rate value using the following data. Service Manager collects data from the node every 5 minutes. Data is collected between 10:00 to 11:00 am.

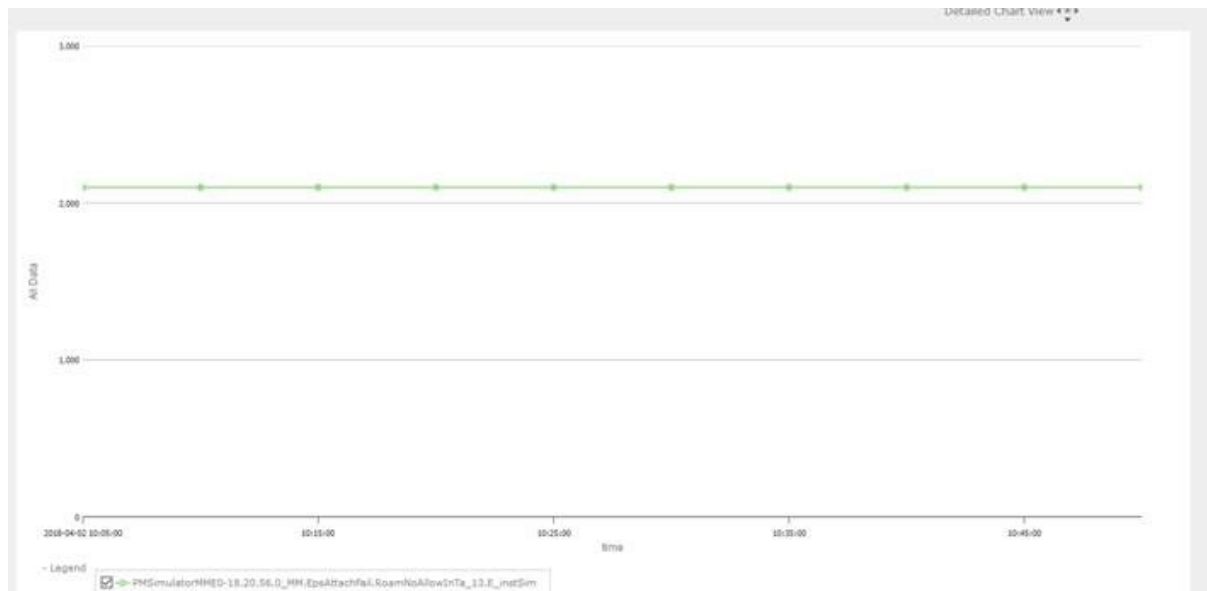
Table 5-2 Example Data

Time of Collection	Collected Value
10:00	2100
10:05	2100
10:10	2100
10:15	2100
10:20	2100

Time of Collection	Collected Value
10:25	2100
10:30	2100
10:35	2100
10:40	2100
10:45	2100
10:50	2100
10:55	2100

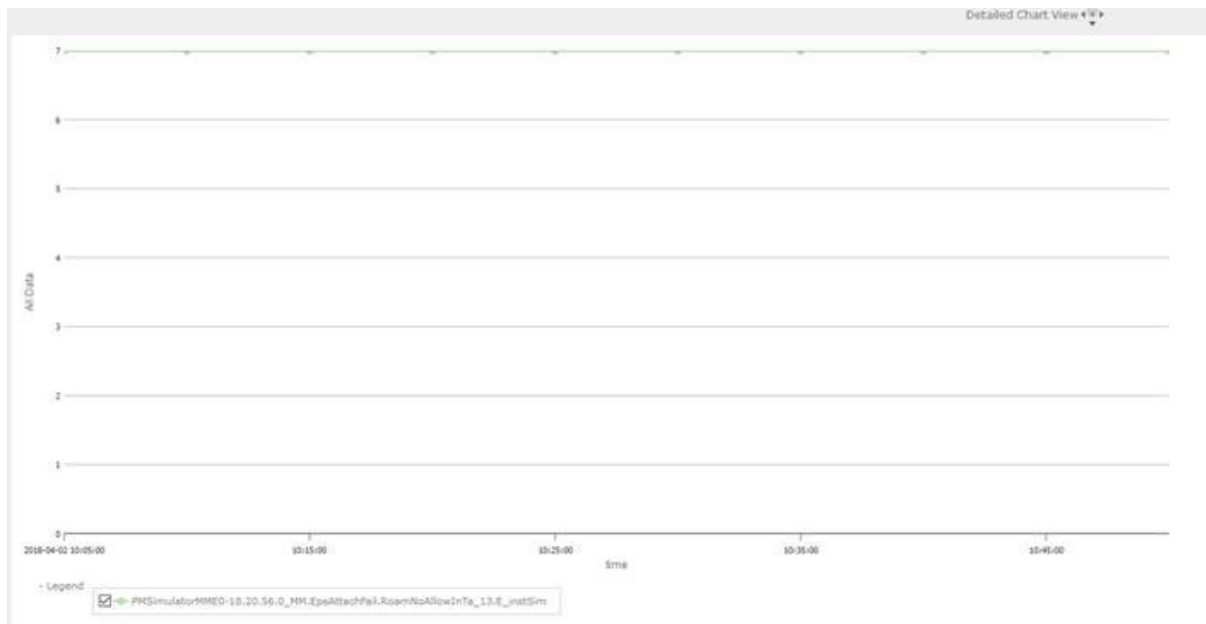
In a **Delta** chart plotting 1 hour of data, each data point represents 5 minutes of data. In this example, each data point shows the value 2100 collected from node.

Figure 5-19 Delta Chart with Collection Time Period 1 Hour



In the **Rate** chart plotting 1 hour of data, each data point represents Delta value / collection time interval in seconds or:

$$(2100 / 300 \text{ seconds}) = 7$$

Figure 5-20 Rate Chart with Collection Time Period 1 Hour

In a **Delta** chart plotting 7 days of data, each data point represents 1 hour of data. At 10:00 on April 2, the Delta value is 25200, which is the summation of collected values from 10:00 to 10:55.

Figure 5-21 Delta Chart with Collection Time Period 7 Days

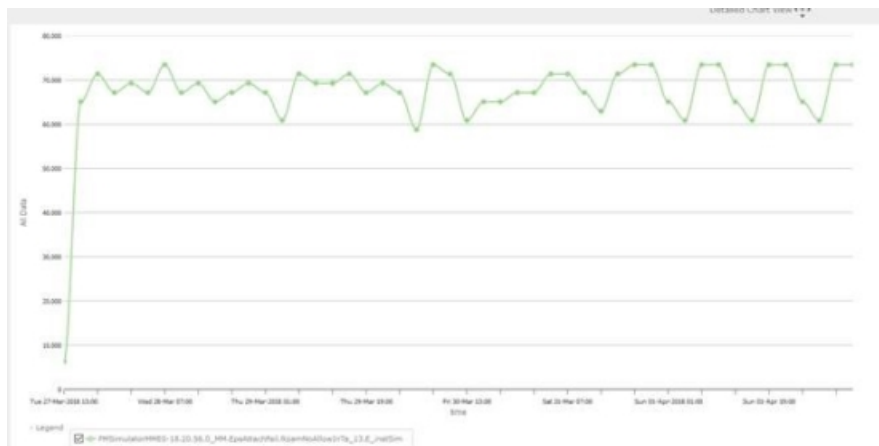
In a **Rate** chart plotting 7 days of data, each data point represents 1 hour of data. At 10:00 on April 2, the Rate value is the Delta value / collection time interval in seconds or:

(the summation of collected values from 10:00 to 10:55 / 3600 seconds) = 7

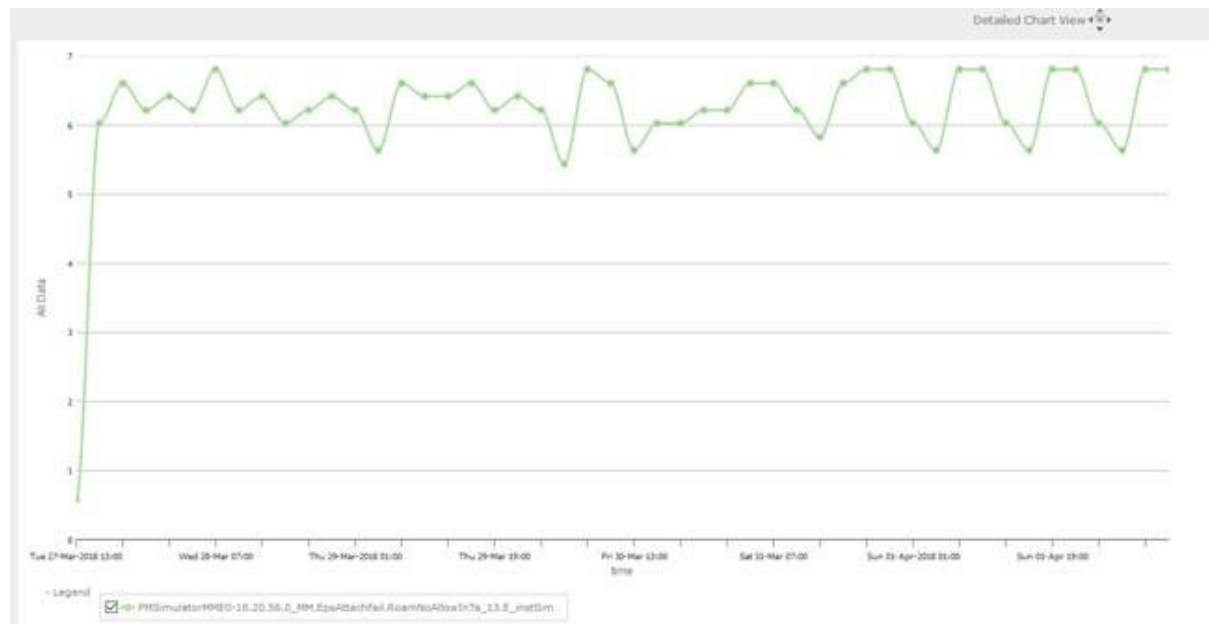
Figure 5-22 Rate Chart with Collection Time Period 7 Days

In a chart plotting more than 7 days of data, each data point represents 3 hours of data. This example illustrates that the collected data are not constant during this 3-hour period.

In a **Delta** chart, each data point represents the summation of 3 hours collected data.

Figure 5-23 Delta Chart with Collection Time Period > 7 Days

In a **Rate** chart, each data point represents Delta value / collection time interval in seconds or:
(the summation of 3 hours collected data / 10800 seconds)

Figure 5-24 Rate Chart with Collection Time Period > 7 Days

Configuring Data Type and Unit Conversion

To configure Data Type at the global level:

1. Select the **Settings**  icon on the toolbar and select **Performance => Reports => Statistics Configuration**.
2. Mark the data type to be configured (Rate Value or Delta Value).
3. Select **OK**.

To configure Unit Conversion at the global level:

1. Select the **Settings**  icon on the toolbar and select **Performance => Reports => Statistics Configuration**.
2. Mark the unit conversion to be configured (Base-2 or Base-10).
3. Select **OK**.

Configuring Number of Data Points to Display in Performance Reports

You can configure Performance Reports to display between 50 and 150 data points in charts. To do so, modify the PerformanceConfiguration.properties in the directory:

/opt/Affirmed/NMS/server/ems/resources

The configurable attribute is:

Randomize Window


```
ChartManager.instancecount.limit=50
```

The default value is 50. It can be set to any value from 50 to 150. Restart the Service Manager server for the new value to take effect.

Viewing Historical Statistics

Service Manager provides the user a view of the historical data of performance statistic attributes for a specified period.

You can configure Performance Reports to display between 50 and 150 data points in charts, as specified by the `ChartManager.instancecount.limit` attribute of the `PerformanceConfiguration.properties` file (see [Configuring Number of Data Points to Display in Performance Reports](#)).

For each interval:

- Counter statistics are displayed as per-second average values. The data is aggregated over the reporting interval. For example, for an hourly reporting interval, data is presented as average values per second over the hour.
- Gauge and average statistics are displayed as average values.
- Minimum and maximum statistics are displayed as actual values.

Configuring Historical Report Data Purging

Service Manager uses a pre-defined range of Purging Interval and a default Purging Interval for each type of historical statistics table. The Purging Interval is the time interval for which the data is retained in the table, with older data being purged.

Table 5-3 Pre-defined Purging Interval Range and Default by Historical Statistics Table Type

Historical Table Type	Data Granularity	Purging Interval Range	Purging Interval Default
BASE	5-minute	6 – 48 hours	24 hours
HOURLY	15-minute	1 – 6 day(s)	3 days
DAILY	Hourly	1 – 3 week(s)	2 weeks
WEEKLY	3-hour	4 – 8 weeks	6 weeks
MONTHLY	Daily	6 – 18 months	~12 months (360 days)
YEARLY	Weekly	2 – 5 years	3 years

You can configure the purging interval for each type of statistics table.

To configure a Purging Interval:

1. Issue the following commands at the command prompt:

```
cd /opt/Affirmed/NMS/bin
```

```
configureserver.sh
```

Service Manager displays the **Configure Server Details** menu.

```
Configure Server Details
```

```
-----
```

```
1.DatabaseDetails
```

```
2.PortDetails
```

```
3.Authentication
```

```
4.ProtocolDetails
```

```
5.PurgeConfigurationDetails
```

```
6.Exit
```

```
Please select your option from (1 | 2 | 3 | 4 | 5 |6) [1] :
```

2. To view the current purge settings, enter **5**.

```
Purge Configuration Details
```

```
-----
```

	Table Type	Current Interval	Default
Interval			
1) BASE	5 hours	24 hours	
2) HOURLY	3 days	5 days	
3) DAILY	Default	2 weeks	
4) WEEKLY	4 weeks	6 weeks	
5) MONTHLY	6 months	12 months	
6) YEARLY	2 years	3 years	

```
Please select table type from (1 | 2 | 3 | 4 | 5 | 6 | 7 - EXIT  
) [7]:
```

3. To manage a purging interval, enter the table number.

```
1) Reset to Default
```

```
2) Configure
```

```
3) Back
```

```
Please select your option from (1 | 2 | 3) [3]:
```

- To reset the purging interval to the default purging interval setting, enter **1**.

For example:

```
WEEKLY Table purge interval set to default value 12 months
```

- To set a new purging interval, enter **2**.

For example:

```
WEEKLY Table (4 - 8 weeks) [4]: 6
```

```
WEEKLY Table purge interval set to 6week(s)
```

```
Please select table type from (1 | 2 | 3 | 4 | 5 | 6 | 7 - EXIT  
) [7]:
```

4. To exit, enter **7**.

Configured Purge Interval Details:

```
BASE Table: 24 hours
```

```
WEEKLY Table: 6 weeks
```

```
HOURLY Table: 4 days
```

Any changes you make take affect when the purging schedule next runs.

To view Historical Statistics:

1. Use any of the following methods to view Historical Statistics:

Settings icon on the toolbar:  Select **Performance => Reports => Historical**.

Pop-up menu: Right-click the Network Element and select **Statistics => Historical**.

Service Manager displays the Historical Statistics screen. The left pane lists all the NEs managed (or the Network Element you right-clicked) and the metrics. The Chart View tab is selected by default.

Figure 5-25 Historical Statistics



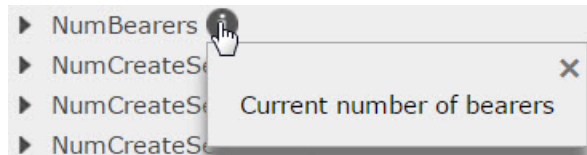
2. In the left panel, to filter the tables and metrics displayed, enter a value in the **Filter by Metric Name** field and select the **Filter Metric** icon. 
3. Expand the tables to list the metrics that match what you filtered.
4. Select the instances you want to view:
 - To select all instances, mark the metric.
 - To select particular instances, expand the metric and mark individual instances.

5. To clear the **Filter by Metric Name** field, select the **Clear Filter**  icon. You can filter multiple times and select multiple matched metrics to plot the charts. Your selections are preserved during the Filter/Clear operations.
6. To view the description of a metric:
 - a. Point the cursor at the metric name.

b. **NumBearers** 

- c. Select the **Detail**  icon.

Service Manager opens a pop-up window that displays the description of the statistic.

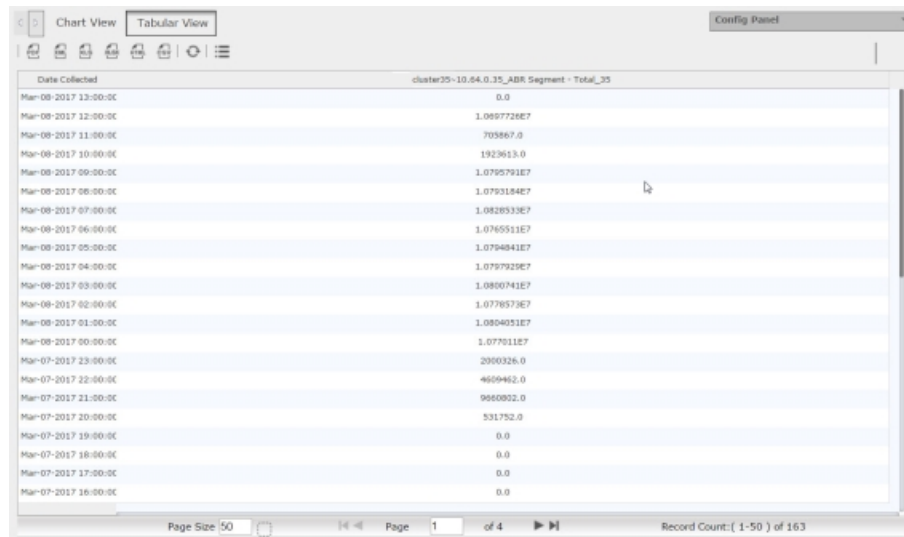


The window remains open until you close it or select another detail icon.

7. Select the **Config Panel** expansion icon to specify:
 - **Period From** – Select the start time
 - **Period To** – Select the end time
 - **X Axis** – Collected Time to appear on the X axis
 - **Show All** – Show statistics for all instances (such as multiple stats slots)
 - **Show Top x** – Show highest statistics for specified number of instances
 - **Show Bottom x** – Show lowest statistics for specified number of instances
 - **Group All** – Mark this checkbox to group the charts for different selected metrics in a single chart. Unmark this checkbox for individual charts for each metric.
8. To view as a graphical chart (Chart View):
 - a. Select the **Chart View** tab from the Historical Statistics window.
 - b. Select the specific chart icons  to view the data in the required graphical format.
9. To view as a table of data (Tabular View):
 - a. Select the **Tabular View** tab from the Historical Statistics window.

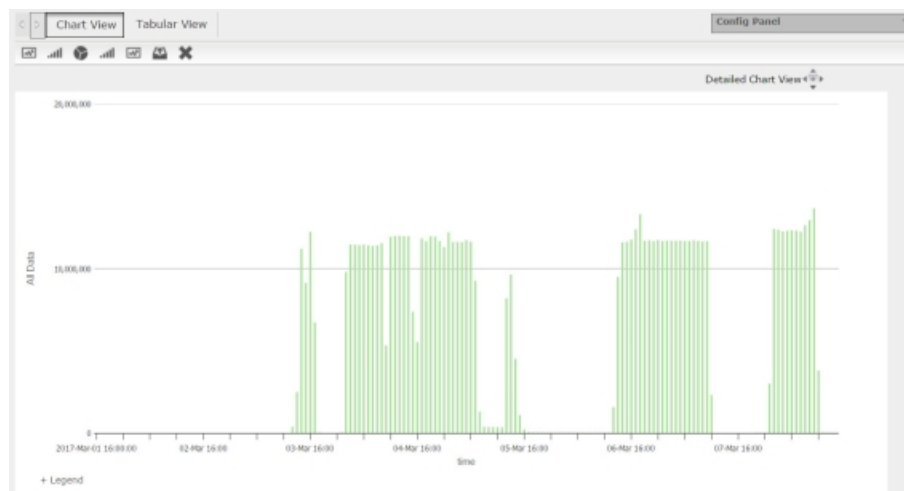
Service Manager retrieves data from the Service Manager database and displays a chart or textual report of the values.

The following example shows a Tabular View.

Figure 5-26 Example Tabular View

The collection time interval is indicated in the Date Collected column.

The following example shows a Chart View.

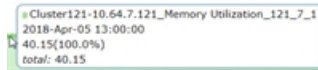
Figure 5-27 Example Chart View

To view the chart legend, select the **Legends** icon. **+ Legends** If necessary, use the scrollbar to display the legend.

Mark/unmark the checkboxes in the legend to select/deselect instances to show/hide the respective data plots in the chart.



To determine the collection time interval for the chart, point the cursor at a particular data point. Service Manager opens a pop-up window that lists the node name, timestamp, and value for that data point.



Point the cursor at an adjacent data point to see its timestamp.

The collection time interval is the ending timestamp minus the beginning timestamp.

To view the detailed chart, select the **Detailed Chart View**  icon.

To clear the displayed chart, select the **Clear**  icon.

Exporting Historical Statistics Reports

You can save the graphical chart as a .pdf file. You can save the textual historical report in various formats, including .csv, .pdf, .xml, .xls, .html.

To save a graphical chart (Chart View):

1. Select the **Chart View** tab from the Historical Statistics window.
2. From the Chart view, select the **Export as PDF**  icon on the tool bar.

From the Detailed Chart view, select the **Take Chart as Snapshot**  icon on the tool bar.

For the Chart view, Service Manager displays the **Export PDF Configuration** window.

For the Detailed Chart view, Service Manager displays the **Take Snapshot** window.

3. For the Chart view, specify the PDF File Name and select **Export**.

For the Detailed Chart view, specify the Zoom Level (either the present zoom level or the default zoom level) and the PDF FileName and select **Apply**.

Note: If all selected instances are from same metric, Service Manager lists a default PDF filename (the metric name) in **Export PDF Configuration** or **Take Snapshot** window. You can modify that filename.

4. Select **Download PDF File**.

Service Manager displays the chart in a separate window, listing the file location on the Service Manager machine and filename (export/<PDF filename><timestamp>.pdf)

5. To save the file elsewhere, right-click the chart and select **Save As**.
Specify where you want the file saved and select **Save**.

6. From the **Download Files** window, select **Close**.

To save as a grid of data (Tabular View):

1. Select the **Tabular View** tab from the Historical Statistics window.
2. Select the icon of the required file format.



Service Manager displays the **Download Files** window.

3. Select **Download Exported File**.
4. From the **Download Files** window, select **Cancel**.

Viewing Real Time Statistics

Service Manager provides the user a quick graphical view of the current values of performance statistic attributes. You can configure Performance Reports to display between 50 and 150 data points in charts, as specified by the `ChartManager.instancecount.limit` attribute of the `PerformanceConfiguration.properties` file (see [Configuring Number of Data Points to Display in Performance Reports](#)).

The data collected as a part of the Real Time Stats is not persisted. Users with sufficient privilege can start an on-demand polling to monitor few performance metrics.

To view Real Time Statistics:

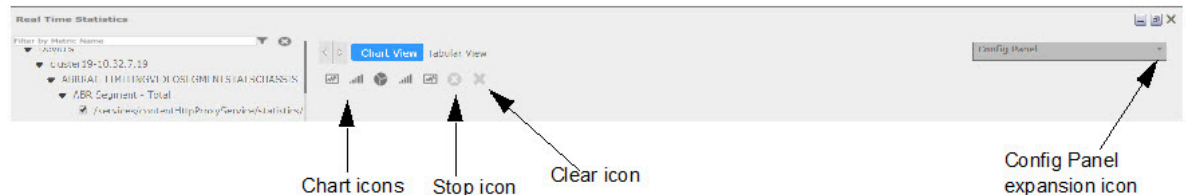
1. Use any of the following methods to view Real Time Statistics:

Settings icon on the toolbar:  Select **Performance => Reports => Real Time**.

Pop-up menu: Right-click the Network Element and select **Statistics => Real Time**.

Service Manager displays the **Real Time Statistics** screen. The left pane lists all the NEs managed (or the Network Element you right-clicked) and the metrics.

Figure 5-28 Real Time Statistics

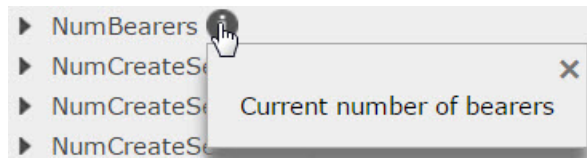


2. In the left panel, to filter the tables and metrics displayed, enter a value in the **Filter by Metric Name** field and select the **Filter Metric**  icon.
3. Expand the tables to list the metrics that match what you filtered.
 - To select all instances, mark the metric.
 - To select particular instances, expand the metric and mark individual instances.
4. To clear the **Filter by Metric Name** field, select the **Clear Filter**  icon. You can filter multiple times and select multiple matched metrics to plot the charts. Your selections are preserved during the Filter/Clear operations.
5. To view the description of a metric:
 - a. Point the cursor at the metric name.

NumBearers 

- b. Select the detail icon. 

Service Manager opens a pop-up window that displays the description of the statistic.



The window remains open until you close it or select another detail icon.

6. Select the **Config Panel** expansion icon to specify:

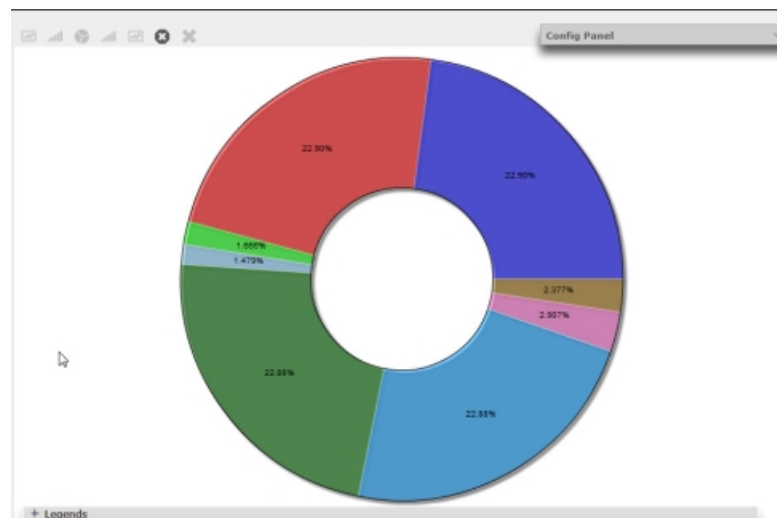
- **X Axis** – Collected Time to appear on the X axis
- **Server Polling** – Enter the number of seconds for the polling period (frequency of chart to be plotted). Range: 10 seconds to 120 seconds.
- **Configure Points** – Enter the number of data points to be plotted in the graph.
- **Group All** – Mark this checkbox to group the charts for different selected metrics in a single chart. Unmark this checkbox for individual charts for each metric.

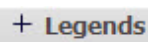
7. Select the specific chart icons  to view the data in the required graphical format.

Service Manager retrieves the data from the Service Manager database and displays the selected chart in the chart area. The chart is refreshed and redrawn whenever new data is collected based on the server polling interval.

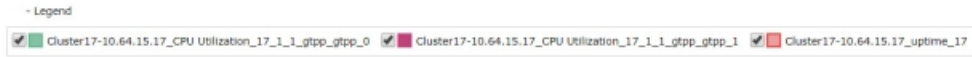
The following example shows a Pie Chart.

Figure 5-29 Example Pie Chart



8. To view the chart legend, select the **Legends** icon.  If necessary, use the scrollbar to display the legend.

Mark/unmark the checkboxes in the legend to select/deselect instances to show/hide the respective data plots in the chart.



9. To clear the displayed chart, select the **Clear** icon.
10. Select the **Stop** icon after viewing the Real Time Statistics.

Viewing Outage Reports

An outage refers to the time during which a network element is unreachable. Service Manager tracks the availability of each network element it manages by recording the start time (Lost Time), end time (Regained Time), and Down Time Period for each unreachable window.

To view an Outage Report:

1. Use any of the following methods to view an Outage Report:

Settings icon on the toolbar: Select **Performance** => **Reports** => **Outage Report**.

Pop-up menu: Right-click the Network Element and select **Statistics** => **Outage Report**.

Service Manager lists the outage details of the devices.

Configuring Export Reports

You can configure Performance Management reports to be exported to CSV files.

To configure an Export Report:

1. Use the following method to configure an Export Report:

Settings icon on the toolbar: Select **Performance** => **Reports** => **Export Report Config**.

Service Manager displays the **Export Report Editor** window, listing the available elements.

2. Select the elements for the historical report by highlighting and selecting the right-arrow.
3. Select **Next**.
4. Expand the metrics tables and mark the metrics you want reported.
5. Select **Save**.

Service Manager exports the metrics to individual files by metric table every midnight, with the exported CSV files containing the last 24 hours data.

For daily Performance Management reports, the export filename is `Historical_<Entity Name>_<Table Name>_<Date>.csv`. The file is exported to the Service Manager server, in the directory `/opt/Affirmed/NMS/server/ems/archive/<Date>/.`

By default, Service Manager purges any file older than 30 days. To configure a different purge interval, see [Configuring Export Purge Interval](#).

Triggering Report Export on Demand

You can trigger the export of Performance Management reports to CSV files on demand.

To trigger report export:

1. Select the **Settings**  icon on the toolbar and select **Performance => Reports => Export Now**.
2. At the confirmation prompt, select **OK**.

Service Manager exports the metrics to individual files by metric table, with the exported CSV file containing the data from 00:00 to the current time.

The files are exported to the Service Manager server, in the directory */opt/Affirmed/NMS/server/ems/archive/reports_export_now*.

For daily Performance Management reports, the export filename is *Historical_<Entity Name>_<Table Name>_<Date>.csv*.

For daily Dashboard reports, the export filename is *<Entity Name>_<Dashboard Title>_<Report Title>_<Date>.csv*.

By default, Service Manager purges any file older than 30 days. To configure a different purge interval, see [Configuring Export Purge Interval](#).

Configuring Export Purge Interval

You can configure the purge interval of exported CSV files.

To configure the Purge Interval:

1. Use the following method to configure the purge interval:

Settings icon on the toolbar:  Select **Performance => Reports => Purge Export Configuration**.

2. Specify the number of days after which the system purges the exported CSV files.
3. Select **OK**.

Thresholds

You can define Thresholds with various severity levels for a performance statistic attribute or a Key Performance Indicator (KPI). When the data collected for an attribute or KPI crosses the threshold level defined, a threshold event is generated with the specified severity.

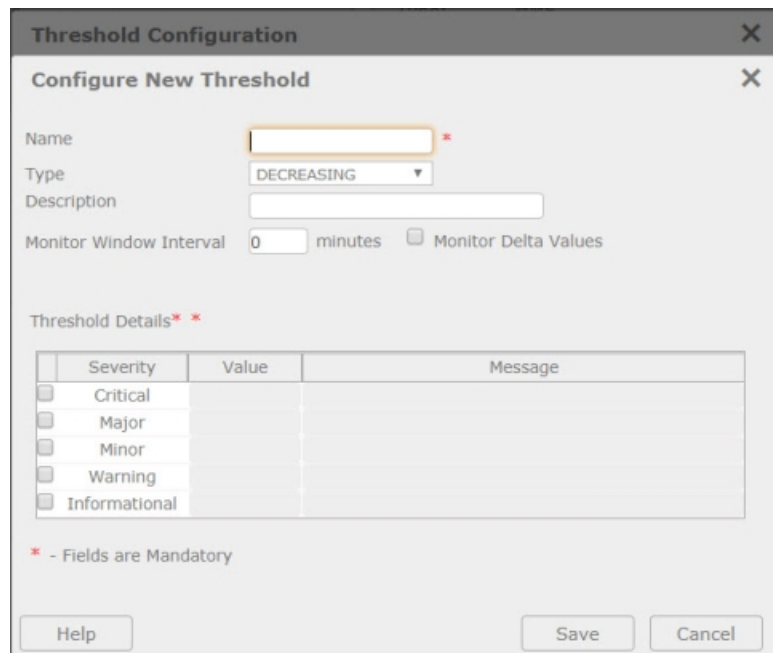
Adding Thresholds

To add a threshold:

1. Select the **Settings**  icon on the toolbar and select **Performance => Configure Thresholds**.

2. Service Manager displays the **Threshold Configuration** window.
3. Select the **Add New Threshold**  icon.

Figure 5-30 Configure New Threshold



Threshold Configuration

Configure New Threshold

Name *

Type

Description

Monitor Window Interval minutes ☐ Monitor Delta Values

Threshold Details* *

	Severity	Value	Message
☐	Critical		
☐	Major		
☐	Minor		
☐	Warning		
☐	Informational		

* - Fields are Mandatory

4. Complete the fields as follows:
 - **Name** – Enter the threshold name.
 - **Type** – Specify whether the threshold applies when a data point exceeds (Increasing) or falls below (Decreasing) the specified level.
 - **Description** – Enter a threshold description.
 - **Monitor Window Interval** – Enter how many minutes must transpire once the threshold has occurred before raising the alarm.
 - **Monitor Delta Values** – Mark the checkbox to have Service Manager monitor the delta value between the previous and current interval counter values.
 - **Threshold Details:**
 - Specify the **Value** for the threshold for configured severity categories.
 - Enter a corresponding **Message** for each configured category. When a threshold event is raised, the configured message is displayed in Alarms/Event Logs tab.
5. Select **Save** to commit the data.
6. Select **Close** to close the **Threshold Configuration** window.

7. You need to associate the threshold with a PM collection task. For instructions, see [Associating a Threshold with a PM Collection Task](#). For the appropriate performance statistic attribute or Key Performance Indicator, select the new threshold from the **Threshold** field.

Viewing Thresholds

To view a threshold:

1. Select the **Settings**  icon on the toolbar and select **Performance => Configure Thresholds**.
2. Service Manager displays the **Threshold Configuration** window, listing the following fields.
 - **Threshold Name**
 - **Type** – Specifies whether the threshold applies when a data point exceeds (Increasing) or falls below (Decreasing) the specified level.
 - **Description**
3. Select a threshold to view its threshold details:
 - **Severity** – Critical, Major, Minor, Warning, Informational
 - **Value** – Value for the threshold
 - **Message** – The configured message to display in Alarms/Event Logs tab when the threshold event is raised.
4. Select **Close** to close **Threshold Configuration** window.

Modifying Thresholds

To modify a threshold:

1. Select the **Settings**  icon on the toolbar and select **Performance => Configure Thresholds**.
2. Service Manager displays the **Threshold Configuration** window.
3. Select a threshold to view then select the **Edit Threshold Details**  icon.
4. Make the necessary changes:
 - **Name** – Enter the threshold name.
 - **Type** – Specify whether the threshold applies when a data point exceeds (Increasing) or falls below (Decreasing) the specified value.
 - **Description** – Enter a threshold description.
 - **Monitor Window Interval** – Enter how many minutes must transpire once the threshold has occurred before raising the alarm.

- **Monitor Delta Values** – Mark the checkbox to have Service Manager monitor the delta value between the previous and current interval counter values.
 - **Threshold Details:**
 - Specify the **Value** for the threshold for configured severity categories.
 - Enter a corresponding **Message** for each configured category. When a threshold event is raised, the configured message is displayed in Alarms/Event Logs tab.
5. Select **Save** to commit the data.
 6. Select **Close** to close **Threshold Configuration** window.

User Tasks Templates

The User Tasks template specifies the metrics to be collected for performance statistics:

***Note:** By default, Service Manager is configured to collect all the appropriate metrics. It is not essential for the administrator to change user task templates.*

Viewing User Tasks Templates

To view a User Tasks Template:

1. Select the **Settings**  icon on the toolbar and select **Performance => User Tasks => Templates**.

Service Manager displays the **Task Template**, listing the user tasks templates.

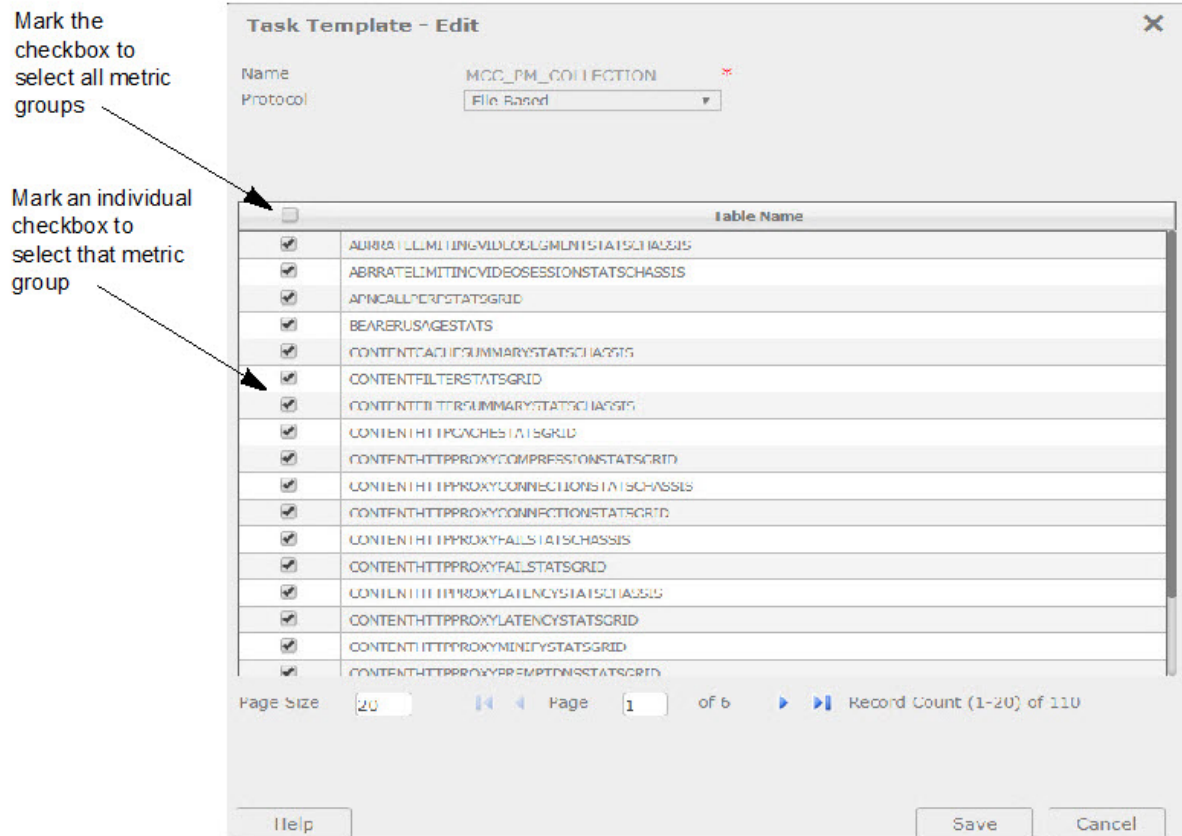
Select a template to view the performance metrics it collects.

Modifying User Tasks Templates

You can modify a data collection task, as long as it is not currently in use by the Task Scheduler.

To Modify a User Tasks Template:

1. Select the **Settings**  icon on the toolbar and select **Performance => User Tasks => Templates**.
Service Manager displays the **Task Template**, listing the user tasks templates.
2. Select a template then the **Edit User Task Template**  icon.
3. For a template with **Task Type** set to File Based, you can configure the data to be collected. Mark the checkboxes to select the metrics groups you want collected. To select all metrics groups, mark the checkbox at the top of the column.

Figure 5-31 User Task Template Details Edit

4. Select **Save** to commit the data.
5. At the confirmation prompt, select **OK**.
6. Select **Close** to close **Task Template** window.

Users Tasks Scheduler

A user tasks scheduler triggers data collection tasks based on task templates to be periodically executed.

Viewing User Tasks Scheduler

To view the User Tasks Scheduler:

1. Select the **Settings**  icon on the toolbar and select **Performance => User Tasks => Schedule**.
2. Use any of the following methods to view a user tasks schedule:
 - Toolbar:** Select the user tasks schedule and select the **View**  icon.
 - Pop-up menu:** Right-click the user tasks schedule and select **View**.
3. Select **Close** to close the window.

Enabling/Disabling User Tasks Schedules

You can enable/disable user tasks schedules using the Task Scheduler.

To enable/disable a user tasks schedule:

1. Select the **Settings** icon on the toolbar  and select **Performance => User Tasks => Schedule**.

Service Manager displays the **Task Scheduler**, listing performance management collections and other scheduled data collections.

2. Use any of the following methods to enable a user tasks schedule:

Toolbar: Select the user tasks schedule then the **Enable Schedule**  icon.

Pop-up menu: Right-click the user tasks schedule and select **Enable Schedule**.

Use any of the following methods to disable a user tasks schedule:

Toolbar: Select the user tasks schedule then the **Disable Schedule**  icon.

Pop-up menu: Right-click the user tasks schedule and select **Disable Schedule**.

3. At the confirmation prompt, select **OK**.

The Admin Status of the user tasks schedule changes to Enabled or Disabled.

Associating a Threshold with a PM Collection Task

For a particular performance statistic attribute or Key Performance Indicator, you associate it with either a metric-based threshold or an instance-based threshold.

For a metric, data is always collected as a list of (instance,value) pairs. Each (instance,value) pair is first matched with a defined instance-based threshold, then a metric-based threshold. When no instance-based threshold exists, only the metric-level threshold is matched. When the data collected for a metric crosses the defined instance-based or metric-based threshold level, a threshold event is generated with the specified severity.

To associate a threshold with a PM collection task, modify a user tasks schedule.

Prerequisite to Configure a Metric-based Threshold

To associate a particular performance statistic attribute or Key Performance Indicator with a metric-based threshold, you first need to create a threshold. For instructions, see [Thresholds](#).

Prerequisites to Configure an Instance-based Threshold

To associate a particular performance statistic attribute or Key Performance Indicator with an instance-based threshold, you first need to create the following entities:

- Threshold. For instructions, see [Thresholds](#).
- Instance filter, see [Instance Filter](#)

Modifying User Tasks Scheduler

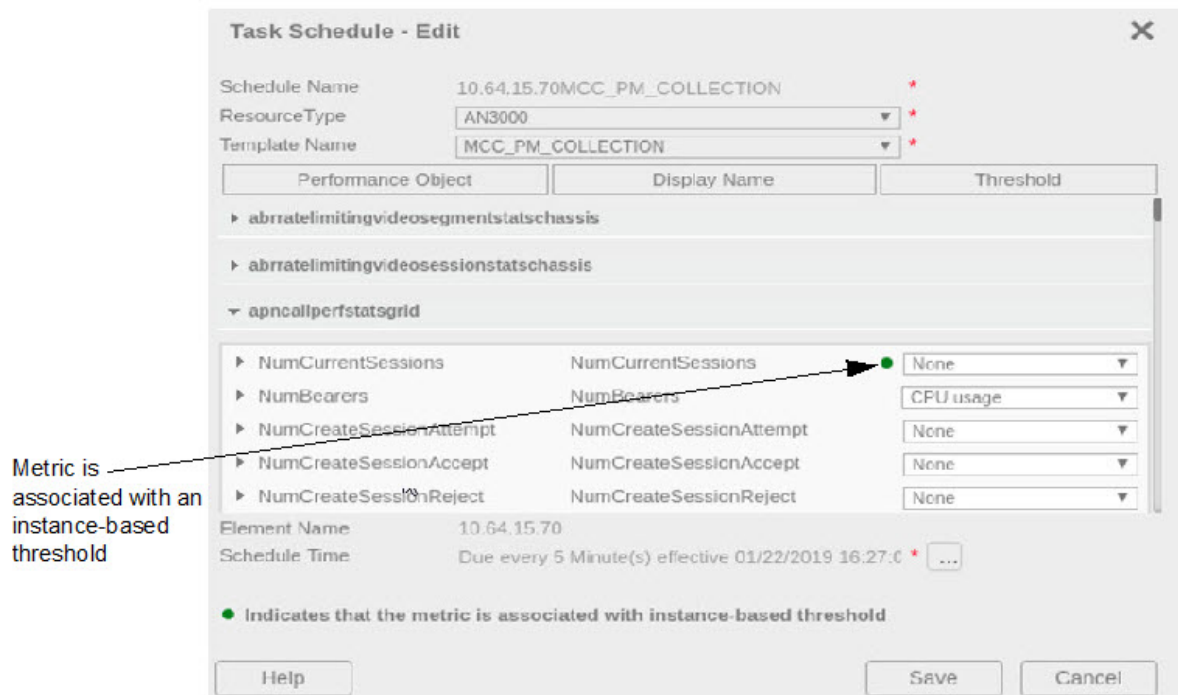
To modify a User Tasks Scheduler:

1. Select the **Settings**  icon on the toolbar and select **Performance => User Tasks => Schedule**.
2. Use any of the following methods to modify a user tasks schedule:

Toolbar: Select the user tasks schedule then select the **Edit User Task Scheduler**  icon.

Pop-up menu: Right-click the user tasks schedule and select **Edit**.

Figure 5-32 Task Schedule Edit



3. Make the necessary changes:
 - **Schedule Name** – Enter the schedule name.
 - **Performance Object** – Expand the Performance object name to list the metrics. An icon  in the Threshold column indicates that the metric is associated with an instance-based threshold. To list the instance-based thresholds, expand the metric.

Figure 5-33 Task Schedule Edit with Expanded Metric

- **Threshold** – Select a threshold to be associated with each performance statistic attribute or Key Performance Indicator.

Note: Reachability task schedules (resourcereachability Performance object) cannot be associated with an instance-based threshold.

4. In the Scheduled Time field, select the **Recurrence Pattern**  icon to configure Recurrence Details.
5. Specify the recurrence details and select **Schedule**.
6. Select **Save**.

Service Manager matches each instance-level value in the collected data with its associated threshold value and based on the results, raises a threshold event.

Quick Start Data Collection

To Quick Start (instantly execute) a data collection task:

1. Select the **Settings**  icon on the toolbar and select **Performance => User Tasks => Schedule**.

Service Manager displays the **Task Scheduler**, listing performance management collections and other scheduled data collections.

2. Use any of the following methods to Quick Start a data collection tasks:

Toolbar: Select the data collection tasks then select the **Quick Start**  icon.

Pop-up menu: Right-click the data collection tasks and select **Quick Start**.

Exporting User Tasks Scheduler

You can export a list of user tasks scheduler instances to a CSV file.

To export user tasks schedules:

1. Select the **Settings**  icon on the toolbar and select **Performance => User Tasks => Schedule**.

Service Manager displays the **Task Scheduler**, listing performance management collections and other scheduled data collections.

2. If desired, use the Filter Manager or Quick Filter to filter the data displayed in the window.

3. Select the **Export**  icon.

Service Manager displays the **Configure Export Parameters** window.

4. Configure the parameters as follows:

- **File Type:** Select the required file format (CSV).
- **Column Configuration:** Select the columns you want exported (Visible Column or All Columns)
- **Export Type:** Select the export type (Whole Data, Current View, Selected Data)
- **Use Current Filter:** Mark the checkbox to restrict data based on the existing filter conditions.

5. Select **Apply**.

Service Manager displays the **Download Files** window.

6. Select **Download Exported File**.

7. From the **Download Files** window, select **Close**.

Network Backups and Upgrades

This section describes how to manage backups and upgrades for your enterprise and private networks.

Database Administration

You can schedule a database backup for automatic execution and can trigger a backup manually on demand. The backup can also be used to restore a database from an earlier date and time.

***Note:** Database backups are blocked and a threshold crossing alarm is raised when the /opt partition free disk space falls below 10%. The alarm severity is 60% (Minor), 75% (Major), and 90% (Critical).*

Adding User Triggered Database Backup

You can trigger a backup on demand.

To configure a user triggered database backup:

1. Select the **Settings**  icon on the toolbar and select **Tools => Database Administration**. Service Manager displays the **Database Backup and Restore** window.
2. Select the **User Triggered DB-Backup**  icon.
3. Configure the parameters as follows:
 - **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as DB_backup.
 - **File Server** – Specify whether you want the backup stored on the local or remote server. It is recommended that you store the backup on the remote server, as a local backup will not be available if the system becomes out of service.

- **Relative Path in File Server** – If you select to have the backup stored on remote server, specify the path where you want it stored. If you select to have the backup stored on the local server, it is stored at: `/opt/Affirmed/NMS/server/databaseBackup`
 - **Full Backup** – Mark this checkbox to trigger a full backup.
4. Select **Save**.
 5. At the confirmation prompt, select **OK**.
 6. Select **Close**.

Service Manager displays the backup in the **Database Backup and Restore** window.

Viewing Database Backup Details

You can view database backups details.

To view database backup details:

1. Select the **Settings**  icon on the toolbar and select **Tools => Database Administration**.
Service Manager displays the **Database Backup and Restore** window.
2. Use any of the following methods to view an backup's details:
 - Toolbar:** Select the backup then the **View DB Backup**  icon.
 - Pop-up menu:** Right-click the backup and select **View DB Backup**.Service Manager displays the **View Database Backup** window.
3. Select **Close** to close the window.

Scheduling Database Backup

You can configure the Service Manager server to backup its database based on a pre-configured schedule. *Do not* schedule backups more than twice a day, as it can impact on system performance (backups cause pauses in Service Manager system and affect performance/scalability of the system).

To schedule a database backup:

1. Select the **Settings**  icon on the toolbar and select **Tools => Database Administration**.
Service Manager displays the **Database Backup and Restore** window.
2. Select the **Scheduled DB-Backup**  icon.
3. Configure the parameters as follows:
 - **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as `DB_backup`.

- **File Server** – Specify whether you want the backup stored on the local or remote server. It is recommended that you store the backup on the remote server, as a local backup will not be available if the system becomes out of service.
- **Relative Path in File Server** – If you select to have the backup stored on remote server, specify the path where you want it stored. If you select to have the backup stored on the local server, it is stored at: `/opt/Affirmed/NMS/server/databaseBackup`
- **Full Backup** – Mark this checkbox to trigger a full backup.
- **Recurrence** – Mark this checkbox to specify a backup recurrence. *Do not* schedule more than twice a day, as it can impact system performance:
 - Select the **Recurrence Pattern** icon and configure Recurrence Details.
 - Select the **Recurrence**.
- **Purge Backup** – Mark this checkbox to specify how many backups to retain.

4. Select **Save**.

Viewing Scheduled Database Backup

You can view a scheduled database backup.

To view a scheduled database backup:

1. Select the **Settings**  icon on the toolbar and select **Tools => Database Administration**.
Service Manager displays the **Database Backup and Restore** window.
2. Select the **Scheduled DB-Backup View**  icon.
3. Select **Close** to close the window.

Deleting Database Backup

You can delete a database backup.

To delete a database backup:

1. Select the **Settings** icon on the toolbar  and select **Tools => Database Administration**.
Service Manager displays the **Database Backup and Restore** window.
2. Use any of the following methods to delete a backup:
 - Toolbar:** Select the backup then select the **Delete DB Backup**  icon.
 - Pop-up menu:** Right-click the backup and select **Delete DB Backup**.
Service Manager displays the View Database Backup window.
3. Select **Close** to close the window.
4. At the confirmation prompt, select **OK**.

5. Select **Close** to close the window.

Service Manager removes the backup from the **Database Backup and Restore** window.

Restoring Database from Backup

You can restore the Service Manager database to a successful backed up version.

Use the appropriate method to restore the database:

- [Standalone Service Manager Server](#)
- [Service Manager Server Cluster \(HA or HA + DR\)](#)

Caution: Database restore is service affecting. During the process, the Service Manager server is brought down.

Standalone Service Manager Server

To restore the database from a backup:

1. Select the **Settings**  icon on the toolbar and select **Tools => Database Administration**.
Service Manager displays the **Database Backup and Restore** window.
2. Use any of the following methods to restore the database to a backup:
Toolbar: Select the backup then the **Restore Database**  icon.
Pop-up menu: Right-click the backup and select **Restore Database**.
3. At the confirmation prompt, select **OK**.
4. Select **Close** to close the window.

Note: If the restore operation fails, Service Manager displays the Database Restore Operation window. Select **Yes** to restore the database from the local machine. Select **No** to restore an existing database.

5. Determine whether the restoration process has finished. To do so, issue the following command from the /opt/Affirmed/NMS/dbrestore directory:

```
[root@qahpems02 dbrestore]# ls -lrt
```

If the command output indicates that the directory contains the sql file, the process *is not* complete.

In the following sample command output, the sql file is shown in red text.

```
total 135096
-rw-r--r-- 1 root root      600 Oct  5 14:19 log4j.xml
drwxr-xr-x 2 root root    4096 Oct 11 10:47 bin
drwxr-xr-x 2 root root    4096 Oct 11 10:47 lib
```

```
-rw-r--r-- 1 root root      103 Oct 11 15:41 restore.txt
-rw-r--r-- 1 root root 138315816 Oct 11 16:31 http3_10-11-2012-
16-30-03.sql
-rw-r--r-- 1 root root      2038 Oct 11 16:31 restore.log
```

If the command output indicates that the directory no longer contains the sql file and contains the restore.txt file and the restore.log file, the restore is successful.

In the following sample command output, the restore.txt and the restore.log files are shown in red text.

```
total 20
-rw-r--r-- 1 root root   600 Oct  5 14:19 log4j.xml
drwxr-xr-x 2 root root 4096 Oct 11 10:47 bin
drwxr-xr-x 2 root root 4096 Oct 11 10:47 lib
-rw-r--r-- 1 root root   103 Oct 11 16:38 restore.txt
-rw-r--r-- 1 root root 2194 Oct 11 16:38 restore.log
```

6. Once the restoration process has finished, you must manually start the server using starttms.sh with the -u option.

Service Manager Server Cluster (HA or HA + DR)

Use the configure_ha script to restore a database from backup. For configure_ha command syntax, see the *Service Manager System Administration Guide*.

To restore the database from a backup:

1. On the HA server, issue the following commands to restore the database:

```
cd /opt/Affirmed/NMS/bin
./configure_ha [--virtualhostname virtual hostname] [--
virtualip virtual IP address] [--primaryhostname primary
hostname] [--primaryip primary IP address] [--secondaryhostname
secondary hostname] [--secondaryip secondary IP address] [--
multicastip multicast IP address] [--backupfile file_path] [--
restore]
```

Please double check the following configuration options:

```
{
  virtualhostname => 'virtual_hostname',
  virtualip => ' virtual_IP_address',
  primaryhostname => 'primary_hostname',
  primaryip => 'primary_IP_address',
  secondaryhostname => 'secondary_hostname',
```

```
secondaryip => 'secondary_IP_address',
primaryslaveid => 'primary_slave_ID',
secondaryslaveid => 'secondary_slave_ID',
primaryipinterface => 'primary_IP_interface',
secondaryipinterface => 'secondary_IP_interface',
ipinterface => 'IP_interface',
multicastip => 'multicast_IP_address'
}
Are you sure you want to continue with this configuration?
(y/n):
```

2. Enter **y** to continue with this configuration.

```
Are you sure you want to use the backup file file_path to
restore the database? (Y/N):
```

3. Enter **y** to restore the database.

Software Schedules

PNSM allows you to view detailed information for Network Upgrades using the Software Schedules screen.

Viewing Software Schedules

To view a software schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Software => Schedule**.

PNSM displays the **Software Schedule** window, listing the software schedules.

2. From the **Software Schedule** window, use any of the following methods to view schedule details:

Toolbar: Select the schedule then the **View**  icon.

Pop-up menu: Right-click the schedule and select **View**.

PNSM displays the **Software Schedule View** window, listing the following information:

- Schedule Name
- Software Release
- File Type
- Upgrade Type
- Last Execution Time

- Next Execution Time
- Operational Status
- Administrative Status

3. Select **Close** to close the window.

Viewing Software Schedule Subtask Status

When multiple Network Elements use a particular software schedule, PNSM creates a separate subtask to run the schedule on each node.

To view software schedule subtask status:

1. From the **Software Schedule** window, use any of the following methods to view subtasks:

Toolbar: Select the schedule then select the **View Sub Tasks**  icon.

Pop-up menu: Right-click the schedule and select **View Sub Tasks**.

PNSM displays the **Sub Tasks Details** window, listing the following information:

- Element Name
- Schedule Name
- Software Release
- Order of Execution
- Status
- Last Attempted Time
- Operational Status

2. To view software schedule status of a subtask, select the subtask then select the **View Status**  icon.

PNSM displays the **Software Schedule View Status** window, listing the software schedule progress status report.

3. Select **Close** to close the window.

4. To close the **Sub Tasks Details** window, select the **Close**  icon.

Software Rollback

PNSM allows you to rollback the software or a DPI signature file of a network element to the previous version.

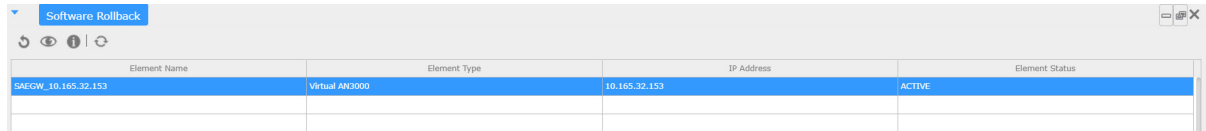
Performing Software Rollback

To perform a software rollback:

1. Select the **Settings** icon on the toolbar  and select **Configuration => Software => Rollback**.

PNSM displays the **Software Rollback** window, listing the network elements.

Figure 6-1 Software Rollback Window



Element Name	Element Type	IP Address	Element Status
SAGOW_10.165.32.153	Virtual AN3000	10.165.32.153	ACTIVE

2. Select the **Refresh**  icon to refresh the list of devices.
 3. Select the **Element Name** column to sort the devices alphabetically by name in ascending or descending order.
- The sort is dynamic: PNSM re-sorts the order every time you refresh the screen.
4. Select the network element to rollback.

5. To perform a software rollback, select the **Rollback**  icon and select the **Software Rollback**.

To perform a DPI signature rollback, select the **Rollback**  icon and select **DPI Rollback**.

6. At the confirmation prompt, select **Yes**.

PNSM verifies that the following conditions are true:

- The `rollback version` is available.
- The `activation-status` is in `Idle` state.

If both are true, PNSM activates the previous release and reboots the Network Element.

Viewing Software Rollback Progress

To view software rollback progress:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Software => Rollback**.

PNSM displays the **Software Rollback** window, listing the network elements.

2. Select the **Refresh**  icon to refresh the list of devices.
 3. Select the **Element Name** column to sort the devices alphabetically by name in ascending or descending order.
- The sort is dynamic: PNSM re-sorts the order every time you refresh the screen.
4. Select the network element then the **View**  icon.

PNSM displays the **Software Rollback View** window, listing the **Software Rollback Details** and the **DPI Rollback Details** for the network element.

Figure 6-2 Software Rollback View Window

5. Select **Close** to close the window.

Viewing Software Rollback Status

To view software rollback status:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Software => Rollback**.

PNSM displays the **Software Rollback** window, listing the network elements.

2. Select the **Refresh**  icon to refresh the list of devices.
3. Select the **Element Name** column to sort the devices alphabetically by name in ascending or descending order.

The sort is dynamic: PNSM re-sorts the order every time you refresh the screen.

4. Select the network element then the **View Rollback Status**  icon.

PNSM displays the **Software Rollback View Status** window, listing the software rollback status.

Figure 6-3 View Status Window for a Software Rollback

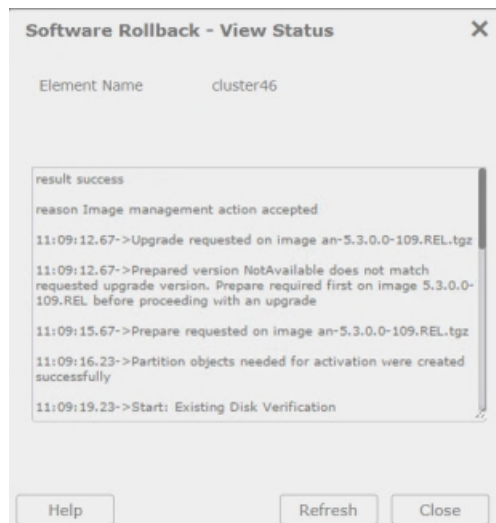
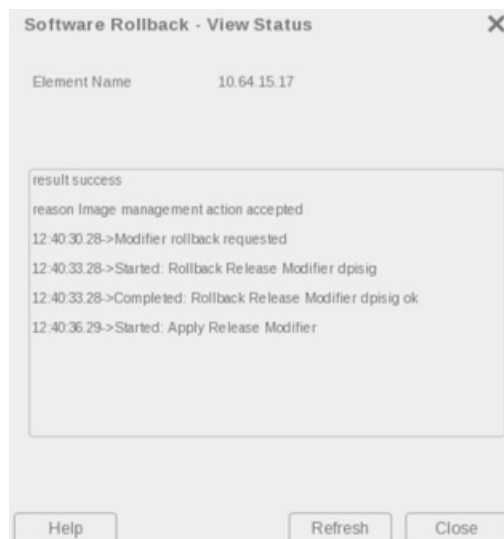


Figure 6-4 View Status Window for a DPI Signature Rollback



5. To refresh the status, select **Refresh**.
6. Select **Close** to close the window.

Network Software Rollbacks

If any network element fails to update during a network upgrade, Service Manager will rollback the network upgrade based on an automatic rollback schedule. You can view the rollback schedule from the Upgrade tab of the private network.

***Note:** If an HSS upgrade fails, Service Manager does not perform the rollback; this will be automatically performed by the HSS.*

Network Software Upgrades

You can perform private network software upgrades directly from the PNSM application. Follow the steps shown below.

Prerequisite

Before you can upgrade your private network from the PNSM, you must create an FTP site using the File Server Settings screen. For detailed instructions, please refer to [File Servers](#).

Note: The software required to upgrade the network are available in the release folder on the FTP site.

Upgrading the Network

The system software must then be imported Using the Network Version Import Feature. A `vnfswimport_apns_X_X.json` file allows you to define the name, version, and file path for each of the parameters to be upgraded.

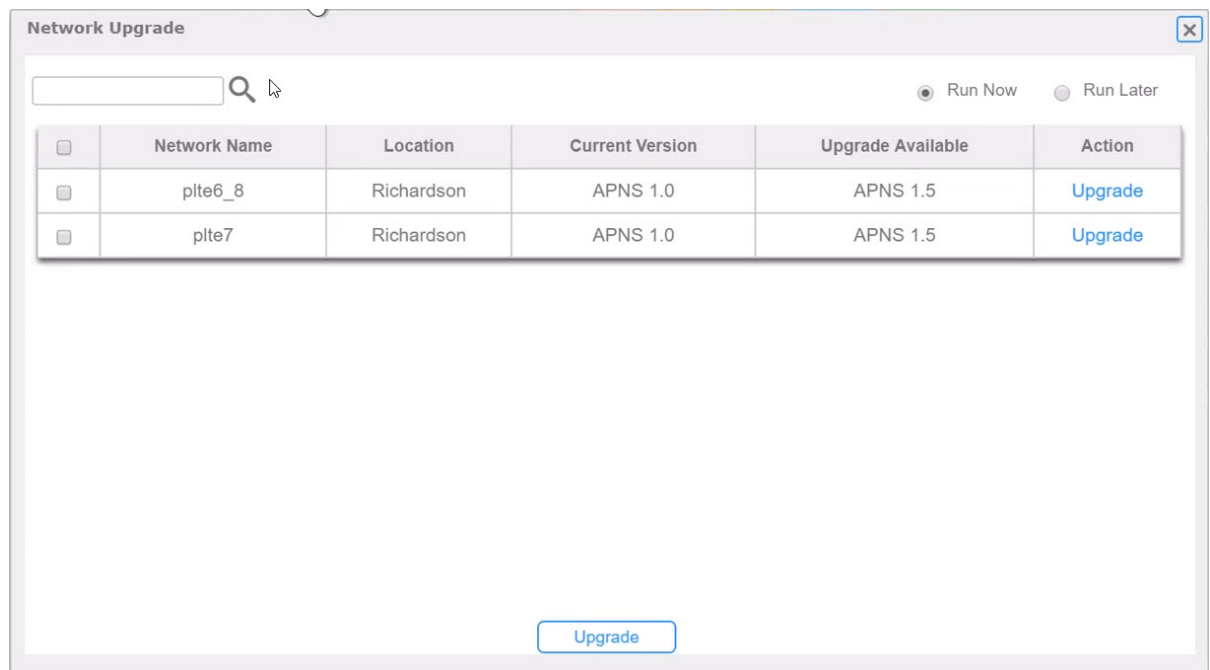
To upgrade the Network from within the PNSM application

1. Open the template `ServiceInstanceSoftware.json` file and update the to reflect the updated names, version numbers, and Image File Path for the components within the Packet Core as applicable.
2. Open the **Network Version Import** dialog by selecting **Configuration => Software => Network Version Import**.
3. Select the **File Server** from the dropdown list of available servers, choose the .json file to import, then select **Import**.

You can view the upgrade release name and status from within the **Software Maintenance** screen.

Note: When a Network Upgrade is available, a red Information () icon appears next to the **Private Network Service Manager** text on the interface. Selecting this icon opens the Network Upgrade dialog, which shows the available upgrade.

4. Select the red information Icon to open the **Network Upgrade** dialog:



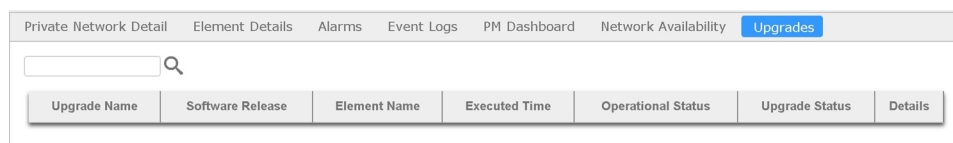
The 'Network Upgrade' dialog box features a search bar at the top left and two radio buttons at the top right: 'Run Now' (selected) and 'Run Later'. Below these is a table with the following data:

☐	Network Name	Location	Current Version	Upgrade Available	Action
☐	plte6_8	Richardson	APNS 1.0	APNS 1.5	Upgrade
☐	plte7	Richardson	APNS 1.0	APNS 1.5	Upgrade

At the bottom center of the dialog is a large 'Upgrade' button.

5. Mark the private network(s) to upgrade, select to **Run Now** or **Run Later**.
6. If you select to **Run Later**, select the date for the software upgrade from the calendar.
7. Select the **Upgrade** button.

High-level information regarding the upgrade is available from the Network-level **Upgrades** tab.



The 'Upgrades' tab in the Network-level view shows a search bar and a table with the following columns:

Upgrade Name	Software Release	Element Name	Executed Time	Operational Status	Upgrade Status	Details
--------------	------------------	--------------	---------------	--------------------	----------------	---------

For information on viewing more detailed information, please refer to [Software Schedules](#).

Note: Service Manager automatically checks if the upgrade image already exists in the VNF, or the specified location on the at Azure Stack Edge box. The upgrade image file is pushed from the Azure cloud only if the image file is not available on the VNF during the upgrade. This prevents unnecessary downloads of data from the portal to the ASE, which does not have backhaul connectivity or operation in low latency networks.

Prior to initiating the network upgrade from Service Manager, ensure the image files are copied to the following directories:

- On the HSS: `/home/hpeadmin/upgrade`
- On the MME: `/usr/local/epc/software`

CSV Stats Recovery

PNSM allows you to perform on-demand statistics recovery in MCC and SCEF devices. For example, in case of an PNSM server outage or a device is unreachable for more than 30 minutes, performance data in the PNSM database may be missing, which would result in gaps in the performance reports. You can launch a statistics recovery operation to save the 5-minute interval CSV files for the specified device(s).

PNSM exports the recovered CSV files to the export directory:

`/opt/Affirmed/NMS/server/ems/data/pm/<node name>`

The export filename is `<PM stats table name>-<interval timestamp>.csv`, where *interval timestamp* represents the starting timestamp of the export interval.

Launching a CSV Stats Recovery

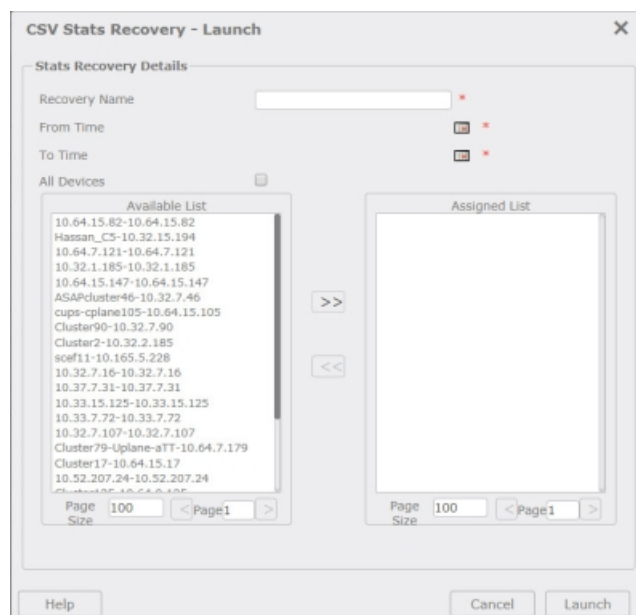
Only a single CSV stats recovery can be launched at a time. The Launch option is disabled if another recovery is running.

To launch a CSV statistics recovery:

1. Select the **Settings**  icon on the toolbar and select **Performance => CSV Stats Recovery**.
PNSM displays the **CSV Stats Recovery** window, listing the statistics recovery schedules.
2. Select the **Refresh**  icon to refresh the list of schedules.
3. To perform a new on-demand statistics recovery, select the **Launch**  icon.

PNSM displays the **CSV Stats Recovery Launch** window.

Figure 6-5 CSV Stats Recovery Launch Window



4. Configure the parameters as follows:

- **Recovery Name** – Enter the recovery name.
- **From Time** – Select the **Calendar**  icon and select a start date and 5-minute interval time (24-hour, based on GMT timezone) for the statistics. Select **Save**.
- **To Time** – Select the **Calendar**  icon and select an end date and 5-minute interval time (24-hour, based on GMT timezone) for the statistics. Select **Save**.

The maximum recovery range is 14 days.

- **All MCC Devices** – Mark the checkbox to recover the statistics from all managed devices. Unmark the checkbox to specify the managed devices.
- **Assigned List** – Select the MCCs by highlighting and selecting the right-arrow.

5. Select **Save**.

PNSM adds the recovery schedule to the **CSV Stats Recovery** window and runs the recovery. If files already exist in the export directory for a particular collection period, the operation *does not* overwrite them.

Viewing Statistics Recovery Schedule Details

To view a statistics recovery schedule details:

1. Select the **Settings**  icon on the toolbar and select **Performance => CSV Stats Recovery**.

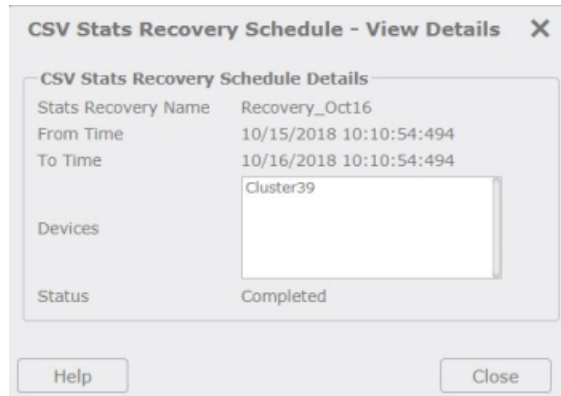
PNSM displays the **CSV Stats Recovery** window, listing the statistics recovery schedules:

- Stats Recovery Name
 - From Time
 - To Time
 - Status (Scheduled, Running, Completed)
2. From the **CSV Stats Recovery** window, use any of the following methods to view schedule details:

Toolbar: Select the schedule then the **View Details**  icon.

Pop-up menu: Right-click the schedule and select **View Details**.

PNSM displays the **View Details** window.



3. Select **Close** to close the window.

Viewing Statistics Recovery Schedule Status

When multiple Network Elements use a particular statistics recovery schedule, PNSM runs the schedule on each node. You can view the status of a recovery schedule on a per-device basis.

To view Statistics Recovery Schedule status:

1. Select the **Settings**  icon on the toolbar and select **Performance => CSV Stats Recovery**.

PNSM displays the **CSV Stats Recovery** window, listing the statistics recovery schedules:

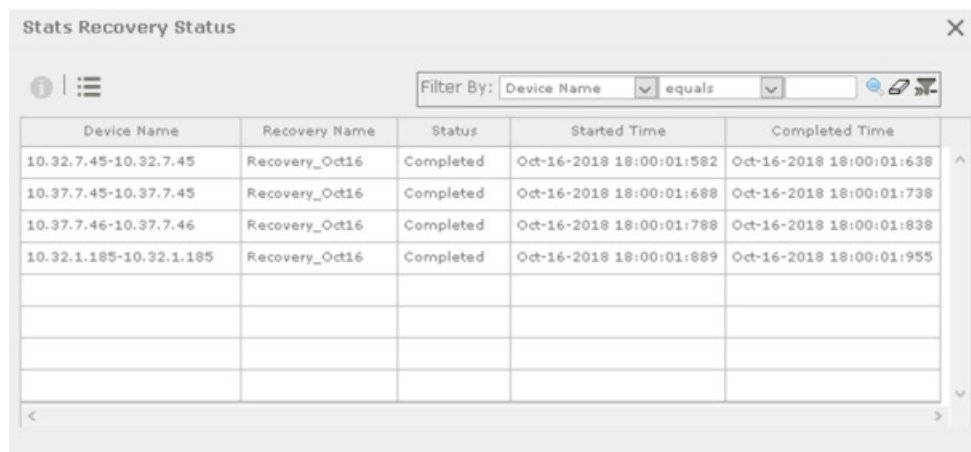
- Stats Recovery Name
 - From Time
 - To Time
 - Status (Scheduled, Running, Completed)
2. From the **CSV Stats Recovery** window, use any of the following methods to view schedule details:

Toolbar: Select the schedule then the **View Status**  icon.

Pop-up menu: Right-click the schedule and select **View Status**.

PNSM displays the **Stats Recovery Status** window, listing the following information:

- Device Name
- Recovery Name
- Software Release
- Status (Scheduled, Running, Completed)
- Started Time
- Completed Time

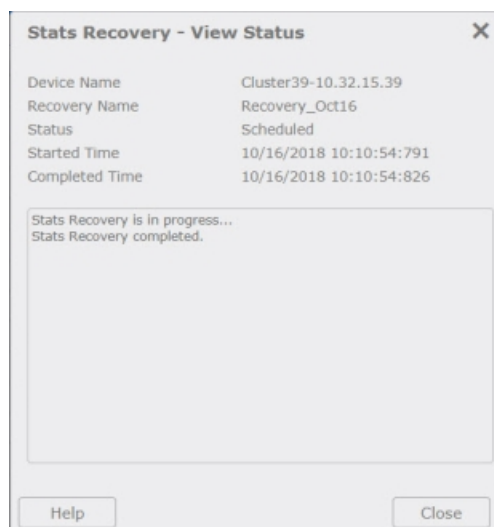
Figure 6-6 Stats Recovery Status Window


The window titled "Stats Recovery Status" contains a table with the following data:

Device Name	Recovery Name	Status	Started Time	Completed Time
10.32.7.45-10.32.7.45	Recovery_Oct16	Completed	Oct-16-2018 18:00:01:582	Oct-16-2018 18:00:01:638
10.37.7.45-10.37.7.45	Recovery_Oct16	Completed	Oct-16-2018 18:00:01:688	Oct-16-2018 18:00:01:738
10.37.7.46-10.37.7.46	Recovery_Oct16	Completed	Oct-16-2018 18:00:01:788	Oct-16-2018 18:00:01:838
10.32.1.185-10.32.1.185	Recovery_Oct16	Completed	Oct-16-2018 18:00:01:889	Oct-16-2018 18:00:01:955

- To view status of a recovery on a per-device basis, select the device and select the **View Status**  icon.

PNSM displays the **Stats Recovery View Status** window, listing the recovery status report for the selected device.



The window titled "Stats Recovery - View Status" displays the following information:

Device Name: Cluster39-10.32.15.39
 Recovery Name: Recovery_Oct16
 Status: Scheduled
 Started Time: 10/16/2018 10:10:54:791
 Completed Time: 10/16/2018 10:10:54:826

Stats Recovery is in progress...
 Stats Recovery completed.

Buttons: Help, Close

Possible scenarios for recovery of a statistics CSV file include:

- No statistics CSV file was recovered because of a connection failure to an MCC. A status message indicates the failure.
 - Partial statistics CSV file was recovered because some data intervals were unavailable on the MCC. A status message lists which intervals were not recovered.
 - Complete statistics CSV file was recovered with all requested intervals. A status message indicates recovery success.
- Select **Close** to close the **Stats Recovery View Status** window.
 - To close the **Stats Recovery Status** window, select the **Close**  icon.

Deleting Statistics Recovery Schedules

You can delete a statistics recovery schedule only if it has the Status Scheduled or Completed.

To delete a statistics recovery schedule:

1. Select the **Settings**  icon on the toolbar and select **Performance => CSV Stats Recovery**.

PNSM displays the **CSV Stats Recovery** window, listing the statistics recovery schedules:

- Stats Recovery Name
 - From Time
 - To Time
 - Status (Scheduled, Running, Completed)
2. Use any of the following methods to delete a statistics recovery schedule:

Toolbar: Select the schedule then the **Delete**  icon.

Pop-up menu: Right-click the schedule and select **Delete**.

3. At the confirmation prompt, select **OK**.

NE Configuration Backup and Restore

From Service Manager, you can backup and restore NE configuration data:

- **Backup** – You can perform a backup of the NE configuration and store it on the local Service Manager server or a remote server. The backup is available to restore the configuration to a previous known state in the event of lost or incorrect configuration.

You can:

- Trigger a backup manually on demand
- Schedule a backup of automatic daily execution

Note: NE Configuration backups are blocked and a threshold crossing alarm is raised when the /opt partition free disk space falls below 10%. The alarm severity is 60% (Minor), 75% (Major), and 90% (Critical).

NE Configuration Backup files are retained for a maximum of 90 days.

Service Manager also supports backing up configuration data and private network inventory reports for each of the reachable managed NEs to individual files. For details, see Network Configuration Backup.

- **Restore** – You can select a successful backed up configuration file and restore it on the NE. The result would be the NE configured exactly as the backup configuration. Be aware that the Restore action impacts service and results in a service restart.

Adding User Triggered NE Configuration Backup

You can trigger an NE backup on demand.

To configure a user triggered NE backup:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Select the **User Triggered NE-Backup**  icon.
3. Configure the parameters as follows:
 - **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as NE_backup.
 - **File Server** – Specify the file server on which you want the backup stored.
 - **Relative Path in File Server** – If you select to have the backup stored on a remote server, specify the path where you want the backup stored. If you select to have the backup stored on the server, it is stored at: *JBOSSHOME/server/ems/data/nebackups*.
 - **Select Elements**– Select the elements by highlighting and selecting the right-arrow.
4. Select **Save**.
5. At the confirmation prompt, select **OK**.
6. Select **Close**.

Service Manager initiates the NE configuration backup on each of the selected nodes. The backups of the individual nodes are stored on the configured file server location in separate folders labeled *MME_<IP-address>_<node name>*, *HSS_<IP-address>_<node name>*, and *SAEGW_<IP-address>_<node name>*. The details of the backup are logged in the configuration Audit Trail on success/failure.

Viewing NE Configuration Backup Details

You can view NE configuration backup details.

To view NE configuration backup details:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Use any of the following methods to view a backup's details:
 - Toolbar:** Select the backup then select the **View**  icon.
 - Pop-up menu:** Right-click the backup and select **View**.

Service Manager displays the **View NE Backup** window.

3. Select **Close** to close the window.

Purging NE Configuration Backup Files

You can specify the purging mechanism to retain only the last number of backup files per MME, HSS, and SAEGW.

To purge NE configuration backup files:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**.
Service Manager displays the **NE Backup and Restore** window.
2. Select the **Purge**  icon.
3. Mark the **Purge Backup** checkbox and specify how many backups to retain.
4. Select **Save**.

Downloading NE Configuration Backup File

You can download an NE configuration backup file only if its status is Success.

To download an NE configuration backup file:

1. Select the **Settings** icon on the toolbar  and select **Tools => NE Configuration Backup**.
Service Manager displays the **NE Backup and Restore** window.
2. Select the NE Configuration backup filename then select the **Download**  icon.
3. At the confirmation prompt, select **OK**.

Restoring NE Configuration Backup File

You can restore an NE configuration backup file only if its status is Success.

To restore an NE configuration backup file:

1. Select the **Settings** icon on the toolbar  and select **Tools => NE Configuration Backup**.
Service Manager displays the **NE Backup and Restore** window.
2. Select the NE Configuration backup filename then select the **Restore**  icon.
Service Manager initiates NE configuration restore process on the specific element (HSS/MME/SAEGW) based on the backup file.

Deleting NE Configuration Backup

You can delete an NE configuration backup.

To delete an NE configuration backup:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Use any of the following methods to delete an NE configuration backup:
Toolbar: Select the backup then the **Delete**  icon.
Pop-up menu: Right-click the backup and select **Delete**.
3. At the confirmation prompt, select **OK**.

Scheduling NE Configuration Backup

You can configure the Service Manager server to backup NE configuration based on a pre-configured schedule.

To schedule an NE configuration backup:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Select the **Scheduled NE-Backup**  icon.
3. Select the **New**  icon.
4. Configure the parameters as follows:
 - **NE Backup Schedule Name** – Specify the schedule name.
 - **Recurrence** – Mark this checkbox to specify a backup recurrence. By default, the scheduler will be scheduled to execute once every 12 hours. If you leave the checkbox unmarked, the schedule is saved in the disabled state:
 - Select the **Recurrence Pattern**  icon and configure the Recurrence Details.
 - Select **Reschedule**.
 - **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as NE_backup.
 - **File Server** – Specify the file server on which you want the backup stored.
 - **Relative Path in File Server** – If you select to have the backup stored on a remote server, specify the path where you want the backup stored. If you select to have the backup stored on the EMS server, it is stored at:
JBOSSHOME/server/ems/data/nebackups.
 - **Select Elements**– Select the elements by highlighting and selecting the right-arrow.
5. Select **Save**.

Service Manager performs the NE configuration backup to execute once in every configured interval. By default, the scheduler executes the backup once in every 12 hours (the minimum interval).

The backups of the individual nodes are stored on the configured file server location in separate folders labeled *MME_<IP-address>_<node name>*, *HSS_<IP-address>_<node name>*, and *SAEGW_<IP-address>_<node name>*.

The details of the backup are logged in the configuration Audit Trail on success/failure.

Viewing Scheduled NE Configuration Backup Details

You can view scheduled NE configuration backup details.

To view NE configuration backup details:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Select the **Scheduled NE-Backup**  icon.
3. Use any of the following methods to view a backup's details:
 - Toolbar:** Select the backup then the **View**  icon.
 - Pop-up menu:** Right-click the backup and select **View**. Service Manager displays the **View NE Backup** window.
4. Select **Cancel** to close the window.

Modifying Scheduled NE Configuration Backup

You can modify a scheduled NE configuration backup only if it does not have the status Complete.

To modify a scheduled NE configuration backup:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Select the **Scheduled NE-Backup**  icon.
3. Select the schedule then select the **Edit**  icon.
4. Configure the parameters as follows:
 - **NE Backup Schedule Name** – Specify the schedule name.
 - **Recurrence** – Mark this checkbox to specify a backup recurrence. By default, the scheduler will be schedule to execute once in every 12 hours. If you leave the checkbox unmarked, the schedule is saved in the disabled state:
 - Select the **Recurrence Pattern** icon and configure Recurrence Details.

— Select the **Recurrence**.

- **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as NE_backup.
- **File Server** – Specify the file server on which you want the backup stored.
- **Relative Path in File Server** – If you select to have the backup stored on a remote server, specify the path where you want the backup stored. If you select to have the backup stored on the EMS server, it is stored at:
JBOSSHOME/server/ems/data/nebackups.
- **Select Elements**– Select the elements by highlighting and selecting the right-arrow.

5. Select **Save**.

Service Manager performs the NE configuration backup to execute once in every configured interval. By default, the scheduler executes the backup once in every 12 hours (the minimum interval).

The details of the backup are logged in the configuration Audit Trail on success/failure.

Enabling/Disabling NE Configuration Backup Schedules

You can enable/disable NE configuration backup schedules.

To enable/disable an NE backup schedule:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Select the **Scheduled NE-Backup**  icon.
3. Use any of the following methods to enable a schedule:

Toolbar: Select the schedule then select the **Enable**  icon.

Pop-up menu: Right-click the schedule and select **Start**.

Use any of the following methods to disable a schedule:

Toolbar: Select the schedule then the **Disable**  icon.

Pop-up menu: Right-click the schedule and select **Stop**.

4. At the confirmation prompt, select **OK**.
The Administrative Status of the schedule changes to Enabled or Disabled.

Quick Start NE Configuration Backup Schedules

To Quick Start (instantly execute) an NE configuration backup schedule:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Select the **Scheduled NE-Backup**  icon.
3. Use any of the following methods to Quick Start a schedule:

Toolbar: Select the schedule then select the **Quick Start**  icon.

Pop-up menu: Right-click the schedule and select **Quick Start**.

Exporting NE Configuration Backup Schedules

You can export an NE configuration backup schedule.

To export an NE configuration backup schedules:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**. Service Manager displays the **NE Backup and Restore** window.
2. Select the **Scheduled NE-Backup**  icon.
3. If desired, use the Filter Manager to filter the data displayed in the window.

4. Select the schedule(s) to be exported. To select multiple schedules, Shift-click or Control-click the schedules in the list. To export the Whole Data or Current View, do not select any schedule.
5. Select the **Export**  icon.
Service Manager displays the **Configure Export Parameters** window.
6. Configure the parameters as follows:
 - **File Type:** Select the required file format (CSV).
 - **Column Configuration:** Select the columns you want exported (Visible Column or All Columns)
 - **Export Type:** Select the export type (Whole Data, Current View, Selected Data)
 - **Use Current Filter:** Mark the checkbox to restrict data based on the existing filter conditions.
7. Select **Apply**.
Service Manager displays the **Download Files** window.
8. Select **Download Exported File**.
9. From the **Download Files** window, select **Close**.

Deleting NE Configuration Backup Schedules

You can delete an NE configuration backup schedule only if its administrative status is Disabled.

To delete an NE configuration backup schedule:

1. Select the **Settings**  icon on the toolbar and select **Tools => NE Configuration Backup**.
Service Manager displays the **NE Backup and Restore** window.
2. Select the **Scheduled NE-Backup**  icon.
3. Use any of the following methods to delete a backup schedule:
 - Toolbar:** Select the schedule then the **Delete**  icon.
 - Pop-up menu:** Right-click the schedule and select **Delete**.
4. At the confirmation prompt, select **OK**.

File Servers

You can use File Servers across a location to maintain network element database backups and network element software.

Adding File Servers

To add a file server:

1. Select the **Settings**  icon on the toolbar and select **Configuration => File Server => Settings**.

PNSM displays the **File Server Settings** window, listing the file servers.

2. Select the **New**  icon.
3. Configure the parameters as follows:
 - **Type** – Select SFTP.
 - **Name**
 - **IP Version** – Select IPv4 or IPv6.
 - **IP Address**
 - **Port**
 - **User Name**
 - **Password**
 - **Confirm Password**
 - **Save File Server, if inactive** – Leave the checkbox unmarked if you want the IP Address validated with the given credentials.
4. Select **Save**.

Viewing File Servers

To view a file server:

1. Select the **Settings**  icon on the toolbar and select **Configuration => File Server => Settings**.

PNSM displays the **File Server Settings** window, listing the file servers.

2. From the **File Server** window, use any of the following methods to view file server details:

Toolbar: Select the file server then select the **View**  icon.

Pop-up menu: Right-click the file server and select **View**.

PNSM displays the **File Server Details** window, listing the following information:

- File Server Type
- File Server Name
- IP Address
- User Name

3. Select **Close** to close the window.

Modifying File Servers

To modify a file server:

1. Select the **Settings** icon on the toolbar  and select **Configuration => File Server => Settings**.

PNSM displays the **File Server Settings** window, listing the file servers.

2. Use any of the following methods to modify a file server:

Toolbar: Select the file server and select the **Edit**  icon.

Pop-up menu: Right-click the file server and select **Edit**.

PNSM displays the **Edit Network Element** window.

3. Make the necessary changes:
 - **File Server Name**
 - **Port**
 - **User Name**
 - **Password**
 - **Confirm Password**
 - **Save File Server, if inactive** – Leave the checkbox unmarked if you want the IP Address validated with the given credentials.
4. Select **Save**.
5. If `SSH_KEY_AUTO_ADD` property is set to false, PNSM prompts to confirm that the file server with the SSH key fingerprint is the trusted host to which to connect. At the confirmation prompt, select **OK**.

If `SSH_KEY_AUTO_ADD` property is set to true, PNSM automatically adds the SSH key of the file server and its details.

For details on `SSH_KEY_AUTO_ADD` property, see [securityconfiguration.properties](#).

Deleting File Servers

You cannot delete the default file server.

To delete a file server:

1. Select the **Settings**  icon on the toolbar and select **Configuration => File Server**.

PNSM displays the **File Server** window, listing the file servers.

2. Use any of the following methods to delete a file server:

Toolbar: Select the file server then select the **Delete**  icon.

Pop-up menu: Right-click the file server and select **Delete**.

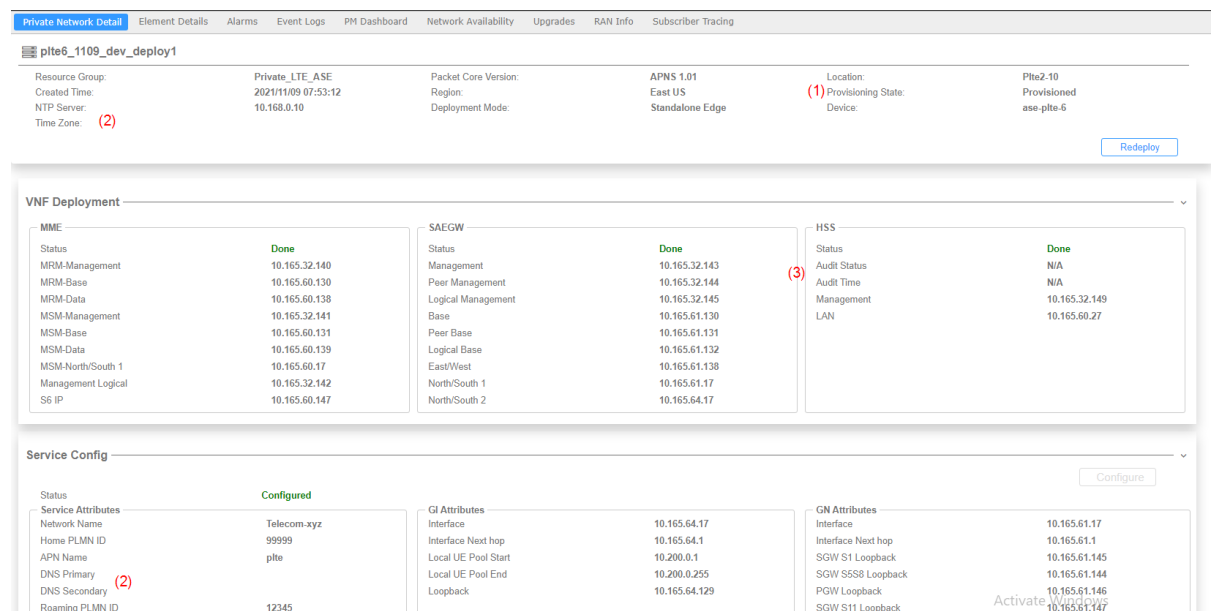
3. At the confirmation prompt, select **OK**.

Verifying a Successful Service Manager Upgrade

Follow the procedure in the Affirmed Private Network ServiceDeployment guide to upgrade the Service Manager application.

To verify a successful upgrade:

1. Open the Private Networks Detail tab.
2. Verify the following field values are as shown in the image below:
 - **Provisioning State** of the Networks will be changed from **Succeeded** to **Provisioned** (1).
 - **TimeZone**, **DNS Primary** and **Secondary** values are Empty (2) for the Networks deployed in PNSM 1.0.
 - HSS last **Audit status** and **Time** is shown in the HSS section.



Private Network Detail | Element Details | Alarms | Event Logs | PM Dashboard | Network Availability | Upgrades | RAN Info | Subscriber Tracing

plte6_1109_dev_deploy1

Resource Group:	Private_LTE_ASE	Packet Core Version:	APNS 1.01	Location:	Plte2-10
Created Time:	2021/11/09 07:53:12	Region:	East US	(1) Provisioning State:	Provisioned
NTP Server:	10.168.0.10	Deployment Mode:	Standalone Edge	Device:	ase-plte-6
Time Zone:	(2)				

[Redeploy](#)

VNF Deployment

MME		SAEGW		HSS	
Status	Done	Status	Done	Status	Done
MRM-Management	10.165.32.140	Management	10.165.32.143	Audit Status	N/A
MRM-Base	10.165.60.130	Peer Management	10.165.32.144	Audit Time	N/A
MRM-Data	10.165.60.138	Logical Management	10.165.32.145	Management	10.165.32.149
MSM-Management	10.165.32.141	Base	10.165.61.130	LAN	10.165.60.27
MSM-Base	10.165.60.131	Peer Base	10.165.61.131		
MSM-Data	10.165.60.139	Logical Base	10.165.61.132		
MSM-North/South 1	10.165.60.17	East/West	10.165.61.138		
Management Logical	10.165.32.142	North/South 1	10.165.61.17		
S6 IP	10.165.60.147	North/South 2	10.165.64.17		

Service Config

Service Attributes		GI Attributes		GN Attributes	
Status	Configured	Interface	10.165.64.17	Interface	10.165.61.17
Network Name	Telecom.xyz	Interface Next hop	10.165.64.1	Interface Next hop	10.165.61.1
Home PLMN ID	99999	Local UE Pool Start	10.200.0.1	SGW S1 Loopback	10.165.61.145
APN Name	plte	Local UE Pool End	10.200.0.255	SGW S588 Loopback	10.165.61.144
DNS Primary	(2)	Loopback	10.165.64.129	PGW Loopback	10.165.61.146
DNS Secondary				SGW S11 Loopback	10.165.61.147
Roaming PLMN ID	12345				

[Configure](#)

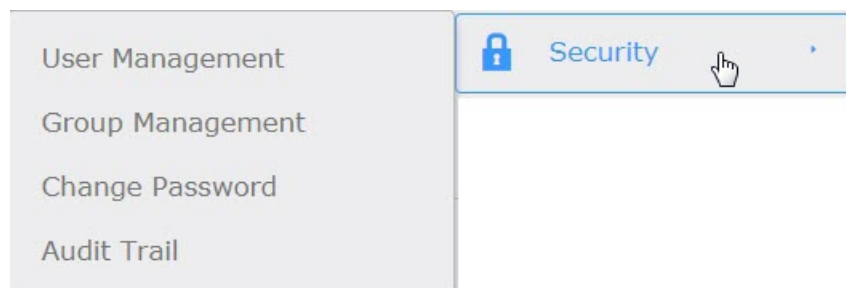
3. Open the File Server Details dialog by selecting Configuration => File Server from the Settings menu.
4. Verify the default server **Name** is **PNSM Server**.

Security Settings

Security management allows you to add a new user and select the user type from those available in the system. You can also use the security settings to view the user-level audit trail and to change the user password used to access the Service Manager application.

To access Security applications, select the applicable option from the **Settings** icon on the toolbar. 

Figure 7-1 Accessing Security Applications



Users

Adding Users

An administrator can add users.

To add a user:

1. Select the **Settings**  icon on the toolbar and select **Security => User Management**.
Service Manager displays the **Security User** window, listing the users in the system.
2. Select the **Add**  icon.
3. Configure the parameters as follows:
 - **User Name** – User account name
 - **Password** – User account password

- **Confirm Password** – User account password
- **Idle Timeout** – Number of minutes of idle time after which the user is logged out. Value 0 indicates the client session does not expire.
- **Maximum Password Age** – Number of days from the first login after which the password expires. A value of 0 indicates that the password does not expire.
- **Maximum Account Age** – Number of days from the first login after which the user account expires. A value of 0 indicates that the account does not expire.
- **Description** – User account description
- **Status** – User account status:
 - Active: User is active and can log in.
 - Inactive: User is inactive and cannot log in
 - Blocked: User is blocked from logging in
 - Password Expired: Maximum password age has expired
 - Account Expired: Maximum Account Age has expired
 - Account Locked: User had consecutive failed logon attempts (default 6), so the user account is disabled and must be activated by the administrator.
 - Change Password on Login: User is forced to change password on next login
 - Dormancy Period Exceeded: User is locked out because the dormancy time was exceeded
- **Supervisor** – Supervisor account (to terminate the user's session, change the user's password, controlling the user's status)
- **Assign Group** – Groups to which the user account belongs. Select a group then select the right-arrow to assign the user to it.
- **Pager** – User account pager address, specified as <PAGERNUMBER>@<SERVICEPROVIDERDOMAIN>.
- **E-mail Address** – User account e-mail address
- **Mobile** – User account phone number
- **Password Change on re-Login** – Mark the checkbox to force the user to change password on next login.

4. Select **Save**.

Viewing Users

To view users:

1. Select the **Settings** icon on the toolbar  and select **Security => User Management**.

Service Manager displays the **Security User** window, listing the users in the system.

Users are listed with the following information:

- **User Name** – User account name
- **Assigned Group(s)** – Groups to which the user account belongs
- **Description** – User account description
- **Status** – User account status:
 - Active: User is active and can log in.
 - Inactive: User is inactive and cannot log in
 - Blocked: User is blocked from logging in
 - Password Expired: Maximum password age has expired
 - Account Expired: Maximum Account Age has expired
 - Account Locked: User had consecutive failed logon attempts (default 6), so the user account is disabled and must be activated by the administrator.
 - Change Password on Login: User is forced to change password on next login
 - Dormancy Period Exceeded: User is locked out because the dormancy time was exceeded
- **E-mail Address** – User account e-mail address
- **Updated Time** – Date and time of last update of the account
- **Account Expiry** – Date and time when the account will expire

2. From the **Security User** window, use any of the following methods to view a user's details:

Toolbar: Select the user then the **View**  icon.

Pop-up menu: Right-click the user and select **View**.

Service Manager displays the User Profile window, listing the settings for the user:

- **Name** – User account name
- **Idle Timeout** – Number of minutes of idle time after which the user is logged out. Value 0 indicates the client session does not expire.
- **Maximum Password Age** – Number of days from the first login after which the password expires. A value of 0 indicates that the password does not expire.
- **Maximum Account Age** – Number of days from the first login after which the user account expires. A value of 0 indicates that the account does not expire.
- **Assign Group** – Groups to which the user account belongs.

- **Pager** – User account pager address, specified as <PAGERNUMBER>@<SERVICEPROVIDERDOMAIN>.
 - **E-mail** – User account e-mail address.
 - **Mobile** – User account phone number.
 - **Supervisor** – Supervisor account (used to terminate the user's session, change the user's password, and control the user's status).
 - **Password Change on re-Login** – Whether to force the user to change password on next login.
3. Select **Close** to close the window.

Modifying Users

An administrator can modify users.

To modify a user:

1. Select the **Settings**  icon on the toolbar and select **Security => User Management**. Service Manager displays the **Security User** window, listing the users in the system.
2. Use any of the following methods to edit the user:

Toolbar: Select the user then the **Edit**  icon.

Pop-up menu: Right-click the user and select **Edit**.

Service Manager displays the Edit User window, listing the settings for the user.

3. Configure the parameters as follows:
 - **User Name** – User account name
 - **Idle Timeout** – Number of minutes of idle time after which the user is logged out. Value 0 indicates the client session does not expire.
 - **Maximum Password Age** – Number of days from the first login after which the password expires. A value of 0 indicates that the password does not expire.

Cannot be modified for admin user.

- **Maximum Account Age** – Number of days from the first login after which the user account expires. A value of 0 indicates that the account does not expire.

Cannot be modified for admin user.

- **Description** – User account description
- **Status** – User account status:

— Active: User is active and can log in.

- Inactive: User is inactive and cannot log in
- Blocked: User is blocked from logging in
- Password Expired: Maximum password age has expired
- Account Expired: Maximum Account Age has expired
- Account Locked: User had consecutive failed logon attempts (default 6), so the user account is disabled and must be activated by the administrator.
- Change Password on Login: User is forced to change password on next login
- Dormancy Period Exceeded: User is locked out because the dormancy time was exceeded
- **Supervisor** – Supervisor account (used to terminate the user's session, change the user's password, and control the user's status)
- **New Password** – User account password
- **Confirm Password** – User account password
- **Send Mail Notification** – Mark the checkbox to notify the user of a password change by e-mail.
- **Assign Group** – Groups to which the user account belongs. Select a group then select the right-arrow to assign the user to it.
- **Pager** – User account pager address, specified as <PAGERNUMBER>@<SERVICEPROVIDERDOMAIN>.
- **E-mail** – User account e-mail address
- **Mobile** – User account phone number
- **Password Change on re-Login** – Mark the checkbox to force the user to change password on next login.

4. Select **Save**.

Exporting Users

You can export a list of users to a CSV file.

To export users:

1. Select the **Settings**  icon on the toolbar and select **Security => User Management**.
Service Manager displays the **Security User** window, listing the users in the system.
2. If desired, use the Filter Manager or Quick Filter to filter the data displayed in the window.
3. Select the **Export**  icon.
Service Manager displays the **Configure Export Parameters** window.

4. Configure the parameters as follows:
 - **File Type:** Select the required file format (CSV).
 - **Column Configuration:** Select the columns you want exported (Visible Column or All Columns)
 - **Export Type:** Select the export type (Whole Data, Current View, Selected Data)
 - **Use Current Filter:** Mark the checkbox to restrict data based on the existing filter conditions.
5. Select **Apply**.

Service Manager displays the **Download Files** window.
6. Select **Download Exported File**.
7. From the **Download Files** window, select **Close**.

Managing Active User Sessions

An administrator can view the list of users who are currently logged in and terminate one or more user session(s).

To manage active user sessions:

1. Select the **Settings**  icon on the toolbar and select **Security => User Management**.

Service Manager displays the **Security User** window, listing the users in the system.
2. Select the **Active User List**  icon.

Service Manager displays the Active User Listing window, listing users currently logged in.
3. To terminate any user sessions, mark the checkboxes for the user(s) and select the **Terminate User Session(s)**  icon.
4. At the confirmation prompt, select **Yes**.

Service Manager forces the termination of the current session of the specified user(s) and takes the user(s) to the login prompt.
5. Select **Close** to close the window.

Unlocking Users

An administrator can unlock a user whose account is locked because the user tried to login with the wrong password six consecutive times (or the number of times specified in the securityconfiguration.properties file).

To unlock a user:

1. Select the **Settings**  icon on the toolbar and select **Security => User Management**.

Service Manager displays the **Security User** window, listing the users in the system.

2. Select the user then the **Unlock User**  icon.
 3. At the confirmation prompt, select **OK**.
- Service Manager unlocks the account so that the user can log in.

Deleting Users

An administrator can delete users. A currently logged-in user cannot be deleted. The default user admin cannot be deleted.

To delete a user:

1. Select the **Settings**  icon on the toolbar and select **Security => User Management**.
Service Manager displays the **Security User** window, listing the users in the system.

Toolbar: Select the user then select the **Delete**  icon.

Pop-up menu: Right-click the user and select **Delete**.

2. At the confirmation prompt, select **OK**.

Changing Your User Password

You can change your user password.

To change your user password:

1. Select the **Settings**  icon on the toolbar and select **Security => Change Password**.
PNSM displays the Change Password window.
2. Enter the new password in the **New Password** and **Confirm Password** fields.
3. Mark the **Send mail notification** checkbox to send an email notification.
4. To force a password change on next login, mark the **Password Change on Re Login** checkbox.
5. Select **Apply**.

User-Level Audit Trail

The security Audit Trail is a record of users' Service Manager operations/activities. The following table lists the operation to the Service Manager system and the corresponding message logged to the audit trail:

Table 7-1 Audit Trail Operations

Category	Event	Logged Message
Authentication	User attempts to log into Service Manager	Login
	User attempts to log into Service Manager using the credentials of the blocked account	Login: User Blocked
	User tries to login into Service Manager using the credentials of account present in inactive status.	Login: User Inactive
	User tries to login into Service Manager using the credentials of the account present in temporary locked status.	Login: User Locked
	User tries to login into Service Manager using the wrong password for more than 3 times.	Login: Maximum failed login count exceeded. <i>Note: Also raises a Critical alarm</i>
	User tries to login into Service Manager using credentials of expired account.	Login: User Account Expired
	User tries to login into Service Manager using expired password	Login: Password Expired
	User tries to logout of Service Manager	Logout
User Security Profile Update	Modified Idle time out	Update: <i>username</i> 's idle time out period
	Update minimum Password Age	Update: <i>username</i> 's minimum password age
	Update maximum Password Age	Update: <i>username</i> 's maximum password age
	Update the account description	Update: <i>username</i> 's description
	Update the user account status	Update: <i>username</i> 's account status
	Update the account's supervisor	Update: <i>username</i> 's supervisor
	Assign or remove group associated with user	Update: <i>username</i> 's group assignment
	Update personal information email, pager	Update: <i>username</i> 's personal information email, pager

Note: In the case of multiple updates to the user profile, a single message listing all changes is logged to the Audit Trail

Category	Event	Logged Message
Security Configuration Operations	Create a user account: 1. Security => User Management 2. Select the Add icon and create the user.	Add User: <i>username</i>
	Terminate a user session: 1. Security => User Management 2. Select the Active User List icon. 3. Mark the checkbox for the user and select the Terminate User Session(s) icon.	Terminate User Session: <i>username</i>
	Delete a user account: 1. Security => User Management 2. Select the user then select the Delete icon.	Delete User: <i>username</i> deleted
	Change a user password: 1. Security => User Management 2. Select the user then select the Edit icon. 3. Enter the new password in the New Password and Confirm Password fields.	Update: <i>username</i> changed the password
User Account Changes made by Service Manager	Password expired: When the user's maximum password age exceeds the number of days after first login, the Service Manager system updates the status of user profile to 'password expired.'	System: User <i>username</i> 's password expired <i>Note: Also raises a Major alarm</i>
	Enable User: Enable user to active status from the 'account blocked' status after waiting for the lockout time. Current lockout time is 30 minutes.	System: User <i>username</i> activated from blocked state
	Account expired: When the user's maximum account age exceeds the number of days after first login, the Service Manager system updates the status of user profile to 'Account expired.'	System: User <i>username</i> 's account expired <i>Note: Also raises a Major alarm</i>
	Dormancy period: Service Manager system marks the user's account status as 'Dormancy period exceeded' when either of following events takes place: ▪ User attempts to login for first time, but the number of days between the account creation and the attempted login is greater than the dormancy period. ▪ User attempts to login, but the number of days between the last successful login and the current login is greater than system's dormancy period. Existing dormancy period is configured as 30 days.	System: User <i>username</i> moved to dormant state <i>Note: Also raises a Major alarm</i>

To view the Audit Trail for all users:

1. Select the **Settings**  icon on the toolbar and select **Security => Audit Trail**.
Service Manager displays the Audit Trail window, listing the Security Audit Trail, with the following details:
 - Date and time of the event
 - User name of the user that performed the operation
 - IP address of the host performing the event.
 - Module on which the operation was performed
 - Operation description
 - Element name
 - Status of the event indicating the success or failure of that event.
2. To view all Audit Trails (Configuration, Fault, Performance, Platform, Security, and Topology), in the Filter Manager, select **All Modules**. If desired, use the Quick Filter to filter the data displayed in the window.

You can export the Audit Trail to a CSV/PDF/XLS/XML/HTML file.

To view the Audit Trail for a particular user:

1. Select the **Settings**  icon on the toolbar and select **Security => User Management**.
Service Manager displays the **Security User** window, listing the users in the system.
2. Select the user then select the **Audit Trail**  icon.
Service Manager displays the Audit Trail window, listing the activities for the user, with the following details:
 - Module on which the operation was performed
 - Description of the operation
 - User name of the user that performed the operation
 - IP address of the host performing the event.
 - Date and time of the event
 - Status of the event indicating the success or failure of that event.
3. If desired, use the Filter Manager or Quick Filter to filter the data displayed in the window.

You can export the Audit Trail to a CSV/PDF/XLS/XML/HTML file.

To export the Audit Trail:

1. From the **Audit Trail** window, select the **Export**  icon.
Service Manager displays the **Configure Export Parameters** window.
2. Configure the parameters as follows:
 - **File Type:** Select the required file format (CSV, PDF, XLS, XML, HTML).
 - **Column Configuration:** Select the columns you want exported (Visible Column or All Columns)
 - **Export Type:** Select the export type (Whole Data, Current View, Selected Data)
 - **Use Current Filter:** Mark the checkbox to restrict data based on the existing filter conditions.
3. Select **Apply**.
Service Manager displays the **Download Files** window.
4. Select **Download Exported File**.
5. From the **Download Files** window, select **Cancel**.

Administrative Tasks

This section details tasks to configure administrative features of the PNSM software, including configuring alarm color settings and external communications. Additionally, it describes how to set filters for viewing alarms and events for your private network service.

Changing Screen Accessibility Settings

You can make changes to the PNSM screen display to improve readability.

Changing Alarm Severity Color Settings

You can configure alarm severity colors settings. And, you can restore alarm severity colors settings to the system default settings.

To configure alarm severity color settings:

1. Select the **Settings**  icon on the toolbar and select **Tools => Alarm Severity Color Settings**.

PNSM displays the **Color Configuration** window.

2. Select the color to be changed. Choose any color from the RGB or Color Slider tab.
3. Select **Apply**.
4. At the confirmation prompt, select **OK**.

To restore alarm severity colors to the system default settings:

1. Select the **Settings**  icon on the toolbar and select **Tools => Alarm Severity Color Settings**.

PNSM displays the **Color Configuration** window.

2. Select **Restore to Defaults**.
3. Select **Apply**.

Changing GUI Theme

To change the GUI theme:

1. Select the **Settings**  icon on the toolbar and select **Change Theme** and the theme you want:
 - **Change Theme => Light Theme**
 - **Change Theme => Dark Theme**
 - **Change Theme => High Contrast**

Configuring Alarm Severity Color Settings

You can configure alarm severity colors settings. And, you can restore alarm severity colors settings to the system default settings.

To configure alarm severity color settings:

1. Select the **Settings** icon on the toolbar  and select **Tools => Alarm Severity Color Settings**.
Service Manager displays the **Color Configuration** window.
2. Select the color to be changed. Choose any color from the RGB or Color Slider tab.
3. Select **Apply**.
4. At the confirmation prompt, select **OK**.

To restore alarm severity colors to the system default settings:

1. Select the **Settings** icon on the toolbar  and select **Tools => Alarm Severity Color Settings**.
Service Manager displays the **Color Configuration** window.
2. Select **Restore to Defaults**.
3. Select **Apply**.

Configuring the Mail Server

You can set the parameters that specify how the application sends messages to the mail server, including SMTP access and authentication settings. If you configure alarm e-mail notification, Service Manager forwards alarms as e-mail to the mail server.

To configure the mail server:

1. Select the **Settings**  icon on the toolbar and select **Topology => Application Settings => Mail Server**.
Service Manager displays the **Mail Server** window.

2. Configure the parameters as follows:

- **Primary SMTP Host Name** – Host name of the Simple Mail Transfer Protocol (SMTP) server. The host name can contain the special characters . and -.
- **Secondary SMTP Host Name** – Another host name of the SMTP server, used in case of any failure in the Primary SMTP Host name. The host name can contain the special characters . and -.
- **SMTP Port** – Port ID for the SMTP:
 - If you specify port 25, Service Manager creates an unauthenticated session with the mail server, so that no user or password should be configured when using that port.
 - If you need Service Manager to create an authenticated session, specify a port other than port 25.
- **SMTP User** – User name of the SMTP server.
- **SMTP Password** – Password of the SMTP server.
- **Confirm Password** – Confirmation of the entered password.
- **Sender Name** – User name used for sending e-mail by the Service Manager server.
- **Sender Mail ID** – Mail ID used for sending e-mail by the Service Manager server.

3. Select **Save**.

Configuring SNMP Settings

SNMP parameters are used as global SNMP settings.

***Note:** These settings are not generally required for configuration.*

To configure SNMP Settings:

1. Select the **Settings**  icon on the toolbar and select **Topology => Application Settings => SNMP Configuration**.

Service Manager displays the **SNMP Configuration** window.

2. To modify the default configuration, select the **Edit**  icon.
3. Configure the parameters as follows:

Southbound SNMP Parameters:

- **No. of Retries** – Number of retries to send SNMP requests to a device.
- **Port** – Port number to which the SNMP request is sent for a device.

- **Timeout** – Timeout in milliseconds after an SNMP request to get a response from the device.

Northbound SNMP Parameters:

- **Local Engine ID** – Auto-generated unique Engine ID based on IP address and random bytes.
- **Trap Community** – (SNMPv2c) Trap Community string used to forward alarms to trap receivers.

4. Select **Save**.

Configuring ICMP Communication

Use this option to enable/disable ICMP in any scenarios for communication.

When disabled, Service Manager does not use ICMP for any communication with the devices. Instead Service Manager uses other supported protocols (if any). For example, to retrieve the reachability status of the device, when ICMP is disabled, Service Manager uses only SNMP to check the status.

For devices types that support ICMP only (such as Siemens-PCRF and HPESSE), when ICMP is enabled, Service Manager can successfully add these devices only if they are ICMP reachable. When ICMP is disabled, these devices are added in Service Manager without checking ICMP status.

To configure ICMP communication:

1. Select the **Settings**  icon on the toolbar and select **Topology => Application Settings => ICMP Configuration**.
2. Mark **Yes** to enable ICMP communication. Mark **No** to disable ICMP communication.
3. Select **OK**.

Configuring Message Prompt on Locked User Accounts

The Administrator can configure the email ID to be included in the system-level message prompt to any user attempting to log into a locked or expired account. For example:

Please contact J.Doe@XYZCorp with your account request

To configure the email ID:

1. Select the **Settings**  icon on the toolbar and select **Topology => Application Settings => Administrator Settings**.
Service Manager displays the **Administrator Details** window.
2. Configure the parameter as follows:

- **Email ID** – Mail ID to be included in the system-level message prompt to any user attempting to log into a locked or expired account.
3. Select **Save**.

Instance Filter

The instance filter can contain a single index value or a set of index values. For example, a single index can be a specified value for the index zone_name, apn_name, or so on. A set of index values can be a specified value for multiple indices. When you configure the instance filter, include the indices, each separated by a period, in the order to match the target metric.

To create an instance filter:

1. Select the **Settings**  icon on the toolbar and select **Performance => Instance Filter Configuration**.

Service Manager displays the **Instance Filter Configuration**.

2. Select the **New Filter**  icon.

In the **Add Filter** dialog, enter the Filter Name. In the case of multiple indices, include the indices values in the name, each separated by a period, in the order to match the target metric. Select **Add**.

A filter with that name is created.

3. In the **Filter Manager**, select the filter then select the **Add Condition**  icon:

- a. Complete the Conditions fields by selecting Instance Name for **Parameter** and appropriate values for **Operator**, **Value**, and **Condition**.

For example, to configure an instance filter for the metric NumCurrentSessions for which data is collected for each zone and apn, specify the following set of indices:

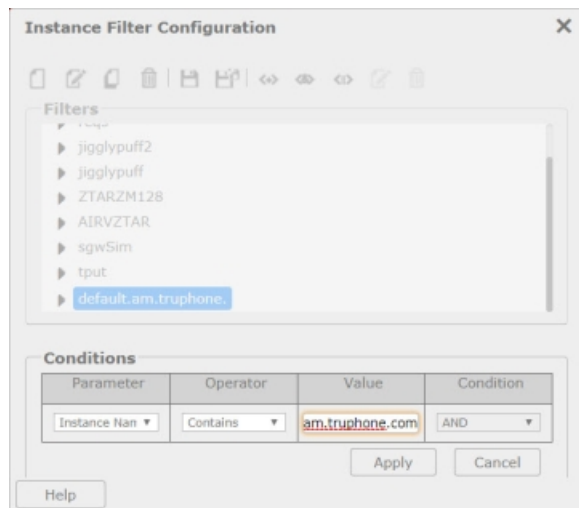
- The index zone_name with the value default.
- The index apn_name with the value am.truphone.com.

Specify the **Operator** and **Value**. For this filter, the **Value** you enter must include the multiple indices values, each separated by a period, in the order to match (that is, zone_name first and apn_name second):

default.am.truphone.com

For example, specify any of the following settings:

- **Starts with** default
- **Ends with** truphone.com or am.truphone.com
- **Equals** default.am.truphone.com
- **Not Equals** default.bm.truphone.com
- **Contains** default or am.truphone.com

Figure 8-1 Instance Filter Configuration

- b. Select **Apply**.
4. To add additional Conditions, repeat [step 3](#) . Specify **Condition** = AND or OR.
5. If the filter needs a combination of AND and OR conditions to achieve the desired result:
 - a. In the **Filter Manager**, select the filter then select the **Create AND group**  icon (or **Create OR Group**  icon).
 - b. In the **Filter Manager**, select the AND... (or OR...) group, the **Add Condition** icon  , and complete the Conditions for the group.
 - c. Select **Apply**.
6. Select the **Save Filter**  icon.

Application Level Audit Trail

The application level Audit Trail records all configuration operations and activities.

For information on user-specific Audit Trails, see [User-Level Audit Trail](#).

Viewing the Audit Trail

To view the Audit Trail:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Audit Trail**.
PNSM displays the **Audit Trail** window, listing the configuration operations.
2. From the **Audit Trail** window, use any of the following methods to view configuration operation details:

Toolbar: Select the operation then select the **View**  icon.

Pop-up menu: Right-click the operation and select **View**.

PNSM displays the **Audit Trail View** window, listing the following information:

- Resource Name
 - Description
 - User Name
 - Date/Time
 - Status
 - Operation
 - CLI Execution status
3. Select **Close** to close the window.

Exporting Audit Trail

You can export the Audit Trail to a CSV/PDF/XLS/XML/HTML file.

To export the Audit Trail:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Audit Trail**.
2. If desired, use the Filter Manager or Quick Filter to filter the data displayed in the window.
3. Select the **Export**  icon.
PNSM displays the **Configure Export Parameters** window.
4. Configure the parameters as follows:
 - **File Type:** Select the required file format (CSV, PDF, XLS, XML, HTML).
 - **Column Configuration:** Select the columns you want exported (Visible Column or All Columns)
 - **Export Type:** Select the export type (Whole Data, Current View, Selected Data)
 - **Use Current Filter:** Mark the checkbox to restrict data based on the existing filter conditions.
5. Select **Apply**.
PNSM displays the **Download Files** window.
6. Select **Download Exported File**.
7. From the **Download Files** window, select **Close**.

Viewing Product Version Information

To view version information for the PNSM application:

1. Select **About** from the **Settings**  icon on the toolbar.

PNSM displays a window listing the version number, release date, last server start date, and copyright information.

2. Select **Close** to close the window.

Additional Configuration Settings

The following sections discuss Configuration settings available in PNSM.

Template Schedules

You can schedule templates to be periodically executed.

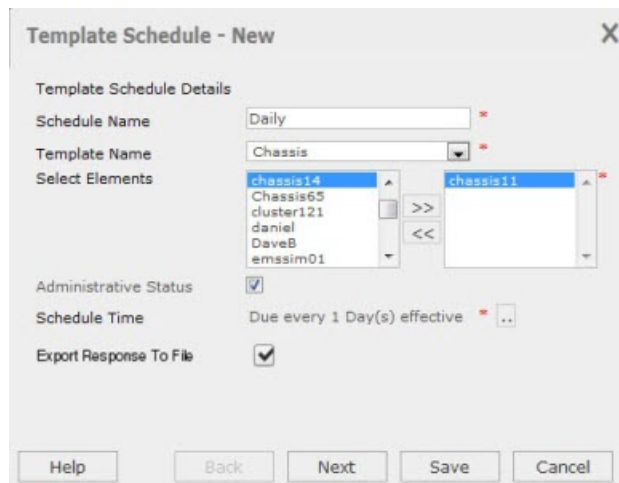
***Note:** If a template executes a large number of CLI commands, the session may timeout before the commands commit. To avoid a timeout, increase the milliseconds value of the CLI Time Out parameter at the Edit Network Element window.*

Adding Template Schedules

To add a template schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Templates => Schedule**.
PNSM displays the **Template Schedule** window, listing the template schedules.
2. Select the **New**  icon.

Figure 9-1 New Template Schedule Window



3. Configure the parameters as follows:

- **Schedule Name** – Enter a schedule name
- **Template Name** – Select the configuration template.
- **Select Elements** – Select the elements by highlighting and selecting the right-arrow.
- **Administrative Status** – Mark the checkbox to set the Admin Status of the schedule to Enabled.
- **Schedule Time:**

— Select **Recurrence Pattern**  icon.

— Specify the recurrence details and select **Schedule**.

- **Export Response To File** – Mark the checkbox to export an individual text file containing template execution status and respective command response for each execution of this template schedule. When enabled, every time the template schedule executes, PNSM creates a file in /data/affirmed/ems/data/config-template-response/<schedule-name> named <schedule-name>_<timestamp>.txt. To purge template schedule response files, see [Purging Template Schedule Response Files](#).

4. Select **Next**.

Figure 9-2 Configuration Command List

Template Schedule - New

Chassis

Slot

Preview CLI Commands

Configuration Details after inserting user input tokens

```
show chassis 7 inventory
show chassis 7 summary
show chassis 7 statistics 2
```

Help Back Next Save Cancel

5. If the specified configuration template contains user input token(s), specify the value(s) in the **Token Details** field. In the example above, the value of the Chassis token is specified as 7 and the value of the Slot token is specified as 2.
6. Select **Preview CLI Commands** to see the preview of the CLI command(s) that were provided in the Configuration Command List of the specified configuration template. If the configuration template contained user input token(s), the specified values are inserted in the CLI command(s).
7. Select **Save**.

To view the results of the template execution, see [Viewing Template Schedule Execution Status](#).

Copying Template Schedules

You can make a copy of an existing template schedule and then modify and save it with a new name.

To copy a template schedule:

1. Select the **Settings** icon on the toolbar  and select **Configuration => Templates => Schedule**.

PNSM displays the **Template Schedule** window, listing the template schedules.

2. Use any of the following methods to copy a template schedule:

Toolbar: Select the template schedule then select the **Copy**  icon.

Pop-up menu: Right-click the template schedule and select **Copy**.

3. Make the necessary changes:
 - **Schedule Name** – Enter a schedule name

- **Template Name** – Select the configuration template.
- **Select Elements** – Select the elements by highlighting and selecting the right-arrow.
- **Administrative Status** – Mark the checkbox to set the Admin Status of the schedule to Enabled.
- **Schedule Time:**

— Select the **Recurrence Pattern** icon. 

— Specify the recurrence details and select **Schedule**.

- **Export Response To File** – Mark the checkbox to export an individual text file containing template execution status and respective command response for each execution of this template schedule. When enabled, every time the template schedule executes, PNSM creates a file in /data/affirmed/ems/data/config-template-response/<schedule-name> named <schedule-name>_<timestamp>.txt. To purge template schedule response files, see [Purging Template Schedule Response Files](#).

4. Select **Next**.
5. If the specified configuration template contains user input token(s), specify the value(s) in the **Token Details** field.
6. Select **Preview CLI Commands** to see the preview of the CLI command(s) that were provided in the Configuration Command List of the specified configuration template. If the configuration template contained user input token(s), the specified values are inserted in the CLI command(s).
7. Select **Save**.

To view the results of the template execution, see [Viewing Template Schedule Execution Status](#).

Viewing Template Schedules

To view a template schedule:

1. Select the **Settings** icon on the toolbar  and select **Configuration => Templates => Schedule**.

PNSM displays the **Template Schedule** window, listing the template schedules.

2. From the **Template Schedule** window, use any of the following methods to view template schedule details:

Toolbar: Select the template schedule and then the **View**  icon.

Pop-up menu: Right-click the template schedule and select **View**.

PNSM displays the **Template Schedule View** window, listing the following information:

- Schedule Name
- Template Name

- Element Name
- Last Execution Time
- Next Execution Time
- Schedule Interval
- Operational Status
- Administrative Status
- Export Response to File
- Configuration Command List

3. Select **Close** to close window.

To view the results of the template execution, see [Viewing Template Schedule Execution Status](#).

Viewing Template Schedule Execution Status

To view a template schedule execution status:

1. Select the **Settings** icon on the toolbar  and select **Configuration => Templates => Schedule**.

PNSM displays the **Template Schedule** window, listing the template schedules.

2. Right-click the template schedule and select **Execution Status**.

PNSM displays the **Template Execution Status View** window, listing the various execution times for the previous 24 hours.

3. Select an execution time to view the details:

- Schedule Name
- Element
- Execution Status
- Commands executed for this schedule:

— Type

— Command

— Response

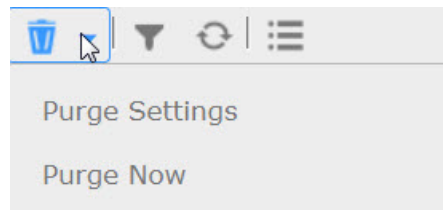
— Status

4. Select **Close** to close window.

Purging Template Schedule Response Files

To purge template schedule response files:

1. Select the **Settings** icon on the toolbar  and select **Configuration => Templates => Schedule**.
PNSM displays the **Template Schedule** window, listing the template schedules.
2. To immediately purge all template response files for a particular template schedule:
 - a. Select the template schedule.
 - b. Select the **Purge** icon on the toolbar.



- c. Select **Purge Now**.
 - d. At the confirmation prompt, select **OK**.
3. To configure purging of template response files for all template schedules, based on template schedule age:
 - a. Select the **Purge** icon on the toolbar.
 - b. Select **Purge Settings**.
 - c. in the **Purge response files older than** box, specify the number of days.
 - d. Select **Save**.

Enabling/Disabling Template Schedules

You can enable/disable template schedules.

To enable/disable a template schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Templates => Schedule**.
PNSM displays the **Template Schedule** window, listing the template schedules.
2. Use any of the following methods to enable a template schedule:

Toolbar: Select the template schedule then the **Enable Schedule**  icon.

Pop-up menu: Right-click the template schedule and select **Enable Schedule**.

Use any of the following methods to disable a template schedule:

Toolbar: Select the template schedule then select the **Disable Schedule**  icon.

Pop-up menu: Right-click the template schedule and select **Disable Schedule**.

3. At the confirmation prompt, select **OK**.

The Administrative Status of the template schedule changes to Enabled or Disabled.

Quick Start Template Schedules

To Quick Start (instantly execute) a template schedule:

1. Select the **Settings** icon on the toolbar  and select **Configuration => Templates => Schedule**.

PNSM displays the **Software Schedule** window, listing the template schedules.

2. Use any of the following methods to Quick Start a template schedule:

Toolbar: Select the template schedule then select the **Quick Start**  icon.

Pop-up menu: Right-click the template schedule and select **Quick Start**.

To view the results of the template execution, see [Viewing Template Schedule Execution Status](#).

Modifying Template Schedules

To modify a template schedule:

1. Select the **Settings** icon on the toolbar  and select **Configuration => Templates => Schedule**.

PNSM displays the **Template Schedule** window, listing the template schedules.

2. Use any of the following methods to modify a template schedule:

Toolbar: Select the template schedule then the **Edit**  icon.

Pop-up menu: Right-click the template schedule and select **Edit**.

3. Make the necessary changes:

- **Schedule Name** – Enter a schedule name
- **Template Name** – Select the configuration template.
- **Select Elements** – Select the elements by highlighting and selecting the right-arrow.
- **Administrative Status** – Mark the checkbox to set the Admin Status of the schedule to Enabled.
- **Schedule Time:**

— Select the **Recurrence Pattern**  icon.

— Specify the recurrence details and select **Schedule**.

- **Export Response To File** – Mark the checkbox to export an individual text file containing template execution status and respective command response for each execution of this template schedule. When enabled, every time the template schedule executes, PNSM creates a file in /data/affirmed/ems/data/config-template-response/<schedule-name> named <schedule-name>_<timestamp>.txt. To purge template schedule response files, see [Purging Template Schedule Response Files](#).
4. Select **Next**.
 5. If the specified configuration template contains user input token(s), specify the value(s) in the **Token Details** field.
 6. Select **Preview CLI Commands** to see the preview of the CLI command(s) that were provided in the Configuration Command List of the specified configuration template. If the configuration template contained user input token(s), the specified values are inserted in the CLI command(s).
 7. Select **Save**.

To view the results of the template execution, see [Viewing Template Schedule Execution Status](#).

Deleting Template Schedules

You can delete a template schedule only if its operational status is Completed or its administrative status is Disabled.

To delete a template schedule:

1. Select the **Settings** icon on the toolbar  and select **Configuration => Templates => Schedule**.
PNSM displays the **Template Schedule** window, listing the template schedules.
2. Use any of the following methods to delete a template schedule:
 - Toolbar:** Select the template schedule then select the **Delete**  icon.
 - Pop-up menu:** Right-click the template schedule and select **Delete**.
3. At the confirmation prompt, select **OK**.

Alarm Triggered Configurations

Alarm Triggered Configuration Templates allow you to apply configurations to elements based on alarm triggers. Configuration Templates designated as type Alarm Triggered can be specified to be executed when the alarm is received. For instructions on creating Alarm Triggered Configuration Templates, see [Adding Configuration Templates](#).

Adding Alarm Triggered Configuration

To add an alarm triggered configuration:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Alarm Triggers**.

PNSM displays the **Alarm Triggered Configuration** window.

The left pane lists the configured Alarm Triggered configurations. The right pane lists the alarm filters.

2. Select the **New**  icon.
3. Configure the parameters as follows:
 - **Configuration Name** – Enter a configuration name.
 - **Template Name** – Select an Alarm Triggered Configuration Template.
 - **Run as** – By default, PNSM executes the configuration using the credentials of the default CLI user. To use a different credential in the device or to run it as a different user, mark this checkbox and provide the necessary credentials to run this configuration task:
 - **User Name**
 - **Password**
 - **Confirm Password**
 - **Delay Time Period** – Select a time (0 - 10 minutes), for which the alarm trigger should delay. The configuration is triggered on the source device only if the alarm is still active (not CLEARED) after the delay. Setting a delay avoids potential cyclic behavior in alarm triggers.
4. Select **Save**. The newly created Alarm Triggered Configuration is listed in the **Alarm Triggered Configuration** window.
5. To associate the configuration with alarm filters, select the configuration in the left pane and mark the required check boxes in the right pane.

***Note:** It is not recommended that you mark the All checkbox, because it associates the template with any alarm (which could be a large number of alarms).*

6. Select **OK** to associate the configuration with alarm filters.

Modifying Alarm Triggered Configurations

To modify an Alarm Triggered Configuration:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Alarm Triggers**. PNSM displays the **Alarm Triggered Configuration** window, listing the Alarm Triggered configurations.
2. Use any of the following methods to modify an Alarm Triggered Configuration:

Toolbar: Select the Alarm Triggered Configuration then select the **Edit**  icon.

Pop-up menu: Right-click the Alarm Triggered Configuration and select **Edit**.

PNSM displays the **Edit Alarm Triggered Configuration** window.

3. Make the necessary changes:
 - **Configuration Name** – Enter a configuration name.
 - **Template Name** – Select an Alarm Triggered Configuration Template.
 - **Run as** – By default, PNSM executes the configuration using the credentials of the default CLI user. To use a different credential in the device or to run it as a different user, mark this checkbox and provide the necessary credentials to run this configuration task:
 - **User Name**
 - **Password**
 - **Confirm Password**
 - **Delay Time Period** – Select a time (0 - 10 minutes), for which the alarm trigger should delay. The configuration is triggered on the source device only if the alarm is still active (not CLEARED) after the delay. Setting a delay avoids potential cyclic behavior in alarm triggers.
4. Select **Save**. The modified Alarm Triggered Configuration is listed in the **Alarm Triggered Configuration** window.
5. To associate the configuration with alarm filters, select the configuration in the left pane and mark the required check boxes in the right pane.

***Note:** It is not recommended that you mark the All checkbox, because it associates the template with any alarm (which could be a large number of alarms).*

6. Select **OK** to associate the configuration with alarm filters.

Deleting Alarm Triggered Configurations

To delete an Alarm Triggered Configuration:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Alarm Triggers**. PNSM displays the **Alarm Triggered Configuration** window, listing the Alarm Triggered configurations.
2. Use any of the following methods to delete a Alarm Triggered Configuration:
 - Toolbar:** Select the Alarm Triggered Configuration then select the **Delete**  icon.
 - Pop-up menu:** Right-click the Alarm Triggered Configuration and select **Delete**.
3. At the confirmation prompt, select **OK**.

Use Case: Alarm Triggered Configuration

This use case demonstrates how to use an alarm triggered configuration to disable transcoding in the event of widespread transcoding failures. The following tasks are described:

- Create an Configuration Maintenance Template
- Create a decreasing threshold of 80%
- Create an alarm filter for the decreasing threshold
- Set up Alarm Triggered Configuration with filter
- Modify the MCC's schedule to add the threshold alarm to the specific KPI
- View execution status

Create an Configuration Maintenance Template

To add a configuration template:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Templates => Maintenance**.
PNSM displays the **Template Maintenance** window, listing the configuration templates.
2. Select the **New**  icon.
3. Configure the parameters as follows:
 - **Template Name** – Enter a template name
 - **Protocol Type** – CLI
 - **Template Type** – Select Alarm Triggered, which configures an element based on the alarm of the element
 - **Applicable Element Type** – MCC or MME

Figure 9-3 Template Maintenance Window

4. Select **Next**.
5. In the Configuration Command list, enter the sequence of commands to be executed on the device. Enter each command on a separate line. The command list can contain a maximum of 1000 lines (including empty lines). Each line can contain a maximum of 1200 characters.

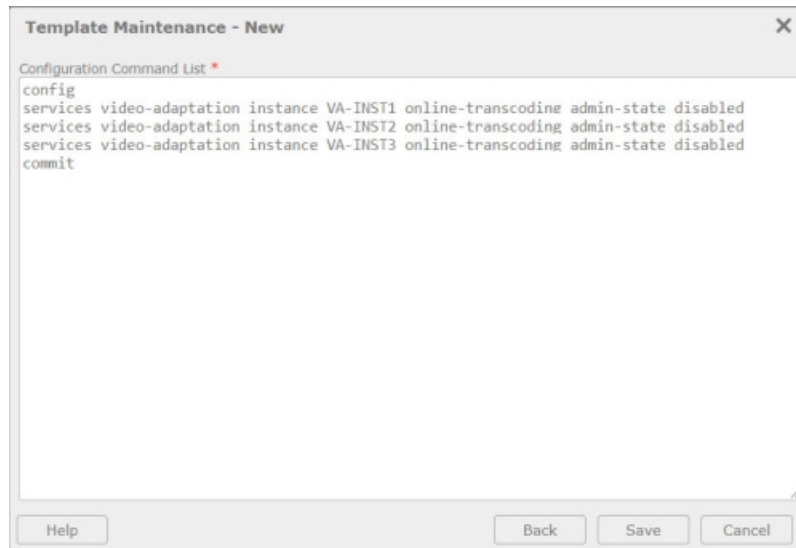
Use the following commands:

config

```
services video-adaptation instance [Video Adaptation Instance]
online-transcoding admin-state disabled
```

commit

Enter a separate command for each [**Video Adaptation Instance**]. In this example, the MCC has three video adaptation instances; each requires its own command to disable the instance. Keep in mind that if changes are later made to the MCC's video adaptation instances, you need to adjust this command list to reflect the changes.

Figure 9-4 Template Maintenance Window

6. Select **Save**.
7. Close the **Template Maintenance** window.

Create a Decreasing Threshold of 80%

To add a threshold:

1. Select the **Settings**  icon on the toolbar and select **Performance => Configure Thresholds**.
2. PNSM displays the **Threshold Configuration** window.
3. Select the **Add New Threshold**  icon.
4. Complete the fields as follows:
 - **Name** – Enter the threshold name.
 - **Type** – Specify Decreasing (the threshold applies when a data point falls below the specified level).
 - **Description** – Enter a threshold description.
 - **Monitor Window Interval** – Leave the value 0.
 - **Monitor Delta Values** – Leave the checkbox unmarked.
 - **Threshold Details:**
 - Mark the **Critical** checkbox.
 - Enter the **Value** 80.

- Enter the following **Message**: “Video Adaptation - Online Transactions Processed Percentage” has reached below 80%.

When a threshold event is raised, this message is displayed in Alarms/Event Logs tab.

Figure 9-5 Configure New Threshold Window

Configure New Threshold

Name: *

Type:

Description:

Monitor Window Interval: minutes ☐ Monitor Delta Values

Threshold Details* *

	Severity	Value	Message
☒	Critical	80	"Video Adaptation - Online Transactions Processed Pe
☐	Major		
☐	Minor		
☐	Warning		
☐	Informational		

* - Fields are Mandatory

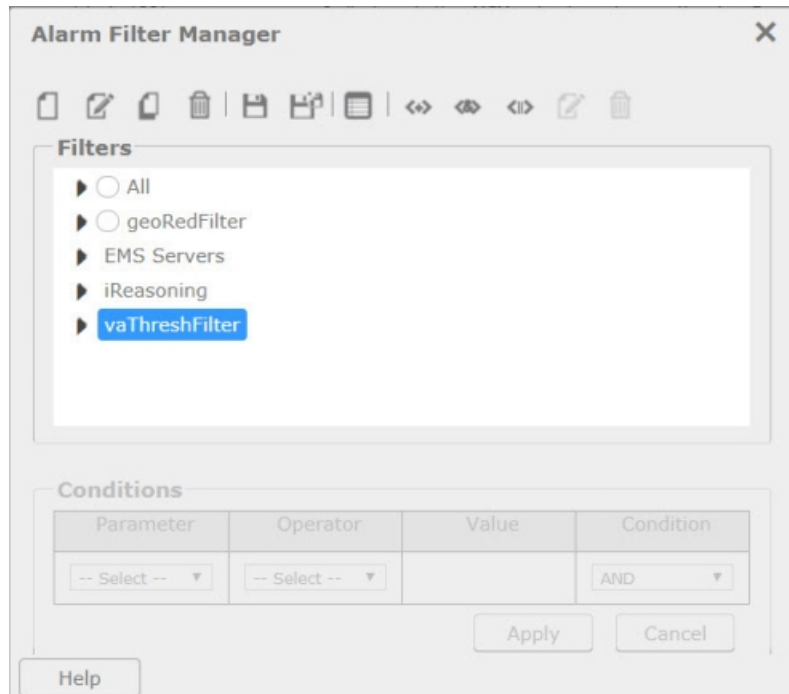
Help Save Cancel

5. Select **Save** to commit the data.
6. Select **Close** to close the **Threshold Configuration** window.

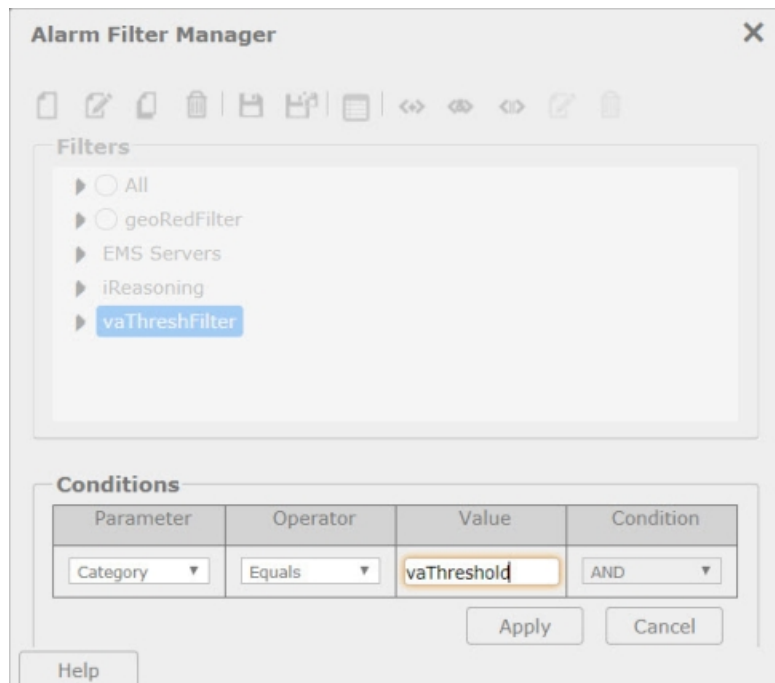
Create an Alarm Filter for the Decreasing Threshold

To create a filter:

1. From the Alarms tab, select the **Filter**  icon.
PNSM displays the **Filter Manager**.
2. Select the **New Filter**  icon.
3. In the **Add Filter** dialog, enter the Filter Name and select **Add**. A filter with that name is created.

Figure 9-6 Filter Manager

4. In the **Filter Manager**, select the filter then select the **Add Condition**  icon:
 - a. Complete the Conditions fields.
 - In the **Parameter** field, select **Category**.
 - In the **Operator** field, select **Equals**.
 - In the **Value** field, enter the name of the threshold created in Create a Decreasing Threshold of 80%.
 - b. Select **Apply**.

Figure 9-7 Filter Manager

5. Select the **Save Filter**  icon.
6. Close the **Alarm Filter Manager**.

Create an Alarm Triggered Configuration

To create an alarm triggered configuration:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Alarm Triggers**. PNSM displays the **Alarm Triggered Configuration** window.
The left pane lists the configured Alarm Triggered configurations. The right pane lists the alarm filters.
2. Select the **New**  icon.
3. Configure the parameters as follows:
 - **Configuration Name** – Enter a configuration name.
 - **Template Name** – Select the Alarm Triggered Configuration Template created in [Create an Configuration Maintenance Template](#).
 - **Run as** – Mark this checkbox and provide the necessary credentials to run this configuration task:
 - **User Name**
 - **Password**

- **Confirm Password**
- **Delay Time Period** – Select a time (0 - 10 minutes), for which the alarm trigger should delay. The configuration is triggered on the source device only if the alarm is still active (not CLEARED) after the delay. Setting a delay avoids potential cyclic behavior in alarm triggers.

Figure 9-8 Alarm Triggered Configuration

Alarm Triggered Configuration - New

Configuration Name: vaThreshConfig *

Template Name: disableVATrans *

Run as: ☒

Username: emsadmin *

Password: ***** *

Confirm Password: ***** *

Delay Time Period: 1 minute(s) *

Buttons: Help, Save, Cancel

4. Select **Save**. The newly created Alarm Triggered Configuration is listed in the **Alarm Triggered Configuration** window.
5. To associate the configuration with the alarm filter, select the configuration in the left pane and mark the check box in the right pane for the alarm filter created in [Create an Alarm Filter for the Decreasing Threshold](#).

Figure 9-9 Alarm Triggered Configuration

Alarm Triggered Configuration

Select a Configuration Template

Configuration Name	Template Name
geoRedConfig	geoRedSwitchoverTemplate
TestAlarm	testalarm
vaThreshConfig	disableVATrans

Associate Filters

Associate	Filter
☐	All
☐	geoRedFilter
☐	EMS Servers
☐	iReasoning
☒	vaThreshFilter

Buttons: Help, OK, Cancel

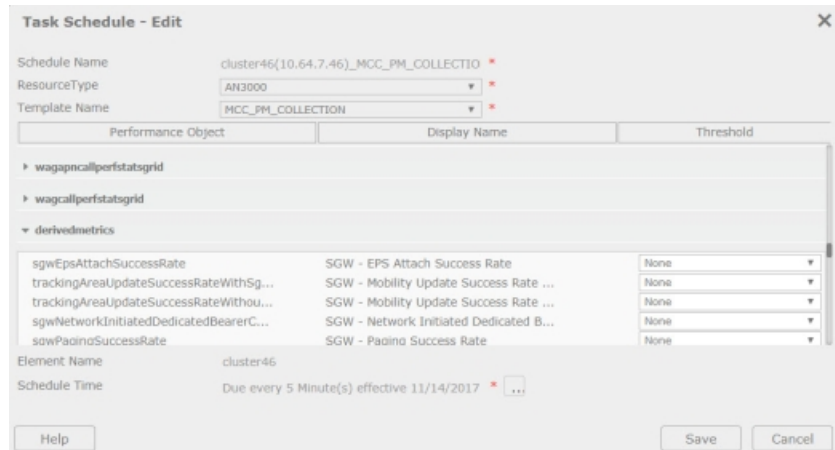
6. Select **OK** to associate the configuration with alarm filter.

Modify the Performance Schedule to Add the Threshold Alarm to the Specific KPI

To modify a User Tasks Scheduler:

1. Select the **Settings**  icon on the toolbar and select **Performance => User Tasks => Schedule**.
2. Select the MCC collection schedule to monitor the Online Transcoding. The scheduler name is <MCC name>_MCC_PM_COLLECTION.
3. Select the **Edit User Task Scheduler**  icon.
4. When the schedule is fully loaded, scroll through the entire Performance Object table and expand the **derivedmetrics** list.

Figure 9-10 Edit Task Scheduler



Task Schedule - Edit

Schedule Name: cluster46(10.64.7.46)_MCC_PM_COLLECTION

ResourceType: AN3000

Template Name: MCC_PM_COLLECTION

Performance Object	Display Name	Threshold
wagapncallperfstatsgrid		
wagcallperfstatsgrid		
derivedmetrics		
sgwEpsAttachSuccessRate	SGW - EPS Attach Success Rate	None
trackingAreaUpdateSuccessRateWithSg...	SGW - Mobility Update Success Rate ...	None
trackingAreaUpdateSuccessRateWithou...	SGW - Mobility Update Success Rate ...	None
sgwNetworkInitiatedDedicatedBearerC...	SGW - Network Initiated Dedicated B...	None
sowPagingSuccessRate	SGW - Paging Success Rate	None

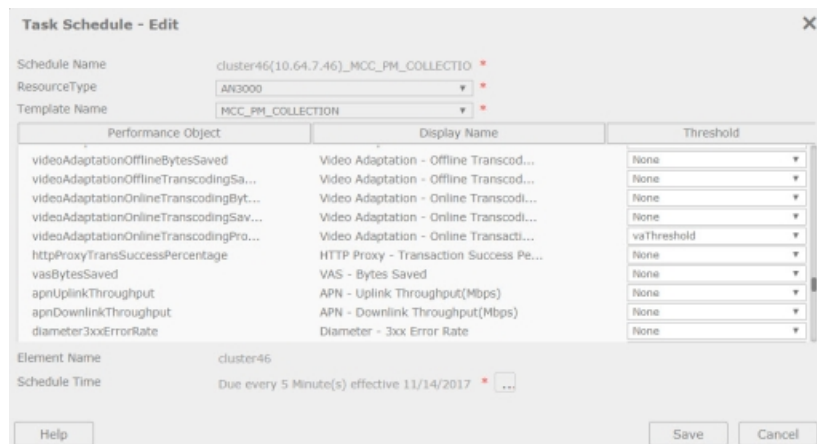
Element Name: cluster46

Schedule Time: Due every 5 Minute(s) effective 11/14/2017

Buttons: Help, Save, Cancel

5. Scroll through the **derivedmetrics** list and select the Performance Object videoAdaptationOnlineTranscodingProcessedPercentage.
6. In the **Threshold** field, select the threshold that was created in [Create a Decreasing Threshold of 80%](#).

Figure 9-11 Edit Task Scheduler



Task Schedule - Edit

Schedule Name: cluster46(10.64.7.46)_MCC_PM_COLLECTION

ResourceType: AN3000

Template Name: MCC_PM_COLLECTION

Performance Object	Display Name	Threshold
videoAdaptationOfflineBytesSaved	Video Adaptation - Offline Transcod...	None
videoAdaptationOfflineTranscodingSa...	Video Adaptation - Offline Transcod...	None
videoAdaptationOnlineTranscodingByt...	Video Adaptation - Online Transcodi...	None
videoAdaptationOnlineTranscodingSav...	Video Adaptation - Online Transcodi...	None
videoAdaptationOnlineTranscodingPro...	Video Adaptation - Online Transacti...	vaThreshold
httpProxyTransSuccessPercentage	HTTP Proxy - Transaction Success Pe...	None
vasBytesSaved	VAS - Bytes Saved	None
apnUplinkThroughput	APN - Uplink Throughput(Mbps)	None
apnDownlinkThroughput	APN - Downlink Throughput(Mbps)	None
diameter3xxErrorRate	Diameter - 3xx Error Rate	None

Element Name: cluster46

Schedule Time: Due every 5 Minute(s) effective 11/14/2017

Buttons: Help, Save, Cancel

7. Select **Save**.

View Execution Status

View Triggered Alarm

If a threshold alarm has triggered, you can view it.

To view alarms:

1. Select the **Alarms** tab.
2. To filter the alarms list to only the threshold alarms, select the filter created in [Create an Alarm Filter for the Decreasing Threshold](#) from the filter list to apply it.

Figure 9-12 Select the Filter



3. Use any of the following methods to view/edit an alarm's details:

Toolbar: Select the alarm then the **Alarm Details**  icon.

Pop-up menu: Right-click the alarm and select **Alarm Details**.

Or: Double-click the alarm.

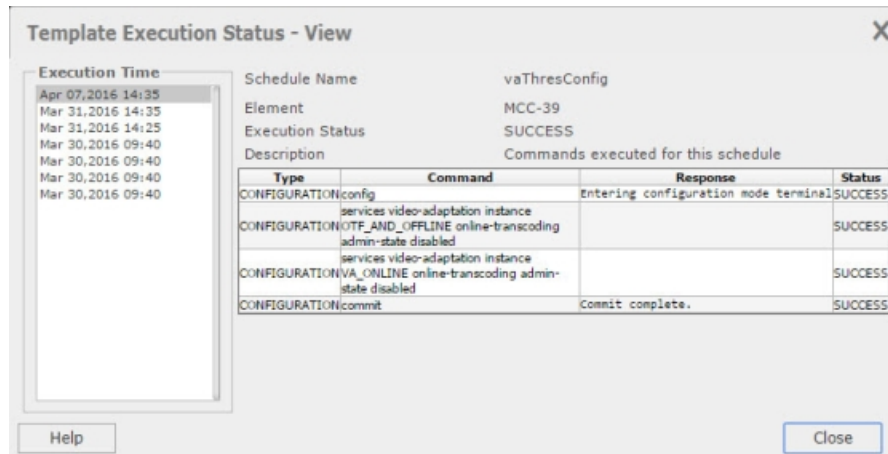
PNSM displays the **Alarm Details** window.

View Triggered Configuration Execution Status

Once the alarm has triggered, you can view the Alarm Triggered Configuration execution status.

To view an alarm triggered configuration execution status:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Alarm Triggers**. PNSM displays the **Alarm Triggered Configuration** window.
2. Select the alarm triggered configuration created for the threshold then select the **Execution Status**  icon.
3. Select an Execution Time to view the CLI response from the MCC. The correct response should be as shown in this example.

Figure 9-13 Template Execution Status*Verify the Video Adaptation Instances were Disabled*

On the MCC, you can verify whether the video adaptation instances were disabled.

To verify video adaptation instances admin-state:

1. Issue the following CLI command on the MCC:

```
show running-config services video-adaptation instance online-transcoding
```

The output should show that the admin-state for each instance is disabled:

```
an3000-mcm-slot7cpu1# show running-config services video-
adaptation instance online-transcoding

services video-adaptation instance [Video Adaptation Instance]
online transcoding admin-state disabled
```

Re-enable the Video Adaptation Instances

Once the admin-state for each instance is disabled, the instances remain in that state until you explicitly re-enable them.

To re-enable video adaptation instances admin-state:

1. Issue the following CLI commands on the MCC for each video adaptation instance. Each instance requires its own command to enable the instance:

```
config

services video-adaptation instance [Video Adaptation Instance]
online-transcoding admin-state enabled

commit

exit
```

For example:

```
an3000-mcm-slot7cpu1# config
an3000-mcm-slot7cpu1(config)# services video-adaptation
instance OTF_AND_OFFLINE online-transcoding admin-state enabled
an3000-mcm-slot7cpu1(config-instance-OTF_AND_OFFLINE)# commit
Commit complete.
an3000-mcm-slot7cpu1(config-instance-OTF_AND_OFFLINE)# exit
an3000-mcm-slot7cpu1(config)# services video-adaptation
instance VA_ONLINE online-transcoding admin-state enabled
an3000-mcm-slot7cpu1(config-instance-VA_ONLINE)# commit
Commit complete.
an3000-mcm-slot7cpu1(config-instance-VA_ONLINE)# exit
an3000-mcm-slot7cpu1(config)# exit
```

Configuration Export Manager

Use the Configuration Export Manager to save an NE's configuration to raw CLI output. You can provision an NE and save its configuration as a "golden" configuration to be input into a configuration template and applied to other NEs.

For instructions on creating configuration templates, see Configuration Templates.

You can store up to three configuration files for an NE.

PNSM also supports backing up configuration data for each of the reachable managed NEs to individual files. For details, see Network Configuration Backup.

Viewing Configuration Files

To view configuration files:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Configuration Export**.

PNSM displays the Configuration Export Manager, listing all configuration files. The filename is `<timestamp>.<version>cgf.cli`, where *timestamp* is the date and time when the file was saved and *version* is the software version running on the NE.

In the case of a large number of files, PNSM displays them over multiple pages.

To specify how many files you want displayed per page, select the value in the Show Entries box. Use the page navigational icons at the bottom of the window to page through the entries.

Use the **Search** text box at the top on the window to search for a particular file based on its node name, filename, or size.

Select **Refresh** to refresh the list.

Viewing Configuration Details

You can view a configuration file's details.

To view a configuration file's details:

1. Select the file then select **View**.

PNSM displays the CLI commands in the configuration file.

Printing Configuration File

You can print a configuration file.

To print a configuration file:

1. Select the file then select **View**.

PNSM displays the CLI commands in the configuration file.

2. Select **Print**.
3. Select the desired printer.
4. Select **OK**.

Saving Configuration Files

You can save an active and manageable NE's current configuration to a file. You can save up to three configuration files for an NE. If three configuration files already exist for the NE, PNSM prompts you to delete one before proceeding.

To save a configuration file:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Configuration Export**.

PNSM displays the Configuration Export Manager, listing all configuration files.

2. Select the NE for which you want to save the configuration.
3. Select the **Save**  icon.

The configuration file is added to the list. You may have to page through the entries to view it.

Downloading Configuration Files

You can download a configuration file.

To download a configuration file:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Configuration Export**.

PNSM displays the Configuration Export Manager, listing all configuration files.

2. Select the file then select **Download**.
3. At the Firefox prompt, select **Save File**.
4. Select **OK**.

Comparing Configuration Files

You can compare two configuration files. For example, you can use the compare tool to identify any configurations changes that occurred between the times the configuration file was saved for a particular node.

To compare configuration files:

1. Select the **Settings**  icon on the toolbar and select **Configuration => Configuration Export**.

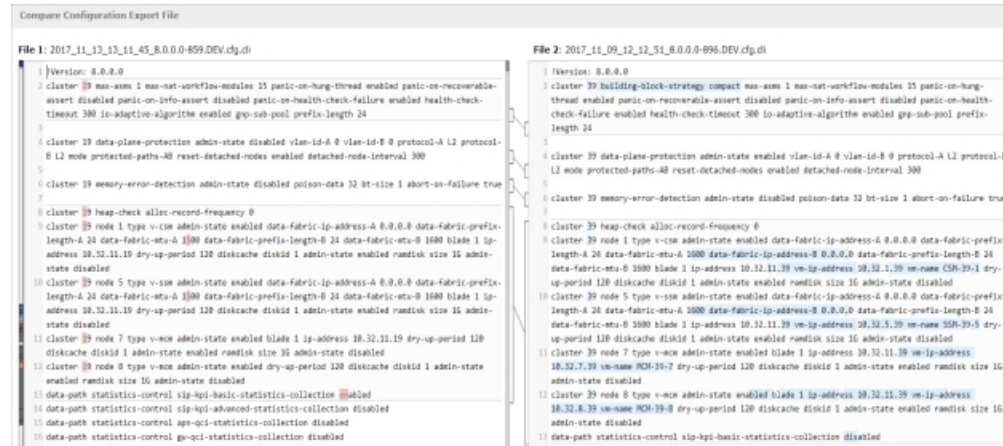
PNSM displays the Configuration Export Manager, listing all configuration files.

2. Select the first file you want to compare and select **Compare**.

- At the selection prompt, select the second file to compare and select **Compare**.
PNSM displays the Configuration Comparison report for the files.

Colored markers indicate where mismatches exist.

Figure 9-14 Configuration Comparison Report



- Scroll down to the colored markers to view the mismatch.
- To close the report, select **Close**.

Deleting Configuration Files

To delete a configuration file:

- Select the **Settings**  icon on the toolbar and select **Configuration => Configuration Export**.
PNSM displays the Configuration Export Manager, listing all configuration files.
- Select the file you want to delete. Select **Delete**.
- At the confirmation prompt, select **OK**.

URL List Management

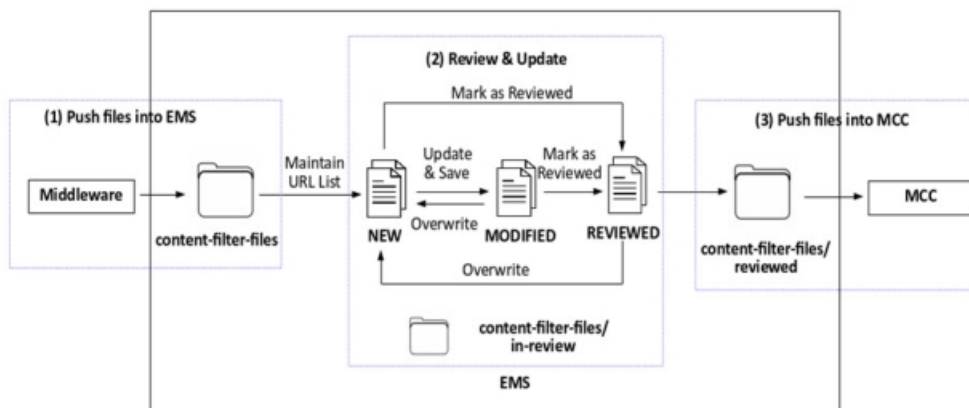
Use URL List Management to manage files of URLs and avoid input errors. A single URL file can contain a maximum of 5 million individual URLs.

You can use URL List Management to:

- View URL file contents
- Edit URL file contents
- Import contents from a local URL file
- Compare current to previous URL file contents
- Push the URL file onto single or multiple network elements

The following figure illustrates the directory structure and lifecycle management of URL files in the /opt/Affirmed/NMS/server/ems/data/content-filter-files directory.

Figure 9-15 URL List File Lifecycle Management



Viewing URL List Files

You can view the current set of URL list files in the /opt/Affirmed/NMS/server/ems/data/content-filter-files directory.

To view URL list files:

1. Select the **Settings**  icon on the toolbar and select **Configuration => URL List => Maintenance**.

PNSM displays the **URL List Maintenance** window, listing the URL list files.

Figure 9-16 URL List Maintenance Window

URL File Name	File Type	File Size	State	Last Reviewed Time	Auto Push	Operations
USCM-URL	CLEAR TEXT	1.56 GB	MODIFIED	N/A	☐	
ESB-MANUAL-URL	CLEAR TEXT	875.45 MB	NEW	N/A	☒	
ESKA-VASAKL3-URL	CLEAR TEXT	372.96 MB	NEW	N/A	☒	
ESB-VASAKL3-COMPANY	HASHED	.8 B	NEW	N/A	☒	

The window lists the following information:

- URL File Name
- File Type
- File Size
- State: NEW, MODIFIED, REVIEWED
- Last Reviewed Time
- Auto Push: checkbox to automatically push a clear-text file to destination network elements.
- Operations:
 - Review and Edit
 - Overwrite
 - Mark As Reviewed
 - Compare

These Operations are described in the following sections.

Specifying to Auto-Push File to Destination Network Elements

You can enable Auto-Push to automatically push a clear-text file to destination network elements. By default, Auto-Push pushes the files to all managed network elements every hour. The existing files on the network elements are overwritten with each Auto-Push.

To enable Auto-Push for a file:

1. From the **URL List Maintenance** window, mark the **Auto Push** checkbox for the URL List.
2. At the confirmation prompt, select **OK**.

When Auto-Push is enabled, the other operations (Review and Edit, Overwrite, Mark As Reviewed, and Compare) are disabled for the file.

You must explicitly enable the Auto-Push schedule.

To enable the Auto-Push schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => URL List => Schedule**.

PNSM displays the **URL Config Schedule** window, listing the URL list schedules.

The pre-defined AUTO-PUSH-HASHED schedule pushes all hashed files to destination network elements. The pre-defined AUTO-PUSH-CLEAR-TEXT schedule pushes specified clear-text files to destination network elements.

2. Use any of the following methods to enable the pre-defined AUTO-PUSH-HASHED or AUTO-PUSH-CLEAR-TEXT schedule:

Toolbar: Select the URL list schedule then select the **Enable Schedule**  icon.

Pop-up menu: Right-click the URL list schedule and select **Enable Schedule**.

Specifying Auto-Push Settings

You can modify the default Auto-Push settings, including push frequency and destination network elements.

To modify Auto-Push settings:

1. Select the **Settings**  icon on the toolbar and select **Configuration => URL List => Schedule**.

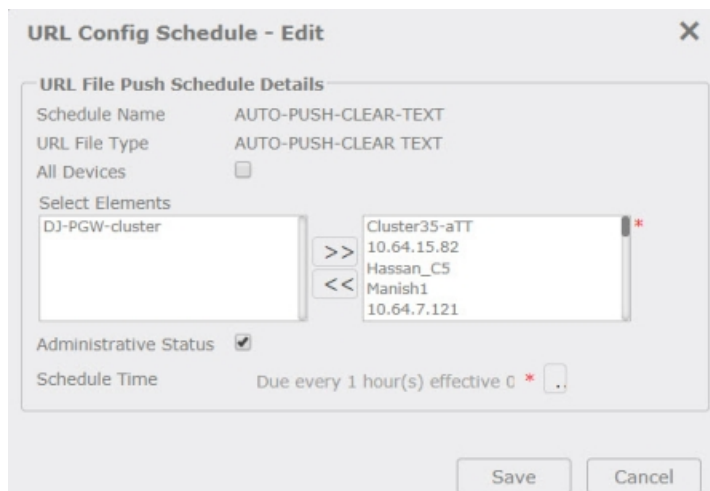
PNSM displays the **URL Config Schedule** window, listing the URL list schedules.

2. Use any of the following methods to modify the pre-defined AUTO-PUSH-HASHED or AUTO-PUSH-CLEAR-TEXT schedule:

Toolbar: Select the URL list schedule then select the **Edit**  icon.

Pop-up menu: Right-click the URL list schedule and select **Edit**.

Figure 9-17 Edit URL List Schedule Window



3. Make the necessary changes:

- **Schedule Name** – *Read-only*.
- **URL File Type** – *Read-only*.
- **All Devices** – Mark the checkbox to push the files to all destination network elements. Unmark the checkbox to specify the destination network elements.
- **Select Elements** – Select the network elements by highlighting and selecting the right-arrow.
- **Administrative Status** – Mark the checkbox to set the Admin Status of the schedule to Enabled.
- **Schedule Time:**

— Select **Recurrence Pattern**  icon.

— Specify the recurrence details and select **Schedule**.

4. Select **Save**.

Review and Edit

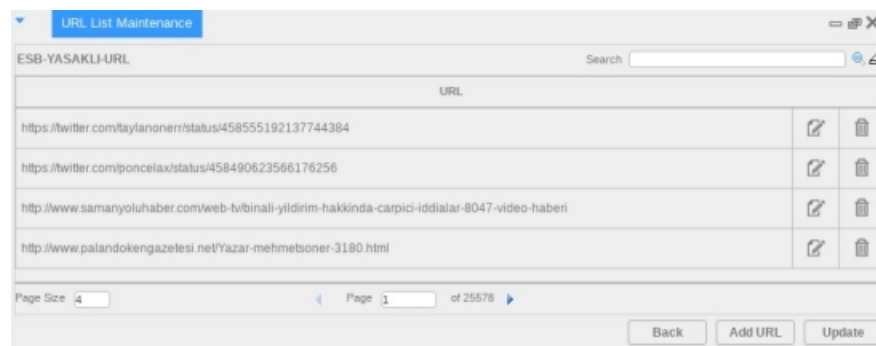
You can review and edit URLs in a clear-text URL file.

To edit URLs in a file:

1. From the **URL List Maintenance** window, select the **Review and Edit**  icon of the URL list you want to review.

PNSM lists the URLs in the file.

Figure 9-18 URLs in the File



2. Use the **Search** text box to search for URL instance(s) by string. Enter the string (minimum 5 characters) contained in the URL and select the **Search** icon. For example, enter twitter to list all URLs containing twitter.

Editing a Single URL

To edit a single URL in a file:

1. Select the **Edit**  icon of the URL you want to modify.
2. Change the URL string in the text box. When you have finished, select the **OK**  icon. The URL is highlighted in yellow to indicate it has been changed.

http://www.0002911.top		
http://www.000webhostapp.com-123		
http://www.004b0bdd1.c11ixsecure.com		

To undo the change, select the **Undo**  icon.

3. You can make up to 50 edit and deletion changes to a file during a single update session. To save all changes for this update session, select **Update**. The state changes to MODIFIED. If you attempt to make more than 50 changes, the system prompts you to save the current set of changes. Select **OK**.
4. To return to the **URL List Maintenance** window, select **Back**.

Deleting a Single URL

To delete a single URL in a file:

1. Select the **Delete**  icon of the URL you want to delete. The URL is highlighted in red to indicate it is deleted.

http://www.0002911.top		
http://www.000webhostapp.com		
http://www.004b0bdd1.c11ixsecure.com		

To undo the change, select the **Undo**  icon.

2. You can make up to 50 edit and deletion changes to a file during a single update session. To save all changes for this update session, select **Update**. The state changes to MODIFIED. If you attempt to make more than 50 changes, the system prompts you to save the current set of changes. Select **OK**.
3. To return to the **URL List Maintenance** window, select **Back**.

Adding URL(s) to a File

You can add URL(s) to a file either by:

- Entering them as text: Maximum of 50 changes to a file during a single update session
- Uploading a text file: Maximum uploaded file size: 9 MB.

To add URL(s) to a file:

1. Select **Add URL**.
2. In the **Type** field, select **Text** or **File**.
 - If you select **Text**, enter the URL(s) in the text box, each on its own line.

- If you select **File**, select **Choose File** to upload a text file of URLs. Select the text file and select **Upload**.
- 3. Select **Add**.
The added URL(s) are appended to the end of the URL file. The file state changes to **MODIFIED**.
- 4. To return to the **URL List Maintenance** window, select **Back**.

Overwrite

You can overwrite a clear-text URL file that is in the **MODIFIED** or **REVIEWED** state, as long as it contains the same content in both the `/opt/Affirmed/NMS/server/ems/data/content-filter-files/` directory and the `/opt/Affirmed/NMS/server/ems/data/content-filter-files/in-reviewed` directory. When you do so, the file in the `/in-review` directory is overwritten by the file in the `content-filter-files/` directory.

To overwrite a file:

1. From the **URL List Maintenance** window, select the **Overwrite**  icon of the URL list you want to overwrite.

Mark As Reviewed

You can mark a clear-text URL file that is in the **MODIFIED** state as reviewed.

To mark a file as reviewed:

1. From the **URL List Maintenance** window, select the **Mark As Reviewed**  icon of the URL list.

The file is copied from the `/opt/Affirmed/NMS/server/ems/data/content-filter-files/in-review` directory to the `/opt/Affirmed/NMS/server/ems/data/content-filter-files/reviewed` directory. The state changes to **REVIEWED**.

Compare

You can compare a clear-text URL file's current contents to its earlier file contents to identify any URLs added or removed since the last time the file was marked as reviewed. The operation compares the current version of file in `/opt/Affirmed/NMS/server/ems/data/content-filter-files/in-review` to its previous reviewed version of file in `/opt/Affirmed/NMS/server/ems/.data/content-filter-files/reviewed`.

To perform the Compare operation, the system copies and separately maintains the current available file from the `/reviewed` and `/in-review` directories until the operation completes. Any subsequent changes made by Review and Edit, Overwrite, or Mark As Reviewed are not taken into account in the comparison. The resulting Comparison file lists the timestamps from the file versions from the `/reviewed` and `/in-review` folders.

You can perform only one Compare operation at a time.

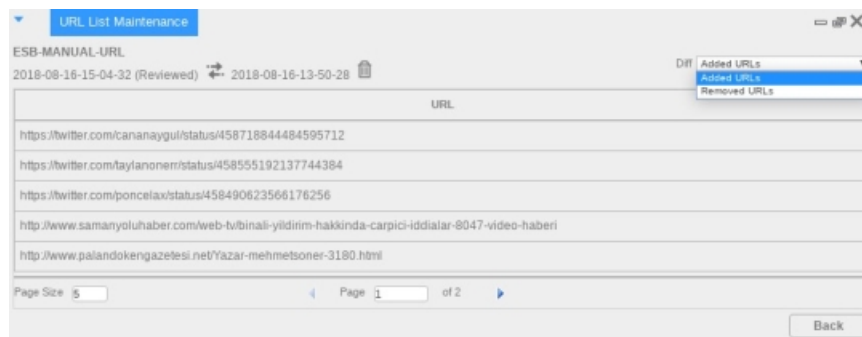
To compare a file:

1. From the **URL List Maintenance** window, select the **Compare**  icon of the URL list you want to compare.
2. If the files are identical, select **OK** at the confirmation prompt.
3. If the files differ, the **Compare** icon changes to a **Comparison File**  icon.
4. Select the icon to view the changes.

File Type	File Size	Status	Last Reviewed Time	Auto Push	Operations
CLEAR TEXT	875.45 MB	NEW	Sep/17/2018 10:33:09:000		   
CLEAR TEXT	531.0 B	REVIEWED	Sep/17/2018 14:36:52:000		   

Click the Comparison File icon 

5. In the **Diff** field, select **Added URLs** or **Removed URLs** to view the list.

Figure 9-19 Comparison File

6. You cannot perform another comparison on these files until the comparison file is deleted. To delete it, select the **Delete**  icon.
7. To return to the **URL List Maintenance** window, select **Back**.

URL List Schedules

PNSM allows you to manage the schedules to push a clear-text URL list onto a single or multiple network elements. For example, you can schedule to upload a URL list for a set of Network Elements as part of a typical maintenance cycle. The operation to push a URL list to a network element is logged to the Audit Trail.

As an alternative, you can enable Auto-Push, which automatically pushes a clear-text file to destination network elements. See [Specifying to Auto-Push File to Destination Network Elements](#).

Adding URL List Schedules

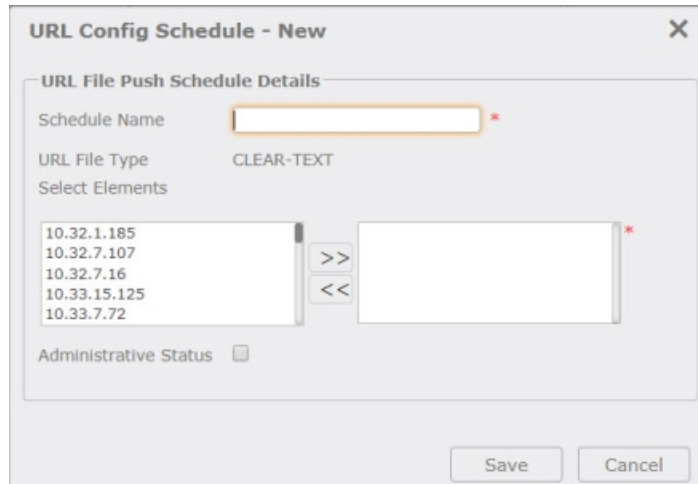
To add a URL list schedule:

1. Select the **Settings** icon on the toolbar  and select **Configuration => URL List => Schedule**.

PNSM displays the **URL Config Schedule** window, listing the URL list schedules.

2. Select the **New**  icon.

Figure 9-20 New URL List Schedule Window



3. Configure the parameters as follows:
 - **Schedule Name** – Enter a schedule name.
 - **Select Elements** – Select the elements by highlighting and selecting the right-arrow.
 - **Administrative Status** – Mark the checkbox to set the Admin Status of the schedule to Enabled.
 - **Schedule Time** – Select the **Calendar**  icon and select a date and time.
4. Select **Save**.

Viewing URL List Schedules

To view a software schedule:

1. Select the **Settings** icon on the toolbar  and select **Configuration => URL List => Schedule**.

PNSM displays the **URL Config Schedule** window, listing the URL list schedules.

2. From the **URL Config Schedule** window, use any of the following methods to view software schedule details:

Toolbar: Select the URL list schedule and select the **View**  icon.

Pop-up menu: Right-click the URL list schedule and select **View**.

PNSM displays the **URL Config Schedule View** window, listing the following information:

- Schedule Name

- URL List Name
 - Element Name(s)
 - Last Execution Time
 - Operational Status
 - Administrative Status
3. Select **Close** to close the window.

Viewing URL List Schedule Subtask Status

When multiple Network Elements use a particular URL list schedule, PNSM creates a separate subtask to run the schedule on each node. You can view the status of a each subtask.

To view URL list schedule subtask status:

1. Select the **Settings** icon on the toolbar  and select **Configuration => URL List => Schedule**.

PNSM displays the **URL List Config Schedule** window, listing the URL list schedules.

2. From the **URL List Config Schedule** window, use any of the following methods to view subtasks:

Toolbar: Select the URL list schedule and select the **View Sub Tasks**  icon.

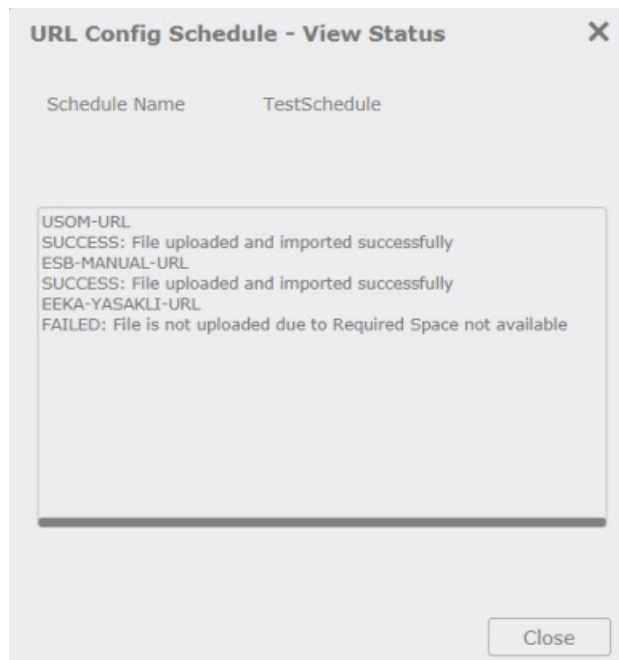
Pop-up menu: Right-click the URL list schedule and select **View Sub Tasks**.

PNSM displays the **Sub Tasks Details** window, listing the following information:

- Element Name
 - Schedule Name
 - URL File Type
 - Order of Execution
 - Status
 - Last Attempted Time
 - Operational Status
3. To view software schedule status of a subtask, select the subtask and select the **View Status**  icon.

PNSM displays the **URL Config Schedule View Status** window, listing the URL list schedule progress status report.

Figure 9-21 URL Config Schedule View Status Window



4. Select **Close** to close the window.
5. To close the **Sub Tasks Details** window, select the **Close**  icon.

Copying URL List Schedules

You can make a copy of an existing URL list schedule and then modify and save it with a new name. You cannot copy the pre-defined AUTO-PUSH-HASHED and AUTO-PUSH-CLEAR-TEXT schedules.

To copy a URL list schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => URL List => Schedule**.
PNSM displays the **URL Config Schedule** window, listing the URL list schedules.
2. To copy a URL list schedule, select the URL list schedule and select the **Copy**  icon.
3. Make the necessary changes:
 - **Schedule Name** – Enter a schedule name.
 - **Select Elements** – Select the elements by highlighting and selecting the right-arrow.
 - **Administrative Status** – Mark the checkbox to set the Admin Status of the schedule to Enabled.
 - **Schedule Time** – Select the **Calendar**  icon and select a date and time.
4. Select **Save**.

Enabling/Disabling URL List Schedules

To enable/disable a URL list schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => URL List => Schedule**.

PNSM displays the **URL Config Schedule** window, listing the URL list schedules.

2. Use any of the following methods to enable a URL list schedule:

Toolbar: Select the URL list schedule then select the **Enable Schedule**  icon.

Pop-up menu: Right-click the URL list schedule and select **Enable Schedule**.

Use any of the following methods to disable a URL list schedule:

Toolbar: Select the URL list schedule and select the **Disable Schedule**  icon.

Pop-up menu: Right-click the URL list schedule and select **Disable Schedule**.

3. At the confirmation prompt, select **OK**.

The Administrative Status of the URL list schedule changes to Enabled or Disabled.

Quick Start Software Schedules

To Quick Start (instantly execute) a URL list schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => URL List => Schedule**.

PNSM displays the **URL Config Schedule** window, listing the URL list schedules.

2. Use any of the following methods to Quick Start a URL list schedule:

Toolbar: Select the URL list schedule and select the **Quick Start**  icon.

Pop-up menu: Right-click the URL list schedule and select **Quick Start**.

Modifying URL List Schedules

To modify a URL list schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => URL List => Schedule**.

PNSM displays the **URL Config Schedule** window, listing the URL list schedules.

2. Use any of the following methods to modify a URL list schedule:

Toolbar: Select the URL list schedule then select the **Edit**  icon.

Pop-up menu: Right-click the URL list schedule and select **Edit**.

3. Make the necessary changes:

- **Schedule Name** – *Read-only*.

- **URL File Type** – *Read-only*.
- **All Devices** – For the pre-defined AUTO-PUSH-HASHED and AUTO-PUSH-CLEAR-TEXT schedules, mark the checkbox to push the files to all destination network elements. Unmark the checkbox to specify the destination network elements.
- **Select Elements** – Select the elements by highlighting and selecting the right-arrow.
- **Administrative Status** – Mark the checkbox to set the Admin Status of the schedule to Enabled.
- **Schedule Time:**

— Select **Recurrence Pattern**  icon.

— Specify the recurrence details and select **Schedule**.

4. Select **Save**.

Deleting URL List Schedules

You can delete a URL list schedule only if it has the Operational state Scheduled, Success, or Failure. You cannot delete the pre-defined AUTO-PUSH-HASHED and AUTO-PUSH-CLEAR-TEXT schedules.

To delete a URL list schedule:

1. Select the **Settings**  icon on the toolbar and select **Configuration => URL List => Schedule**.

PNSM displays the **URL Config Schedule** window, listing the URL list schedules.

2. Use any of the following methods to delete a URL list schedule:

Toolbar: Select the URL list schedule then select the **Delete**  icon.

Pop-up menu: Right-click the URL list schedule and select **Delete**.

3. At the confirmation prompt, select **OK**.

End-to-End Security Overview for APNS

This section discusses information regarding the security principles utilized in the Affirmed Private Network Service, and specifically defines how each component of the APNS solution is secured. This section includes:

- [Private Network Service Manager \(PNSM\) Security](#)
- [Azure Stack Edge \(ASE\) Security](#)
- [NetFoundry Security](#)
- [Virtual Network Function \(VNF\) Security](#)

Private Network Service Manager (PNSM) Security

The Private Network Service Manager resides on the Azure public cloud which deploys, configures, and monitors the packet core network and the network elements or VNFs in the Azure Stack Edge.

Refer to the following sections for information on how the PNSM securely allows access to the application and communication between the internal modules, and provides data security.

PNSM Access (Web Security)

To ensure only authorized clients can connect, external access is allowed only through secure HTTP (HTTPS) connections

- Inbound traffic is allowed based on an inbound port rule (port 443), so only specific, verified public IP addresses can connect to the PNSM server.
- Data transmission between the browser clients and service manager is encoded and secured by the Secure Socket Layer (SSL).

Note: PNSM currently uses self-signed certificate for the HTTPs communication. Customers can update the default certificates with signed certificates for SSL certificate verification.

- Basic authentication is used for PNSM GUI clients, who must provide a username and password. Two-factor authentication will be supported in the upcoming Service Manager release.

PNSM Multi-Factor Authentication

PNSM supports multi-factor authentication (MFA) to log into the software. If your system administrator has enabled MFA for PNSM, in addition to User name and Password, you also need a One-Time password (OTP) code. The OTP code is a Time-based, One-Time password (TOTP) generated from the Microsoft Authenticator app, which is installed on your mobile device. The OTP code is a regularly changing code based on a secret key that is shared between the PNSM server and the Authenticator app.

Prerequisites

Installing Authenticator App

Before logging in to Service Manager, you must install Microsoft Authenticator on your mobile device.

Enabling/Disabling Multi-Factor Authentication

By default, PNSM installs with MFA enabled. This property may be changed in the `authentication.properties` file. To disable MFA, set ensure that the property is set as **local-mfa-authentication=0** the `authentication.properties` file.

Registering the Authenticator App for Multi-factor Authentication

When your system administrator has enabled MFA for PNSM, and you log into PNSM using your User name and Password for the first time (or following a secret key reset by Administrator), ASAP displays a secret key. Register the secret key into the Authenticator app. To do so, from the Authenticator app, click the + or ... option in the top right corner of the app and select **Add account > Other account**. Either scan the QR code or manually enter the account name and secret key displayed on the PNSM login screen. Select the **Complete** button to complete the setup on your mobile device and continue to login to PNSM.

Logging in with Multi-factor Authentication

Once registration is complete, the Authenticator app generates a time-based OTP based on time and secret key. This mechanism *does not* require an active Internet or cellular connection for your mobile device. When logging in, enter the time-based OTP listed in the app in the **Token** field for the Username displayed on the Login screen.

Resetting a Secret Key

A system administrator can reset the secret key shared between the PNSM server and the Authenticator app. You need to reset the secret key if, for example, you have uninstalled the Authenticator app or no longer have your mobile device. During your next login, PNSM prompts you to enter the secret key in the Authenticator app as described in the section [Registering the Authenticator App for Multi-factor Authentication](#).

Note: Administrators cannot bypass the Multi-factor Authentication requirements for any user . Each Admin only has privileges to reset the MFA status to **Not Registered**.

To reset MFA access for a user:

1. Navigate to the **User and Group Management** screen in the Service Manager software.
2. Select the user for whom you are resetting access.
3. Change the selection in the MFA dropdown to **Not Registered**.

The user must then follow the steps outlined in the previous sections to be granted access to the APNS software.

Using Certificates Issued by Trusted Certificate Authorities

To use certificates issued by trusted Certificate Authorities (CA), use the following procedure. Perform the steps from this directory:

/opt/Affirmed/NMS/server/ems/conf

To use certificates issued by trusted Certificate Authorities:

1. Issue the following command to generate a keystore with public and private keys (use *affirmed* for storepass):


```
# keytool -genkeypair -alias affirmed -keyalg RSA -keystore newTestServer.keystore -storepass affirmed --dn "CN=bgoenka,OU=Engineering,O=Affirmed Test,L=Acton,S=MA,C=US"
```
2. Issue the following command to verify the key works properly:


```
# keytool -list -v -keystore ./newTestServer.keystore
```
3. Issue the following command to generate a certificate signing request using the public key from the keystore you created in [step 1](#) .


```
# keytool -certreq -keyalg RSA -alias affirmed -keystore newTestServer.keystore -file server.csr
```
4. Issue the following command to test the newly generated certificate:


```
# openssl req -in server.csr -noout -text
```
5. *Optional:* Issue the following command to export a self-signed certificate from the keystore:


```
# keytool -export -alias affirmed -keystore newTestServer.keystore -file server.crt
```
6. Import the signed certificate, along with any intermediate certificates. Import each certificate, in the order that you are instructed by the CA. For each certificate to import, replace *intermediate.ca* or *server.crt* with the actual file name. If your certificates are not provided as separate files, create a separate file for each certificate, and paste its contents into the file.


```
# keytool -import -alias affirmed -keystore newTestServer.keystore -file server.crt
```

Note: Skip this step if using a self-signed certificate, as it was already generated in [step 1](#) .

7. Issue the following command to verify all the certificates were correctly imported:


```
# keytool -list -keystore newTestServer.keystore
```
8. Issue the following commands to move the original keystore and save newly created keystore as *affirmedhttps.keystore*:


```
# mv affirmedhttps.keystore affirmedhttps.keystore.orig
# mv newTestServer.keystore affirmedhttps.keystore
```
9. Restart the Service Manager Server.

Data Storage

MySQL is used to store all PNSM application data. All passwords are encrypted based on the following:

- An MD5 hashing algorithm is used to encrypt the GUI login password. This is a one-way encryption technique, and the password can never be decrypted.
- Device credentials are encrypted/decrypted using the JBoss vault mechanism, a RedHat encryption mechanism that encrypts all plain-text passwords. The vault stores the passwords in an AES-256 encrypted KeyStore.
- All passwords stored in the configuration files are encrypted/decrypted using the JBoss value mechanism .

PNSM VM Hardening

PNSM vulnerabilities are significantly reduced through VM hardening. This is achieved in the following ways:

- OS security patches are installed while deploying the PNSM instance on the Cloud through ARM templates.
- Unnecessary OS packages are removed and weak algorithms are disabled during the patch installation.
- The Azure Security Pack is installed on the VM to meet the Azure security principles.
- Outdated OS packages are monitored through a security scanning tool. Any vulnerabilities are addressed as part of the security patches during the installation or upgrade.

Penetration Test

A security scanning tool (Nessus) detects any security vulnerabilities as part of the testing phase and addresses critical issues that have high CVS scores. All Microsoft security guidelines are followed to ensure compliance with Microsoft Security standards.

The following high-security vulnerabilities are addressed in the Private Network Service Manager application:

- Cross-site scripting (XSS)
- Remote Code Execution (REC)
- Injections including SQL Injection, HTTP response header injection
- Cross Site Request Forgery (CSRF)
- Man-in-the-Middle attacks
- Weak passwords, encryption algorithms
- Arbitrary File Upload, Directory Traversal Attack

- Privilege elevation to the root user

PNSM is continuously scanned to detect any security vulnerabilities and to ensure all high-severity security issues are addressed.

Private Network Service Manager and Azure Network Function Manager Communication

Secure communication between the Private Network Service Manager and the Azure Network Function Manager (ANFM) occurs in the following ways:

- All Azure APIs are invoked using secure HTTP (HTTPS)
- Auth Tokens are renewed with Azure every 50 minutes
- Subscription details, such as subscription id, secret, and client ID are stored in the database as a blob object

PNSM to VNF Communication

NetFoundry facilitates fast, safe, and encrypted communication between the Private Network Service Manager and Network Functions. Both the PNSM server IPs and the VNF management IPs are configured using NetFoundry service rules. The NetFoundry VM has two gateways; one on the Azure public cloud and the other on the Azure Stack Edge.

PNSM to ASAP communication

ASAP communicates only with the Service Manager application using private IP addresses. No external communication is allowed. All communication between the applications is controlled by the Cloud security group and port inbound rules on the ASAP Virtual Machine.

ASAP Service Models are imported through SFTP using SSH keys for Day1 push. SSH keys are generated dynamically on each PNSM/ASAP installation.

ASAP to VNF communication

Communication between the Affirmed Service Automation Platform application and the VNF follows the same security principles as [PNSM to VNF Communication](#). NetFoundry services, including the necessary open ports for each VNF and other services for VNF communication, are enabled in NetFoundry both on the Cloud edge and on the Azure Stack Edge.

PNSM Cluster Communication

The PNSM cluster uses UNICAST (TCP) mode for cluster communication. Cluster communication occurs only between the Private IPs of the two Service Manager nodes, and does not allow any external security threats. Additionally, each cluster node authenticates the other using the password-based authentication for messaging service. This password is a static password stored securely in the JBoss vault using AES-256 encryption.

North Bound Interface

The Northbound REST API client access follows the security principles described in the [PNSM Access \(Web Security\)](#) section.

Azure Stack Edge (ASE) Security

For additional information regarding the Azure security baseline for Azure Stack Edge, please refer to the following Microsoft document:

[Azure security baseline for Azure Stack Edge](#)

NetFoundry Security

NetFoundry allows you to manage zero-trust networking by managing each function as a service:

- Secure identity and authentication, based on bi-directional certificate verification, replaces IP addresses
- Least privileged access replaces wide open VPNs, providing segmentation and isolation
- End-to-end control, visibility and security are enabled by the NetFoundry Fabric routers

For additional information on how the APNS solution is implemented using NetFoundry security measures, please refer to their website:

[NetFoundry Information](#)

Virtual Network Function (VNF) Security

VNF Access

Virtual Network Functions deployed in the Azure Stack Edge box are secured through NetFoundry, and use a private VPN connection from the Service Manager application.

The VNF is accessed through SSH keys, with NetFoundry acting as a gatekeeper to access the VNF on the ASE box.

***Note:** SSH keys are generated dynamically per Location where the network is deployed. Each VNF created for a different location in an enterprise generates a unique SSH key.*

VNF Data security

VNF access is controlled by NetFoundry VPN, which prevents any unauthorized access. To secure VNF data:

- All data is encrypted using AES-256-GCM encryption during transmission from the VNF to Cloud.

- No sensitive information is stored in the VNF disk as plain text.
- No sensitive information is written to the log files.
- Local VNF User account use the Linux default method of storing SHA-512 hash strings in /etc/shadow.
- SSH keys are stored in directories with strict read permissions.

Northbound APIs

This section provides the JSON syntax and Resources necessary for to run Northbound APIs in the Service Manager application.

Resources

Table B-1 Resources and Descriptions

Name	Path	Methods	Description
Configuration Management API	/rest/configurations/pnsm	GET PUT	Configuration Management APIs to Create, Read, Update and Delete the PNSM Configurations.
File Server Management API	/rest/fileservers /rest/fileservers/sftp /rest/fileservers/{fileServerID}	GET POST GET DELETE GET PUT	File Server Management APIs to Create, Read, Update and Delete the File Server(s).
Location Management API	/rest/locations /rest/locations/{locationID} /rest/locations/{locationID}/ips	GET POST DELETE GET PUT GET	Location Management APIs to Create, Read, Update and Delete the Location(s).

Name	Path	Methods	Description
Network Element Backup and Restore Management API	/rest/nebackup /rest/nebackupschedule /rest/nebackup/purge /rest/nebackup/{backupID} /rest/nerestore/{backupID} /rest/nebackupschedule/{scheduleID}/disable /rest/nebackupschedule/{scheduleID}/enable /rest/nebackupschedule/{scheduleID}/start	GET POST POST POST DELETE GET POST POST POST	Backup and Restore APIs to perform Network Elements Backup and Restore.
Network Management API	/rest/networks /rest/networks/devices /rest/networks/globalhssresources /rest/networks/globalpgwresources /rest/networks/packetcoreversions /rest/networks/regions /rest/networks/resources /rest/networks/{networkID} /rest/networks/packetcoreversions/active /rest/networks/resources/{resourceID} /rest/networks/{networkID}/day1 /rest/networks/{networkID}/nat /rest/networks/{networkID}/resources /rest/networks/{networkID}/retry /rest/networks/{networkID}/upgradestatus /rest/networks/{resourceID}/siminstances /rest/networks/resources/{resourceID}/sync /rest/networks/{networkID}/day0/status	GET POST GET GET POST GET POST GET GET GET DELETE GET GET DELETE GET POST PUT GET POST GET POST GET GET POST GET	Network Management APIs to Create, Read, Configure and Upgrade the Network(s).
RAN Info Management API	/rest/raninfo	GET	
SIM Management API	/rest/siminstances /rest/siminstances/activate /rest/siminstances/deactivate /rest/siminstances/delete /rest/siminstances/uploadSIMFile /rest/siminstances/{simInstanceID} /rest/siminstances/{simInstanceID}/activate /rest/siminstances/{simInstanceID}/status	GET POST POST POST POST POST GET POST GET	SIM Management APIs to Add, Activate, Deactivate and Delete the SIM (s).

Name	Path	Methods	Description
Software Management API	/rest/software/import	POST	Software Management APIs to Import Software and track the Software Upgrade.
	/rest/software/upgrade	GET POST	
	//rest/software/troubleshootbundle	GET POST	
	/rest/software/upgrade {upgradeScheduleDetailsID}	GET	

Configuration Management API

Configuration Management APIs to Create, Read, Update and Delete the PNSM Configurations.

GET /rest/configurations/pnsm

Get the Global PNSM Configuration for the ASAP Server and Azure.

Response Codes

Code	Condition
200	SUCCESS: Azure Configuration has been fetched.
500	FAILURE: Error while fetching the Azure Configuration.

Response Body

Media type	Data Type	Description
application/json	AzureConfig (JSON)	PNSM Configuration. Empty object will be returned if Azure Configuration is not added.

Example

Request

```
GET /rest/configurations/pnsm
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "subscriptionId" : "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
  "vendorName" : "AffirmedVendor",
  "apiVersion" : "2020-01-01-preview",
  "clientId" : "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
```

```

    "authority" : "https://login.microsoftonline.com/aaaaaaa-
bbbb-cccc-dddd-eeeeeeeeeeeeee/oauth2/token", ,
    "clientSecret" : "~i97-0W-7q.aaaaaaaaaaaaaaaaaaaa.G",
    "asapServerIpPort" : "10.168.2.16",
    "asapUsername" : "admin",
    "asapPassword" : "****",
    "hlrnerfEntityKey" : "6/19999990000",
    "hlrnerfAddressDigits" : "19999990000",
    "gsysHlrNumber" : "1/1/1/19999990000",
    "opValue" : "CDC202D5123E20F62B6D676AC72CB318",
    "provisioningType" : "OP"
  }

```

PUT /rest/configurations/pnsm

Update the Azure Configuration.

Request Body

Media Type	Data Type
application/json	AzureConfig (JSON)

Response Codes

Code	Condition
200	SUCCESS: Azure Configuration updated successfully.
500	FAILURE: Error while updating the Azure Configuration.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Azure Configuration UPDATE status.

Example

Request

```

PUT /rest/configurations/pnsm
Content-Type: application/json
Accept: application/json

{
  "subscriptionId" : "aaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
  "vendorName" : "AffirmedVendor",
  "apiVersion" : "2020-01-01-preview",
  "clientId" : "aaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
  "authority" : "https://login.microsoftonline.com/aaaaaaa-
bbbb-cccc-dddd-eeeeeeeeeeee/oauth2/token",

```

```
"clientSecret" : "~i97-0W-7q.xxxxxxxxxxxxxxxxxxxxxx.G",
"asapServerIpPort" : "10.168.2.16",
"opValue" : "CDC202D5123E20F62B6D676AC72CB318",
"provisioningType" : "OPC"}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : null,
  "objectName" : null,
  "successMessage" : "Azure Configuration updated
successfully"
}
```

File Server Management API

File Server Management APIs to Create, Read, Update and Delete the File Server(s).

GET /rest/fileservers

Get all the configured Filer Servers.

Response Codes

Code	Condition
200	SUCCESS: All the configured File Servers has been fetched.
500	FAILURE: Error while fetching configured File Servers.

Response Body

Media Type	Data Type	Description
application/json	FileServer (JSON)	List of File Servers. Empty list will be returned if no File Servers are configured.

Example

Request

```
GET /rest/fileservers
Content-Type: application/json
Accept: application/json

...
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "id" : "1",
  "protocol" : "FTP",
  "name" : "PNSM Server",
  "ipAddress" : "10.20.15.167",
  "port" : "21",
  "username" : "admin",
  "password" : "*****",
  "status" : true
}, {
  "id" : "2",
  "protocol" : "SFTP",
  "name" : "SFTP_Server",
  "ipAddress" : "10.20.15.167",
  "port" : "21",
  "username" : "admin",
  "password" : "*****",
  "status" : false
} ]
```

POST /rest/fileservers

Create a File Server to import Software Packages, perform Network Upgrades and perform Backup/Restore.

Request Body

Media Type	Data Type	Description
application/json	FileServer (JSON)	File Server Object

Response Codes

Code	Condition
200	SUCCESS: File Server created successfully.
500	FAILURE: Error while creating File Server.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	File Server CREATE status.

Example

Request

```
POST /rest/fileservers
Content-Type: application/json
Accept: application/json
```

```
{
  "protocol" : "SFTP",
  "name" : "SFTP_Server",
  "ipAddress" : "10.20.15.167",
  "port" : "21",
  "username" : "sftpuser",
  "password" : "*****",
  "confirmPassword" : "*****",
  "saveInactiveServer" : true
}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 2,
  "objectName" : "SFTP_Server",
  "successMessage" : "File Server created successfully"
}
```

GET /rest/fileservers/sftp

Get all the configured SFTP Filer Servers.

Response Codes

Code	Condition
200	SUCCESS: All the configured SFTP File Servers has been fetched.
500	FAILURE: Error while fetching configured SFTP File Servers.

Response Body

Media Type	Data Type	Description
application/json	FileServer (JSON)	List of SFTP File Servers. Empty list will be returned if no SFTP File Servers are configured.

Example

Request

```
GET /rest/fileservers/sftp
Content-Type: application/json
```

Accept: application/json

...

Response

HTTP/1.1 200 OK

Content-Type: application/json

```
[ {
  "id" : "2",
  "protocol" : "SFTP",
  "name" : "SFTP_Server",
  "ipAddress" : "10.20.15.167",
  "port" : "21",
  "username" : "sftpuser",
  "password" : "*****",
  "status" : false
} ]
```

DELETE /rest/fileservers/{fileServerID}

Delete a File Server by File Server ID.

Request Parameters

Name	Type	Description	Constraints
fileServerID	path	File Server ID	required

Response Codes

Code	Condition
200	SUCCESS: File Server deleted successfully.
500	FAILURE: Error while deleting File Server.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	File Server DELETE status.

Example

Request

```
DELETE /rest/fileservers/2
Content-Type: application/json
Accept: application/json
```

...

Response**HTTP/1.1 200 OK****Content-Type: application/json**

```
{
  "objectID" : 2,
  "objectName" : null,
  "successMessage" : "File Server deleted successfully"
}
```

GET /rest/fileservers/{fileServerID}

Get a File Server by File Server ID.

Request Parameters

Name	Type	Description	Constraints
fileServerID	path	File Server ID	required

Response Codes

Code	Condition
200	SUCCESS: File Server for the given File Server ID has been fetched.
500	FAILURE: Error while fetching the File Server for the given File Server ID.

Response Body

Media Type	Data Type	Description
application/json	FileServer (JSON)	File Server Object.

*Example***Request**

```
GET /rest/fileservers/1
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "id" : "1",
  "protocol" : "FTP",
  "name" : "PNSM Server",
  "ipAddress" : "10.20.15.167",
  "port" : "21",
  "username" : "admin",
  "password" : "*****",
  "status" : true
}
```

PUT /rest/fileservers/{fileServerID}

Update a File Server by File Server ID.

Request Parameters

Name	Type	Description	Constraints
fileServerID	path		required

Request Body

Media Type	Data Type	Description
application/json	FileServer (JSON)	File Server Object

Response Codes

Code	Condition
200	SUCCESS: File Server updated successfully.
500	FAILURE: Error while updated File Server.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	File Server UPDATE status.

Example

Request

```
PUT /rest/fileservers/2
Content-Type: application/json
Accept: application/json
```



```
{
  "protocol" : "SFTP",
  "name" : "SFTP_Server",
  "port" : "22",
  "username" : "sftpuser",
  "password" : "*****",
  "confirmPassword" : "*****"
  "saveInactiveServer" : true
}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 2,
  "objectName" : "SFTP_Server",
  "successMessage" : "File Server updated successfully"
}
```

Location Management API

Location Management APIs to Create, Read, Update and Delete the Location(s).

GET /rest/locations

Get the list of Locations.

Response Codes

Code	Condition
200	SUCCESS: All the Locations have been fetched.
500	FAILURE: Error while fetching Locations.

Response Body

Media Type	Data Type	Description
application/json	LocationEntity (JSON)	List of Locations. Empty list will be returned if no Locations are found.

Example

Request

```
GET /rest/locations
Content-Type: application/json
Accept: application/json
```

...

Response

HTTP/1.1 200 OK

Content-Type: application/json

```
[ {
  "version" : 0,
  "locationid" : 1,
  "name" : "Pltel2-13",
  "mgmtsubnet" : "10.165.31.0/27",
  "mgmtgateway" : null,
  "lansubnet" : "10.165.68.0/25",
  "langateway" : null,
  "wansubnet" : "10.165.68.128/25",
  "wangateway" : null,
  "publickey" : ... pnsnm@pnsnm",
  "privatekey" : "-----BEGIN RSA PRIVATE KEY-----\\n-----END
RSA PRIVATE KEY-----",
  "networkConfig" : "{\\"mgmtSubnet\\":{\\"10.165.31.0/27\\":
{\\"BaseSubnetIndex\\":\\"All\\",\\"NSSubnetIndex\\":\\"All\\"}},\\"1
anSubnet\\":{\\"10.165.68.0/25\\":
{\\"BaseSubnetIndex\\":\\"All\\",\\"NSSubnetIndex\\":\\"All\\"}},\\"w
anSubnet\\":{\\"10.165.68.128/26\\":
{\\"BaseSubnetIndex\\":\\"All\\",\\"NSSubnetIndex\\":\\"All\\"}},\\"10
.165.68.192/26\\":
{\\"BaseSubnetIndex\\":\\"All\\",\\"NSSubnetIndex\\":\\"All\\"}}}",
  "useSystemGeneratedIPs" : null
} ]
```

POST /rest/locations

Create a Location to deploy Networks.

Request Body

Media Type	Data Type	Description
application/json	LocationEntity (JSON)	Location Object

Response Codes

Code	Condition
200	SUCCESS: Location created successfully.
500	FAILURE: Error while creating Location.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Location CREATE status.

Example

Request

```
POST /rest/locations
Content-Type: application/json
Accept: application/json
```

```
{
  "name" : "Plte12-13",
  "mgmtsubnet" : "10.165.31.0/27",
  "lansubnet" : "10.165.68.0/25",
  "wansubnet" : "10.165.68.128/25"}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 1,
  "objectName" : "Acton",
  "successMessage" : "Location created successfully"}
```

DELETE /rest/locations/{locationID}

Delete a Location by Location ID.

Request Parameters

Name	Type	Description	Constraints
locationID	path	Location ID	required

Response Codes

Code	Condition
200	SUCCESS: Location deleted successfully.
500	FAILURE: Error while deleting Location.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Location DELETE status.

Example

Request

```
DELETE /rest/locations/1
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 1,
  "objectName" : null,
  "successMessage" : "Location deleted successfully"}
```

GET /rest/locations/{locationID}

Get a Location by Location ID.

Request Parameters

Name	Type	Description	Constraints
locationID	path	Location ID	required

Response Codes

Code	Condition
200	SUCCESS: Location for the given Location ID has been fetched.
500	FAILURE: Error while fetching the Location for the given Location ID.

Response Body

Media Type	Data Type	Description
application/json	LocationEntity (JSON)	Location Object.

*Example***Request**

```
GET /rest/locations/1
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "version" : 0,
  "locationid" : 1,
  "name" : "Plte12-13",
  "mgmtsubnet" : "10.165.31.0/27",
  "mgmtgateway" : null,
  "lansubnet" : "10.165.68.0/25",
  "langateway" : null,
  "wansubnet" : "10.165.68.128/25",
  "wangateway" : null,
  "publickey" : "ssh-rsa...pnsmpnsmpns",
  "privatekey" : "-----BEGIN RSA PRIVATE KEY-----\...\n-----
END RSA PRIVATE KEY-----",
  "networkConfig" : "{ \"mgmtSubnet\": { \"10.165.31.0/27\":
{ \"BaseSubnetIndex\": \"All\", \"NSSubnetIndex\": \"All\" } }, \"lanSubnet\": { \"10.165.68.0/25\":
{ \"BaseSubnetIndex\": \"All\", \"NSSubnetIndex\": \"All\" } }, \"wanSubnet\": { \"10.165.68.128/26\":
```

```
{\"BaseSubnetIndex\": \"All\", \"NSSubnetIndex\": \"All\"}, \"10
.165.68.192/26\":
{\"BaseSubnetIndex\": \"All\", \"NSSubnetIndex\": \"All\"}}\",
  \"useSystemGeneratedIPs\" : null
}
```

PUT /rest/locations/{locationID}

Update a Location by Location ID.

Request Parameters

Name	Type	Description	Constraints
locationID	path	Location ID	required

Request Body

Media Type	Data Type	Description
application/json	LocationEntity (JSON)	Location Object

Response Codes

Code	Condition
200	SUCCESS: Location updated successfully.
500	FAILURE: Error while updating Location.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Location UPDATE status.

Example

Request

```
PUT /rest/locations/1
Content-Type: application/json
Accept: application/json
```

```
{
  \"name\" : \"Plte12-13\",
  \"mgmtsubnet\" : \"10.165.31.0/27\",
  \"lansubnet\" : \"10.165.68.0/25\",
  \"wansubnet\" : \"10.165.68.128/25\"}
```

Response

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "objectID" : 1,
  "objectName" : "Acton",
  "successMessage" : "Location updated successfully"}
```

GET /rest/locations/{locationID}/ips

Get Next Available IP Addresses from the given Location

Request Parameters

name	type	description	constraints
locationID	path	Location ID	required
isHADeployment	query	HA Deployment Flag	required
node0	query	ASE Device Node 0	
node1	query	ASE Device Node 1	
resourceGroupName	query	Resource Group Name	

Response Codes

Code	Condition
200	SUCCESS: IP Addresses available in the given Location have been fetched.
500	FAILURE: Error while fetching IP Addresses available in the given Location.

Response Body

media type	data type	description
application/json	NetworkDeploymentConfig (JSON)	Network Deployment Configuration Object.

Example

Request

```
GET /rest/locations/1/ips?isHADeployment=true&node0=ase-plte-12&node1-
1=ase-plte-13&resourceGroupName=Private_LTE_ASE
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "networkIndex" : 1,
  "aseDeviceName" : "ase-plte-12",
  "subnetIndex" : "1",
  "mgmtSubnet" : "10.165.31.0/27",
  "lanSubnet" : "10.165.68.0/25",
  "wanSubnet" : "10.165.68.128/26",
  "managementIPAddress" : {
    "mgmtCIDR" : "10.165.31.0/27",
    "mgmtSubnetPrefix" : "/27",
    "mgmtGatewayIPAddress" : "10.165.31.1",
    "mgmtNetMaskIPAddress" : "255.255.255.224",
    "mrmManagementIPAddress" : "10.165.31.8",
    "msmManagementIPAddress" : "10.165.31.9",
    "mmeLogicalManagementIPAddress" : "10.165.31.10",
    "mccManagement0IPAddress" : "10.165.31.11",
    "mccManagement1IPAddress" : "10.165.31.12",
    "mccManagement2IPAddress" : "10.165.31.13",
    "hssManagementNic1IPAddress" : "10.165.31.14",
    "nfManagementIPAddress" : "10.165.31.7"  },
  "lanIPAddress" : {
    "lanCIDR" : "10.165.68.0/25",
    "lanSubnetPrefix" : "/25",
    "lanGatewayIPAddress" : "10.165.68.1",
    "lanNetMaskIPAddress" : "255.255.255.128",
    "nfLANIPAddress" : "10.165.68.24"  },
  "hadeployment" : true,
  "primaryNode" : true
}, {
  "networkIndex" : 1,
  "aseDeviceName" : "ase-plte-13",
```



```

"subnetIndex" : "2",
"mgmtSubnet" : "10.165.31.0/27",
"lanSubnet" : "10.165.68.0/25",
"wanSubnet" : "10.165.68.128/26",
"managementIPAddress" : {
  "mgmtCIDR" : "10.165.31.0/27",
  "mgmtSubnetPrefix" : "/27",
  "mgmtGatewayIPAddress" : "10.165.31.1",
  "mgmtNetMaskIPAddress" : "255.255.255.224",
  "mrmManagementIPAddress" : "10.165.31.24",
  "msmManagementIPAddress" : "10.165.31.25",
  "mmeLogicalManagementIPAddress" : "10.165.31.10",
  "mccManagement0IPAddress" : "10.165.31.27",
  "mccManagement1IPAddress" : "10.165.31.28",
  "mccManagement2IPAddress" : "10.165.31.29",
  "hssManagementNic1IPAddress" : "10.165.31.30",
  "nfManagementIPAddress" : "10.165.31.23" },
"lanIPAddress" : {
  "lanCIDR" : "10.165.68.0/25",
  "lanSubnetPrefix" : "/25",
  "lanGatewayIPAddress" : "10.165.68.1",
  "lanNetMaskIPAddress" : "255.255.255.128",
  "nfLANIPAddress" : "10.165.68.28" },
"hadeployment" : true,
"primaryNode" : false
} ]

```

Network Element Backup and Restore Management API

Backup and Restore APIs to perform Network Elements Backup and Restore.

GET /rest/nebackup

Get all NE Backups.

Response Codes

Code	Condition
200	SUCCESS: NE backups have been fetched.
500	FAILURE: Error while fetching the NE backup list.

Response Body

Media Type	Data Type	Description
application/json	NEBackupRestore (JSON)	List of NE backups.

Example

Request

```
GET /rest/nebackup
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "version" : 0,
  "nebackuprestoreid" : 1,
  "resourceid" : 7,
  "backuptime" : 1626960228643,
  "backuptriggtypeid" : 2,
  "statusid" : 4,
  "ftpprotocolid" : 1,
  "backupfilename" : "backup-ConfigBackup",
  "relativepath" :
"/data/affirmed/pnsm/anroot/Affirmed/output/Affirmed/NMS/ser
ver/ems/data/nebackups/MME_10.165.31.24-[node1]",
  "ftpprotocol" : null
}, {
  "version" : 0,
  "nebackuprestoreid" : 2,
  "resourceid" : 6,
  "backuptime" : 1626960279881,
  "backuptriggtypeid" : 2,
  "statusid" : 3,
  "ftpprotocolid" : 1,
  "backupfilename" : "backup-ConfigBackup-Jul_22_13_24_40-
15.0.20.2-0.tar",
  "relativepath" :
"/data/affirmed/pnsm/anroot/Affirmed/output/Affirmed/NMS/ser
ver/ems/data/nebackups/MME_10.165.31.8-[node0]",
  "ftpprotocol" : null
}, {
  "version" : 0,
  "nebackuprestoreid" : 3,
  "resourceid" : 7,
  "backuptime" : 1626960281724,
```

```

    "backuptriggtypeid" : 2,
    "statusid" : 4,
    "ftpprotocolid" : 1,
    "backupfilename" : "backup-ConfigBackup",
    "relativepath" :
"/data/affirmed/pnsm/anroot/Affirmed/output/Affirmed/NMS/ser
ver/ems/data/nebackups/MME_10.165.31.24-[node1]",
    "ftpprotocol" : null
} ]

```

POST /rest/nebackup

Create an NE Backup.

Request Body

Media Type	Data Type	Description
application/json	NEBackup (JSON)	NE Backup Object

Response Codes

Code	Condition
200	SUCCESS: NE backup has been initiated for the given resources.
500	FAILURE: Error while initiating NE backup for the given resources.

Response Body

Media Type	Data Type	Description
application/json	ConfigBackupResponseVO (JSON)	Backup status. If Resource for given Resource ID is not found, empty Backup object with error message will be returned.

Example

Request

```

POST /rest/nebackup
Content-Type: application/json
Accept: application/json

```

```

{
  "backupFilePrefix" : "backup",
  "relativeFilePath" : "ems/data/nebackups",
  "ftpServerId" : 1,
  "resourceIdList" : [ 6, 7, 8 ]
}

```

```
}
```

Response

HTTP/1.1 200 OK

Content-Type: application/json

```
[ {
  "status" : "Success",
  "elementName" : "MME_10.165.31.8-[node0]",
  "ipaddress" : "10.165.31.8",
  "message" : "Backup from this node."}, {
  "status" : "Failure",
  "elementName" : "MME_10.165.31.24-[node1]",
  "ipaddress" : "10.165.31.24",
  "message" : "Failed to backup on MME_10.165.31.24-[node1]
since it is in standby."}, {
  "status" : null,
  "elementName" : null,
  "ipaddress" : null,
  "message" : "Resource 8 not found"} ]
```

POST /rest/nebackupschedule

Create NE backup schedule.

Request Body

Media Type	Data Type	Description
application/json	NEBackupSchedule (JSON)	NE Backup Schedule Object

Response Codes

Code	Conditions
200	SUCCESS: NE backup schedule has been created for the given resources.
500	FAILURE: Error while creating NE backup for the given resources.

Response Body

media type	data type	description
application/json	object (JSON)	NE backup schedule CREATE status.

*Example***Request**

```
POST /rest/nebackupschedule
Content-Type: application/json
Accept: application/json
```

```
{
  "schdeuleName" : "schedule",
  "backupFilePrefix" : "backup",
  "relativeFilePath" : "ems/data/nebackups",
  "ftpServerId" : 1,
  "resourceIdList" : [ 6, 7, 8 ],
  "recurrence" : {
    "startTime" : 193864849230,
    "endTime" : 193864849230,
    "repeatCount" : 5,
    "minute" : 21,
    "hour" : 11,
    "day" : 27,
    "week" : 4,
    "month" : 1,
    "year" : 2
  }
}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "status" : "Success",
  "elementName" : "MME_10.165.31.8-[node0]",
  "ipaddress" : "10.165.31.8",
  "message" : "Backup from this node."}, {
  "status" : "Failure",
  "elementName" : "MME_10.165.31.24-[node1]",
  "ipaddress" : "10.165.31.24",
  "message" : "Failed to backup on MME_10.165.31.24-[node1]
since it is in standby."}, {
  "status" : null,
  "elementName" : null,
  "ipaddress" : null,
  "message" : "Resource 8 not found"} ]
```

POST /rest/nebackup/purge

Configure NE Backup purge.

Request Body

Media Type	Data Type	Description
application/json	NEBackupPurge (JSON)	NE Backup Purge Object

Response Codes

Code	Condition
200	SUCCESS: NE backup file purge has been configured.
500	FAILURE: Error while configuring the NE backup file purge.

Response Body

media type	data type	description
application/json	NEBackupPurge (JSON)	NE backup purge status.

Example**Request**

```
POST /rest/nebackup/purge
Content-Type: application/json
Accept: application/json
```

```
{
  "count" : 1,
  "status" : true
}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 1,
  "objectName" : null,
  "successMessage" : "Backup purge configured successfully"}
```

DELETE /rest/nebackup/{backupID}

Delete NE Backup details by ID.

Request Parameters

Name	Type	Description	Constraints
backupID	path	Backup id of the NE backup to be deleted	required

Response Codes

Code	Condition
200	SUCCESS: NE backup has been deleted.
500	FAILURE: Error while deleting the NE backup.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	NE backup DELETE status.

*Example***Request**

```
DELETE /rest/nebackup/2
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 1,
  "objectName" : null,
  "successMessage" : "Backup deleted successfully"}
```

GET /rest/nebackup/{backupID}

Get NE Backup details by ID.

Request Parameters

Name	Type	Description	Constraints
backupID	path	Backup id of the NE backup to be fetched	required

Response Codes

Code	Condition
200	SUCCESS: NE backup has been fetched.
500	FAILURE: Error while fetching the NE backup.

Response Body

media type	data type	description
application/json	NEBackupRestore (JSON)	NE backup.

Example

Request

```
GET /rest/nebackup/2
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "version" : 0,
  "nebackuprestoreid" : 2,
  "resourceid" : 6,
  "backuptime" : 1626960279881,
  "backuptriggertypeid" : 2,
  "statusid" : 3,
  "ftpprotocolid" : 1,
  "backupfilename" : "backup-ConfigBackup-Jul_22_13_24_40-
```



```

15.0.20.2-0.tar",
  "relativepath" :
"/data/affirmed/pnsm/anroot/Affirmed/output/Affirmed/NMS/ser
ver/ems/data/nebackups/MME_10.165.31.8-[node0]",
  "ftpprotocol" : null
}

```

POST /rest/nerestore/{backupID}

Restore an NE Backup.

Request Parameters

Name	Type	Description	Constraints
backupID	path	Backup id of the NE backup to be restored	required

Response Codes

Code	Condition
200	SUCCESS: NE backup has been restored for the given backup id.
500	FAILURE: Error while restoring NE backup for the given backup id.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Restore message object of the selected network element.

Example

Request

```

POST /rest/nerestore/2
Content-Type: application/json
Accept: application/json

```

...

Response

```

HTTP/1.1 200 OK
Content-Type: application/json

```

```
{
  "objectID" : null,
  "objectName" : null,
  "successMessage" : "Restore operation successfully
initiated on node MME_10.165.31.8-[node0] with backup file
backup-ConfigBackup-Jul_22_13_24_40-15.0.20.2-0.tar"}
```

POST /rest/nebackupschedule/{scheduleId}/disable

Disable NE backup schedule by ID

Request Parameters

Name	Type	Description	Constraints
scheduleId	path	NE Backup Schedule ID	required

Response Codes

Code	Condition
200	SUCCESS: NE backup schedule has been disabled for the given resources.
500	FAILURE: Error while disabling NE backup for the given resources.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	NE backup schedule disable status.

Example

Request

```
POST /rest/nebackupschedule/1/disable
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
```

```

    "objectID" : 1,
    "objectName" : null,
    "successMessage" : "Backup schedule disabled
successfully"}

```

POST /rest/nebackupschedule/{scheduleId}/enable

Enable NE backup schedule by ID

Request Parameters

Name	Type	Description	Constraints
scheduleId	path	NE Backup Schedule ID	required

Response Codes

Code	Condition
200	SUCCESS: NE backup schedule has been enabled for the given resources.
500	FAILURE: Error while enabling NE backup for the given resources.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	NE backup schedule enable status.

Example

Request

```

POST /rest/nebackupschedule/1/enable
Content-Type: application/json
Accept: application/json

```

...

Response

```

HTTP/1.1 200 OK
Content-Type: application/json

```

```

{
  "objectID" : 1,
  "objectName" : null,

```

```
"successMessage" : "Backup schedule enabled successfully"}
```

POST /rest/nebackupschedule/{scheduleId}/start

Start NE backup schedule by ID

Request Parameters

Name	Type	Description	Constraints
scheduleId	path	NE Backup Schedule ID	required

Response Codes

Code	Condition
200	SUCCESS: NE backup schedule has been started for the given resources.
500	FAILURE: Error while starting NE backup for the given resources.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	NE backup schedule disable status.

Example

Request

```
POST /rest/nebackupschedule/1/start
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "objectID" : 1,
  "objectName" : null,
  "successMessage" : "Backup schedule started successfully"}
```

Network Management API

Network Management APIs to Create, Read, Configure and Upgrade the Network(s).

GET /rest/networks

Get all available Networks.

Response Codes

Condition	Condition
200	SUCCESS: All the Networks have been fetched.
500	FAILURE: Error while fetching Networks.

Response Body

media type	data type	description
application/json	ServiceInstanceConfig (JSON)	List of Networks. Empty list will be returned if no Networks are found.

Example

Request

```
GET /rest/networks
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "serviceInstanceId" : 1,
  "parentResourceGroupId" : 1,
  "nfsProvisioningState" : "Succeeded",
  "provisioningState" : "Provisioned",
  "deploymentMode" : "ONPREMHA",
  "locations" : [ {
    "name" : "Plte12-13",
    "displayName" : null,
    "mgmtsubnet" : "10.165.31.0/27",
    "mgmtgw" : null,
```

```

    "lansubnet" : "10.165.68.0/25",
    "langw" : null,
    "wansubnet" : "10.165.68.128/25",
    "wangw" : null,
    "locationNetworkConfig" : null,
    "asedeviceContainers" : [ {
      "regionName" : null,
      "regionId" : 0,
      "node0" : {
        "mme" : {
          "networkFunction" : {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
2_13_2506_HA_deploy1-MME-node0",
            "name" : "pltel2_13_2506_HA_deploy1-MME-node0",
            "type" :
"microsoft.hybridnetwork/networkfunctions",
            "location" : "eastus",
            "properties" : {
              "provisioningState" : "Succeeded",
              "device" : {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-
dddd-eeeeeeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-12"
              },
              "skuName" : "apns_mme_1_0",
              "skuType" : "EvolvedPacketCore",
              "vendorName" : "AffirmedNetworks",
              "serviceKey" : "aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeeeeeee",
              "vendorProvisioningState" : "Provisioned",
              "managedApplication" : null,
              "managedApplicationParameters" : null,
              "networkFunctionUserConfigurations" : [ {
                "roleName" : "mrm-0",
                "userDataParameters" : null,
                "networkInterfaces" : [ {
                  "networkInterfaceName" : "mrm-0-mgmt",
                  "macAddress" : null,
                  "ipConfigurations" : [ {
                    "ipAllocationMethod" : "Static",
                    "ipAddress" : "10.165.31.8",
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[illegible]

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[illegible]

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```

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```

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    }
  } ]
}
},
"provisioningState" : null,
"detailedStatusURL" : null,
"createCorrelationId" : null,
"nfPortalName" : "plte12_13_2506_HA_deploy1-SAEGW-
node1",
"smName" : "SAEGW_10.165.31.26-[node1]",
"asapName" : "SAEGW-10.165.31.26",
"day0Conf" : {
  "clusterIndex" : 22,
  "managementCIDRSuffix" : "/27",
  "lanCIDRSuffix" : "/25",
  "wanCIDRSuffix" : "/25",
  "management0" : "10.165.31.26",
  "management1" : "10.165.31.27",
  "management2" : "10.165.31.28",
  "managementNetworkAddress" : "10.165.31.0/27",

```

```

    "managementGatewayAddress" : "10.165.31.1",
    "base0" : "10.165.68.90",
    "base1" : "10.165.68.91",
    "base2" : "10.165.68.92",
    "baseNetworkAddress" : "10.165.68.0/25",
    "baseGatewayAddress" : "10.165.68.1",
    "ew" : "10.165.68.98",
    "ewNetworkAddress" : "10.165.68.0/25",
    "ewGatewayAddress" : "10.165.68.1",
    "ns1" : "10.165.68.18",
    "ns1NetworkAddress" : "10.165.68.0/25",
    "ns1GatewayAddress" : "10.165.68.1",
    "ns2" : "10.165.68.145",
    "ns2NetworkAddress" : "10.165.68.128/25",
    "ns2GatewayAddress" : "10.165.68.129",
    "cloudBaseInternalCIDR" : "10.165.68.90/29"
  },
  "day1Conf" : {
    "networkName" : "Telecom-xyz",
    "shortNetworkName" : "xyz",
    "apnName" : "plte",
    "homePlmnId" : "99999",
    "pgwIpAddress" : "10.165.68.112",
    "sgwS11IpAddress" : "10.165.68.114",
    "giLoopbackIp" : "10.165.68.176",
    "giInterfaceIp" : "10.165.68.145",
    "giInterfaceIpRouteDest" : "0.0.0.0",
    "giInterfaceIpRoutePrefix" : "0",
    "giInterfaceIpRouteNextHop" : "10.165.68.129",
    "giIpSubPoolRangeStart" : "200.130.0.1",
    "giIpSubPoolRangeEnd" : "200.130.0.150",
    "gnSgwS1" : "10.165.68.113",
    "gnSgwLoopbackIp" : "10.165.68.115",
    "gnInterfaceIp" : "10.165.68.18",
    "gnInterfaceIpRouteDest" : "0.0.0.0",
    "gnInterfaceIpRoutePrefix" : "0",
    "gnInterfaceIpRouteNextHop" : "10.165.68.1",
    "scRoamingPlmnId" : "12345",
    "zoneDefaultQosProfile" : "QOSPROFDEFAULT",
    "bgpRouterGN" : {
      "routerId" : "10.165.68.18",
      "localAs" : "65321",
      "remoteAs" : "64512",
      "neighborIP" : "10.165.68.1",
      "fallOverBFD" : "multi-hop",
      "bfdMultiHopPeer" : "10.165.68.1",
      "bgpGEOLoopIP" : "10.165.68.118",
      "bgpGNLoopIP" : "10.165.68.119"
    },
    "bgpRouterGI" : {
      "routerId" : "10.165.68.145",

```

```

        "localAs" : "65321",
        "remoteAs" : "64512",
        "neighborIP" : "10.165.68.129",
        "fallOverBFD" : "single-hop",
        "bfdMultiHopPeer" : "10.165.68.129",
        "bgpGILoopIP" : "10.165.68.179"      },
    "giV4IpSubPoolRangeStart" : "10.202.0.1",
    "giV4IpSubPoolRangeEnd" : "10.202.0.255",
    "giRemoteIpSubPoolRangeStart" : "10.201.0.1",
    "giRemoteIpSubPoolRangeEnd" : "10.201.0.255",
    "dataLoopIP" : "10.165.68.176",
    "dataFabricIpAddress" : "10.165.68.93",
    "dnsPrimaryIp" : "200.5.0.30",
    "dnsSecondaryIp" : "200.5.0.20"      },
    "natProfile" : {
        "configStatus" : "PROVISIONED",
        "status" : "Success",
        "message" : "Configured",
        "natSubPoolRangeStart" : "10.200.0.1",
        "natSubPoolRangeEnd" : "10.200.0.255",
        "profileName" : "NAT"      },
    "edrProfile" : null,
    "mccStatus" : {
        "day0Status" : "Success",
        "day0message" : "Done",
        "day1Status" : "Success",
        "day1Message" : "Configured",
        "day1GIStatus" : null,
        "day1GImessage" : null,
        "day1GNStatus" : null,
        "day1GNmessage" : null,
        "day1SCStatus" : null,
        "day1SCmessage" : null,
        "day1ServiceStatus" : null,
        "day1Servicemessage" : null,
        "dataFabricConfigured" : true
    }
},
    "hss" : {
        "networkFunction" : {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
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ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
2_13_2506_HA_deploy1-HSS-node1",
            "name" : "pltel2_13_2506_HA_deploy1-HSS-node1",
            "type" :
"microsoft.hybridnetwork/networkfunctions",
            "location" : "eastus",
            "properties" : {
                "provisioningState" : "Succeeded",

```

```

        "device" : {
            "id" : "/subscriptions/aaaaaaa-bbbb-cccc-
            dddd-eeeeeeeeeeee/resourceGroups/Private_LTE_
            ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-13"
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        "skuName" : "apns_hss_1_0",
        "skuType" : "EvolvedPacketCore",
        "vendorName" : "AffirmedNetworks",
        "serviceKey" : "aaaaaaa-bbbb-cccc-dddd-
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        "vendorProvisioningState" : "Provisioned",
        "managedApplication" : null,
        "managedApplicationParameters" : null,
        "networkFunctionUserConfigurations" : [ {
            "roleName" : "hpehss",
            "userDataParameters" : null,
            "networkInterfaces" : [ {
                "networkInterfaceName" :
                "mrmmmanagementnic1",
                "macAddress" : null,
                "ipConfigurations" : [ {
                    "ipAllocationMethod" : "Static",
                    "ipAddress" : "10.165.31.29",
                    "subnet" : "10.165.31.0/27",
                    "gateway" : "10.165.31.1",
                    "ipVersion" : "IPv4",
                    "dnsServers" : null
                } ],
                "vmSwitchType" : "Management"
            }, {
                "networkInterfaceName" : "mrmlannic1",
                "macAddress" : null,
                "ipConfigurations" : [ {
                    "ipAllocationMethod" : "Static",
                    "ipAddress" : "10.165.68.19",
                    "subnet" : "10.165.68.0/25",
                    "gateway" : "10.165.68.1",
                    "ipVersion" : "IPv4",
                    "dnsServers" : null
                } ],
                "vmSwitchType" : "Lan"
            } ],
            "osProfile" : {
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                OIjoicGx0ZTIyaHNzLmFmZmlybWVhLmNvbSIsCiAgICAgICAgICAgICAgICAgICA
            }
        }
    ]
}

```


[illegible]

```

        "useSystemGeneratedIPs" : false,
        "hadeploymentMode" : true
    } ],
    "cloudContainers" : null,
    "networkName" : "plte12_13_2506_HA_deploy1",
    "networkId" : 2,
    "regionId" : 2,
    "regionName" : "eastus",
    "resourceGroupName" : "Private_LTE_ASE",
    "mmePrivateKey" : "-----BEGIN RSA PRIVATE KEY-----\...\n--
---END RSA PRIVATE KEY-----",
    "skuName" : "APNS 1.0",
    "serviceManagerSkuId" : 1,
    "createdTime" : 1626698701851,
    "useSystemGeneratedIPs" : false,
    "mccPart" : "999",
    "mncPart" : "099",
    "ntpIP" : "10.168.0.10",
    "optionalNtpIP1" : "10.168.0.11",
    "optionalNtpIP2" : "10.168.0.12",
    "timeZone" : "Asia/Kolkata",
    "resourcesAvailableForDay1" : true
} ]

```

POST /rest/networks

Create a Standalone Edge / Distributed Edge / All in Cloud Network.

Request Body

Media Type	Data Type	Description
application/json	ServiceInstanceConfig (JSON)	Network Object

Response Codes

Code	Condition
200	SUCCESS: Network created successfully.
500	FAILURE: Error while creating Network.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Network CREATE status.

*Example***Request**

```

POST /rest/networks
Content-Type: application/json
Accept: application/json

{
  "parentResourceGroupId" : 1,
  "deploymentMode" : "ONPREMHA",
  "locations" : [ {
    "name" : "Pltel2-13",
    "asedeviceContainers" : [ {
      "node0" : {
        "name" : "ase-plte-12",
        "mme" : {
          "day0Conf" : {
            "mrmManagement" : "10.165.31.8",
            "mrmBase" : "10.165.68.34",
            "mrmData" : "10.165.68.42",
            "msmManagement" : "10.165.31.9",
            "msmBase" : "10.165.68.35",
            "msmData" : "10.165.68.43",
            "ns1" : "10.165.68.9",
            "logicalManagement" : "10.165.31.10",
            "s6IP" : "10.165.68.49"
          }
        },
        "mcc" : {
          "day0Conf" : {
            "management0" : "10.165.31.11",
            "management1" : "10.165.31.12",
            "management2" : "10.165.31.13",
            "base0" : "10.165.68.58",
            "base1" : "10.165.68.59",
            "base2" : "10.165.68.60",
            "ew" : "10.165.68.66",
            "ns1" : "10.165.68.10",
            "ns2" : "10.165.68.137"
          }
        },
        "hss" : {
          "day0Conf" : {
            "managementNic1" : "10.165.31.14",
            "lanNic1" : "10.165.68.11"
          }
        }
      },
      "node1" : {
        "name" : "ase-plte-13",
        "mme" : {

```

```

        "day0Conf" : {
            "mrmManagement" : "10.165.31.24",
            "mrmBase" : "10.165.68.36",
            "mrmData" : "10.165.68.44",
            "msmManagement" : "10.165.31.25",
            "msmBase" : "10.165.68.37",
            "msmData" : "10.165.68.45",
            "ns1" : "10.165.68.17",
            "logicalManagement" : "10.165.31.10"
        },
        "mcc" : {
            "day0Conf" : {
                "management0" : "10.165.31.26",
                "management1" : "10.165.31.27",
                "management2" : "10.165.31.28",
                "base0" : "10.165.68.90",
                "base1" : "10.165.68.91",
                "base2" : "10.165.68.92",
                "ew" : "10.165.68.98",
                "ns1" : "10.165.68.18",
                "ns2" : "10.165.68.145"
            },
            "hss" : {
                "day0Conf" : {
                    "managementNic1" : "10.165.31.29",
                    "lanNic1" : "10.165.68.19"
                }
            }
        }
    ],
    "networkName" : "plte12_13_2506_HA_deploy1",
    "regionId" : "2",
    "resourceGroupName" : "Private_LTE_ASE",
    "serviceManagerSkuId" : "1",
    "mccPart" : "999",
    "mncPart" : "099",
    "ntpIP" : "10.168.0.10",
    "optionalNtpIP1" : "10.168.0.11",
    "optionalNtpIP2" : "10.168.0.12",
    "timeZone" : "Asia/Kolkata",
    "useSystemGeneratedIPs" : false
}

```

Response

```

HTTP/1.1 200 OK
Content-Type: application/json

```

```
{
  "objectID" : 2,
  "objectName" : "plte12_13_2506_HA_deploy1",
  "successMessage" : "Network created successfully"}
```

GET /rest/networks/devices

Get all the ASE Devices. Devices are listed based on the Service Configuration available in PNSM configuration.

Response Codes

Code	Condition
200	SUCCESS: All the ASE Devices have been fetched.
500	FAILURE: Error while fetching ASE Devices.

Response Body

Media Type	Data Type	Description
application/json	Device (JSON)	List of ASE Devices. Empty list will be returned if no ASE Device found.

Example

Request

```
GET /rest/networks/devices
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "id" : "/subscriptions/aaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-4",
  "name" : "ase-plte-4",
  "type" : "microsoft.hybridnetwork/devices",
```

```

    "properties" : {
      "status" : "Registered",
      "provisioningState" : "Succeeded",
      "deviceType" : "AzureStackEdge",
      "azureStackEdge" : {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE4-G"    },
        "networkFunctions" : [ {
          "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/PLTE
4-NF-VM"    }, {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
4_2103_deploy01-MCC"    } ]
        }
      }, {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-7",
        "name" : "ase-plte-7",
        "type" : "Microsoft.HybridNetwork/devices",
        "properties" : {
          "status" : "Registered",
          "provisioningState" : "Succeeded",
          "deviceType" : "AzureStackEdge",
          "azureStackEdge" : {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/dataBoxEdgeDevices/Stack
Edge-Private-LTE7"    },
            "networkFunctions" : [ {
              "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210701140258/providers/Microsoft.HybridNetwork/networkFunc
tions/NFASEPLTE7"    } ]
            }
          }, {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-5",
            "name" : "ase-plte-5",
            "type" : "Microsoft.HybridNetwork/devices",
            "properties" : {
              "status" : "Registered",
              "provisioningState" : "Succeeded",

```

```

    "deviceType" : "AzureStackEdge",
    "azureStackEdge" : {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE5-G"    },
        "networkFunctions" : [ {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210709143039/providers/Microsoft.HybridNetwork/networkFunc
tions/NFASEPLTE5"    }, {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
5_0715_deploy1-HSS"    }, {
                    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
5_0715_deploy1-SAEGW"    }, {
                        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
5_0715_deploy1-MME"    } ]
        }
    }, {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-11",
        "name" : "ase-plte-11",
        "type" : "microsoft.hybridnetwork/devices",
        "properties" : {
            "status" : "Registered",
            "provisioningState" : "Succeeded",
            "deviceType" : "AzureStackEdge",
            "azureStackEdge" : {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE11"    },
                "networkFunctions" : [ {
                    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
1_0715_deploy1-HSS"    }, {
                        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
1_0715_deploy1-SAEGW"    }, {
                            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-

```

```

eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte1
1_0715_deploy1-MME"    } ]
    }
  }, {
    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-12",
    "name" : "ase-plte-12",
    "type" : "microsoft.hybridnetwork/devices",
    "properties" : {
      "status" : "Registered",
      "provisioningState" : "Succeeded",
      "deviceType" : "AzureStackEdge",
      "azureStackEdge" : {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE12"    },
        "networkFunctions" : [ {
          "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210619192556/providers/Microsoft.HybridNetwork/networkFunc
tions/NFASEPLTE12"    }, {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte1
2_13_2506_HA_deploy1-HSS-node0"    }, {
              "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte1
2_13_2506_HA_deploy1-SAEGW-node0"    }, {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte1
2_13_2506_HA_deploy1-MME-node0"    } ]
            }
          }, {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-13",
            "name" : "ase-plte-13",
            "type" : "microsoft.hybridnetwork/devices",
            "properties" : {
              "status" : "Registered",
              "provisioningState" : "Succeeded",
              "deviceType" : "AzureStackEdge",
              "azureStackEdge" : {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-

```



```

eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE13"    },
    "networkFunctions" : [ {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210621152348/providers/Microsoft.HybridNetwork/networkFunc
tions/NFASEPLTE13"    }, {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
2_13_2506_HA_deploy1-HSS-node1"    }, {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
2_13_2506_HA_deploy1-SAEGW-node1"    }, {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
2_13_2506_HA_deploy1-MME-node1"    } ]
    }
}, {
    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-6",
    "name" : "ase-plte-6",
    "type" : "microsoft.hybridnetwork/devices",
    "properties" : {
        "status" : "Registered",
        "provisioningState" : "Succeeded",
        "deviceType" : "AzureStackEdge",
        "azureStackEdge" : {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE6-G"    },
            "networkFunctions" : [ {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210630142619/providers/Microsoft.HybridNetwork/networkFunc
tions/NFASEPLTE6"    }, {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
6_0630_dep1-HSS"    }, {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_

```

```

ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
6_0630_depl-SAEGW"    }, {
    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
6_0630_depl-MME"    } ]
    }
}, {
    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-8",
    "name" : "ase-plte-8",
    "type" : "microsoft.hybridnetwork/devices",
    "properties" : {
        "status" : "Registered",
        "provisioningState" : "Succeeded",
        "deviceType" : "AzureStackEdge",
        "azureStackEdge" : {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE8-G"    },
            "networkFunctions" : [ {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210630155656/providers/Microsoft.HybridNetwork/networkFunc
tions/NFASEPLTE8"    }, {
                    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
8_depl-HSS"    }, {
                        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
8_depl-SAEGW"    }, {
                            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
8_depl-MME"    } ]
                }
            }, {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-3",
                "name" : "ase-plte-3",
                "type" : "microsoft.hybridnetwork/devices",
                "properties" : {
                    "status" : "Registered",
                    "provisioningState" : "Succeeded",

```

```

    "deviceType" : "AzureStackEdge",
    "azureStackEdge" : {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE03"    },
        "networkFunctions" : [ {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210622110739/providers/Microsoft.HybridNetwork/networkFunc
tions/plte3nfjun22"    } ]
    }
}, {
    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-2",
    "name" : "ase-plte-2",
    "type" : "microsoft.hybridnetwork/devices",
    "properties" : {
        "status" : "Registered",
        "provisioningState" : "Succeeded",
        "deviceType" : "AzureStackEdge",
        "azureStackEdge" : {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-PrivateLTE02"    },
            "networkFunctions" : [ {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
2_0716_deploy1-HSS"    }, {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
2_0716_deploy1-SAEGW"    }, {
                "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
2_0716_deploy1-MME"    } ]
            }
    }, {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-10",
        "name" : "ase-plte-10",
        "type" : "microsoft.hybridnetwork/devices",
        "properties" : {
            "status" : "Registered",

```

```

    "provisioningState" : "Succeeded",
    "deviceType" : "AzureStackEdge",
    "azureStackEdge" : {
      "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE-10"    },
    "networkFunctions" : [ {
      "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210708083158/providers/Microsoft.HybridNetwork/networkFunc
tions/NFASEPLTE10"    }, {
      "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
9_10_1907_HA_deploy1-HSS-node1"    }, {
      "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
9_10_1907_HA_deploy1-SAEGW-node1"    }, {
      "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
9_10_1907_HA_deploy1-MME-node1"    } ]
    }
  }, {
    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-9",
    "name" : "ase-plte-9",
    "type" : "microsoft.hybridnetwork/devices",
    "properties" : {
      "status" : "Registered",
      "provisioningState" : "Succeeded",
      "deviceType" : "AzureStackEdge",
      "azureStackEdge" : {
        "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.DataBoxEdge/DataBoxEdgeDevices/Stack
Edge-Private-LTE09"    },
        "networkFunctions" : [ {
          "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/mrg-application-ziti-private-
edge-
20210708224207/providers/Microsoft.HybridNetwork/networkFunc
tions/NFASEPLTE9"    }, {
          "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_

```

```

ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
9_10_1907_HA_deploy1-HSS-node0"    }, {
    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
9_10_1907_HA_deploy1-SAEGW-node0"    }, {
    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte
9_10_1907_HA_deploy1-MME-node0"    } ]
  }
} ]

```

GET /rest/networks/globalhssresources

Get all the Global HSS Resources

Response Codes

Code	Condition
200	SUCCESS: All the Global HSS Resources fetched successfully.
500	FAILURE: Error while fetching the Global HSS Resources.

Response Body

Media Type	Data Type	Description
application/json	Resourcedetails (JSON)	List of Global HSS Resources.

Example

Request

```

GET /rest/networks/globalhssresources
Content-Type: application/json
Accept: application/json

```

...

Response

```

HTTP/1.1 200 OK
Content-Type: application/json

```

```
[ {
```

```

"version" : 0,
"resourceType" : {
  "version" : 1,
  "resourcetypeid" : 17,
  "type" : "HPEHSSEMS",
  "description" : "HPE-HSS-EMS",
  "iconname" : "HPEHSSEMS.png",
  "resyncInterval" : 15,
  "isdefault" : true,
  "primaryKey" : "17" },
"resourceComponent" : null,
"resourceid" : 5,
"ipaddress" : "10.165.32.139",
"displayname" : "HSS-10.165.32.139",
"snmpCommunity" : null,
"resyncInterval" : 0,
"componentid" : null,
"resourceVersion" : null,
"preProvisioned" : false,
"resourceGroupID" : 1,
"resourceStatusID" : 1,
"subnetMask" : "",
"resourceTypeID" : 17,
"resourceName" : "HSS-10.165.32.139",
"identifier" : 5
} ]

```

POST /rest/networks/globalhssresources

Add a Global HSS

Request Body

Media Type	Data Type	Description
application/json	GlobalNF (JSON)	Global HSS Object

Response Codes

Code	Condition
200	SUCCESS: Global HSS added successfully.
500	FAILURE: Error while adding Global HSS.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Global HSS ADD status.

*Example***Request**

```
POST /rest/networks/globalhssresources
Content-Type: application/json
Accept: application/json
```

```
{
  "resourceName" : "HSS-10.165.31.46",
  "resourceIpAddress" : "10.165.31.46"}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 8,
  "objectName" : "HSS-10.165.31.46",
  "successMessage" : "Global HSS added successfully"}
```

GET /rest/networks/globalpgwresources

Get all the Global PGW Resources

Response Codes

Code	Condition
200	SUCCESS: All the Global PGW Resources fetched successfully.
500	FAILURE: Error while fetching the Global PGW Resources.

Response Body

Media Type	Data Type	Description
application/json	Resourcedetails (JSON)	List of Global PGW Resources.

*Example***Request**

```
GET /rest/networks/globalpgwresources
Content-Type: application/json
Accept: application/json
```

...

Response

HTTP/1.1 200 OK
Content-Type: application/json

```
[ {
  "version" : 1,
  "resourceType" : {
    "version" : 1,
    "resourcetypeid" : 6,
    "type" : "AN3000",
    "description" : "MCC",
    "iconname" : "affirmed.png",
    "resyncInterval" : 2,
    "isdefault" : true,
    "primaryKey" : "6"  },
  "resourceComponent" : null,
  "resourceid" : 6,
  "ipaddress" : "10.165.32.133",
  "displayName" : "SAEGW-10.165.32.133",
  "snmpCommunity" : null,
  "resyncInterval" : 0,
  "componentid" : null,
  "resourceVersion" : null,
  "preProvisioned" : false,
  "resourceGroupID" : 1,
  "resourceStatusID" : 2,
  "subnetMask" : "",
  "resourceTypeID" : 6,
  "resourceName" : "SAEGW-10.165.32.133",
  "identifier" : 6
} ]
```

POST /rest/networks/globalpgwresources

Add a Global Gateway

Request Body

Media Type	Data Type	Description
application/json	GlobalNF (JSON)	Global Gateway Object

Response Codes

Code	Condition
200	SUCCESS: Global PGW added successfully.
500	FAILURE: Error while adding Global Gateway.

Response Body

media type	data type	description
application/json	object (JSON)	Global Gateway ADD status.

Example

Request

```
POST /rest/networks/globalpgwresources
Content-Type: application/json
Accept: application/json
```

```
{
  "resourceName" : "SAEGW-10.165.31.43",
  "resourceIpAddress" : "10.165.31.43"}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 9,
  "objectName" : "SAEGW-10.165.31.43",
  "successMessage" : "Global PGW added successfully"}
```

GET /rest/networks/packetcoreversions

Get all the available Packet Core Version.

Response Codes

Code	Condition
200	SUCCESS: All the Packet Core Versions have been fetched
500	FAILURE: Error while fetching Packet Core Version.

Response Body

Media Type	Data Type	Description
application/json	ServiceManagerSKU (JSON)	List of available Packet Core Version.

Example

Request

```
GET /rest/networks/packetcoreversions
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "version" : 1,
  "servicemanagerskuid" : 1,
  "name" : "APNS 1.0",
  "type" : "packetcore",
  "skuVersion" : "1.0",
  "active" : false
}, {
  "version" : 1,
  "servicemanagerskuid" : 2,
  "name" : "APNS 1.1",
  "type" : "packetcore",
  "skuVersion" : "1.1",
  "active" : true
} ]
```

GET /rest/networks/regions

Get all the Regions.

Response Codes

Code	Condition
200	SUCCESS: All the Regions have been fetched.
500	FAILURE: Error while fetching Regions.

Response Body

media type	data type	description
application/json	Region (JSON)	List of Regions. Empty list will be returned if no Regions are configured.

Example

Request

```
GET /rest/networks/regions
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "version" : 1,
  "regionid" : 1,
  "regionName" : "westeurope"}, {
  "version" : 1,
  "regionid" : 2,
  "regionName" : "eastus"} ]
```

GET /rest/networks/resources

Get all available Resources.

Response Codes

Code	Condition
200	SUCCESS: All the Resources have been fetched.
500	FAILURE: Error while fetching Resources.

Response Body

Media Type	Data Type	Description
application/json	Resourcedetails (JSON)	List of Resources. Empty list will be returned if no Resources are found.

Example

Request

```
GET /rest/networks/resources
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "version" : 1,
  "resourceType" : {
    "version" : 1,
    "resourcetypeid" : 17,
    "type" : "HPEHSSEMS",
    "description" : "HPE-HSS-EMS",
    "iconname" : "HPEHSSEMS.png",
    "resyncInterval" : 15,
    "isdefault" : true,
    "primaryKey" : "17"  },
  "resourceComponent" : null,
  "resourceid" : 3,
  "ipaddress" : "10.165.31.14",
  "displayName" : "HSS_10.165.31.14-[node0]",
  "snmpCommunity" : null,
  "resyncInterval" : 0,
  "componentid" : null,
```

```

    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 17,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : false,
    "resourceName" : "HSS_10.165.31.14-[node0]",
    "identifier" : 3
  }, {
    "version" : 1,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 17,
      "type" : "HPEHSSEMS",
      "description" : "HPE-HSS-EMS",
      "iconname" : "HPEHSSEMS.png",
      "resyncInterval" : 15,
      "isdefault" : true,
      "primaryKey" : "17" },
    "resourceComponent" : null,
    "resourceid" : 5,
    "ipaddress" : "10.165.31.29",
    "displayName" : "HSS_10.165.31.29-[node1]",
    "snmpCommunity" : null,
    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 17,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : false,
    "resourceName" : "HSS_10.165.31.29-[node1]",
    "identifier" : 5
  }, {
    "version" : 2,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 10,
      "type" : "MMEHvEPC",
      "description" : "MME",
      "iconname" : "MME.png",
      "resyncInterval" : 2,
      "isdefault" : true,
      "primaryKey" : "10" },
    "resourceComponent" : null,
    "resourceid" : 7,
    "ipaddress" : "10.165.31.24",
    "displayName" : "MME_10.165.31.24-[node1]",
    "snmpCommunity" : null,

```

```

    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 10,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : true,
    "resourceName" : "MME_10.165.31.24-[node1]",
    "identifier" : 7
  }, {
    "version" : 2,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 10,
      "type" : "MMEHvEPC",
      "description" : "MME",
      "iconname" : "MME.png",
      "resyncInterval" : 2,
      "isdefault" : true,
      "primaryKey" : "10" },
    "resourceComponent" : null,
    "resourceid" : 6,
    "ipaddress" : "10.165.31.8",
    "displayName" : "MME_10.165.31.8-[node0]",
    "snmpCommunity" : null,
    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 10,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : true,
    "resourceName" : "MME_10.165.31.8-[node0]",
    "identifier" : 6
  }, {
    "version" : 1,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 6,
      "type" : "AN3000",
      "description" : "MCC",
      "iconname" : "affirmed.png",
      "resyncInterval" : 2,
      "isdefault" : true,
      "primaryKey" : "6" },
    "resourceComponent" : null,
    "resourceid" : 2,
    "ipaddress" : "10.165.31.11",

```

```

    "displayName" : "SAEGW_10.165.31.11-[node0]",
    "snmpCommunity" : null,
    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 6,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : false,
    "resourceName" : "SAEGW_10.165.31.11-[node0]",
    "identifier" : 2
  }, {
    "version" : 1,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 6,
      "type" : "AN3000",
      "description" : "MCC",
      "iconname" : "affirmed.png",
      "resyncInterval" : 2,
      "isdefault" : true,
      "primaryKey" : "6" },
    "resourceComponent" : null,
    "resourceid" : 4,
    "ipaddress" : "10.165.31.26",
    "displayName" : "SAEGW_10.165.31.26-[node1]",
    "snmpCommunity" : null,
    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 6,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : false,
    "resourceName" : "SAEGW_10.165.31.26-[node1]",
    "identifier" : 4
  } ]

```

DELETE /rest/networks/{networkID}

Delete a Network by Network ID

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	required

Response Codes

Code	Condition
200	SUCCESS: Network delete triggered successfully for the given Network ID.
500	FAILURE: Error while triggering Network delete for the given Network ID.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Network DELETE status.

Example

Request

```
DELETE /rest/networks/2
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 2,
  "objectName" : null,
  "successMessage" : "Network delete triggered
successfully"}
```

GET /rest/networks/{networkID}

Get a Network by Network ID.

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	long

Response Codes

Code	Condition
200	SUCCESS: Network for the given Network ID has been fetched.
500	FAILURE: Error while fetching the Network for the given Network ID.

Response Body

media type	data type	description
application/json	ServiceInstanceConfig (JSON)	Network Object.

Example

Request

```
GET /rest/networks/2
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "serviceInstanceId" : 1,
  "parentResourceGroupId" : 1,
  "nfsProvisioningState" : "Succeeded",
  "provisioningState" : "Provisioned",
  "deploymentMode" : "ONPREMHA",
  "locations" : [ {
    "name" : "Plte12-13",
    "displayName" : null,
    "mgmtsubnet" : "10.165.31.0/27",
    "mgmtgw" : null,
    "lansubnet" : "10.165.68.0/25",
    "langw" : null,
    "wansubnet" : "10.165.68.128/25",
    "wangw" : null,
    "locationNetworkConfig" : null,
    "asedeviceContainers" : [ {
      "regionName" : null,
```

```

    "regionId" : 0,
    "node0" : {
      "mme" : {
        "networkFunction" : {
          "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
2_13_2506_HA_deploy1-MME-node0",
          "name" : "pltel2_13_2506_HA_deploy1-MME-node0",
          "type" :
"microsoft.hybridnetwork/networkfunctions",
          "location" : "eastus",
          "properties" : {
            "provisioningState" : "Succeeded",
            "device" : {
              "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-
dddd-eeeeeeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-12"
            },
            "skuName" : "apns_mme_1_0",
            "skuType" : "EvolvedPacketCore",
            "vendorName" : "AffirmedNetworks",
            "serviceKey" : "aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeeeeeee",
            "vendorProvisioningState" : "Provisioned",
            "managedApplication" : null,
            "managedApplicationParameters" : null,
            "networkFunctionUserConfigurations" : [ {
              "roleName" : "mrm-0",
              "userDataParameters" : null,
              "networkInterfaces" : [ {
                "networkInterfaceName" : "mrm-0-mgmt",
                "macAddress" : null,
                "ipConfigurations" : [ {
                  "ipAllocationMethod" : "Static",
                  "ipAddress" : "10.165.31.8",
                  "subnet" : "10.165.31.0/27",
                  "gateway" : "10.165.31.1",
                  "ipVersion" : "IPv4",
                  "dnsServers" : null
                } ] ,
                "vmSwitchType" : "Management"
              }, {
                "networkInterfaceName" : "mrm-0-base",
                "macAddress" : null,
                "ipConfigurations" : [ {
                  "ipAllocationMethod" : "Static",
                  "ipAddress" : "10.165.68.34",
                  "subnet" : "10.165.68.0/25",
                  "gateway" : "10.165.68.1",

```

[illegible]

[illegible]

TqkFhdUQ4cHREVDJreWttZ3pQS3pxRXRZcFRVS0hFOUN3dlBGAlFHV1YzNTU
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 BcVhcbm9GbzRiS0FIUFJZTzNPWG5KVC9RTFRFQ1ZkTDJsNjv3T2JxSmhwOGV
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 5THkwaUJhUnN2SjNhB0xaY3JhajkrndVbHhFRG1pNCtWdVJDRkc4dzhVQkl
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 3dVA5d1VyZ29FVknZU1JZ05xc21aTnovQkdQd2VwcTRTU2xZd2oyYVQvS1J
 EOuHIn1JWae93UUIzMTFDazNTEhZ0Z0QvTTZIOENTZFdmZVBkSGVISWZudU1
 ReVpoTGpvZ21DZVZiT0ZFWS9Iei9lUENTTXN5bk5wdEpLZUZIY1I2Y2tqeKc
 xM1NiWUtscER2aDFyr3YrSXR5dk5Od0dRWn1RWGxxZU92bkROWjJEVXBTS0w
 yZnhDb0g4bGVCOE9KM1LEYTBWAdGzakxhOU11RmN5aEJYRDZjTHNKeUkrcnZ
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 ySFpueEtVrkJLeTRTOWhVb0xacmhWd0dXRExTS2pYSEtGa01DMUG5V2NPRSt
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zbU9pczNyUlXubWYrTjZQRGxxYzVwOTZFZkhSM1NrR211cnB5eThyMHNIOwt
RYWFhajRScTdMWF1OVkxGbXZaUkVZTTZaT2pzb1xubVNCQWF1RDhwdERUMmt
5a21nelBLenFFdFlwVFLSEU5Q3d2UEZqUUDXVjM1NStUNSSyaGE2Ym1xcUg
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zKzQ5dWtrOEVMMXpLajNGN2lWwW5KTTdTZ1xuaW1sTl1MTFUyQU2WHI4RFU
xRTB2OT1zTVZTT2Z0Y21BU2UxbHdvMW9ubV1DeXR0eGniU09IVVpURHRSdEN
vM1xuOEZpNVpSMW50SmFIa1FMZjNkEMxQjZZam90Z1BBMUxZM2tGY2FHUF1
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XUDJJBWtiSCtnL2c1WCsxWGtCb1NGVDFueGZ5eXByenB4eEhRbUgxdkViZVx
uMGtNMjJHNT1SS0NjS3F0ZmRTd3dRMk00WDZ1ZFRhVm9lOWpVae1rR1NpNFd
BYTdmNnFzbe91SHE1OFRvZTNoSVxuQnJZQngxanBQTlJWMLFHcGFKcXVMEg5
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FTFVKRUdLb05zNzRib05MznZBN0pnV0pBVfh1ZDRldVVS10dOOULWYVMwUXJ
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FWS0tLS0tIiwKICAgICAgICAgICJQdWJsaWNlZXkxIiA6ICJzc2gtcnNhIEF
BQUFCM056YUMxeWMyRUFBUFEQVFBQkFBQUJBUUNsNnd1UD13VXJnb0VWQ11
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xMUNrM1N4dnRnRC9NNkg4Q1NkV2Z1UGRIZUhJZm51SVF5WmhMam9nbUN1VmJ
PRkVZL0h6L2VQQ21Nc3luTnB0Sk1RkhiUjZja2p6RzEzU2JZS2xwRHZOMXJ
HditJdH12Tk53R1FaeVFYbHF1T3ZuRE5aMkRVcFNLTDJmeENvSDhsZUI4T0o
yWURhMFZ0ODNqTGE5QjVGY3loQ1hENmNmC0p5SStydnFqTmVjdTdiaXJGRnl
QODdYCHrIR1JNd0gyTTQrZ0NoL20xYkoxbjVjRf1EdTJIWm54S1VGQkt5NFM
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EejdDN3VaOfhqaiBwbnNtQHBuc20iLAogICAgICAgICAgI1B1YmXpY0tleTI
iIDogI1BVQktFWTJfYWRtaW4iLAogICAgICAgICAgI1Bhc3N3b3JkIiA6ICI
kNiRxVXpS3Z5T3h4ZDQ3YmokSUwzV0JXQXZTZWJUZGZaZkNNcHVMemExWXV
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gICAiVXNlciIgOiAibXRjIiwKICAgICAgICAgICJQcm12YXR1S2V5IiA6ICI
tLS0tLUJFR0l0IEVDIFBSSVZBVEUgS0VZLS0tLS1cbk1IY0NBUUVFSUQrSzB
jTW9kaTZIdUVsbGp0eldudUZ2d3pJLzA1cEJHMMNoTVdjYy94RkVvQW9HQ0N
xR1NNND1cbkF3RUhVvVFEUWdBRUNwdDJ3SG53VVVVNTVOcjd0bG5OV1VzUHV
TaDB0KzJhenFGbmcxcFV0Zk14SHJ5WkRhdlZcbmVmc3pMcWtiS2o1Tm13eUR
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xMUNrM1N4dnRnRC9NNnkq4Q1NkVZ2lUGRIzUHJZm51SVF5WmhMam9nbUnbUN1VmJ
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    },
    "day1Conf" : {
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        "shortNetworkName" : "xyz",
        "apnName" : "plte",
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        "pgwIpAddress" : "10.165.68.112",
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        "giLoopbackIp" : "10.165.68.176",
        "giInterfaceIp" : "10.165.68.137",
        "giInterfaceIpRouteDest" : "0.0.0.0",
        "giInterfaceIpRoutePrefix" : "0",
        "giInterfaceIpRouteNextHop" : "10.165.68.129",
        "giIpSubPoolRangeStart" : "200.120.0.1",
        "giIpSubPoolRangeEnd" : "200.120.0.150",
        "gnSgwS1" : "10.165.68.113",
        "gnSgwLoopbackIp" : "10.165.68.115",
        "gnInterfaceIp" : "10.165.68.10",
        "gnInterfaceIpRouteDest" : "0.0.0.0",

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"gnInterfaceIpRoutePrefix" : "0",
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"scRoamingPlmnId" : "12345",
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"bgpRouterGN" : {
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  "localAs" : "65321",
  "remoteAs" : "64512",
  "neighborIP" : "10.165.68.1",
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  "bfdMultiHopPeer" : "10.165.68.1",
  "bgpGEOLoopIP" : "10.165.68.116",
  "bgpGNLoopIP" : "10.165.68.117" },
"bgpRouterGI" : {
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  "localAs" : "65321",
  "remoteAs" : "64512",
  "neighborIP" : "10.165.68.129",
  "fallOverBFD" : "single-hop",
  "bfdMultiHopPeer" : "10.165.68.129",
  "bgpGILoopIP" : "10.165.68.178" },
"giV4IpSubPoolRangeStart" : "10.202.0.1",
"giV4IpSubPoolRangeEnd" : "10.202.0.255",
"giRemoteIpSubPoolRangeStart" : "10.201.0.1",
"giRemoteIpSubPoolRangeEnd" : "10.201.0.255",
"dataLoopIP" : "10.165.68.176",
"dataFabricIpAddress" : "10.165.68.61"
},

"natProfile" : {
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"edrProfile" : null,
"mmcStatus" : {
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  "day0message" : "Done",
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  "day1GNStatus" : null,
  "day1GNmessage" : null,
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  "day1ServiceStatus" : null,
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```

```

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        "serviceKey" : "aaaaaaaa-bbbb-cccc-dddd-
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            "ipConfigurations" : [ {
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},
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    "gateway" : "10.165.68.1",
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    "dnsServers" : null
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  "vmSwitchType" : "Lan" } ],
  "osProfile" : {
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    "mrmData" : "10.165.68.44",
    "msmData" : "10.165.68.45",
    "dataNetworkAddress" : "10.165.68.0/25",
    "dataGatewayAddress" : "10.165.68.1",
    "ns1" : "10.165.68.17",
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    "ns1GatewayAddress" : "10.165.68.1",
    "cloudMMEManagementNetMask" : "255.255.255.224",
    "cloudMMEBaseNetMask" : "255.255.255.248",
    "cloudMMEDataNetMask" : "255.255.255.248",
    "dataTunnelingSupernode0" : "10.165.68.42",
    "dataTunnelingSupernode1" : "10.165.68.44",
    "baseTunnelingSupernode0" : "10.165.68.34",
    "baseTunnelingSupernode1" : "10.165.68.36",
    "s6IP" : "10.165.68.49"      },
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    "mmeStatus" : {
        "day0Status" : "Success",
        "day0message" : "Done",
        "day1Status" : null,
        "day1message" : null
    }
},
    "mcc" : {
        "networkFunction" : {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/plte1
2_13_2506_HA_deploy1-SAEGW-node1",
            "name" : "plte12_13_2506_HA_deploy1-SAEGW-
node1",
            "type" :
"microsoft.hybridnetwork/networkfunctions",
            "location" : "eastus",
            "properties" : {
                "provisioningState" : "Succeeded",
                "device" : {
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dddd-eeeeeeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-13"
                }
            },
            "skuName" : "apns_saegw_1_0",
            "skuType" : "EvolvedPacketCore",
            "vendorName" : "AffirmedNetworks",

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        "serviceKey" : "aaaaaaa-bbbb-cccc-dddd-
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        "vendorProvisioningState" : "Provisioned",
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            "userDataParameters" : null,
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                "macAddress" : null,
                "ipConfigurations" : [ {
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                    "ipAddress" : "10.165.31.26",
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                    "gateway" : "10.165.31.1",
                    "ipVersion" : "IPv4",
                    "dnsServers" : null
                }, {
                    "ipAllocationMethod" : "Static",
                    "ipAddress" : "10.165.31.27",
                    "subnet" : "10.165.31.0/27",
                    "gateway" : "10.165.31.1",
                    "ipVersion" : "IPv4",
                    "dnsServers" : null
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                    "ipAddress" : "10.165.31.28",
                    "subnet" : "10.165.31.0/27",
                    "gateway" : "10.165.31.1",
                    "ipVersion" : "IPv4",
                    "dnsServers" : null
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            ],
            "vmSwitchType" : "Management"
        }, {
            "networkInterfaceName" : "mcc-0-base",
            "macAddress" : null,
            "ipConfigurations" : [ {
                "ipAllocationMethod" : "Static",
                "ipAddress" : "10.165.68.90",
                "subnet" : "10.165.68.0/25",
                "gateway" : "10.165.68.1",
                "ipVersion" : "IPv4",
                "dnsServers" : null
            }, {
                "ipAllocationMethod" : "Static",
                "ipAddress" : "10.165.68.91",
                "subnet" : "10.165.68.0/25",
                "gateway" : "10.165.68.1",
            }
        ]
    ]
}

```

```

        "ipVersion" : "IPv4",
        "dnsServers" : null
    }, {
        "ipAllocationMethod" : "Static",
        "ipAddress" : "10.165.68.92",
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        "gateway" : "10.165.68.1",
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    "vmSwitchType" : "Lan" }, {
    "networkInterfaceName" : "mcc-0-ew",
    "macAddress" : null,
    "ipConfigurations" : [ {
        "ipAllocationMethod" : "Static",
        "ipAddress" : "10.165.68.98",
        "subnet" : "10.165.68.0/25",
        "gateway" : "10.165.68.1",
        "ipVersion" : "IPv4",
        "dnsServers" : null
    } ],
    "vmSwitchType" : "Lan" }, {
    "networkInterfaceName" : "mcc-0-ns1",
    "macAddress" : null,
    "ipConfigurations" : [ {
        "ipAllocationMethod" : "Static",
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        "subnet" : "10.165.68.0/25",
        "gateway" : "10.165.68.1",
        "ipVersion" : "IPv4",
        "dnsServers" : null
    } ],
    "vmSwitchType" : "Lan" }, {
    "networkInterfaceName" : "mcc-0-ns2",
    "macAddress" : null,
    "ipConfigurations" : [ {
        "ipAllocationMethod" : "Static",
        "ipAddress" : "10.165.68.145",
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        "gateway" : "10.165.68.129",
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        "dnsServers" : null
    } ],
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    "osProfile" : {
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```

[illegible]

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```

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    }
  }
}
},
"provisioningState" : null,
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node1",
"smName" : "SAEGW_10.165.31.26-[node1]",
"asapName" : "SAEGW-10.165.31.26",
"day0Conf" : {
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  "lanCIDRSuffix" : "/25",
  "wanCIDRSuffix" : "/25",
  "management0" : "10.165.31.26",
  "management1" : "10.165.31.27",
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  "managementGatewayAddress" : "10.165.31.1",
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  "base1" : "10.165.68.91",
  "base2" : "10.165.68.92",
  "baseNetworkAddress" : "10.165.68.0/25",
  "baseGatewayAddress" : "10.165.68.1",

```

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    "ew" : "10.165.68.98",
    "ewNetworkAddress" : "10.165.68.0/25",
    "ewGatewayAddress" : "10.165.68.1",
    "ns1" : "10.165.68.18",
    "ns1NetworkAddress" : "10.165.68.0/25",
    "ns1GatewayAddress" : "10.165.68.1",
    "ns2" : "10.165.68.145",
    "ns2NetworkAddress" : "10.165.68.128/25",
    "ns2GatewayAddress" : "10.165.68.129",
    "cloudBaseInternalCIDR" : "10.165.68.90/29"
  },
  "day1Conf" : {
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    "homePlmnId" : "99999",
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    "sgwS11IpAddress" : "10.165.68.114",
    "giLoopbackIp" : "10.165.68.176",
    "giInterfaceIp" : "10.165.68.145",
    "giInterfaceIpRouteDest" : "0.0.0.0",
    "giInterfaceIpRoutePrefix" : "0",
    "giInterfaceIpRouteNextHop" : "10.165.68.129",
    "giIpSubPoolRangeStart" : "200.130.0.1",
    "giIpSubPoolRangeEnd" : "200.130.0.150",
    "gnSgwS1" : "10.165.68.113",
    "gnSgwLoopbackIp" : "10.165.68.115",
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      "localAs" : "65321",
      "remoteAs" : "64512",
      "neighborIP" : "10.165.68.1",
      "fallOverBFD" : "multi-hop",
      "bfdMultiHopPeer" : "10.165.68.1",
      "bgpGEOLoopIP" : "10.165.68.118",
      "bgpGNLoopIP" : "10.165.68.119"
    },
    "bgpRouterGI" : {
      "routerId" : "10.165.68.145",
      "localAs" : "65321",
      "remoteAs" : "64512",
      "neighborIP" : "10.165.68.129",
      "fallOverBFD" : "single-hop",
      "bfdMultiHopPeer" : "10.165.68.129",
      "bgpGILoopIP" : "10.165.68.179"
    }
  }
}

```

```

    "giV4IpSubPoolRangeStart" : "10.202.0.1",
    "giV4IpSubPoolRangeEnd" : "10.202.0.255",
    "giRemoteIpSubPoolRangeStart" : "10.201.0.1",
    "giRemoteIpSubPoolRangeEnd" : "10.201.0.255",
    "dataLoopIP" : "10.165.68.176",
    "dataFabricIpAddress" : "10.165.68.93",
    "dnsPrimaryIp" : "200.5.0.30",
    "dnsSecondaryIp" : "200.5.0.20"           },
    "natProfile" : {
        "configStatus" : "PROVISIONED",
        "status" : "Success",
        "message" : "Configured",
        "natSubPoolRangeStart" : "10.200.0.1",
        "natSubPoolRangeEnd" : "10.200.0.255",
        "profileName" : "NAT"           },
    "edrProfile" : null,
    "mccStatus" : {
        "day0Status" : "Success",
        "day0message" : "Done",
        "day1Status" : "Success",
        "day1Message" : "Configured",
        "day1GIStatus" : null,
        "day1GImessage" : null,
        "day1GNStatus" : null,
        "day1GNmessage" : null,
        "day1SCStatus" : null,
        "day1SCmessage" : null,
        "day1ServiceStatus" : null,
        "day1Servicemessage" : null,
        "dataFabricConfigured" : true
    }
},
    "hss" : {
        "networkFunction" : {
            "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-dddd-
eeeeeeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/networkFunctions/pltel
2_13_2506_HA_deploy1-HSS-node1",
            "name" : "pltel2_13_2506_HA_deploy1-HSS-node1",
            "type" :
"microsoft.hybridnetwork/networkfunctions",
            "location" : "eastus",
            "properties" : {
                "provisioningState" : "Succeeded",
                "device" : {
                    "id" : "/subscriptions/aaaaaaaa-bbbb-cccc-
dddd-eeeeeeeeeeeeeeee/resourceGroups/Private_LTE_
ASE/providers/Microsoft.HybridNetwork/devices/ase-plte-13"
                },
                "skuName" : "apns_hss_1_0",

```

```
"skuType" : "EvolvedPacketCore",  
  "vendorName" : "AffirmedNetworks",  
  "serviceKey" : "aaaaaaaa-bbbb-cccc-dddd-  
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  "managedApplicationParameters" : null,  
  "networkFunctionUserConfigurations" : [ {  
    "roleName" : "hpehss",  
    "userDataParameters" : null,  
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"mrmmmanagementnic1",  
      "macAddress" : null,  
      "ipConfigurations" : [ {  
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        "subnet" : "10.165.31.0/27",  
        "gateway" : "10.165.31.1",  
        "ipVersion" : "IPv4",  
        "dnsServers" : null  
      } ],  
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    }, {  
  
      "networkInterfaceName" : "mrmlannic1",  
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        "subnet" : "10.165.68.0/25",  
        "gateway" : "10.165.68.1",  
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        "dnsServers" : null  
      } ],  
      "vmSwitchType" : "Lan"  
    } ],  
    "osProfile" : {  
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"I2Nsb3VklWNvbmZpZWp3cm10ZV9maWxlczokLSBwYXRoOiAvdmFyL2xpYi9jbG9lZC9paHNzYT29uZmllnLmpzb24KICBwZXJtaXNzaW9uczogJzA2NDQnCiAgb3duZXI6IHJvb3Q6cm9vdAoogIGNVbnRlbnQ6IHwKICAgIAHsKICAgICAgICAggICAiRGhhbWV0ZXJHVyI6ewogICAgICAgICAgICAgICAgICAISe9TVElQQURkUkVTUyI6IjEwLjE2NS42OC4xOSIsCiAgICAgICAgICAgICAgICAgICAgICJCJUUROIjoicGx0ZTIyaHNzMmEmZmlybWVkbWVkbSIscCIAGICAgICAgICAgICAgICAgICAgICJSRUFTSII6Imhzcy5lcGMubW5jMDk5Lm1jYzk5OS4zZ3BwbmV0d29yay5vcnciCiAgICAgICAgICAgfSwKICAgICAgICAgICAIeRedXQmluZEfkZHIIonsKICAgICAgICAgICAgICAgICAgIkFEREFJFU1MiOIlxMC4xNjUUNjguMTkiLAogICAgICAgICAgICAgICAgICAIvFBTlNQTFJUIjoiuU0NUUCIsCiAgICAgICAgICAgICAgICAgICJCQTlJUIjozuODY4CiAgICAgICAgICAgfSwKICAgICAgICAgICAIU05NUFRhcmdldCI6ewogICAgICAgICAgICAgICAgICAISe9TVCI6Ije
```

[illegible]

```

    "regionId" : 2,
    "regionName" : "eastus",
    "resourceGroupName" : "Private_LTE_ASE",
    "mmePrivateKey" : "-----BEGIN RSA PRIVATE KEY-----\n...\n-
-----END RSA PRIVATE KEY-----",
    "skuName" : "APNS 1.0",
    "serviceManagerSkuId" : 1,
    "createdTime" : 1626698701851,
    "useSystemGeneratedIPs" : false,
    "mccPart" : "999",
    "mncPart" : "099",
    "ntpIP" : "10.168.0.10",
    "optionalNtpIP1" : "10.168.0.11",
    "optionalNtpIP2" : "10.168.0.12",
    "timeZone" : "Asia/Kolkata",
    "resourcesAvailableForDay1" : true
  }

```

GET /rest/networks/packetcoreversions/active

Get all the Active Packet Core Version.

Response Codes

Code	Condition
200	SUCCESS: All the Active Packet Core Versions have been fetched
500	FAILURE: Error while fetching Active Packet Core Version.

Response Body

Media Type	Data Type	Description
application/json	ServiceManagerSKU (JSON)	List of Active Packet Core Version.

Example

Request

```

GET /rest/networks/packetcoreversions/active
Content-Type: application/json
Accept: application/json

```

...

Response

HTTP/1.1 200 OK
 Content-Type: application/json

```
[ {
  "version" : 1,
  "servicemanagerskuid" : 2,
  "name" : "APNS 1.1",
  "type" : "packetcore",
  "skuVersion" : "1.1",
  "active" : true
} ]
```

DELETE /rest/networks/resources/{resourceID}

Delete a Resource by Resource ID

Request Parameters

Name	Type	Description	Constraints
resourceID	path	Resource ID	required

Response Codes

Code	Condition
200	SUCCESS: Resource delete triggered successfully for the given Resource ID.
500	FAILURE: Error while triggering Resource delete for the given Resource ID.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Resource DELETE status.

Example

Request

```
DELETE /rest/networks/resources/5
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 5,
  "objectName" : null,
  "successMessage" : "Resource deleted successfully"}
```

GET /rest/networks/resources/{resourceID}

Get a Resource by Resource ID.

Request Parameters

Name	Type	Description	Constraints
resourceID	path	Resource ID	long

Response Codes

Code	Condition
200	SUCCESS: Resource for the given Resource ID has been fetched.
500	FAILURE: Error while fetching the Resource for the given Resource ID.

Response Body

Media Type	Data Type	Description
application/json	Resourcedetails (JSON)	Resource Object.

Example

Request

```
GET /rest/networks/resources/3
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```



```

{
  "version" : 1,
  "resourceType" : {
    "version" : 1,
    "resourcetypeid" : 17,
    "type" : "HPEHSSEMS",
    "description" : "HPE-HSS-EMS",
    "iconname" : "HPEHSSEMS.png",
    "resyncInterval" : 15,
    "isdefault" : true,
    "primaryKey" : "17"  },
  "resourceComponent" : null,
  "resourceid" : 3,
  "ipaddress" : "10.165.31.14",
  "displayName" : "HSS_10.165.31.14-[node0]",
  "snmpCommunity" : null,
  "resyncInterval" : 0,
  "componentid" : null,
  "resourceVersion" : null,
  "resourceGroupID" : 2,
  "resourceStatusID" : 1,
  "resourceTypeID" : 17,
  "subnetMask" : "255.0.0.0",
  "preProvisioned" : false,
  "resourceName" : "HSS_10.165.31.14-[node0]",
  "identifier" : 3
}

```

POST /rest/networks/{networkID}/day1

Set Day1 configuration to the Network.

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	long

Request Body

Media Type	Data Type	Description
application/json	ServiceInstanceConfig (JSON)	Network Object with Day1 configurations

Response Codes

Code	Condition
200	SUCCESS: Service configuration updated successfully to the Network.
500	FAILURE: Error while configuring service to the Network.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Day1 Configuration status.

Example

Request

```
POST /rest/networks/2/day1
Content-Type: application/json
Accept: application/json
```

```
{
  "deploymentMode" : "ONPREMHA",
  "locations" : [ {
    "name" : "Plte12-13",
    "asedeviceContainers" : [ {
      "node0" : {
        "mme" : {
          "day1Conf" : {
            "networkName" : "Telecom-xyz",
            "shortNetworkName" : "xyz",
            "apnName" : "plte",
            "homePlmnId" : "99999",
            "pgwIpAddress" : "10.165.68.112",
            "sgwS11IpAddress" : "10.165.68.114",
            "mmeRealm" :
              "epc.mnc099.mcc999.3gppnetwork.org",
            "diameterPeer" : "plte21hss.affirmed.com",
            "diameterPeerIpAddress" : "10.165.68.11",
            "realmPeerMap" :
              "hss.epc.mnc099.mcc999.3gppnetwork.org",
            "diameterClientInstanceIpAddress" :
              "10.165.68.49",
            "gtpProfileIpAddress" : "10.165.68.50",
            "s1IfIpAddress" : "10.165.68.51",
            "domainName1" :
              "plte.apn.epc.mnc099.mcc999.3gppnetwork.org",
            "domainName2" : "tac-lb00.tac-
              hb01.tac.epc.mnc099.mcc999.3gppnetwork.org",
            "ntpServerIp" : "10.168.0.10",
```

```

    "externalNs1" : "10.165.68.9/27",
    "mmeVmName" : "plte21-msm0-0",
    "node1DiameterPeer" : "plte22hss.affirmed.com",
    "node1DiameterPeerIpAddress" : "10.165.68.19",
    "node1RealmPeerMap" :
    "hss.epc.mnc099.mcc999.3gppnetwork.org",
    "node1ExternalNs1" : "10.165.68.17/27",
    "node1MmeVmName" : null,
    "node1MrmVmName" : null,
    "nextHopGWIp" : "10.165.68.1",
    "routingInstancesBfd" : "enable",
    "tacLowByte" : "00",
    "tacHighByte" : "b9"
  },
  "mcc" : {
    "day1Conf" : {
      "networkName" : "Telecom-xyz",
      "shortNetworkName" : "xyz",
      "apnName" : "plte",
      "homePlmnId" : "99999",
      "pgwIpAddress" : "10.165.68.112",
      "sgwS11IpAddress" : "10.165.68.114",
      "giLoopbackIp" : "10.165.68.176",
      "giInterfaceIp" : "10.165.68.137",
      "giInterfaceIpRouteDest" : "0.0.0.0",
      "giInterfaceIpRoutePrefix" : "0",
      "giInterfaceIpRouteNextHop" : "10.165.68.129",
      "giIpSubPoolRangeStart" : "200.120.0.1",
      "giIpSubPoolRangeEnd" : "200.120.0.150",
      "gnSgwS1" : "10.165.68.113",
      "gnSgwLoopbackIp" : "10.165.68.115",
      "gnInterfaceIp" : "10.165.68.10",
      "gnInterfaceIpRouteDest" : "0.0.0.0",
      "gnInterfaceIpRoutePrefix" : "0",
      "gnInterfaceIpRouteNextHop" : "10.165.68.1",
      "scRoamingPlmnId" : "12345",
      "zoneDefaultQosProfile" : "QOSPROFDEFAULT",
      "bgpRouterGN" : {
        "routerId" : "10.165.68.10",
        "localAs" : "65321",
        "remoteAs" : "64512",
        "neighborIP" : "10.165.68.1",
        "fallOverBFD" : "multi-hop",
        "bfdMultiHopPeer" : "10.165.68.1",
        "bgpGEOLoopIP" : "10.165.68.116",
        "bgpGNLoopIP" : "10.165.68.117"
      },
      "bgpRouterGI" : {
        "routerId" : "10.165.68.137",
        "localAs" : "65321",
        "remoteAs" : "64512",

```

```

        "neighborIP" : "10.165.68.129",
        "fallOverBFD" : "single-hop",
        "bfdMultiHopPeer" : "10.165.68.129",
        "bgpGILoopIP" : "10.165.68.178"      },
    "giV4IpSubPoolRangeStart" : "10.202.0.1",
    "giV4IpSubPoolRangeEnd" : "10.202.0.255",
    "giRemoteIpSubPoolRangeStart" : "10.201.0.1",
    "giRemoteIpSubPoolRangeEnd" : "10.201.0.255",
    "dataLoopIP" : "10.165.68.176",
    "dataFabricIpAddress" : "10.165.68.61"
}

}

},
"node1" : {
    "mcc" : {
        "day1Conf" : {
            "networkName" : "Telecom-xyz",
            "shortNetworkName" : "xyz",
            "apnName" : "plte",
            "homePlmnId" : "99999",
            "pgwIpAddress" : "10.165.68.112",
            "sgwS11IpAddress" : "10.165.68.114",
            "giLoopbackIp" : "10.165.68.176",
            "giInterfaceIp" : "10.165.68.145",
            "giInterfaceIpRouteDest" : "0.0.0.0",
            "giInterfaceIpRoutePrefix" : "0",
            "giInterfaceIpRouteNextHop" : "10.165.68.129",
            "giIpSubPoolRangeStart" : "200.130.0.1",
            "giIpSubPoolRangeEnd" : "200.130.0.150",
            "gnSgwS1" : "10.165.68.113",
            "gnSgwLoopbackIp" : "10.165.68.115",
            "gnInterfaceIp" : "10.165.68.18",
            "gnInterfaceIpRouteDest" : "0.0.0.0",
            "gnInterfaceIpRoutePrefix" : "0",
            "gnInterfaceIpRouteNextHop" : "10.165.68.1",
            "scRoamingPlmnId" : "12345",
            "zoneDefaultQosProfile" : "QOSPROFDEFAULT",
            "bgpRouterGN" : {
                "routerId" : "10.165.68.18",
                "localAs" : "65321",
                "remoteAs" : "64512",
                "neighborIP" : "10.165.68.1",
                "fallOverBFD" : "multi-hop",
                "bfdMultiHopPeer" : "10.165.68.1",
                "bgpGEOLoopIP" : "10.165.68.118",
                "bgpGNLoopIP" : "10.165.68.119"      },
            "bgpRouterGI" : {
                "routerId" : "10.165.68.145",
                "localAs" : "65321",
                "remoteAs" : "64512",

```

```

        "neighborIP" : "10.165.68.129",
        "fallOverBFD" : "single-hop",
        "bfdMultiHopPeer" : "10.165.68.129",
        "bgpGILoopIP" : "10.165.68.179"      },
    "giV4IpSubPoolRangeStart" : "10.202.0.1",
    "giV4IpSubPoolRangeEnd" : "10.202.0.255",
    "giRemoteIpSubPoolRangeStart" : "10.201.0.1",
    "giRemoteIpSubPoolRangeEnd" : "10.201.0.255",
    "dataLoopIP" : "10.165.68.176",
    "dataFabricIpAddress" : "10.165.68.93",
    "dnsPrimaryIp" : "200.5.0.30",
    "dnsSecondaryIp" : "200.5.0.20"      }
    }
  } ]
} ]
}

```

Response

HTTP/1.1 200 OK
 Content-Type: application/json

```

{
  "objectID" : 2,
  "objectName" : "plte12_13_2506_HA_deploy1",
  "successMessage" : "Day1 configured successfully"}

```

PUT /rest/networks/{networkID}/day1

Edit Day1 configuration to the Network.

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	long

Request Body

Media Type	Data Type	Description
application/json	ServiceInstanceConfig (JSON)	Network Object with Day1 configurations

Response Codes

Code	Condition
200	SUCCESS: Service configuration updated successfully to the Network.
500	FAILURE: Error while configuring service to the Network.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Day1 Configuration status.

Example

Request

```

PUT /rest/networks/2/modifyServiceConfig
Content-Type: application/json
Accept: application/json

{
  "deploymentMode" : "ONPREMHA",
  "locations" : [ {
    "name" : "Plte12-13",
    "asedeviceContainers" : [ {
      "node0" : {
        "mme" : {
          "day1Conf" : {
            "networkName" : "Telecom-xyz",
            "shortNetworkName" : "xyz",
            "apnName" : "plte",
            "homePlmnId" : "99999",
            "pgwIpAddress" : "10.165.68.112",
            "sgwS11IpAddress" : "10.165.68.114",
            "mmeRealm" :
"epc.mnc099.mcc999.3gppnetwork.org",
            "diameterPeer" : "plte21hss.affirmed.com",
            "diameterPeerIpAddress" : "10.165.68.11",
            "realmPeerMap" :
"hss.epc.mnc099.mcc999.3gppnetwork.org",
            "diameterClientInstanceIpAddress" :
"10.165.68.49",
            "gtpProfileIpAddress" : "10.165.68.50",
            "s1IfIpAddress" : "10.165.68.51",
            "domainName1" :
"plte.apn.epc.mnc099.mcc999.3gppnetwork.org",
            "domainName2" : "tac-lb00.tac-
hb01.tac.epc.mnc099.mcc999.3gppnetwork.org",
            "ntpServerIp" : "10.168.0.10",

```

```

        "externalNs1" : "10.165.68.9/27",
        "mmeVmName" : "plte21-msm0-0",
        "node1DiameterPeer" : "plte22hss.affirmed.com",
        "node1DiameterPeerIpAddress" : "10.165.68.19",
        "node1RealmPeerMap" :
    "hss.epc.mnc099.mcc999.3gppnetwork.org",
        "node1ExternalNs1" : "10.165.68.17/27",
        "node1MmeVmName" : null,
        "node1MrmVmName" : null,
        "nextHopGWIp" : "10.165.68.1",
        "routingInstancesBfd" : "enable",
        "tacLowByte" : "00",
        "tacHighByte" : "b9"
    },
    "mcc" : {
        "day1Conf" : {
            "networkName" : "Telecom-xyz",
            "shortNetworkName" : "xyz",
            "apnName" : "plte",
            "homePlmnId" : "99999",
            "pgwIpAddress" : "10.165.68.112",
            "sgwS11IpAddress" : "10.165.68.114",
            "giLoopbackIp" : "10.165.68.176",
            "giInterfaceIp" : "10.165.68.137",
            "giInterfaceIpRouteDest" : "0.0.0.0",
            "giInterfaceIpRoutePrefix" : "0",
            "giInterfaceIpRouteNextHop" : "10.165.68.129",
            "giIpSubPoolRangeStart" : "200.120.0.1",
            "giIpSubPoolRangeEnd" : "200.120.0.150",
            "gnSgwS1" : "10.165.68.113",
            "gnSgwLoopbackIp" : "10.165.68.115",
            "gnInterfaceIp" : "10.165.68.10",
            "gnInterfaceIpRouteDest" : "0.0.0.0",
            "gnInterfaceIpRoutePrefix" : "0",
            "gnInterfaceIpRouteNextHop" : "10.165.68.1",
            "scRoamingPlmnId" : "12345",
            "zoneDefaultQosProfile" : "QOSPROFDEFAULT",
            "bgpRouterGN" : {
                "routerId" : "10.165.68.10",
                "localAs" : "65321",
                "remoteAs" : "64512",
                "neighborIP" : "10.165.68.1",
                "fallOverBFD" : "multi-hop",
                "bfdMultiHopPeer" : "10.165.68.1",
                "bgpGEOLoopIP" : "10.165.68.116",
                "bgpGNLoopIP" : "10.165.68.117"
            },
            "bgpRouterGI" : {
                "routerId" : "10.165.68.137",
                "localAs" : "65321",
                "remoteAs" : "64512",

```

```

        "neighborIP" : "10.165.68.129",
        "fallOverBFD" : "single-hop",
        "bfdMultiHopPeer" : "10.165.68.129",
        "bgpGILoopIP" : "10.165.68.178"    },
    "giV4IpSubPoolRangeStart" : "10.202.0.1",
    "giV4IpSubPoolRangeEnd" : "10.202.0.255",
    "giRemoteIpSubPoolRangeStart" : "10.201.0.1",
    "giRemoteIpSubPoolRangeEnd" : "10.201.0.255",
    "dataLoopIP" : "10.165.68.176",
    "dataFabricIpAddress" : "10.165.68.61"
}

}

},
"node1" : {
    "mcc" : {
        "day1Conf" : {
            "networkName" : "Telecom-xyz",
            "shortNetworkName" : "xyz",
            "apnName" : "plte",
            "homePlmnId" : "99999",
            "pgwIpAddress" : "10.165.68.112",
            "sgwS11IpAddress" : "10.165.68.114",
            "giLoopbackIp" : "10.165.68.176",
            "giInterfaceIp" : "10.165.68.145",
            "giInterfaceIpRouteDest" : "0.0.0.0",
            "giInterfaceIpRoutePrefix" : "0",
            "giInterfaceIpRouteNextHop" : "10.165.68.129",
            "giIpSubPoolRangeStart" : "200.130.0.1",
            "giIpSubPoolRangeEnd" : "200.130.0.150",
            "gnSgwS1" : "10.165.68.113",
            "gnSgwLoopbackIp" : "10.165.68.115",
            "gnInterfaceIp" : "10.165.68.18",
            "gnInterfaceIpRouteDest" : "0.0.0.0",
            "gnInterfaceIpRoutePrefix" : "0",
            "gnInterfaceIpRouteNextHop" : "10.165.68.1",
            "scRoamingPlmnId" : "12345",
            "zoneDefaultQosProfile" : "QOSPROFDEFAULT",
            "bgpRouterGN" : {
                "routerId" : "10.165.68.18",
                "localAs" : "65321",
                "remoteAs" : "64512",
                "neighborIP" : "10.165.68.1",
                "fallOverBFD" : "multi-hop",
                "bfdMultiHopPeer" : "10.165.68.1",
                "bgpGEOLoopIP" : "10.165.68.118",
                "bgpGNLoopIP" : "10.165.68.119"    },
            "bgpRouterGI" : {
                "routerId" : "10.165.68.145",
                "localAs" : "65321",
                "remoteAs" : "64512",

```



```

        "neighborIP" : "10.165.68.129",
        "fallOverBFD" : "single-hop",
        "bfdMultiHopPeer" : "10.165.68.129",
        "bgpGILoopIP" : "10.165.68.179"      },
    "giV4IpSubPoolRangeStart" : "10.202.0.1",
    "giV4IpSubPoolRangeEnd" : "10.202.0.255",
    "giRemoteIpSubPoolRangeStart" : "10.201.0.1",
    "giRemoteIpSubPoolRangeEnd" : "10.201.0.255",
    "dataLoopIP" : "10.165.68.176",
    "dataFabricIpAddress" : "10.165.68.93",
    "dnsPrimaryIp" : "200.5.0.30",
    "dnsSecondaryIp" : "200.5.0.20"      }
    }
  } ]
} ]
}

```

Response

HTTP/1.1 200 OK

Content-Type: application/json

```

{
  "objectID" : 2,
  "objectName" : "plte12_13_2506_HA_deploy1",
  "successMessage" : "Day1 configured edited successfully"}

```

GET /rest/networks/{networkID}/nat

Get the NAT configuration by the given Network ID.

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	long

Response Codes

Code	Conditions
200	SUCCESS: NAT Profile configuration for the given Network ID has been fetched.
500	FAILURE: Error while fetching NAT Profile configuration for the given Network ID.

Response Body

Media Type	Data Type	Description
application/json	NATProfile (JSON)	NAT Profile Object

Example

Request

```
GET /rest/networks/2/nat
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "configStatus" : "PROVISIONED",
  "status" : "Success",
  "message" : "Configured",
  "profileName" : "NAT",
  "natSubPoolRangeStart" : "10.200.0.1",
  "natSubPoolRangeEnd" : "10.200.0.255"}
```

POST /rest/networks/{networkID}/nat

Set NAT configuration to the Network.

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	long
performProvision	query	Perform Network Provisioning	required

Request Body

Media Type	Data Type	Description
application/json	NATProfile (JSON)	NAT Profile Object

Response Codes

Code	Condition
200	SUCCESS: NAT Profile configured successfully.
500	FAILURE: Error while configuring NAT to the Network.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	NAT Configuration status.

Example

Request

```
POST /rest/networks/2/nat?performProvision=true
Content-Type: application/json
Accept: application/json
```

```
{
  "natSubPoolRangeStart" : "10.200.0.1",
  "natSubPoolRangeEnd" : "10.200.0.255"}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 2,
  "objectName" : "plte12_13_2506_HA_deploy1",
  "successMessage" : "NAT Profile configured successfully"}
```

GET /rest/networks/{networkID}/resources

Get all Resources by Network ID.

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	long

Response Codes

code	condition
200	SUCCESS: Resources for the given Network ID have been fetched.
500	FAILURE: Error while fetching the Resources for the given Network ID.

Response Body

Media Type	Data Type	Description
application/json	Resourcedetails (JSON)	Resources Object.

Example

Request

```
GET /rest/networks/2/resources
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "version" : 1,
  "resourceType" : {
    "version" : 1,
    "resourcetypeid" : 17,
    "type" : "HPEHSSEMS",
    "description" : "HPE-HSS-EMS",
    "iconname" : "HPEHSSEMS.png",
    "resyncInterval" : 15,
    "isdefault" : true,
    "primaryKey" : "17"  },
  "resourceComponent" : null,
  "resourceid" : 3,
  "ipaddress" : "10.165.31.14",
  "displayName" : "HSS_10.165.31.14-[node0]",
  "snmpCommunity" : null,
  "resyncInterval" : 0,
  "componentid" : null,
  "resourceVersion" : null,
```

```

    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 17,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : false,
    "resourceName" : "HSS_10.165.31.14-[node0]",
    "identifier" : 3
  }, {
    "version" : 1,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 17,
      "type" : "HPEHSSEMS",
      "description" : "HPE-HSS-EMS",
      "iconname" : "HPEHSSEMS.png",
      "resyncInterval" : 15,
      "isdefault" : true,
      "primaryKey" : "17" },
    "resourceComponent" : null,
    "resourceid" : 5,
    "ipaddress" : "10.165.31.29",
    "displayName" : "HSS_10.165.31.29-[node1]",
    "snmpCommunity" : null,
    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 17,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : false,
    "resourceName" : "HSS_10.165.31.29-[node1]",
    "identifier" : 5
  }, {
    "version" : 2,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 10,
      "type" : "MMEHvEPC",
      "description" : "MME",
      "iconname" : "MME.png",
      "resyncInterval" : 2,
      "isdefault" : true,
      "primaryKey" : "10" },
    "resourceComponent" : null,
    "resourceid" : 7,
    "ipaddress" : "10.165.31.24",
    "displayName" : "MME_10.165.31.24-[node1]",
    "snmpCommunity" : null,
    "resyncInterval" : 0,

```

```

    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 10,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : true,
    "resourceName" : "MME_10.165.31.24-[node1]",
    "identifier" : 7
  }, {
    "version" : 2,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 10,
      "type" : "MMEHvEPC",
      "description" : "MME",
      "iconname" : "MME.png",
      "resyncInterval" : 2,
      "isdefault" : true,
      "primaryKey" : "10" },
    "resourceComponent" : null,
    "resourceid" : 6,
    "ipaddress" : "10.165.31.8",
    "displayName" : "MME_10.165.31.8-[node0]",
    "snmpCommunity" : null,
    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 10,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : true,
    "resourceName" : "MME_10.165.31.8-[node0]",
    "identifier" : 6
  }, {
    "version" : 1,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 6,
      "type" : "AN3000",
      "description" : "MCC",
      "iconname" : "affirmed.png",
      "resyncInterval" : 2,
      "isdefault" : true,
      "primaryKey" : "6" },
    "resourceComponent" : null,
    "resourceid" : 2,
    "ipaddress" : "10.165.31.11",
    "displayName" : "SAEGW_10.165.31.11-[node0]",

```

```

    "snmpCommunity" : null,
    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 6,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : false,
    "resourceName" : "SAEGW_10.165.31.11-[node0]",
    "identifier" : 2
  }, {
    "version" : 1,
    "resourceType" : {
      "version" : 1,
      "resourcetypeid" : 6,
      "type" : "AN3000",
      "description" : "MCC",
      "iconname" : "affirmed.png",
      "resyncInterval" : 2,
      "isdefault" : true,
      "primaryKey" : "6" },
    "resourceComponent" : null,
    "resourceid" : 4,
    "ipaddress" : "10.165.31.26",
    "displayName" : "SAEGW_10.165.31.26-[node1]",
    "snmpCommunity" : null,
    "resyncInterval" : 0,
    "componentid" : null,
    "resourceVersion" : null,
    "resourceGroupID" : 2,
    "resourceStatusID" : 1,
    "resourceTypeID" : 6,
    "subnetMask" : "255.0.0.0",
    "preProvisioned" : false,
    "resourceName" : "SAEGW_10.165.31.26-[node1]",
    "identifier" : 4
  } ]

```

POST /rest/networks/{networkID}/retry

Retry failed Network.

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	long

Request Body

Media Type	Data Type
application/json	ServiceInstanceConfig (JSON)

Response Codes

Code	Condition
200	SUCCESS: Retry Network Deployment started.
500	FAILURE: Error while retrying network deployment.

Response Body

Media Type	Data Type	Description
application/json	Resourcedetails (JSON)	Resources Object.

Example

Request

```
POST /rest/networks/2/retry
Content-Type: application/json
Accept: application/json
```

```
{
  "locations" : [ {
    "asedeviceContainers" : [ {
      "node0" : {
        "mme" : {
          "day0Conf" : {
            "mrmManagement" : "10.165.31.8",
            "mrmBase" : "10.165.68.34",
            "mrmData" : "10.165.68.42",
            "msmManagement" : "10.165.31.9",
            "msmBase" : "10.165.68.35",
            "msmData" : "10.165.68.43",
            "ns1" : "10.165.68.9",
            "logicalManagement" : "10.165.31.10",
            "s6IP" : "10.165.68.49"
          }
        },
        "mcc" : {
          "day0Conf" : {
            "management0" : "10.165.31.11",
            "management1" : "10.165.31.12",
            "management2" : "10.165.31.13",
            "base0" : "10.165.68.58",
            "base1" : "10.165.68.59",
```



```

        "base2" : "10.165.68.60",
        "ew" : "10.165.68.66",
        "ns1" : "10.165.68.10",
        "ns2" : "10.165.68.137"
    },
    "hss" : {
        "day0Conf" : {
            "managementNic1" : "10.165.31.14",
            "lanNic1" : "10.165.68.11"
        }
    },
    "node1" : {
        "mme" : {
            "day0Conf" : {
                "mrmManagement" : "10.165.31.24",
                "mrmBase" : "10.165.68.36",
                "mrmData" : "10.165.68.44",
                "msmManagement" : "10.165.31.25",
                "msmBase" : "10.165.68.37",
                "msmData" : "10.165.68.45",
                "ns1" : "10.165.68.17",
                "logicalManagement" : "10.165.31.10"
            },
            "mcc" : {
                "day0Conf" : {
                    "management0" : "10.165.31.26",
                    "management1" : "10.165.31.27",
                    "management2" : "10.165.31.28",
                    "base0" : "10.165.68.90",
                    "base1" : "10.165.68.91",
                    "base2" : "10.165.68.92",
                    "ew" : "10.165.68.98",
                    "ns1" : "10.165.68.18",
                    "ns2" : "10.165.68.145"
                },
                "hss" : {
                    "day0Conf" : {
                        "managementNic1" : "10.165.31.29",
                        "lanNic1" : "10.165.68.19"
                    }
                }
            }
        }
    }
}

```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 2,
  "objectName" : "plte12_13_2506_HA_deploy1",
  "successMessage" : "Network re-deployment started"}
```

GET /rest/networks/{networkID}/upgradestatus

Get Software Upgrade details by Network ID.

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	required

Response Codes

Code	Condition
200	SUCCESS: Software Upgrade details have been fetched for the given Network ID.
500	FAILURE: Error while fetching Software Upgrade details for the given Network ID.

Response Body

Media Type	Data Type	Description
application/json	SoftwareUpgradeDetails (JSON)	Software Upgrade Details.

Example

Request

```
GET /rest/networks/2/upgradestatus
Content-Type: application/json
Accept: application/json
```

...

Response

HTTP/1.1 200 OK
 Content-Type: application/json

```
[ {
  "scheduleId" : "1",
  "scheduleDetailsId" : "1",
  "upgradeName" : "plte12_13_2506_HA_deploy1_hss_17.0",
  "upgradeVersion" : "APNS 1.1",
  "softwareRelease" : "Affirmed-HSS-0213",
  "elementName" : "HSS_10.165.31.14-[node0]",
  "executedTime" : "1626701503025",
  "nextScheduleTime" : "-1",
  "operationalStatus" : "Completed",
  "status" : "Succeeded"}, {
  "scheduleId" : "1",
  "scheduleDetailsId" : "2",
  "upgradeName" : "plte12_13_2506_HA_deploy1_hss_17.0",
  "upgradeVersion" : "APNS 1.1",
  "softwareRelease" : "Affirmed-HSS-0213",
  "elementName" : "HSS_10.165.31.29-[node1]",
  "executedTime" : "1626701503025",
  "nextScheduleTime" : "-1",
  "operationalStatus" : "Completed",
  "status" : "Succeeded"}, {
  "scheduleId" : "2",
  "scheduleDetailsId" : "3",
  "upgradeName" : "plte12_13_2506_HA_deploy1_mme_18.0",
  "upgradeVersion" : "APNS 1.1",
  "softwareRelease" : "Affirmed-MME-0214",
  "elementName" : "MME_10.165.31.8-[node0]",
  "executedTime" : "1626701503026",
  "nextScheduleTime" : "-1",
  "operationalStatus" : "Completed",
  "status" : "Failed"}, {
  "scheduleId" : "2",
  "scheduleDetailsId" : "4",
  "upgradeName" : "plte12_13_2506_HA_deploy1_mme_18.0",
  "upgradeVersion" : "APNS 1.1",
  "softwareRelease" : "Affirmed-MME-0214",
  "elementName" : "MME_10.165.31.24-[node1]",
  "executedTime" : "1626701503026",
  "nextScheduleTime" : "-1",
  "operationalStatus" : "Completed",
  "status" : "Failed"}, {
  "scheduleId" : "3",
  "scheduleDetailsId" : "5",
  "upgradeName" : "plte12_13_2506_HA_deploy1_mcc_12.0",
  "upgradeVersion" : "APNS 1.1",
  "softwareRelease" : "Affirmed-MCC-0214",
```

```

"elementName" : "SAEGW_10.165.31.11-[node0]",
"executedTime" : "1626701503027",
"nextScheduleTime" : "-1",
"operationalStatus" : "Completed",
"status" : "Failed"}, {
"scheduleId" : "3",
"scheduleDetailsId" : "6",
"upgradeName" : "plte12_13_2506_HA_deploy1_mcc_12.0",
"upgradeVersion" : "APNS 1.1",
"softwareRelease" : "Affirmed-MCC-0214",
"elementName" : "SAEGW_10.165.31.26-[node1]",
"executedTime" : "1626701503027",
"nextScheduleTime" : "-1",
"operationalStatus" : "Completed",
"status" : "Failed"} ]

```

GET /rest/networks/resources/{resourceID}/siminstances

Get list of SIM Instances deployed to Network.

Request Parameters

Name	Type	Description
resourceID	path	Resource ID

Response Codes

Code	Condition
200	SUCCESS: All the SIM Instances deployed to the given Network have been fetched.
500	FAILURE: Error while fetching SIM Instances deployed to the given Network.

Response Body

Media Type	Data Type	Description
application/json	ResourceIMSIStatus (JSON)	Resource IMSI status.

Example

Request

```

GET /rest/networks/resources/3/siminstances
Content-Type: application/json
Accept: application/json

```

...

Response

HTTP/1.1 200 OK

Content-Type: application/json

```
[ {
  "deletetime" : "",
  "imsino" : "1000000000000002",
  "status" : null,
  "rstatus" : "SUCCESS",
  "comment" : "HSS_EPS_Subscriber Activation Success"}, {
  "deletetime" : "",
  "imsino" : "1000000000000001",
  "status" : null,
  "rstatus" : "SUCCESS",
  "comment" : "HSS_EPS_Subscriber Activation Success"}, {
  "deletetime" : "",
  "imsino" : "1000000000000000",
  "status" : null,
  "rstatus" : "SUCCESS",
  "comment" : "HSS_EPS_Subscriber Activation Success"} ]
```

POST /rest/networks/resources/{resourceID}/sync

Perform Resource Sync

Request Parameters

Name	Type	Description	Constraints
resourceID	path	Resource ID	required

Response Codes

Code	Condition
200	SUCCESS: Resource Sync Triggered successfully for the given Resource ID.
500	FAILURE: Error while performing Resource Sync for the given Resource ID.

Response Body

media type	data type	description
application/json	object (JSON)	Resource Sync status.

*Example***Request**

```
POST /rest/networks/resources/3/sync
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 3,
  "objectName" : null,
  "successMessage" : "Resource Sync Triggered successfully"}
```

GET /rest/networks/{networkID}/day0/status

Get the Day0 configuration status by Network ID. In-Progress: Day0 Configuration is In-Progress, Succeeded: Day0 Configuration completed successfully, Failed: Day0 Configuration Failed, Deleting: Network is in Deleting state

Request Parameters

Name	Type	Description	Constraints
networkID	path	Network ID	long

Response Codes

Code	Condition
200	SUCCESS: Day0 Configuration Status has been fetched successfully for the given Network ID.
500	FAILURE: Error while fetching Day0 Configuration Status.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Day0 Configuration status.

*Example***Request**

```
GET /rest/networks/2/day0/status
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "objectID" : 2,
  "objectName" : null,
  "successMessage" : "Succeeded"}
```

RAN Info Management API***GET /rest/raninfo***

Get the list of RAN info from database.

Request Parameters

Name	Type	Description	Constraints
fetchfrommme	query		required
networkId	query		required

Response Codes

Code	Condition
200	SUCCESS: All the RAN info have been fetched.
500	FAILURE: Error while fetching RAN info.

Response Body

Media Type	Data Type	Description
application/json	MmeEnbOperEntity (JSON)	List of RAN info.

Example

Request

```
GET /ranInfo
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "plmn-id" : "...",
  "enodeb-id" : 12345,
  "enodeb-type" : "...",
  "ip-address1-role" : "...",
  "ip-address2-role" : "...",
  "enodeb-name" : "...",
  "process-pid" : 12345,
  "overload-start-sent" : "...",
  "connection-state" : "...",
  "cell-access-mode" : "...",
  "local-ip-address1-role" : "...",
  "local-ip-address2-role" : "...",
  "local-port" : 12345,
  "connection-time" : "...",
  "raninford" : 12345,
  "plmnId" : "...",
  "enodebId" : 12345,
  "enodebType" : "...",
  "ipAddress1Role" : "...",
  "ipAddress2Role" : "...",
  "port" : 12345,
  "enodebName" : "...",
  "processPid" : 12345,
  "service" : "...",
  "overloadStartSent" : "...",
  "connectionState" : "...",
  "cellAccessMode" : "...",
  "localIpAddress1Role" : "...",
  "localIpAddress2Role" : "...",
  "localPort" : 12345,
  "connectionTime" : "...",
```



```
"networkId" : 12345,
"mmeResourceIP" : "...",
"networkname" : "..."}

```

SIM Management API

SIM Management APIs to Add, Activate, Deactivate and Delete the SIM(s).

GET/rest/siminstances

Get all SIM Instances added in PNSM and its configuration details.

Response Codes

Code	Condition
200	SUCCESS: All the SIM Instances has been fetched.
500	FAILURE: Error while fetching SIM Instances.

Response Body

Media Type	Data Type	Description
application/json	SimInstance (JSON)	List of SIM Instances. Empty list will be returned if no SIM Instances are added.

Example

Request

```
GET /rest/siminstances
Content-Type: application/json
Accept: application/json

...
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json

[ {
  "version" : 1,
  "simFileMaster" : {
    "version" : 0,
    "simfileid" : 1,
    "simFile" : null,
    "fileName" : "simInstances.json",
```

```

    "centralHssEnabled" : false
  },
  "siminstanceid" : 3,
  "imsiNo" : "1000000000000002",
  "status" : "ACTIVATED",
  "simfileid" : 1,
  "userId" : null,
  "phoneNumber" : null,
  "deletetime" : null,
  "opcValue" : "1234567890ABCDEF1234567890AAAAAA",
  "kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
  "version" : 1,
  "simFileMaster" : {
    "version" : 0,
    "simfileid" : 1,
    "simFile" : null,
    "fileName" : "simInstances.json",
    "centralHssEnabled" : false
  },
  "siminstanceid" : 2,
  "imsiNo" : "1000000000000001",
  "status" : "ACTIVATED",
  "simfileid" : 1,
  "userId" : null,
  "phoneNumber" : null,
  "deletetime" : null,
  "opcValue" : "1234567890ABCDEF1234567890AAAAAA",
  "kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
  "version" : 1,
  "simFileMaster" : {
    "version" : 0,
    "simfileid" : 1,
    "simFile" : null,
    "fileName" : "simInstances.json",
    "centralHssEnabled" : false
  },
  "siminstanceid" : 1,
  "imsiNo" : "1000000000000000",
  "status" : "ACTIVATED",
  "simfileid" : 1,
  "userId" : "1",
  "phoneNumber" : "9876543210",
  "deletetime" : null,
  "opcValue" : "1234567890ABCDEF1234567890AAAAAA",
  "kiValue" : "1234567890ABCDEF1234567890ABCDEF"} ]

```

POST/rest/siminstances

Create SIM Instances to PNSM.

Request Body

Media Type	Data Type	Description
application/json	SimFileInstance (JSON)	SIM Instances Object

Response Codes

Code	Condition
200	SUCCESS: SIM Instances created successfully.
500	FAILURE: Error while creating SIM Instances.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	SIM Instances CREATE status.

Example

Request

```
POST /rest/siminstances
Content-Type: application/json
Accept: application/json
```

```
{
  "simFileFileName" : "simInstances.json",
  "addToCentralHSS" : false,
  "instanceList" : [ {
    "imsiNo" : 1000000000000000,
    "kiValue" : "1234567890ABCDEF1234567890ABCDEF",
    "opcValue" : "1234567890ABCDEF1234567890AAAAAA" }, {
    "imsiNo" : 1000000000000001,
    "kiValue" : "1234567890ABCDEF1234567890ABCDEF",
    "opcValue" : "1234567890ABCDEF1234567890AAAAAA" }, {
    "imsiNo" : 1000000000000002,
    "kiValue" : "1234567890ABCDEF1234567890ABCDEF",
    "opcValue" : "1234567890ABCDEF1234567890AAAAAA" } ]
}
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
```

```

"version" : 0,
"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 1,
"imsiNo" : "1000000000000000",
"status" : "DEACTIVATED",
"simfileid" : 1,
"userId" : null,
"phoneNumber" : null,
"deletetime" : null,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
"version" : 0,
"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 2,
"imsiNo" : "1000000000000001",
"status" : "DEACTIVATED",
"simfileid" : 1,
"userId" : null,
"phoneNumber" : null,
"deletetime" : null,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
"version" : 0,
"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 3,
"imsiNo" : "1000000000000002",
"status" : "DEACTIVATED",
"simfileid" : 1,
"userId" : null,
"phoneNumber" : null,
"deletetime" : null,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",

```

```
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"} } ]
```

POST/rest/siminstances/activate

Activate the SIM Instance on the PNSM database and initiate the process of activation on all the configured HSS Devices asynchronously.

Request Body

Media Type	Data Type	Description
application/json	array of number (JSON)	List of SIM Instance IDs

Response Codes

Code	Condition
200	SUCCESS: SIM Instances activated successfully.
500	FAILURE: Error while activating SIM Instances.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Activated SIM Instances. If SIM Instance for given SIM Instance ID is not found, empty SIM Instance object with error message will be returned.

Example

Request

```
POST /rest/siminstances/activate
Content-Type: application/json
Accept: application/json
```

```
[ 2, 3, 4 ]
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "version" : 1,
  "simFileMaster" : {
```

```

    "version" : 0,
    "simfileid" : 1,
    "simFile" : null,
    "fileName" : "simInstances.json",
    "centralHssEnabled" : false
  },
  "siminstanceid" : 2,
  "imsiNo" : "1000000000000001",
  "status" : "ACTIVATED",
  "simfileid" : 1,
  "userId" : null,
  "phoneNumber" : null,
  "deletetime" : null,
  "opcValue" : "1234567890ABCDEF1234567890AAAAAA",
  "kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
  "version" : 1,
  "simFileMaster" : {
    "version" : 0,
    "simfileid" : 1,
    "simFile" : null,
    "fileName" : "simInstances.json",
    "centralHssEnabled" : false
  },
  "siminstanceid" : 3,
  "imsiNo" : "1000000000000002",
  "status" : "ACTIVATED",
  "simfileid" : 1,
  "userId" : null,
  "phoneNumber" : null,
  "deletetime" : null,
  "opcValue" : "1234567890ABCDEF1234567890AAAAAA",
  "kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
  "version" : 0,
  "simFileMaster" : null,
  "siminstanceid" : 4,
  "imsiNo" : null,
  "status" : "SIM Instances 4 not found",
  "simfileid" : null,
  "userId" : null,
  "phoneNumber" : null,
  "deletetime" : null,
  "opcValue" : null,
  "kiValue" : null
} ]

```

POST/rest/siminstances/deactivate

Deactivate the SIM Instance on the PNSM database and initiate the process of deactivation on all the configured HSS Devices asynchronously.

Request Body

Media Type	Data Type
application/json	array of number (JSON)

Response Codes

Code	Condition
200	SUCCESS: SIM Instances deactivated successfully.
500	FAILURE: Error while deactivating SIM Instances.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Deactivated SIM Instances. If SIM Instance for given SIM Instance ID is not found, empty SIM Instance object with error message will be returned.

Example**Request**

```
POST /rest/siminstances/deactivate
```

```
Content-Type: application/json
```

```
Accept: application/json
```

```
[ 1, 2, 3, 4 ]
```

Response

```
HTTP/1.1 200 OK
```

```
Content-Type: application/json
```

```
[ {
  "version" : 2,
  "simFileMaster" : {
    "version" : 0,
    "simfileid" : 1,
    "simFile" : null,
    "fileName" : "simInstances.json",
    "centralHssEnabled" : false
  },
}
```

```

"siminstanceid" : 1,
"imsiNo" : "1000000000000000",
"status" : "DEACTIVATED",
"simfileid" : 1,
"userId" : "1",
"phoneNumber" : "9876543210",
"deletetime" : null,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
"version" : 2,
"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 2,
"imsiNo" : "1000000000000001",
"status" : "DEACTIVATED",
"simfileid" : 1,
"userId" : null,
"phoneNumber" : null,
"deletetime" : null,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
"version" : 2,
"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 3,
"imsiNo" : "1000000000000002",
"status" : "DEACTIVATED",
"simfileid" : 1,
"userId" : null,
"phoneNumber" : null,
"deletetime" : null,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
"version" : 0,
"simFileMaster" : null,
"siminstanceid" : 4,
"imsiNo" : null,
"status" : "SIM Instances 4 not found",
"simfileid" : null,
"userId" : null,

```



```

    "phoneNumber" : null,
    "deletetime" : null,
    "opcValue" : null,
    "kiValue" : null
  } ]

```

POST/rest/siminstances/delete

Delete SIM Instances.

Request Body

Media Type	Data Type
application/json	array of number (JSON)

Response Codes

Code	Condition
200	SUCCESS: SIM Instances marked for delete successfully.
500	FAILURE: Error while marking SIM Instances for delete.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	SIM Instance marked for Delete. If SIM Instance for given SIM Instance ID is not found, empty SIM Instance object with error message will be returned.

Example

Request

```

POST /rest/siminstances/delete
Content-Type: application/json
Accept: application/json

[ 1, 2, 3, 4 ]

```

Response

```

HTTP/1.1 200 OK
Content-Type: application/json

[ {
  "version" : 3,

```

```

"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 1,
"imsiNo" : "1000000000000000",
"status" : "DEACTIVATEDFORDELETE",
"simfileid" : 1,
"userId" : "1",
"phoneNumber" : "9876543210",
"deletetime" : 1626949163366,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
"version" : 3,
"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 2,
"imsiNo" : "1000000000000001",
"status" : "DEACTIVATEDFORDELETE",
"simfileid" : 1,
"userId" : null,
"phoneNumber" : null,
"deletetime" : 1626949163366,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {
"version" : 3,
"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 3,
"imsiNo" : "1000000000000002",
"status" : "DEACTIVATEDFORDELETE",
"simfileid" : 1,
"userId" : null,
"phoneNumber" : null,
"deletetime" : 1626949163366,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"}, {

```

```

    "version" : 0,
    "simFileMaster" : null,
    "siminstanceid" : 4,
    "imsiNo" : null,
    "status" : "SIM Instances 4 not found",
    "simfileid" : null,
    "userId" : null,
    "phoneNumber" : null,
    "deletetime" : null,
    "opcValue" : null,
    "kiValue" : null
  } ]

```

POST /rest/siminstances/uploadSIMFile

Request Parameters

Name	Type	Description	Constraints
uploadedInputStream	formdata		
addToCentralHSS	query		required
fileName	query		

Request Body

Media Type	Data Type
multipart/form-data	(custom)

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	

Example

Request

```

POST /rest/siminstances/uploadSIMFile
Content-Type: multipart/form-data
Accept: application/json

```

...

Response

HTTP/1.1 201 Created
 Content-Type: application/json

...

GET /rest/siminstances/{simInstanceID}

Get SIM Instance by SIM Instance ID.

Request Parameters

Name	Type	Description	Constraints
simInstanceID	path	SIM Instance ID	required

Response Codes

Code	Condition
200	SUCCESS: SIM Instance has been fetched.
500	FAILURE: Error while fetching SIM Instance.

Response Body

Media Type	Data Type	Description
application/json	SimInstance (JSON)	SIM Instances by SIM Instance ID

Example

Request

GET /rest/siminstances/1
 Content-Type: application/json
 Accept: application/json

...

Response

HTTP/1.1 200 OK
 Content-Type: application/json

```
{
  "version" : 1,
```

```

"simFileMaster" : {
  "version" : 0,
  "simfileid" : 1,
  "simFile" : null,
  "fileName" : "simInstances.json",
  "centralHssEnabled" : false
},
"siminstanceid" : 1,
"imsiNo" : "1000000000000000",
"status" : "ACTIVATED",
"simfileid" : 1,
"userId" : "1",
"phoneNumber" : "9876543210",
"deletetime" : null,
"opcValue" : "1234567890ABCDEF1234567890AAAAAA",
"kiValue" : "1234567890ABCDEF1234567890ABCDEF"
}

```

POST /rest/siminstances/{simInstanceID}/activate

Activate the SIM Instance on the PNSM database and initiate the process of activation on all the configured HSS Devices asynchronously.

Request Parameters

Name	Type	Description	Constraints
simInstanceID	path	SIM Instance ID	required
phoneNumber	query	Phone Number to Activate the SIM Instance	
userID	query	User ID to Activate the SIM Instance	

Response Codes

Code	Condition
200	SUCCESS: SIM Instances activated successfully.
500	FAILURE: Error while activating SIM Instances.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Activated SIM Instance.

Example

Request

```

POST
/rest/siminstances/1/activate?userID=1&phoneNumber=987654321
0
Content-Type: application/json
Accept: application/json

...

```

Response

```

HTTP/1.1 200 OK
Content-Type: application/json

```

```

{
  "version" : 1,
  "simFileMaster" : {
    "version" : 0,
    "simfileid" : 1,
    "simFile" : null,
    "fileName" : "simInstances.json",
    "centralHssEnabled" : false
  },
  "siminstanceid" : 1,
  "imsiNo" : "1000000000000000",
  "status" : "ACTIVATED",
  "simfileid" : 1,
  "userId" : "1",
  "phoneNumber" : "9876543210",
  "deletetime" : null,
  "opcValue" : "1234567890ABCDEF1234567890AAAAAA",
  "kiValue" : "1234567890ABCDEF1234567890ABCDEF"}

```

GET/rest/siminstances/{simInstanceID}/status

Get SIM Instance status on configured resources by SIM Instance ID.

Request Parameters

Name	Type	Description	Constraints
simInstanceID	path	SIM Instance ID	required

Response Codes

Code	Condition
200	SUCCESS: Resource SIM Instance status has been fetched.
500	FAILURE: Error while fetching Resource SIM Instance status.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Resource SIM status.

Example

Request

```
GET /rest/siminstances/{simInstanceID}/status
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "resourceName" : "HSS_10.165.31.29-[node1]",
  "status" : "SUCCESS",
  "comment" : "HSS_EPS_Subscriber Activation Success"}, {
  "resourceName" : "HSS_10.165.31.14-[node0]",
  "status" : "SUCCESS",
  "comment" : "HSS_EPS_Subscriber Activation Success"} ]
```

Software Management API

Software Management APIs to Import Software and track the Software Upgrade.

POST /rest/software/import

Import Software Package that contains the Software Images Path for MME, MCC and HSS with required Templates and Signature File Paths.

Request Parameters

Name	Type	Description	Constraints
fileServerID	query	File Server ID	required

Request Body

Media Type	Data Type
application/json	ServiceInstanceSoftware (JSON)

Response Codes

Code	Condition
200	SUCCESS: Software Package imported successfully.
500	FAILURE: Error while importing Software Package.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Software Import status.

Example

Request

POST /rest/software/import?fileServerID=2

Content-Type: application/json

Accept: application/json

```
{
  "name" : "APNS 1.1",
  "version" : "1.1",
  "skuDetails" : [ {
    "skuType" : "mme",
    "skuName" : "Affirmed-MME-0214",
    "relVersion" : "18.0",
    "imageFilePath" : "/home/affirmed/software_packages/epc-16.0.20.5.tar",
    "signatureFilePath" : "NA",
    "armTempFilePath" : "/home/affirmed/software_packages/mme16.0-arm-template.tgz",
    "skuVhdName" : "NA",
    "skuImageName" : "NA"  }, {
    "skuType" : "mcc",
    "skuName" : "Affirmed-MCC-0214",
    "relVersion" : "12.0",
    "imageFilePath" : "/home/affirmed/software_packages/an-12.0.0.1-26.REL.tgz",
    "signatureFilePath" : "/home/affirmed/software_packages/an-12.0.0.1-26.REL.tgz.sig",
    "armTempFilePath" : "/home/affirmed/software_packages/mcc12.0-arm-template.tgz",
    "skuVhdName" : "NA",
    "skuImageName" : "NA"  }, {
    "skuType" : "hss",
    "skuName" : "Affirmed-HSS-0213",
    "relVersion" : "17.0",
    "imageFilePath" : "/home/affirmed/software_packages/hpe-
```



```

pnw-lte-upgrade-04.01.00.202105012023.705e4aa8c9d.zip",
  "signatureFilePath" : "/home/affirmed/software_
packages/hpe-pnw-lte-upgrade-
04.01.00.202105012023.705e4aa8c9d.zip.SHA256SUM",
  "armTempFilePath" : "/home/affirmed/software_
packages/hss15.0-template.tgz",
  "skuVhdName" : "NA",
  "skuImageName" : "NA" } ]
}

```

Response

```

HTTP/1.1 200 OK
Content-Type: application/json

```

```

{
  "objectID" : null,
  "objectName" : null,
  "successMessage" : "Software Package imported
successfully"}

```

GET /rest/software/upgrade

Get the list of Networks that need Upgrade.

Response Codes

Code	Condition
200	SUCCESS: All the Networks that needs Upgrade has been fetched.
500	FAILURE: Error while fetching the Networks that needs Upgrade.

Response Body

Media Type	Data Type	Description
application/json	NUDetails (JSON)	List of Networks that needs Upgrade.

Example

Request

```

GET /rest/software/upgrade
Content-Type: application/json
Accept: application/json

```

...

Response

HTTP/1.1 200 OK

Content-Type: application/json

```
[ {
  "networkId" : 2,
  "networkName" : "plte12_13_2506_HA_deploy1",
  "networkLocation" : "Plte12-13",
  "currentVersion" : "APNS 1.0",
  "availableVersion" : "APNS 1.1"} ]
```

POST /rest/software/upgrade

Perform Software Upgrade for the given list of Network IDs.

Request Parameters

Name	Type	Description	Constraints
scheduleTime	query	Upgrade Schedule Time in Milliseconds. Default Upgrade Schedule Time will be current time, if Schedule Time is not set or set to 0	required

Request Body

Media Type	Data Type	Description
application/json	array of number (JSON)	List of Network IDs

Response Codes

Code	Condition
200	SUCCESS: Software Upgrade triggered successfully for the given Network IDs.
500	FAILURE: Error while triggering Software Upgrade for the given Network IDs.

Response Body

Media Type	Data Type	Description
application/json	object (JSON)	Software Upgrade status. If Network for given Network ID is not found, empty Schedule Details object with error message will be returned.

*Example***Request**

```
POST /rest/software/upgrade?scheduleTime=1626781631027
Content-Type: application/json
Accept: application/json
```

```
[ 1, 2 ]
```

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[ {
  "scheduleID" : 0,
  "scheduleName" : null,
  "networkID" : 1,
  "resourceID" : 0,
  "scheduleDetailsID" : 0,
  "scheduleStatus" : "Network 1 not found"}, {
  "scheduleID" : 1,
  "scheduleName" : "plte12_13_2506_HA_deploy1_hss_17.0",
  "networkID" : 2,
  "resourceID" : 3,
  "scheduleDetailsID" : 1,
  "scheduleStatus" : "Scheduled"}, {
  "scheduleID" : 1,
  "scheduleName" : "plte12_13_2506_HA_deploy1_hss_17.0",
  "networkID" : 2,
  "resourceID" : 5,
  "scheduleDetailsID" : 2,
  "scheduleStatus" : "Scheduled"}, {
  "scheduleID" : 2,
  "scheduleName" : "plte12_13_2506_HA_deploy1_mme_18.0",
  "networkID" : 2,
  "resourceID" : 6,
  "scheduleDetailsID" : 3,
  "scheduleStatus" : "Scheduled"}, {
  "scheduleID" : 2,
  "scheduleName" : "plte12_13_2506_HA_deploy1_mme_18.0",
  "networkID" : 2,
  "resourceID" : 7,
  "scheduleDetailsID" : 4,
  "scheduleStatus" : "Scheduled"}, {
  "scheduleID" : 3,
```

```

    "scheduleName" : "plte12_13_2506_HA_deploy1_mcc_12.0",
    "networkID" : 2,
    "resourceID" : 2,
    "scheduleDetailsID" : 5,
    "scheduleStatus" : "Scheduled"}, {
    "scheduleID" : 3,
    "scheduleName" : "plte12_13_2506_HA_deploy1_mcc_12.0",
    "networkID" : 2,
    "resourceID" : 4,
    "scheduleDetailsID" : 6,
    "scheduleStatus" : "Scheduled"} ]

```

GET /rest/software/apns/troubleshootbundle

Get the list recent Troubleshoot bundles.

Response Codes

Code	Condition
200	SUCCESS: Fetched the latest available troubleshoot bundles.
500	FAILURE: Error while fetching troubleshoot bundle.

Response Body

Media Type	Data Type	Description
application/json	TroubleshootBundleDetails (JSON)	List of Troubleshoot bundles.

Example

Request

```

GET /rest/software/apns/troubleshootbundle
Content-Type: application/json
Accept: application/json

```

...

Response

```

HTTP/1.1 200 OK
Content-Type: application/json

```

```
[ {
```

```

    "creationTime" : 1628591381000,
    "filename" : "pnsm-troubleshootlogs-1628588059.tar.gz",
    "filePath" : "{Server_HOME_
DIRECTORY}/logcollector/./server/ems/troubleshootlogs/pnsm-
troubleshootlogs-1628588059.tar.gz", {
    "creationTime" : 1628591367000,
    "filename" : "pnsm-troubleshootlogs-1628587969.tar.gz",
    "filePath" : "{Server_HOME_
DIRECTORY}/logcollector/./server/ems/troubleshootlogs/pnsm-
troubleshootlogs-1628587969.tar.gz", {
    "creationTime" : 1628591356000,
    "filename" : "pnsm-troubleshootlogs-1628587872.tar.gz",
    "filePath" : "{Server_HOME_
DIRECTORY}/logcollector/./server/ems/troubleshootlogs/pnsm-
troubleshootlogs-1628587872.tar.gz"} ]

```

POST /rest/software/apns/troubleshootbundle

Collect Troubleshoot Information of PNSM deployment.

Request Body

Media Type	Data Type	Description
application/json	TroubleshootStorageDetails (JSON)	Object

Response Codes

Code	Condition
200	SUCCESS: Troubleshoot bundle created successfully.
500	FAILURE: Error while creating troubleshoot bundle.

Response Body

Media Type	Data Type	Description
application/json	TroubleshootScriptResponse (JSON)	Software Import status.

Example

Request

```

POST /rest/software/apns/troubleshootbundle
Content-Type: application/json
Accept: application/json

```

```
{
  "storageMode" : "local"}
```

Response

HTTP/1.1 200 OK
Content-Type: application/json

```
{
  "collectionStatus" : true,
  "message" : null,
  "creationTime" : 1628839434,
  "filename" : "pnsm-troubleshootlogs-1628839433.tar.gz",
  "path" : "{Server_HOME_
DIRECTORY}/logcollector/./server/ems/troubleshootlogs/pnsm-
troubleshootlogs-1628839433.tar.gz",
  "storageMode" : "local"}
```

GET /rest/software/upgrade/{upgradeScheduleDetailsID}

Get Software Upgrade details by Schedule ID.

Request Parameters

Name	Type	Description	Constraints
upgradeScheduleDetailsID	path	Upgrade Schedule Details ID	required

Response Codes

Code	Condition
200	SUCCESS: Software Upgrade details has been fetched for the given Schedule ID.
500	FAILURE: Error while fetching Software Upgrade details for the given Schedule ID.

Response Body

Media Type	Data Type	Description
application/json	SoftwareScheduleDetails (JSON)	Software Upgrade Details.

Example

Request

```
GET /rest/software/upgrade/1
Content-Type: application/json
Accept: application/json
```

...

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "scheduleID" : 1,
  "scheduleName" : "plte12_13_2506_HA_deploy1_hss_17.0",
  "networkID" : 2,
  "resourceID" : 3,
  "scheduleDetailsID" : 1,
  "scheduleStatus" : "Software schedule progress status
report started at 2021-07-22 09:09:06...\nCurrent HPE HSS
Version Info :{HSS Version=04.01.00, Build Date=20:23 on
05/01/2021, Build Hash=705e4aa8c9d}\nHSS 10.165.31.14 is
already in latest version, Upgrade is not required."}
```

Data Types

JSON

Table B-2 JSON Types and Descriptions

Type	Description
ASAPOperationStatus	ASAP Operation Status. All ASAP related operations are handled based on this status. Values: PROVISIONED, UNPROVISIONED
ASEDevice	Node Resources configuration
ASEDeviceContainer	Nodes configurations deployed in a Network. HA deployment will have one Node and Non-HA will have two Nodes.
ActionStatus	Task Status Values: INPROGRESS, SUCCESS, FAIL
AzureConfig	PNSM Configurations for Service Manager communication with Azure and ASAP servers

Type	Description
BGPRouterConfig	BGP configurations for Node's Day 1 configuration
BGPRouterGI	BGP configurations for Node's Day 1 GI Interface configuration
BGPRouterGN	BGP configurations for Node's Day 1 GN Interface configuration
CloudContainer	Nodes configurations deployed in a Cloud. HA deployment will have one Node and Non-HA will have two Nodes
CloudDevice	Node Resources configuration in Cloud
CommonResources	
ConfigBackupResponseVO	Backup task response details
ConfigurationFileEntity	PNSM Configuration Details
Day0Base	Node Day 0 Base Configurations which are common for MME, MCC and HSS Resources Day 0 Configurations
Day1Base	Node Day 1 Base Configurations which are common for MME and MCC Resources Day 1 Configurations
Day2InitialConfigStatus	Node Day 2 Initial Configuration Status
DeploymentMode	Network deployment mode Values: ONPREM, ONPREMHA, CUPS, CUPSHA, CLOUD, CLOUDHA
Device	Device Details
Device Properties	Device properties
DeviceTypeEnum	Device Type Values: UNKNOWN, AZURESTACKEDGE
EJBBaseEntity	
FileServer	File Server to import Software Packages, perform Network Upgrades and perform Backup/Restore. By default Service Manager will have its own local FTP Server which cannot be Updated/Deleted. User can Create SFTP servers to Import the Software Packages. User created SFTP servers can be Updated/Deleted
Ftpprotocol	
GlobalNF	Global Network Function Details
HSS	HSS resource configurations
HSSDay0Conf	HSS Day 0 configurations
HSSStatus	HSS Day 0 status and comment

Type	Description
IpAllocationMethodEnum	IP Address allocation method Values: UNKNOWN , STATIC , DYNAMIC
IpVersionEnum	IP Address version Values: UNKNOWN , IPV4
LANIpAddress	LANIPAddresses reserved/assigned for the network
Location	Location to deploy Networks. Location's Management, LAN, WAN Subnets and Gateway IPs will be used for the Networks deployed in the Location
LocationEntity	Location to deploy Networks. Location's Management, LAN, WAN Subnets and Gateway IPs will be used for the Networks deployed in the Location
LocationNetworkConfig	List of deployed Networks and assigned Subnets mapping. List of assigned and available Subnet slots for Network deployments
MCC	MCC resource configurations
MCCDay0Conf	MCC Day 0 configurations
MCCDay1Conf	MCC Day 1 configurations
MCCEDRProfile	EDR Profile configurations. This is and Optional Day 1 configuration
MCCStatus	MCC Day 0 and Day 1 status.
MME	MME resource configurations
MMEDay0Conf	MME Day 0 configurations
MMEDay1Conf	MME Day 1 configurations
MMEStatus	MME Day 0 and Day 1 status
ManagedApplicationParameters	Managed Applications Parameters
ManagementIPAddress	Management IP Addresses reserved/assigned for the network
MmeEnbOperEntity Data Type	
NATProfile	NAT Profile configurations. This is and Optional Day 1 configuration
NEBackup	Data input to trigger the Network Element Backup. Backup File Name, File Path, for which Resources the Backup has to be taken and in which FTP Server the Backup to be stored
NEBackupPurge Data Type	NE Backup Purge Configuration

Type	Description
NEBackupRestore	Network Element Backup Details. Backup ID, File Name, File Path, Status, from which Resource the Backup was taken and in which FTP Server the Backup is stored
NEBackupSchedule	The Class NEBackup Schedule
NFBase	Resource configuration base for MME, MCC and HSS resources
NFProvisioningState	Network Function Provisioning State Values: INPROGRESS, SUCCEEDED, PROVISIONED, FAILED, DELETING
NFServiceInstance	Network Instance configurations
NUDetails	Network Element details that runs in the older versions of Software and the latest version to Upgrade
NetworkDeploymentConfig	Location Subnet and the set of Management IP Addresses reserved/assigned for the ASE Devices
NetworkFunction	Network Function resource response
NetworkFunctionPropertiesFormat	Network Function properties
NetworkFunctionSKU	Network Function SKU Details
NetworkFunctionUserConfiguration	The Network Function User configuration
NetworkFunctionUserConfigurationOsProfile	Specifies the operating system settings for the role instance
NetworkInterface	Network Interface properties
NetworkInterfaceIPConfiguration	Network Interface IP configurations
NetworkRetryConfig	Retry Configuration Information. * Will be used during network retry
ProvisioningState	Current Provisioning State Values: UNKNOWN, SUCCEEDED, ACCEPTED, DELETING, FAILED, CANCELED, DELETED, PROVISIONING
ProvisioningState	Current Provisioning State Values: UNKNOWN, SUCCEEDED, ACCEPTED, DELETING, FAILED, CANCELED, DELETED
Recurrencepattern	
Region	Network deployment Region details
ResourceIMSISStatus	SIM Instance Status
ResourceSimManagementStatus	SIM Instance Status in the Activated Resource Instances

Type	Description
Resourcecomponent	
Resourcedetails	
Resourceparams	
Resourcetype	
SMDevice	Node Resources configuration
ServiceInstanceConfig	Configurations for all the DeploymentMode .
ServiceInstanceSoftware	Software Package Details. Software Package Name, Version and SKU Details of each Node Type
ServiceManagerSKU	Service Manager SKU Details
SimFileInstance	SIM File Details and the List of SIM Instances
SimFileMaster	SIM File Entity which holds the details of SIM file and Instances added as part of it
SimInstance	SIM Instance Entity which holds the details of SIM (IMSI Number, User Id, Phone Number, Profile) and it's status
SkuDetails	SKU Details for MME, MCC and HSS Nodes. SKU Name, Version, Image File Path, Signature File Path and ARM Template File Path
SkuMapping	Mapping between Network Function SKU and Service Manager SKU
SkuType	SKU Type
SoftwareScheduleDetails	Software Upgrade Schedule Details
SoftwareUpgradeDetails	Software Upgrade Schedule Details
StatusEnum	Device Status Values: UNKNOWN, NOTREGISTERED, REGISTERED, DELETED
SubResource	Reference to another Sub Resource
SubResource	Reference to another Sub Resource
SubscriberTracingEntity	Network Subscriber Tracing Details
SubscriberTracingTemplateEntity	Network Subscriber Tracing Template Details
TroubleshootBundleDetails	Troubleshoot Bundle Details provides detail of its creation, location and name.

Type	Description
TroubleshootScriptResponse	Troubleshoot Script Response is the representation of the output provided by the troubleshootInformationCollector script. It has the collection status and relevant details based on the status
TroubleshootStorageDetails	This class represents the request body to create the troubleshoot bundle
UserDataParameters	User Data Parameters
VendorProvisioningState	Vendor Provisioning State Values: UNKNOWN, NOTPROVISIONED, PROVISIONING, PROVISIONED, DEPROVISIONED, USERDATAVALIDATIONFAILED
VmSwitchTypeEnum	VM Switch Type Values: UNKNOWN, MANAGEMENT, WAN, LAN

ActionStatus

Task Status.

Values: **INPROGRESS, SUCCESS, FAIL**

Values

Value	Description
INPROGRESS	Task is currently Running
SUCCESS	Task completed successfully
FAIL	Task Failed

ASAPOperationStatus

ASAP Operation Status. All ASAP related operations are handled based on this status.

Values: **PROVISIONED, UNPROVISIONED**

Values

Value	Description
PROVISIONED	Profiles are configured
UNPROVISIONED	Profiles are not configured

ASEDevice

Node Resources configuration

Properties

Name	Data Type	Description
Properties inherited from SMDevice		
name	string	Unique Node Name.
mme	MME	MME resource configuration.
mcc	MCC	MCC resource configuration.
hss	HSS	HSS resource configuration.
clusterIndex	number	Unique Cluster Index. Internal Service Manager usage.

Example

```
{
  "name" : "...",
  "mme" : {
    "day0Conf" : {
      "msmVMServiceName" : "...",
      "mrmVMServiceName" : "...",
      "mrmManagement" : "...",
      "msmManagement" : "...",
      "logicalManagement" : "...",
      "managementNetworkAddress" : "...",
      "managementGatewayAddress" : "...",
      "mrmBase" : "...",
      "msmBase" : "...",
      "baseNetworkAddress" : "...",
      "baseGatewayAddress" : "...",
      "mrmData" : "...",
      "msmData" : "...",
      "dataNetworkAddress" : "...",
      "dataGatewayAddress" : "...",
      "ns1" : "...",
      "ns1NetworkAddress" : "...",
      "ns1GatewayAddress" : "...",
      "cloudMMEManagementNetMask" : "...",
      "cloudMMEBaseNetMask" : "...",
      "cloudMMEDataNetMask" : "...",
      "dataTunnelingSupernode0" : "...",
      "dataTunnelingSupernode1" : "...",
      "baseTunnelingSupernode0" : "...",
      "baseTunnelingSupernode1" : "...",
      "s6IP" : "...",
      "clusterIndex" : 12345,
      "managementCIDRSuffix" : "...",
      "lanCIDRSuffix" : "...",
      "wanCIDRSuffix" : "..."}
  },
}
```

```

"day1Conf" : {
  "mmeRealm" : "...",
  "diameterPeer" : "...",
  "diameterPeerIpAddress" : "...",
  "realmPeerMap" : "...",
  "diameterClientInstanceIpAddress" : "...",
  "gtpProfileIpAddress" : "...",
  "s1IfIpAddress" : "...",
  "domainName1" : "...",
  "domainName2" : "...",
  "ntpServerIp" : "...",
  "externalNs1" : "...",
  "mmeVmName" : "...",
  "node1DiameterPeer" : "...",
  "node1DiameterPeerIpAddress" : "...",
  "node1RealmPeerMap" : "...",
  "node1ExternalNs1" : "...",
  "node1MmeVmName" : "...",
  "node1MrmVmName" : "...",
  "nextHopGWIp" : "...",
  "routingInstancesBfd" : "...",
  "tacLowByte" : "...",
  "tacHighByte" : "...",
  "networkName" : "...",
  "shortNetworkName" : "...",
  "apnName" : "...",
  "homePlmnId" : "...",
  "pgwIpAddress" : "...",
  "sgwS11IpAddress" : "..." },
"mmeStatus" : {
  "day0Status" : "FAIL",
  "day0message" : "...",
  "day1Status" : "INPROGRESS",
  "day1message" : "...",
  "optionalNTPServer1Configured" : true,
  "optionalNTPServer2Configured" : true,
  "requiredConfigurationsCompleted" : true
},
"networkFunction" : {
  "id" : "...",
  "name" : "...",
  "type" : "...",
  "location" : "...",
  "properties" : { }
},
"provisioningState" : "ACCEPTED",
"detailedStatusURL" : "...",
"nfPortalName" : "...",
"smName" : "...",
"asapName" : "..."

```

```

    "createCorrelationId" : "...",
  },
  "mcc" : {
    "day0Conf" : {
      "management0" : "...",
      "management1" : "...",
      "management2" : "...",
      "managementNetworkAddress" : "...",
      "managementGatewayAddress" : "...",
      "base0" : "...",
      "base1" : "...",
      "base2" : "...",
      "baseNetworkAddress" : "...",
      "baseGatewayAddress" : "...",
      "ew" : "...",
      "ewNetworkAddress" : "...",
      "ewGatewayAddress" : "...",
      "ns1" : "...",
      "ns1NetworkAddress" : "...",
      "ns1GatewayAddress" : "...",
      "ns2" : "...",
      "ns2NetworkAddress" : "...",
      "ns2GatewayAddress" : "...",
      "cloudBaseInternalCIDR" : "...",
      "clusterIndex" : 12345,
      "managementCIDRSuffix" : "...",
      "lanCIDRSuffix" : "...",
      "wanCIDRSuffix" : "...",
    },
    "day1Conf" : {
      "scRoamingPlmnId" : "...",
      "zoneDefaultQosProfile" : "...",
      "gnInterfaceIpRouteNextHop" : "...",
      "gnInterfaceIpRoutePrefix" : "...",
      "gnInterfaceIpRouteDest" : "...",
      "gnSgwS1" : "...",
      "gnSgwLoopbackIp" : "...",
      "giInterfaceIp" : "...",
      "giInterfaceIpRoutePrefix" : "...",
      "giInterfaceIpRouteDest" : "...",
      "giInterfaceIpRouteNextHop" : "...",
      "giIpSubPoolRangeStart" : "...",
      "giIpSubPoolRangeEnd" : "...",
      "giLoopbackIp" : "...",
      "bgpRouterGN" : { },
      "bgpRouterGI" : { },
      "giV4IpSubPoolRangeStart" : "...",
      "giV4IpSubPoolRangeEnd" : "...",
      "giRemoteIpSubPoolRangeStart" : "...",
      "giRemoteIpSubPoolRangeEnd" : "...",
      "dataLoopIP" : "...",
      "gnInterfaceIp" : "...",
    }
  }
}

```

```

    "dataFabricIpAddress" : "...",
    "dnsPrimaryIp" : "...",
    "dnsSecondaryIp" : "...",
    "pfcPPeerIp" : "...",
    "networkName" : "...",
    "shortNetworkName" : "...",
    "apnName" : "...",
    "homePlmnId" : "...",
    "pgwIpAddress" : "...",
    "sgwS11IpAddress" : "..." },
  "mccStatus" : {
    "day0Status" : "INPROGRESS",
    "day0message" : "...",
    "day1GISStatus" : "FAIL",
    "day1GImessage" : "...",
    "day1GNStatus" : "INPROGRESS",
    "day1GNmessage" : "...",
    "day1SCStatus" : "SUCCESS",
    "day1SCmessage" : "...",
    "day1ServiceStatus" : "INPROGRESS",
    "day1Servicemessage" : "...",
    "day1Status" : "SUCCESS",
    "day1Message" : "...",
    "dataFabricConfigured" : true,
    "optionalNTPServer1Configured" : true,
    "optionalNTPServer2Configured" : true,
    "requiredConfigurationsCompleted" : true,
    "pfcPConfigurationsCompleted" : true,
    "dnsPrimaryConfigured" : true,
    "dnsSecondaryConfigured" : true
  },
  "natProfile" : {
    "configStatus" : "PROVISIONED",
    "status" : "FAIL",
    "message" : "...",
    "natSubPoolRangeStart" : "...",
    "natSubPoolRangeEnd" : "...",
    "profileName" : "..." },
  "edrProfile" : {
    "configStatus" : "UNPROVISIONED",
    "status" : "FAIL",
    "message" : "...",
    "ipAddress" : "...",
    "profileName" : "..." },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : { }
  }

```



```

    },
    "provisioningState" : "CANCELED",
    "detailedStatusURL" : "...",
    "nfPortalName" : "...",
    "smName" : "...",
    "asapName" : "...",
    "createCorrelationId" : "..." },
    "hss" : {
        "day0Conf" : {
            "managementNic1" : "...",
            "managementGateway" : "...",
            "managementNetworkAddress" : "...",
            "lanNic1" : "...",
            "lanGateway" : "...",
            "lanNetworkAddress" : "...",
            "s6MMEIp" : "...",
            "s6MMEGateway" : "...",
            "clusterIndex" : 12345,
            "managementCIDRSuffix" : "...",
            "lanCIDRSuffix" : "...",
            "wanCIDRSuffix" : "..." },
        "hssStatus" : {
            "day0Status" : "SUCCESS",
            "day0message" : "...",
            "auditStatus" : "SUCCESS",
            "auditTime" : 12345
        },
    },
    "networkFunction" : {
        "id" : "...",
        "name" : "...",
        "type" : "...",
        "location" : "...",
        "properties" : { }
    },
    "provisioningState" : "SUCCEEDED",
    "detailedStatusURL" : "...",
    "nfPortalName" : "...",
    "smName" : "...",
    "asapName" : "...",
    "createCorrelationId" : "..." },
    "clusterIndex" : 12345
}

```

ASEDeviceContainer

Nodes configurations deployed in a Network. HA deployment will have one Node and Non-HA will have two Nodes.

Properties

Name	Data Type	Constraints	Description
node0	ASEDevice		Node 0 configurations
regionName	string		Unique Region Name in which the Resources are deployed.
node1	ASEDevice		Node 1 configurations
regionId	number	required	Unique Region Identifier in which the Resources are deployed.

Example

```
{
  "node0" : {
    "name" : "...",
    "mme" : {
      "day0Conf" : { },
      "day1Conf" : { },
      "mmeStatus" : { },
      "networkFunction" : { },
      "provisioningState" : "DELETING",
      "detailedStatusURL" : "...",
      "nfPortalName" : "...",
      "smName" : "...",
      "asapName" : "...",
      "createCorrelationId" : "..."
    },
    "mcc" : {
      "day0Conf" : { },
      "day1Conf" : { },
      "mccStatus" : { },
      "natProfile" : { },
      "edrProfile" : { },
      "networkFunction" : { },
      "provisioningState" : "CANCELED",
      "detailedStatusURL" : "...",
      "nfPortalName" : "...",
      "smName" : "...",
      "asapName" : "...",
      "createCorrelationId" : "..."
    },
    "hss" : {
      "day0Conf" : { },
      "hssStatus" : { },
      "networkFunction" : { },
      "provisioningState" : "CANCELED",
      "detailedStatusURL" : "...",
      "nfPortalName" : "...",

```

```

        "smName" : "...",
        "asapName" : "...",
        "createCorrelationId" : "..."
    },
    "clusterIndex" : 12345
},
"regionName" : "...",
"node1" : {
    "name" : "...",
    "mme" : {
        "day0Conf" : { },
        "day1Conf" : { },
        "mmeStatus" : { },
        "networkFunction" : { },
        "provisioningState" : "FAILED",
        "detailedStatusURL" : "...",
        "nfPortalName" : "...",
        "smName" : "...",
        "asapName" : "...",
        "createCorrelationId" : "..."
    },
    "mcc" : {
        "day0Conf" : { },
        "day1Conf" : { },
        "mccStatus" : { },
        "natProfile" : { },
        "edrProfile" : { },
        "networkFunction" : { },
        "provisioningState" : "FAILED",
        "detailedStatusURL" : "...",
        "nfPortalName" : "...",
        "smName" : "...",
        "asapName" : "...",
        "createCorrelationId" : "..."
    },
    "hss" : {
        "day0Conf" : { },
        "hssStatus" : { },
        "networkFunction" : { },
        "provisioningState" : "DELETING",
        "detailedStatusURL" : "...",
        "nfPortalName" : "...",
        "smName" : "...",
        "asapName" : "...",
        "createCorrelationId" : "..."
    },
    "clusterIndex" : 12345
},
"regionId" : 12345
}

```

AzureConfig

PNSM Configurations for Service Manager communication with Azure and ASAP servers

Properties

Name	Data Type	Constraints	Description
subscriptionId	string	required	Azure Subscription Identifier
vendorName	string	required	Azure Vendor Name
apiVersion	string	required	Azure API version
clientId	string	required	Azure Client Identifier
authority	string	required	Azure Authority
clientSecret	string	required	Azure Client Secret
asapServerIpPort	string	required	ASAP Server IP Address
asapUsername	string	read-only	ASAP Server Username
asapPassword	string	read-only	ASAP Server Password
hlrnerfEntityKey	string	read-only	Hlrnerf Entity Key
hlrnerfAddressDigits	string	read-only	Hlrnerf Address Digits
gsysHlrNumber	string	read-only	Gsys Hlr Number
opValue	string	required	OP Value
provisioningType	string	required	SIM Provisioning Type - OP/OPC

Example

```
{
  "subscriptionId" : "...",
  "vendorName" : "...",
  "apiVersion" : "...",
  "clientId" : "...",
  "authority" : "...",
  "clientSecret" : "...",
  "asapServerIpPort" : "...",
  "asapUsername" : "...",
  "asapPassword" : "...",
  "hlrnerfEntityKey" : "...",
  "hlrnerfAddressDigits" : "...",
  "gsysHlrNumber" : "...",
  "opValue" : "...",
  "provisioningType" : "..."}

```

BGPRouterConfig

BGP configurations for Node's Day 1 configuration.

Properties

Name	Data Type	Description
routerId	string	BGP Router IP Address. Should be same as GI/GN Interface IP Address.
localAs	string	Local AS Numeric value.
remoteAs	string	Remote AS Number value.
neighborIP	string	Neighbor Resource IP Address. Should be same as GI/GN Interface Next-Hop IP Address.
fallOverBFD	string	Fall Over BFD value. Values -Single-Hop/Multi-Hop
bfdMultiHopPeer	string	BFD Multi-Hop Peer IP Address. Should be same as GI/GN Interface Next-Hop IP Address.

Example

```
{
  "routerId" : "...",
  "localAs" : "...",
  "remoteAs" : "...",
  "neighborIP" : "...",
  "fallOverBFD" : "...",
  "bfdMultiHopPeer" : "..."}

```

BGPRouterGI

BGP configurations for Node's Day 1 GI Interface configuration.

Properties

Name	Data Type	Description
bgpGILoopIP	string	BGP GI Loop IP Address.
Properties inherited from BGPRouterConfig		
routerId	string	BGP Router IP Address. Should be same as GI/GN Interface IP Address.
localAs	string	Local AS Numeric value.
remoteAs	string	Remote AS Number value.
neighborIP	string	Neighbor Resource IP Address. Should be same as GI/GN Interface Next-Hop IP Address.

Name	Data Type	Description
fallOverBFD	string	Fall Over BFD value. Values -Single-Hop/Multi-Hop
bfdMultiHopPeer	string	BFD Multi-Hop Peer IP Address. Should be same as GI/GN Interface Next-Hop IP Address.

Example

```
{
  "bgpGILoopIP" : "...",
  "routerId" : "...",
  "localAs" : "...",
  "remoteAs" : "...",
  "neighborIP" : "...",
  "fallOverBFD" : "...",
  "bfdMultiHopPeer" : "..."}

```

BGPRouterGN

BGP configurations for Node's Day 1 GN Interface configuration.

Properties

Name	Data Type	Description
bgpGEOLoopIP	string	BGP GEO Loop IP Address.
bgpGNLoopIP	string	BGP GN Loop IP Address.

Properties inherited from [BGPRouterConfig](#)

routerId	string	BGP Router IP Address. Should be same as GI/GN Interface IP Address.
localAs	string	Local AS Numeric value.
remoteAs	string	Remote AS Number value.
neighborIP	string	Neighbor Resource IP Address. Should be same as GI/GN Interface Next-Hop IP Address.
fallOverBFD	string	Fall Over BFD value. Values -Single-Hop/Multi-Hop
bfdMultiHopPeer	string	BFD Multi-Hop Peer IP Address. Should be same as GI/GN Interface Next-Hop IP Address.

Example

```
{
  "bgpGEOLoopIP" : "...",
  "bgpGNLoopIP" : "...",
  "routerId" : "...",
  "localAs" : "...",

```

```

    "remoteAs" : "...",
    "neighborIP" : "...",
    "fallOverBFD" : "...",
    "bfdMultiHopPeer" : "..."
  }

```

CloudContainer

Nodes configurations deployed in a Cloud. HA deployment will have one Node and Non-HA will have two Nodes.

Properties

Name	Data Type	Constraints	Descriptions
regionName	string		Unique Region Name in which the Resources are deployed.
storageAccountName	string		the storageAccountName
regionId	number	required	Unique Region Identifier in which the Resources are deployed.
storageAccountContainerName	string		the storageAccountContainerName
resourceGroupName	string		Azure Resource group name for resource deployment.
vnetName	string		the vnetName
node0	CloudDevice		Node 0 configurations in Cloud
node1	CloudDevice		Node 1 configurations in Cloud

Example

```

{
  "regionName" : "...",
  "storageAccountName" : "...",
  "regionId" : 12345,
  "storageAccountContainerName" : "...",
  "resourceGroupName" : "...",
  "vnetName" : "...",
  "node0" : {
    "commonResources" : {
      "name" : "...",
    },
    "name" : "...",
    "mme" : {
      "day0Conf" : { },
      "day1Conf" : { },
      "mmeStatus" : { },
      "networkFunction" : { },
      "provisioningState" : "DELETED",
    }
  }
}

```

```

    "detailedStatusURL" : "...",
    "nfPortalName" : "...",
    "smName" : "...",
    "asapName" : "...",
    "createCorrelationId" : "..."    },
  "mcc" : {
    "day0Conf" : { },
    "day1Conf" : { },
    "mccStatus" : { },
    "natProfile" : { },
    "edrProfile" : { },
    "networkFunction" : { },
    "provisioningState" : "SUCCEEDED",
    "detailedStatusURL" : "...",
    "nfPortalName" : "...",
    "smName" : "...",
    "asapName" : "...",
    "createCorrelationId" : "..."    },
  "hss" : {
    "day0Conf" : { },
    "hssStatus" : { },
    "networkFunction" : { },
    "provisioningState" : "DELETING",
    "detailedStatusURL" : "...",
    "nfPortalName" : "...",
    "smName" : "...",
    "asapName" : "...",
    "createCorrelationId" : "..."    },
  "clusterIndex" : 12345
},
"node1" : {
  "commonResources" : {
    "name" : "..."    },
  "name" : "...",
  "mme" : {
    "day0Conf" : { },
    "day1Conf" : { },
    "mmeStatus" : { },
    "networkFunction" : { },
    "provisioningState" : "CANCELED",
    "detailedStatusURL" : "...",
    "nfPortalName" : "...",
    "smName" : "...",
    "asapName" : "...",
    "createCorrelationId" : "..."    },
  "mcc" : {
    "day0Conf" : { },
    "day1Conf" : { },
    "mccStatus" : { },
    "natProfile" : { },

```



```

    "edrProfile" : { },
    "networkFunction" : { },
    "provisioningState" : "CANCELED",
    "detailedStatusURL" : "...",
    "nfPortalName" : "...",
    "smName" : "...",
    "asapName" : "...",
    "createCorrelationId" : "..." },
    "hss" : {
        "day0Conf" : { },
        "hssStatus" : { },
        "networkFunction" : { },
        "provisioningState" : "CANCELED",
        "detailedStatusURL" : "...",
        "nfPortalName" : "...",
        "smName" : "...",
        "asapName" : "...",
        "createCorrelationId" : "..." },
    "clusterIndex" : 12345
}
}

```

CloudDevice

Node Resources configuration in Cloud

Properties

Name	Data Type	Description
Properties inherited from SMDevice		
name	string	Unique Node Name.
mme	MME	MME resource configuration.
mcc	MCC	MCC resource configuration.
hss	HSS	HSS resource configuration.
clusterIndex	number	Unique Cluster Index. Internal Service Manager usage.

Example

```

{
  "commonResources" : {
    "name" : "...",
  },
  "name" : "...",
  "mme" : {
    "day0Conf" : {
      "msmVMServiceName" : "...",
      "mrmVMServiceName" : "...",
    }
  }
}

```

```

"mrmManagement" : "...",
"msmManagement" : "...",
"logicalManagement" : "...",
"managementNetworkAddress" : "...",
"managementGatewayAddress" : "...",
"mrmBase" : "...",
"msmBase" : "...",
"baseNetworkAddress" : "...",
"baseGatewayAddress" : "...",
"mrmData" : "...",
"msmData" : "...",
"dataNetworkAddress" : "...",
"dataGatewayAddress" : "...",
"ns1" : "...",
"ns1NetworkAddress" : "...",
"ns1GatewayAddress" : "...",
"cloudMMEManagementNetMask" : "...",
"cloudMMEBaseNetMask" : "...",
"cloudMMEDataNetMask" : "...",
"dataTunnelingSupernode0" : "...",
"dataTunnelingSupernode1" : "...",
"baseTunnelingSupernode0" : "...",
"baseTunnelingSupernode1" : "...",
"s6IP" : "...",
"clusterIndex" : 12345,
"mcc" : "...",
"mnc" : "...",
"managementCIDRSuffix" : "...",
"lanCIDRSuffix" : "...",
"wanCIDRSuffix" : "...",
},
"day1Conf" : {
  "mmeRealm" : "...",
  "diameterPeer" : "...",
  "diameterPeerIpAddress" : "...",
  "realmPeerMap" : "...",
  "diameterClientInstanceIpAddress" : "...",
  "gtpProfileIpAddress" : "...",
  "s1IfIpAddress" : "...",
  "domainName1" : "...",
  "domainName2" : "...",
  "ntpServerIp" : "...",
  "externalNs1" : "...",
  "mmeVmName" : "...",
  "node1DiameterPeer" : "...",
  "node1DiameterPeerIpAddress" : "...",
  "node1RealmPeerMap" : "...",
  "node1ExternalNs1" : "...",
  "node1MmeVmName" : "...",
  "node1MrmVmName" : "...",

```

```

    "nextHopGWIp" : "...",
    "routingInstancesBfd" : "...",
    "tacLowByte" : "...",
    "tacHighByte" : "...",
    "networkName" : "...",
    "shortNetworkName" : "...",
    "apnName" : "...",
    "homePlmnId" : "...",
    "pgwIpAddress" : "...",
    "sgwS11IpAddress" : "...",
  },
  "mmeStatus" : {
    "day0Status" : "INPROGRESS",
    "day0message" : "...",
    "day1Status" : "FAIL",
    "day1message" : "...",
    "optionalNTPServer1Configured" : true,
    "optionalNTPServer2Configured" : true,
    "requiredConfigurationsCompleted" : true
  },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : { }
  },
  "provisioningState" : "FAILED",
  "detailedStatusURL" : "...",
  "nfPortalName" : "...",
  "smName" : "...",
  "asapName" : "...",
  "createCorrelationId" : "...",
},
"mcc" : {
  "day0Conf" : {
    "management0" : "...",
    "management1" : "...",
    "management2" : "...",
    "managementNetworkAddress" : "...",
    "managementGatewayAddress" : "...",
    "base0" : "...",
    "base1" : "...",
    "base2" : "...",
    "baseNetworkAddress" : "...",
    "baseGatewayAddress" : "...",
    "ew" : "...",
    "ewNetworkAddress" : "...",
    "ewGatewayAddress" : "...",
    "ns1" : "...",

```

```

    "ns1NetworkAddress" : "...",
    "ns1GatewayAddress" : "...",
    "ns2" : "...",
    "ns2NetworkAddress" : "...",
    "ns2GatewayAddress" : "...",
    "cloudBaseInternalCIDR" : "...",
    "clusterIndex" : 12345,
    "mcc" : "...",
    "mnc" : "...",
    "managementCIDRSuffix" : "...",
    "lanCIDRSuffix" : "...",
    "wanCIDRSuffix" : "...",
    "ntpIp" : "..." },
    "day1Conf" : {
        "scRoamingPlmnId" : "...",
        "zoneDefaultQosProfile" : "...",
        "gnInterfaceIpRouteNextHop" : "...",
        "gnInterfaceIpRoutePrefix" : "...",
        "gnInterfaceIpRouteDest" : "...",
        "gnSgwS1" : "...",
        "gnSgwLoopbackIp" : "...",
        "giInterfaceIp" : "...",
        "giInterfaceIpRoutePrefix" : "...",
        "giInterfaceIpRouteDest" : "...",
        "giInterfaceIpRouteNextHop" : "...",
        "giIpSubPoolRangeStart" : "...",
        "giIpSubPoolRangeEnd" : "...",
        "giLoopbackIp" : "...",
        "bgpRouterGN" : { },
        "bgpRouterGI" : { },
        "giV4IpSubPoolRangeStart" : "...",
        "giV4IpSubPoolRangeEnd" : "...",
        "giRemoteIpSubPoolRangeStart" : "...",
        "giRemoteIpSubPoolRangeEnd" : "...",
        "dataLoopIP" : "...",
        "gnInterfaceIp" : "...",
        "dataFabricIpAddress" : "...",
        "dnsPrimaryIp" : "...",
        "dnsSecondaryIp" : "...",
        "pfpcPeerIp" : "...",
        "networkName" : "...",
        "shortNetworkName" : "...",
        "apnName" : "...",
        "homePlmnId" : "...",
        "pgwIpAddress" : "...",
        "sgwS11IpAddress" : "..."
    },
    "mccStatus" : {
        "day0Status" : "FAIL",
        "day0message" : "..."
    }

```

```

    "day1GIStatus" : "INPROGRESS",
    "day1GImessage" : "...",
    "day1GNStatus" : "INPROGRESS",
    "day1GNmessage" : "...",
    "day1SCStatus" : "SUCCESS",
    "day1SCmessage" : "...",
    "day1ServiceStatus" : "SUCCESS",
    "day1Servicemessage" : "...",
    "day1Status" : "SUCCESS",
    "day1Message" : "...",
    "dataFabricConfigured" : true,
    "optionalNTPServer1Configured" : true,
    "optionalNTPServer2Configured" : true,
    "requiredConfigurationsCompleted" : true,
    "pfcpcConfigurationsCompleted" : true,
    "dnsPrimaryConfigured": true,
    "dnsSecondaryConfigured" : true
  },
  "natProfile" : {
    "configStatus" : "UNPROVISIONED",
    "status" : "INPROGRESS",
    "message" : "...",
    "natSubPoolRangeStart" : "...",
    "natSubPoolRangeEnd" : "...",
    "profileName" : "..."
  },
  "edrProfile" : {
    "configStatus" : "PROVISIONED",
    "status" : "INPROGRESS",
    "message" : "...",
    "ipAddress" : "...",
    "profileName" : "..."
  },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : { }
  },
  "provisioningState" : "ACCEPTED",
  "detailedStatusURL" : "...",
  "nfPortalName" : "...",
  "smName" : "...",
  "asapName" : "...",
  "createCorrelationId" : "..."
},
"hss" : {
  "day0Conf" : {
    "managementNic1" : "...",

```

```

    "managementGateway" : "...",
    "managementNetworkAddress" : "...",
    "lanNic1" : "...",
    "lanGateway" : "...",
    "lanNetworkAddress" : "...",
    "s6MMEIp" : "...",
    "s6MMEGateway" : "...",
    "clusterIndex" : 12345,
    "managementCIDRSuffix" : "...",
    "lanCIDRSuffix" : "...",
    "wanCIDRSuffix" : "...",
  },
  "hssStatus" : {
    "day0Status" : "FAIL",
    "day0message" : "...",
    "auditStatus" : "SUCCESS",
    "auditTime" : 12345
  },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : { }
  },
  "provisioningState" : "CANCELED",
  "detailedStatusURL" : "...",
  "nfPortalName" : "...",
  "smName" : "...",
  "asapName" : "...",
  "createCorrelationId" : "..."
},
"clusterIndex" : 12345
}

```

CommonResources

Properties

name	data type	description
name	string	the name

Example

```

{
  "name" : "..."}

```

ConfigBackupResponseVO

Backup task response details

Properties

Name	Data Type	Constraints	Description
status	string	read-only	Backup task status
elementName	string	read-only	Network Element Name to which the Backup task has been triggered
ipaddress	string	read-only	Network Element IP Address to which the Backup task has been triggered
message	string	read-only	Backup task status message

Example

```
{
  "status" : "...",
  "elementName" : "...",
  "ipaddress" : "...",
  "message" : "..."}

```

ConfigurationFileEntity

PNSM Configuration Details

Properties

Name	Data Type	Constraints	Description
configurationfileid	number	required	Unique PNSM Configuration Identifier
file	string		PNSM Configuration data in String format

Properties inherited from [EJBBaseEntity](#)

version	number	required	
---------	--------	----------	--

Example

```
{
  "configurationfileid" : 12345,
  "file" : "...",
  "version" : 12345
}
```

Day0Base

Node Day 0 Base Configurations which are common for MME, MCC and HSS Resources Day 0 Configurations

Properties

Name	Data Type	Constraints	Description
clusterIndex	number	required	Unique Cluster Index. Internal Service Manager usage.
managementCIDRSuffix	string		Management Subnet Prefix. Value will be set from the Location selected while Network deployment.
lanCIDRSuffix	string		LAN Subnet Prefix. Value will be set from the Location selected while Network deployment.
wanCIDRSuffix	string		WAN Subnet Prefix. Value will be set from the Location selected while Network deployment.

Example

```
{
  "clusterIndex" : 12345,
  "managementCIDRSuffix" : "...",
  "lanCIDRSuffix" : "...",
  "wanCIDRSuffix" : "...",
}
```

Day1Base

Node Day 1 Base Configurations which are common for MME and MCC Resources Day 1 Configurations

Properties

Name	Data Type	Description
networkName	string	Unique Network Name.
shortNetworkName	string	Shorter Name for the Network.
apnName	string	APN Name.
homePlmnId	string	Home PLMN Identifier.
pgwIpAddress	string	PGateway IP Address
sgwS11IpAddress	string	SGateway S11 IP Address

Example

```
{
  "networkName" : "...",
  "shortNetworkName" : "...",
}
```



```

    "apnName" : "...",
    "homePlmnId" : "...",
    "pgwIpAddress" : "...",
    "sgwS11IpAddress" : "..."
  }

```

Day2InitialConfigStatus

Node Day 2 Initial Configuration Status

Properties

Name	Data Type	Description
day2InitialConfigStatusid	number	Unique Node Day 2 Initial Configuration Identifier
resourceId	number	Unique Resource Identifier
status	string	Node Day 2 Initial Configuration Status
comment	string	Node Day 2 Initial Configuration Comment
Properties inherited from EJBBaseEntity		
version	number	

Example

```

{
  "day2InitialConfigStatusid" : 12345,
  "resourceId" : 12345,
  "status" : "...",
  "comment" : "...",
  "version" : 12345
}

```

DeploymentMode

Network deployment mode.

Values: **ONPREM**, **ONPREMHA**, **CUPS**, **CUPSHA**, **CLOUD**, **CLOUDHA**

Values

Value	Description
ONPREM	Enterprise ON Premises Non HA deployment
ONPREMHA	Enterprise ON Premises HA deployment
CUPS	Control Plane User Plane separation non HA deployment.
CUPSHA	Control Plane User Plane separation HA deployment.
CLOUD	All resources are deployed in cloud as non HA deployment.
CLOUDHA	All resources are deployed in cloud as HA deployment.

Device

Device Details

Properties

Name	Data Type	Description
id	string	
name	string	
type	string	
properties		

Example

```
{
  "id" : "...",
  "name" : "...",
  "type" : "...",
  "properties" : {
    "status" : "UNKNOWN",
    "provisioningState" : "UNKNOWN",
    "deviceType" : "UNKNOWN",
    "azureStackEdge" : {
      "id" : "..."
    },
    "networkFunctions" : [ {
      "id" : "..."
    }, {
      "id" : "..."
    } ]
  }
}
```

Device Properties Format

Device properties.

Properties

Name	Data Type	Description
status	StatusEnum	
provisioningState	ProvisioningState	
deviceType	DeviceTypeEnum	
azureStackEdge	SubResource	
networkFunctions	array of SubResource	

Example

```

{
  "status" : "NOTREGISTERED",
  "provisioningState" : "DELETED",
  "deviceType" : "UNKNOWN",
  "azureStackEdge" : {
    "id" : "..."
  },
  "networkFunctions" : [ {
    "id" : "..."
  }, {
    "id" : "..."
  } ]
}

```

DeviceTypeEnum

Device Type.

Values: **UNKNOWN**, **AZURESTACKEDGE**

Values

Value	Description
UNKNOWN	Device Type is Unknown
AZURESTACKEDGE	Device Type is AzureStackEdge

EJBBBaseEntity

Properties

Name	Data Type	Constraints	Description
version	number	required	

Example

```

{
  "version" : 12345
}

```

FileServer

File Server to import Software Packages, perform Network Upgrades and perform Backup/Restore. By default Service Manager will have its own local FTP Server which cannot be Updated/Deleted. User can Create SFTP servers to Import the Software Packages. User created SFTP servers can be Updated/Deleted

Properties

Name	Data Type	Constraints	Description
id	string	read-only	Unique File Server Identifier. This value cannot be updated
protocol	string	read-only	Communication Protocol used to transfer the files. This value cannot be updated
name	string	required	Unique File Server Name
ipAddress	string	required	Unique File Server IP Address. This value cannot be updated
port	string	required	Communication Endpoint
username	string	required	Username to communicate with the File Server
password	string	required	Password to communicate with the File Server
confirmPassword	string	required	Confirm Password to confirm/validate the given Password
saveInactiveServer	boolean	required	If TRUE, File Server will be Created/Updated despite of the status (Active/Inactive). If FALSE, File Server will be Created/Updated only if the File Server status is Active. Active: File Server is Reachable and the service is UP and Running. Inactive: File Server is Not Reachable or the service is Down
status	boolean	required, read-only	TRUE if the File Server is Reachable and the service is UP and Running. FALSE if the File Server is Not Reachable or the service is Down. This value cannot be updated

Example

```

{
  "id" : "...",
  "protocol" : "...",
  "name" : "...",
  "ipAddress" : "...",
  "port" : "...",
  "username" : "...",
  "password" : "...",
  "confirmPassword" : "...",
  "saveInactiveServer" : true,
  "status" : true
}

```

Ftpprotocol

Properties

Name	Data Type	Constraints	Description
secondaryftpusername	string		
port	number	required	
secondaryftppassword	string		
ftpservername	string		
primaryftpipaddress	string		
primaryftppassword	string		
default	boolean	required	
primaryftpusername	string		
protocolparamsid	number	required	
protocolType	string		
status	boolean	required	
secondaryftpipaddress	string		

Properties inherited from [EJBBaseEntity](#)

version	number	required	
---------	--------	----------	--

Example

```
{
  "secondaryftpusername" : "...",
  "port" : 12345,
  "secondaryftppassword" : "...",
  "ftpservername" : "...",
  "primaryftpipaddress" : "...",
  "primaryftppassword" : "...",
  "default" : true,
  "primaryftpusername" : "...",
  "protocolparamsid" : 12345,
  "protocolType" : "...",
  "status" : true,
  "secondaryftpipaddress" : "...",
  "version" : 12345
}
```

GlobalNF

Global Network Function Details

Properties

Name	Data Type	Description
resourceName	string	Unique Global Network Function Name
resourceIpAddress	string	Unique Global Network Function IP Address

Example

```
{
  "resourceName" : "...",
  "resourceIpAddress" : "..."
}
```

HSS

HSS resource configurations.

Properties

Name	Data Type	Description
day0Conf	HSSDay0Conf	HSS Day 0 configurations.
hssStatus	HSSStatus	HSS Status.

Properties inherited from [NFBase](#)

networkFunction	NetworkFunction	Gets the network function. Complete Azure configuration object. Mainly for service Manager internal use
provisioningState	ProvisioningState	Gets the provisioning state. Provisioning status of individual resources like HSS, MME and MCC. for Service manager internal use
detailedStatusURL	string	Gets the detailed status URL. Url to fetch the detailed error message when the deployment failed in the Azure portal.
nfPortalName	string	Gets the nf portal name. Name of the resource in Azure portal
smName	string	Gets the sm name. Name of the resource in Service Manager left side tree view
asapName	string	Gets the asap name. Name of the resource in ASAP server
createCorrelationId	string	Gets the creates the correlation id.

Example

```
{
  "day0Conf" : {
```

```

    "managementNic1" : "...",
    "managementGateway" : "...",
    "managementNetworkAddress" : "...",
    "lanNic1" : "...",
    "lanGateway" : "...",
    "lanNetworkAddress" : "...",
    "s6MMEIp" : "...",
    "s6MMEGateway" : "...",
    "clusterIndex" : 12345,
    "managementCIDRSuffix" : "...",
    "lanCIDRSuffix" : "...",
    "wanCIDRSuffix" : "...",
  },
  "hssStatus" : {
    "day0Status" : "SUCCESS",
    "day0message" : "...",
    "auditStatus" : "FAIL",
    "auditTime" : 12345
  },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : {
      "provisioningState" : "PROVISIONING",
      "device" : { },
      "skuName" : "...",
      "skuType" : "FIREWALL",
      "vendorName" : "...",
      "serviceKey" : "...",
      "vendorProvisioningState" : "PROVISIONING",
      "managedApplication" : { },
      "managedApplicationParameters" : { },
      "networkFunctionContainerConfiguration" : { },
      "networkFunctionUserConfigurations" : [ { }, { } ]
    }
  },
  "provisioningState" : "DELETING",
  "detailedStatusURL" : "...",
  "nfPortalName" : "...",
  "smName" : "...",
  "asapName" : "...",
  "createCorrelationId" : "..."
}

```

HSSDay0Conf

HSS Day 0 configurations.

Properties

Name	Data Type	Description
managementNic1	string	Gets the HSS management IP address.
managementGateway	string	Gets the HSS management gateway address.
managementNetworkAddress	string	Gets the HSS management network address.
lanNic1	string	Gets the HSS lan interface IP.
lanGateway	string	Gets the HSS lan gateway address.
lanNetworkAddress	string	Gets the HSS lan network address.
s6MMEIp	string	Gets the s6 MME ip. This IP is MME s6 IP.
s6MMEGateway	string	Gets the s6 MME gateway.
Properties inherited from Day0Base		
clusterIndex	number	Unique Cluster Index. Internal Service Manager usage.
managementCIDRSuffix	string	Management Subnet Prefix. Value will be set from the Location selected while Network deployment.
lanCIDRSuffix	string	LAN Subnet Prefix. Value will be set from the Location selected while Network deployment.
wanCIDRSuffix	string	WAN Subnet Prefix. Value will be set from the Location selected while Network deployment.

Example

```
{
  "managementNic1" : "...",
  "managementGateway" : "...",
  "managementNetworkAddress" : "...",
  "lanNic1" : "...",
  "lanGateway" : "...",
  "lanNetworkAddress" : "...",
  "s6MMEIp" : "...",
  "s6MMEGateway" : "...",
  "clusterIndex" : 12345,
  "managementCIDRSuffix" : "...",
  "lanCIDRSuffix" : "...",
  "wanCIDRSuffix" : "...",
}
```

HSSStatus

HSS Day 0 status and comment.

Properties

Name	Data Type	Constraints	Description
day0Status	ActionStatus		Gets the deployment status. ActionStatus refers Success/Fail
day0message	string		Gets the day 0 message. Contains detailed message of the deployment.
auditStatus	ActionStatus		Get HSS Last Audit Status
auditTime	number	required	Get HSS Last Audit Time

Example

```
{
  "day0Status" : "SUCCESS",
  "day0message" : "...",
  "auditStatus" : "INPROGRESS"
  "auditTime" : 12345
}
```

IpAllocationMethodEnum

IP Address allocation method.

Values: UNKNOWN, STATIC, DYNAMIC

Values

Value	Description
UNKNOWN	IP Address allocation Unknown
STATIC	IP Address Static allocation
DYNAMIC	IP Address Dynamic allocation

IpVersionEnum

IP Address version.

Values: UNKNOWN, IPV4

Values

Value	Description
UNKNOWN	IP Address version Unknown
IPV4	IPv4 Address

LANIPAddress

LAN IP Addresses reserved/assigned for the Network

Properties

Name	Data Type	Description
lanCIDR	string	LAN Subnet CIDR
lanSubnetPrefix	string	LAN Subnet Prefix
lanGatewayIPAddress	string	LAN Gateway IP Address
lanNetMaskIPAddress	string	LAN NetMask IP Address
nfLANIPAddress	string	Net Foundry LAN IP Address

Example

```
{
  "lanCIDR" : "...",
  "lanSubnetPrefix" : "...",
  "lanGatewayIPAddress" : "...",
  "lanNetMaskIPAddress" : "...",
  "nfLANIPAddress" : "..."}

```

Location

Location to deploy Networks. Location's Management, LAN, WAN Subnets and Gateway IPs will be used for the Networks deployed in the Location

Properties

Name	Data Type	Constraints	Description
name	string		Unique Location Name.
displayName	string		Unique Location Display Name.
HADeploymentMode	boolean	required	TRUE if the Network deployment is HA Mode. FALSE if the Network deployment is Non-HA Mode.
mgmtsubnet	string		Management Subnet.
lansubnet	string		LAN Subnet.
wansubnet	string		WAN Subnet.
locationNetworkConfig	LocationNetworkConfig		List of deployed Networks and assigned Subnets mapping. List of assigned and available Subnet slots for Network deployments. This value cannot be updated.

Name	Data Type	Constraints	Description
asedeviceContainers	array of ASEDeviceContainer		Nodes configurations deployed in a Network. HA deployment will have one Node and Non-HA will have two Nodes.
useSystemGeneratedIPs	boolean	required	TRUE if the Location has Network deployed with System Generated IPs. FALSE if the Location has Network deployed with Custom IPs. Default to NULL if the Location does not have any Networks deployed. This value cannot be updated
mgmtgw	string		Gateway IP to be used for Management Subnet.
langw	string		Gateway IP to be used for LAN Subnet.
wangw	string		Gateway IP to be used for WAN Subnet.

Example

```
{
  "name" : "...",
  "displayName" : "...",
  "HADeploymentMode" : true,
  "mgmtsubnet" : "...",
  "lansubnet" : "...",
  "wansubnet" : "...",
  "locationNetworkConfig" : {
    "nwIDSubnetMap" : {
      "property1" : { },
      "property2" : { }
    },
    "mgmtSubnet" : {
      "property1" : { },
      "property2" : { }
    },
    "lanSubnet" : {
      "property1" : { },
      "property2" : { }
    },
    "wanSubnet" : {
      "property1" : { },
      "property2" : { }
    }
  },
  "networkList" : {
```

```

        "property1" : "...",
        "property2" : "...",
    }, {
        "property1" : "...",
        "property2" : "...",
    } ],
    "aseNetworkDeploymentConfig" : {
        "property1" : {
            "networkIndex" : 12345
            "aseDeviceName" : "...",
            "HADeployment" : true,
            "primaryNode" : true,
            "subnetIndex" : "...",
            "mgmtSubnet" : "...",
            "lanSubnet" : "...",
            "wanSubnet" : "...",
            "managementIPAddress" : "{ }"
            "lanIPAddress" : { }
        },
        "property2" : {
            "networkIndex" : 12345
            "aseDeviceName" : "...",
            "HADeployment" : true,
            "primaryNode" : true,
            "subnetIndex" : "...",
            "mgmtSubnet" : "...",
            "lanSubnet" : "...",
            "wanSubnet" : "...",
            "managementIPAddress" : "{ }"
            "lanIPAddress" : { }
        }
    }
},
    "asedeviceContainers" : [ {
        "node0" : {
            "name" : "...",
            "mme" : { },
            "mcc" : { },
            "hss" : { },
            "clusterIndex" : 12345
        },
        "regionName" : "...",
        "node1" : {
            "name" : "...",
            "mme" : { },
            "mcc" : { },
            "hss" : { },
            "clusterIndex" : 12345
        },
        "regionId" : 12345
    }

```

```

    }, {
      "node0" : {
        "name" : "...",
        "mme" : { },
        "mcc" : { },
        "hss" : { },
        "clusterIndex" : 12345
      },
      "regionName" : "...",
      "node1" : {
        "name" : "...",
        "mme" : { },
        "mcc" : { },
        "hss" : { },
        "clusterIndex" : 12345
      },
      "regionId" : 12345
    } ],
    "useSystemGeneratedIPs" : true,
    "mgmtgw" : "...",
    "langw" : "...",
    "wangw" : "..."
  }

```

LocationEntity

Location to deploy Networks. Location's Management, LAN, WAN Subnets and Gateway IPs will be used for the Networks deployed in the Location.

Properties

Name	Data Type	Constraints	Description
locationid	number	required, read-only	Unique Location Identifier. This value cannot be updated
name	string	required	Unique Location Name. This value cannot be updated
mgmtsubnet	string	required	Management Subnet
lansubnet	string	required	LAN Subnet
wansubnet	string	required	WAN Subnet
publickey	string	read-only	Public Key to access the Networks deployed in the Location. This value cannot be updated
privatekey	string	read-only	Private Key to access the Networks deployed in the Location. This value cannot be updated

Name	Data Type	Constraints	Description
networkConfig	LocationNetworkConfig	read-only	List of deployed Networks and assigned Subnets mapping. List of assigned and available Subnet slots for Network deployments. This value cannot be updated
useSystemGeneratedIPs	boolean	read-only	TRUE if the Location has Network deployed with System Generated IPs. FALSE if the Location has Network deployed with Custom IPs. Default to NULL if the Location does not have any Networks deployed. This value cannot be updated
mgmtgateway	string		Gateway IP to be used for Management Subnet
langateway	string		Gateway IP to be used for LAN Subnet
wangateway	string		Gateway IP to be used for WAN Subnet
Properties inherited from EJBBaseEntity			
version	number	required	

Properties

Name	Data Type	Constraints	Description
locationid	number	required, read-only	Unique Location Identifier. This value cannot be updated
name	string	required	Unique Location Name. This value cannot be updated
mgmtsubnet	string	required	Management Subnet
lansubnet	string	required	LAN Subnet
wansubnet	string	required	WAN Subnet
publickey	string	read-only	Public Key to access the Networks deployed in the Location. This value cannot be updated
privatekey	string	read-only	Private Key to access the Networks deployed in the Location. This value cannot be updated

Name	Data Type	Constraints	Description
networkConfig	LocationNetworkConfig	read-only	List of deployed Networks and assigned Subnets mapping. List of assigned and available Subnet slots for Network deployments. This value cannot be updated
useSystemGeneratedIPs	boolean	read-only	TRUE if the Location has Network deployed with System Generated IPs. FALSE if the Location has Network deployed with Custom IPs. Default to NULL if the Location does not have any Networks deployed. This value cannot be updated
mgmtgateway	string		Gateway IP to be used for Management Subnet
langateway	string		Gateway IP to be used for LAN Subnet
wangateway	string		Gateway IP to be used for WAN Subnet
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```

{
  "locationid" : 12345,
  "name" : "...",
  "mgmtsubnet" : "...",
  "lansubnet" : "...",
  "wansubnet" : "...",
  "publickey" : "...",
  "privatekey" : "...",
  "networkConfig" : {
    "nwIDSubnetMap" : {
      "property1" : { },
      "property2" : { }
    },
    "mgmtSubnet" : {
      "property1" : { },
      "property2" : { }
    },
    "lanSubnet" : {
      "property1" : { },
      "property2" : { }
    }
  },

```

```

    "wanSubnet" : {
      "property1" : { },
      "property2" : { }
    },
    "networkList" : [ {
      "property1" : "...",
      "property2" : "...",    }, {
      "property1" : "...",
      "property2" : "...",    } ],
    "aseNetworkDeploymentConfig" : {
      "property1" : {
        "networkIndex" : 12345,
        "aseDeviceName" : "...",
        "HADeployment" : true,
        "primaryNode" : true,
        "subnetIndex" : "...",
        "mgmtSubnet" : "...",
        "lanSubnet" : "...",
        "wanSubnet" : "...",
        "managementIPAddress" : { },
        "lanIPAddress" : { }
      },
      "property2" : {
        "networkIndex" : 12345,
        "aseDeviceName" : "...",
        "HADeployment" : true,
        "primaryNode" : true,
        "subnetIndex" : "...",
        "mgmtSubnet" : "...",
        "lanSubnet" : "...",
        "wanSubnet" : "...",
        "managementIPAddress" : { },
        "lanIPAddress" : { }
      }
    }
  },
  "useSystemGeneratedIPs" : true,
  "mgmtgateway" : "...",
  "langateway" : "...",
  "wangateway" : "...",
  "version" : 12345
}

```

LocationNetworkConfig

List of deployed Networks and assigned Subnets mapping. List of assigned and available Subnet slots for Network deployments

Properties

Name	Data Type	Description
nwIDSubnetMap	map of object	List of deployed Networks and assigned Subnets mapping
mgmtSubnet	map of object	List of assigned and available Management Subnet slots for Network deployments
lanSubnet	map of object	List of assigned and available LAN Subnet slots for Network deployments
wanSubnet	map of object	List of assigned and available WAN Subnet slots for Network deployments
networkList	array of map of string	List of HA Networks mapped against the ASE Device Names that can be deployed in the Location
aseNetworkDeploymentConfig	map of NetworkDeploymentConfig	the aseNetworkDeploymentConfig

Example

```

{
  "nwIDSubnetMap" : {
    "property1" : { },
    "property2" : { }
  },
  "mgmtSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "lanSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "wanSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "networkList" : [ {
    "property1" : "...",
    "property2" : "..."
  }, {
    "property1" : "...",
    "property2" : "..."
  } ],
  "aseNetworkDeploymentConfig" : {
    "property1" : {
      "networkIndex" : 12345,
      "aseDeviceName" : "...",
      "HADeployment" : true,

```

```

    "primaryNode" : true,
    "subnetIndex" : "...",
    "mgmtSubnet" : "...",
    "lanSubnet" : "...",
    "wanSubnet" : "...",
    "managementIPAddress" : {
        "mgmtCIDR" : "...",
        "mgmtSubnetPrefix" : "...",
        "mgmtGatewayIPAddress" : "...",
        "mgmtNetMaskIPAddress" : "...",
        "mrmManagementIPAddress" : "...",
        "msmManagementIPAddress" : "...",
        "mmeLogicalManagementIPAddress" : "...",
        "mccManagement0IPAddress" : "...",
        "mccManagement1IPAddress" : "...",
        "mccManagement2IPAddress" : "...",
        "hssManagementNic1IPAddress" : "...",
        "nfManagementIPAddress" : "..."
    },
    "lanIPAddress" : {
        "lanCIDR" : "...",
        "lanSubnetPrefix" : "...",
        "lanGatewayIPAddress" : "...",
        "lanNetMaskIPAddress" : "...",
        "nfLANIPAddress" : "..."
    }
}, "property2" : {
    "networkIndex" : 12345,
    "aseDeviceName" : "...",
    "HADeployment" : true,
    "primaryNode" : true,
    "subnetIndex" : "...",
    "mgmtSubnet" : "...",
    "lanSubnet" : "...",
    "wanSubnet" : "...",
    "managementIPAddress" : {
        "mgmtCIDR" : "...",
        "mgmtSubnetPrefix" : "...",
        "mgmtGatewayIPAddress" : "...",
        "mgmtNetMaskIPAddress" : "...",
        "mrmManagementIPAddress" : "...",
        "msmManagementIPAddress" : "...",
        "mmeLogicalManagementIPAddress" : "...",
        "mccManagement0IPAddress" : "...",
        "mccManagement1IPAddress" : "...",
        "mccManagement2IPAddress" : "...",
        "hssManagementNic1IPAddress" : "...",
        "nfManagementIPAddress" : "..."
    },
    "lanIPAddress" : {

```

```

        "lanCIDR" : "...",
        "lanSubnetPrefix" : "...",
        "lanGatewayIPAddress" : "...",
        "lanNetMaskIPAddress" : "...",
        "nfLANIPAddress" : "..."
    }
}
}
}

```

ManagedApplicationParameters

Managed Applications Parameters.

Properties

Name	Data Type	Description
------	-----------	-------------

Example

```
{ }
```

ManagementIPAddress

Management IP Addresses reserved/assigned for the Network

Properties

Name	Data Type	Description
mgmtCIDR	string	Management Subnet CIDR
mgmtSubnetPrefix	string	Management Subnet Prefix
mgmtGatewayIPAddress	string	Management Gateway IP Address
mgmtNetMaskIPAddress	string	Management NetMask IP Address
mrmManagementIPAddress	string	MME MRM Management IP Address
msmManagementIPAddress	string	MME MSM Management IP Address
mmeLogicalManagementIPAddress	string	MME Logical Management IP Address
mccManagement0IPAddress	string	MCC Management IP Address
mccManagement1IPAddress	string	MCC Peer Management IP Address
mccManagement2IPAddress	string	MCC Logical Management IP address
hssManagementNic1IPAddress	string	HSS Management IP Address
nfManagementIPAddress	string	Net Foundry Management IP Address

Example

```
{
  "mgmtCIDR" : "...",
  "mgmtSubnetPrefix" : "...",
  "mgmtGatewayIPAddress" : "...",
  "mgmtNetMaskIPAddress" : "...",
  "mrmManagementIPAddress" : "...",
  "msmManagementIPAddress" : "...",
  "mmeLogicalManagementIPAddress" : "...",
  "mccManagement0IPAddress" : "...",
  "mccManagement1IPAddress" : "...",
  "mccManagement2IPAddress" : "...",
  "hssManagementNic1IPAddress" : "...",
  "nfManagementIPAddress" : "..."}
}
```

MCC

MCC resource configurations.

Properties

Name	Data Type	Description
day0Conf	MCCDay0Conf	MCC Day 0 configurations.
day1Conf	MCCDay1Conf	MCC Day 1 configurations.
mccStatus	MCCStatus	MCC Status.
natProfile	NATProfile	NAT Profile configurations. This is and Optional Day 1 configuration.
edrProfile	MCCEDRProfile	EDR Profile configurations. This is and Optional Day 1 configuration.
Properties inherited from NFBase		
networkFunction	NetworkFunction	Gets the network function. Complete Azure configuration object. Mainly for service Manager internal use
provisioningState	ProvisioningState	Gets the provisioning state. Provisioning status of individual resources like HSS, MME and MCC. for Service manager internal use
detailedStatusURL	string	Gets the detailed status URL. Url to fetch the detailed error message when the deployment failed in the Azure portal.
nfPortalName	string	Gets the nf portal name. Name of the resource in Azure portal
smName	string	Gets the sm name. Name of the resource in Service Manager left side tree view

Name	Data Type	Description
asapName	string	Gets the asap name.Name of the resource in ASAP server
createCorrelationId	string	Gets the creates the correlation id.

Example

```
{
  "day0Conf" : {
    "management0" : "...",
    "management1" : "...",
    "management2" : "...",
    "managementNetworkAddress" : "...",
    "managementGatewayAddress" : "...",
    "base0" : "...",
    "base1" : "...",
    "base2" : "...",
    "baseNetworkAddress" : "...",
    "baseGatewayAddress" : "...",
    "ew" : "...",
    "ewNetworkAddress" : "...",
    "ewGatewayAddress" : "...",
    "ns1" : "...",
    "ns1NetworkAddress" : "...",
    "ns1GatewayAddress" : "...",
    "ns2" : "...",
    "ns2NetworkAddress" : "...",
    "ns2GatewayAddress" : "...",
    "cloudBaseInternalCIDR" : "...",
    "clusterIndex" : 12345,
    "managementCIDRSuffix" : "...",
    "lanCIDRSuffix" : "...",
    "wanCIDRSuffix" : "...",
  },
  "day1Conf" : {
    "scRoamingPlmnId" : "...",
    "zoneDefaultQosProfile" : "...",
    "gnInterfaceIpRouteNextHop" : "...",
    "gnInterfaceIpRoutePrefix" : "...",
    "gnInterfaceIpRouteDest" : "...",
    "gnSgwS1" : "...",
    "gnSgwLoopbackIp" : "...",
    "giInterfaceIp" : "...",
    "giInterfaceIpRoutePrefix" : "...",
    "giInterfaceIpRouteDest" : "...",
    "giInterfaceIpRouteNextHop" : "...",
    "giIpSubPoolRangeStart" : "...",
    "giIpSubPoolRangeEnd" : "...",
  }
}
```

```

"giLoopbackIp" : "...",
"bgpRouterGN" : {
  "bgpGEOLoopIP" : "...",
  "bgpGNLoopIP" : "...",
  "routerId" : "...",
  "localAs" : "...",
  "remoteAs" : "...",
  "neighborIP" : "...",
  "fallOverBFD" : "...",
  "bfdMultiHopPeer" : "..."
},
"bgpRouterGI" : {
  "bgpGILoopIP" : "...",
  "routerId" : "...",
  "localAs" : "...",
  "remoteAs" : "...",
  "neighborIP" : "...",
  "fallOverBFD" : "...",
  "bfdMultiHopPeer" : "..."
},
"giV4IpSubPoolRangeStart" : "...",
"giV4IpSubPoolRangeEnd" : "...",
"giRemoteIpSubPoolRangeStart" : "...",
"giRemoteIpSubPoolRangeEnd" : "...",
"dataLoopIP" : "...",
"gnInterfaceIp" : "...",
"dataFabricIpAddress" : "...",
"dnsPrimaryIp" : "...",
"dnsSecondaryIp" : "...",
"pfcpPeerIp" : "...",
"networkName" : "...",
"shortNetworkName" : "...",
"apnName" : "...",
"homePlmnId" : "...",
"pgwIpAddress" : "...",
"sgwS11IpAddress" : "..." },
"mccStatus" : {
  "day0Status" : "INPROGRESS",
  "day0message" : "...",
  "day1GIStatus" : "FAIL",
  "day1GImessage" : "...",
  "day1GNStatus" : "SUCCESS",
  "day1GNmessage" : "...",
  "day1SCStatus" : "FAIL",
  "day1SCmessage" : "...",
  "day1ServiceStatus" : "SUCCESS",
  "day1Servicemessage" : "...",
  "day1Status" : "INPROGRESS",
  "day1Message" : "...",
  "dataFabricConfigured" : true,

```

```

    "optionalNTPServer1Configured" : true,
    "optionalNTPServer2Configured" : true,
    "requiredConfigurationsCompleted" : true,
    "pfcPConfigurationsCompleted" : true,
    "dnsPrimaryConfigured" : true,
    "dnsSecondaryConfigured" : true
  },
  "natProfile" : {
    "configStatus" : "UNPROVISIONED",
    "status" : "INPROGRESS",
    "message" : "...",
    "natSubPoolRangeStart" : "...",
    "natSubPoolRangeEnd" : "...",
    "profileName" : "..."
  },
  "edrProfile" : {
    "configStatus" : "PROVISIONED",
    "status" : "SUCCESS",
    "message" : "...",
    "ipAddress" : "...",
    "profileName" : "..."
  },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : {
      "provisioningState" : "ACCEPTED",
      "device" : { },
      "skuName" : "...",
      "skuType" : "SDWAN",
      "vendorName" : "...",
      "serviceKey" : "...",
      "vendorProvisioningState" : "PROVISIONED",
      "managedApplication" : { },
      "managedApplicationParameters" : { },
      "networkFunctionContainerConfigurations" : { },
      "networkFunctionUserConfigurations" : [ { }, { } ]
    }
  },
  "provisioningState" : "DELETED",
  "detailedStatusURL" : "...",
  "nfPortalName" : "...",
  "smName" : "...",
  "asapName" : "...",
  "createCorrelationId" : "..."
}

```

MCCDay0Conf

MCC Day 0 configurations.

Properties

Name	Data Type	Description
management0	string	Gets the first management IP.
management1	string	Gets the second management IP.
management2	string	Gets the second management IP.
managementNetworkAddress	string	Gets the management network address.
managementGatewayAddress	string	Gets the management gateway address.
base0	string	Gets the first base IP.
base1	string	Gets the second base IP.
base2	string	Gets the third base IP.
baseNetworkAddress	string	Gets the base network address.
baseGatewayAddress	string	Gets the base gateway address.
ew	string	Gets the east west IP address.
ewNetworkAddress	string	Gets the east west network address.
ewGatewayAddress	string	Gets the east west gateway address.
ns1	string	Gets the first north south IP.
ns1NetworkAddress	string	Gets the first north south IP network address.
ns1GatewayAddress	string	Gets the first north south IP gateway address.
ns2	string	Gets the second north south IP.
ns2NetworkAddress	string	Gets the second north south IP network address.
ns2GatewayAddress	string	Gets the second north south IP gateway address.
cloudBaseInternalCIDR	string	Gets the cloud base internal CIDR.

Properties inherited from [Day0Base](#)

clusterIndex	number	Unique Cluster Index. Internal Service Manager usage.
managementCIDRSuffix	string	Management Subnet Prefix. Value will be set from the Location selected while Network deployment.
lanCIDRSuffix	string	LAN Subnet Prefix. Value will be set from the Location selected while Network deployment.

Name	Data Type	Description
wanCIDRSuffix	string	WAN Subnet Prefix. Value will be set from the Location selected while Network deployment.

Example

```
{
  "management0" : "...",
  "management1" : "...",
  "management2" : "...",
  "managementNetworkAddress" : "...",
  "managementGatewayAddress" : "...",
  "base0" : "...",
  "base1" : "...",
  "base2" : "...",
  "baseNetworkAddress" : "...",
  "baseGatewayAddress" : "...",
  "ew" : "...",
  "ewNetworkAddress" : "...",
  "ewGatewayAddress" : "...",
  "ns1" : "...",
  "ns1NetworkAddress" : "...",
  "ns1GatewayAddress" : "...",
  "ns2" : "...",
  "ns2NetworkAddress" : "...",
  "ns2GatewayAddress" : "...",
  "cloudBaseInternalCIDR" : "...",
  "clusterIndex" : 12345,
  "managementCIDRSuffix" : "...",
  "lanCIDRSuffix" : "...",
  "wanCIDRSuffix" : "..."
}
```

MCCDay1Conf

MCC Day 1 configurations.

Properties

Name	Data Type	Description
scRoamingPlmnId	string	Gets the sc roaming plmn id.
zoneDefaultQosProfile	string	Gets the zone default qos profile.
gnInterfaceIpRouteNextHop	string	Gets the gn interface ip route next hop.
gnInterfaceIpRoutePrefix	string	Gets the gn interface ip route prefix.
gnInterfaceIpRouteDest	string	Gets the gn interface ip route dest.

Name	Data Type	Description
gnSgwS1	string	Gets the gn sgw S1.
gnSgwLoopbackIp	string	Gets the gn sgw loopback ip.
giInterfaceIp	string	Gets the gi interface ip.
giInterfaceIpRoutePrefix	string	Gets the gi interface ip route prefix.
giInterfaceIpRouteDest	string	Gets the gi interface ip route dest.
giInterfaceIpRouteNextHop	string	Gets the gi interface ip route next hop.
giIpSubPoolRangeStart	string	Gets the gi ip sub pool range start.
giIpSubPoolRangeEnd	string	Gets the gi ip sub pool range end.
giLoopbackIp	string	Gets the gi loopback ip.
bgpRouterGN	BGPRouterGN	Gets the bgp router GN.
bgpRouterGI	BGPRouterGI	Gets the bgp router GI.
giV4IpSubPoolRangeStart	string	Gets the gi V 4 ip sub pool range start.
giV4IpSubPoolRangeEnd	string	Gets the gi V 4 ip sub pool range end.
giRemoteIpSubPoolRangeStart	string	Gets the gi remote ip sub pool range start.
giRemoteIpSubPoolRangeEnd	string	Gets the gi remote ip sub pool range end.
dataLoopIP	string	Gets the data loop IP.
gnInterfaceIp	string	Gets the gn interface ip.
dataFabricIpAddress	string	Gets the data fabric ip address.
dnsPrimaryIp	string	Primary DNS IP for APN name
dnsSecondaryIp	string	Secondary DNS IP for APN name
pfcpPeerIp	string	the pfcpPeerIp
Properties inherited from Day1Base		
networkName	string	Unique Network Name.
shortNetworkName	string	Shorter Name for the Network.
apnName	string	APN Name.
homePlmnId	string	Home PLMN Identifier.
pgwIpAddress	string	PGateway IP Address
sgwS11IpAddress	string	SGateway S11 IP Address

Example

```

{
  "scRoamingPlmnId" : "...",
  "zoneDefaultQosProfile" : "...",
  "gnInterfaceIpRouteNextHop" : "...",
  "gnInterfaceIpRoutePrefix" : "...",
  "gnInterfaceIpRouteDest" : "...",
  "gnSgwS1" : "...",
  "gnSgwLoopbackIp" : "...",
  "giInterfaceIp" : "...",
  "giInterfaceIpRoutePrefix" : "...",
  "giInterfaceIpRouteDest" : "...",
  "giInterfaceIpRouteNextHop" : "...",
  "giIpSubPoolRangeStart" : "...",
  "giIpSubPoolRangeEnd" : "...",
  "giLoopbackIp" : "...",
  "bgpRouterGN" : {
    "bgpGEOLoopIP" : "...",
    "bgpGNLoopIP" : "...",
    "routerId" : "...",
    "localAs" : "...",
    "remoteAs" : "...",
    "neighborIP" : "...",
    "fallOverBFD" : "...",
    "bfdMultiHopPeer" : "..." },
  "bgpRouterGI" : {
    "bgpGILoopIP" : "...",
    "routerId" : "...",
    "localAs" : "...",
    "remoteAs" : "...",
    "neighborIP" : "...",
    "fallOverBFD" : "...",
    "bfdMultiHopPeer" : "..."
  },
  "giV4IpSubPoolRangeStart" : "...",
  "giV4IpSubPoolRangeEnd" : "...",
  "giRemoteIpSubPoolRangeStart" : "...",
  "giRemoteIpSubPoolRangeEnd" : "...",
  "dataLoopIP" : "...",
  "gnInterfaceIp" : "...",
  "dataFabricIpAddress" : "...",
  "dnsPrimaryIp" : "...",
  "dnsSecondaryIp" : "...",
  "pfcPPeerIp" : "...",
  "networkName" : "...",
  "shortNetworkName" : "...",
  "apnName" : "...",
  "homePlmnId" : "...",
  "pgwIpAddress" : "...",
  "sgwS11IpAddress" : "..."
}

```

MCCEDRProfile

EDR Profile configurations. This is an Optional Day 1 configuration.

Properties

Name	Data Type	Description
configStatus	ASAPOperationStatus	Gets the config status.
status	ActionStatus	Gets the status.
message	string	Gets the message.
ipAddress	string	Gets the ip address.
profileName	string	Gets the profile name.

Example

```
{
  "configStatus" : "UNPROVISIONED",
  "status" : "FAIL",
  "message" : "...",
  "ipAddress" : "...",
  "profileName" : "..."}

```

MCCStatus

MCC Day 0 and Day 1 status.

Properties

Name	Data Type	Constraints	Description
day0Status	ActionStatus		Gets the day 0 status. MCC deployment status. Any error during deployment will be captured in this object.
day0message	string		Gets the day 0 message. Detailed information about the deployment will be stored in this object.
day1GIStatus	ActionStatus		Gets the day 1 GI status. Service Configuration information is stored in this object.
day1GImessage	string		Gets the day 1 Gi message.
day1GNStatus	ActionStatus		Gets the day 1 GN status.
day1GNmessage	string		Gets the day 1 Gn message.
day1SCStatus	ActionStatus		Gets the day 1 SC status.
day1SCmessage	string		Gets the day 1 S cmessage.

Name	Data Type	Constraints	Description
day1ServiceStatus	ActionStatus		Gets the day 1 service status.
day1Servicemessage	string		Gets the day 1 servicemessage.
day1Status	ActionStatus		Gets the day 1 status.
day1Message	string		Gets the day 1 message.
dataFabricConfigured	boolean	required	Checks if is data fabric configured.
optionalNTPServer1Configured	boolean	required, read-only	the optionalNTPServer1Configured
optionalNTPServer2Configured	boolean	required, read-only	the optionalNTPServer2Configured
requiredConfigurationsCompleted	boolean	required, read-only	the requiredConfigurationsCompleted
pfcpcConfigurationsCompleted	boolean	required	Is User plane PFCP list configuration completed or not
dnsPrimaryConfigured	boolean	required	Is DNS Primary configured completed or not
dnsSecondaryConfigured	boolean	required	Is DNS Secondary configured completed or not

Example

```
{
  "day0Status" : "FAIL",
  "day0message" : "...",
  "day1GISTatus" : "INPROGRESS",
  "day1GImessage" : "...",
  "day1GNStatus" : "FAIL",
  "day1GNmessage" : "...",
  "day1SCStatus" : "SUCCESS",
  "day1SCmessage" : "...",
  "day1ServiceStatus" : "INPROGRESS",
  "day1Servicemessage" : "...",
  "day1Status" : "FAIL",
  "day1Message" : "...",
  "dataFabricConfigured" : true,
  "optionalNTPServer1Configured" : true,
  "optionalNTPServer2Configured" : true,
  "requiredConfigurationsCompleted" : true,
  "pfcpcConfigurationsCompleted" : true,
  "dnsPrimaryConfigured" : true,
  "dnsSecondaryConfigured" : true
}
```

MME

MME resource configurations

Properties

Name	Data Type	Description
day0Conf	MMEDay0Conf	Gets the day 0 conf. MME deployment configurations like management IP LAN and Wan IP will be stored in this object.
day1Conf	MMEDay1Conf	Gets the day 1 conf. Service Configuration related configurations will be stored in this object.
mmeStatus	MMEStatus	Gets the mme status. Status of Deployment and Service Configurations are stored in this object.

Properties inherited from [NFBase](#)

networkFunction	NetworkFunction	Gets the network function. Complete Azure configuration object. Mainly for service Manager internal use
provisioningState	ProvisioningState	Gets the provisioning state. Provisioning status of individual resources like HSS, MME and MCC. for Service manager internal use
detailedStatusURL	string	Gets the detailed status URL. Url to fetch the detailed error message when the deployment failed in the Azure portal.
nfPortalName	string	Gets the nf portal name. Name of the resource in Azure portal
smName	string	Gets the sm name. Name of the resource in Service Manager left side tree view
asapName	string	Gets the asap name. Name of the resource in ASAP server
createCorrelationId	string	Gets the creates the correlation id.

Example

```
{
  "day0Conf" : {
    "msmVMServiceName" : "...",
    "mrmVMServiceName" : "...",
    "mrmManagement" : "...",
    "msmManagement" : "...",
    "logicalManagement" : "...",
    "managementNetworkAddress" : "...",
    "managementGatewayAddress" : "...",
    "mrmBase" : "...",
    "msmBase" : "...",
    "baseNetworkAddress" : "...",
    "baseGatewayAddress" : "...",
    "mrmData" : "...",
    "msmData" : "...",
  }
}
```

```

    "dataNetworkAddress" : "...",
    "dataGatewayAddress" : "...",
    "ns1" : "...",
    "ns1NetworkAddress" : "...",
    "ns1GatewayAddress" : "...",
    "cloudMMEManagementNetMask" : "...",
    "cloudMMEBaseNetMask" : "...",
    "cloudMMEDataNetMask" : "...",
    "dataTunnelingSupernode0" : "...",
    "dataTunnelingSupernode1" : "...",
    "baseTunnelingSupernode0" : "...",
    "baseTunnelingSupernode1" : "...",
    "s6IP" : "...",
    "clusterIndex" : 12345,
    "managementCIDRSuffix" : "...",
    "lanCIDRSuffix" : "...",
    "wanCIDRSuffix" : "...",
  },
  "day1Conf" : {
    "mmeRealm" : "...",
    "diameterPeer" : "...",
    "diameterPeerIpAddress" : "...",
    "realmPeerMap" : "...",
    "diameterClientInstanceIpAddress" : "...",
    "gtpProfileIpAddress" : "...",
    "s1IfIpAddress" : "...",
    "domainName1" : "...",
    "domainName2" : "...",
    "ntpServerIp" : "...",
    "externalNs1" : "...",
    "mmeVmName" : "...",
    "node1DiameterPeer" : "...",
    "node1DiameterPeerIpAddress" : "...",
    "node1RealmPeerMap" : "...",
    "node1ExternalNs1" : "...",
    "node1MmeVmName" : "...",
    "node1MrmVmName" : "...",
    "nextHopGWIp" : "...",
    "routingInstancesBfd" : "...",
    "tacLowByte" : "...",
    "tacHighByte" : "...",
    "networkName" : "...",
    "shortNetworkName" : "...",
    "apnName" : "...",
    "homePlmnId" : "...",
    "pgwIpAddress" : "...",
    "sgwS11IpAddress" : "...",
  },
  "mmeStatus" : {
    "day0Status" : "SUCCESS",
  },

```

```

    "day0message" : "...",
    "day1Status" : "FAIL",
    "day1message" : "...",
    "optionalNTPServer1Configured" : true,
    "optionalNTPServer2Configured" : true,
    "requiredConfigurationsCompleted" : true
  },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : {
      "provisioningState" : "UNKNOWN",
      "device" : { },
      "skuName" : "...",
      "skuType" : "UNKNOWN",
      "vendorName" : "...",
      "serviceKey" : "...",
      "vendorProvisioningState" :
"USERDATAVALIDATIONFAILED",
      "managedApplication" : { },
      "managedApplicationParameters" : { },
      "networkFunctionContainerConfigurations" : { },
      "networkFunctionUserConfigurations" : [ { }, { } ]
    }
  },
  "provisioningState" : "DELETING",
  "detailedStatusURL" : "...",
  "nfPortalName" : "...",
  "smName" : "...",
  "asapName" : "...",
  "createCorrelationId" : "..."
}

```

MMEDay0Conf

MME Day 0 configurations.

Properties

Name	Data Type	Description
msmVMServiceName	string	Gets the msm VM service name. Service Name is used for the iner communication between MRM and MSM
mrmVMServiceName	string	Gets the mrm VM service name.
mrmManagement	string	Gets the mrm management.

Name	Data Type	Description
msmManagement	string	Gets the msm management.
logicalManagement	string	Gets the logical management.
managementNetworkAddress	string	Gets the management network address.
managementGatewayAddress	string	Gets the management gateway address.
mrnBase	string	Gets the mrn base.
msmBase	string	Gets the msm base.
baseNetworkAddress	string	Gets the base network address.
baseGatewayAddress	string	Gets the base gateway address.
mrnData	string	Gets the mrn data.
msmData	string	Gets the msm data.
dataNetworkAddress	string	Gets the data network address.
dataGatewayAddress	string	Gets the data gateway address.
ns1	string	Gets the ns 1.
ns1NetworkAddress	string	Gets the ns 1 network address.
ns1GatewayAddress	string	Gets the ns 1 gateway address.
cloudMMEManagementNetMask	string	Gets the cloud MME management net mask.
cloudMMEBaseNetMask	string	Gets the cloud MME base net mask.
cloudMMEDataNetMask	string	Gets the cloud MME data net mask.
dataTunnelingSupernode0	string	Gets the data tunneling supernode 0.
dataTunnelingSupernode1	string	Gets the data tunneling supernode 1.
baseTunnelingSupernode0	string	Gets the base tunneling supernode 0.
baseTunnelingSupernode1	string	Gets the base tunneling supernode 1.
s6IP	string	Gets the s6ip.
Properties inherited from Day0Base		
clusterIndex	number	Unique Cluster Index. Internal Service Manager usage.
managementCIDRSuffix	string	Management Subnet Prefix. Value will be set from the Location selected while Network deployment.

Name	Data Type	Description
lanCIDRSuffix	string	LAN Subnet Prefix. Value will be set from the Location selected while Network deployment.
wanCIDRSuffix	string	WAN Subnet Prefix. Value will be set from the Location selected while Network deployment.

Example

```
{
  "msmVMServiceName" : "...",
  "mrmVMServiceName" : "...",
  "mrmManagement" : "...",
  "msmManagement" : "...",
  "logicalManagement" : "...",
  "managementNetworkAddress" : "...",
  "managementGatewayAddress" : "...",
  "mrmBase" : "...",
  "msmBase" : "...",
  "baseNetworkAddress" : "...",
  "baseGatewayAddress" : "...",
  "mrmData" : "...",
  "msmData" : "...",
  "dataNetworkAddress" : "...",
  "dataGatewayAddress" : "...",
  "ns1" : "...",
  "ns1NetworkAddress" : "...",
  "ns1GatewayAddress" : "...",
  "cloudMMEManagementNetMask" : "...",
  "cloudMMEBaseNetMask" : "...",
  "cloudMMEDataNetMask" : "...",
  "dataTunnelingSupernode0" : "...",
  "dataTunnelingSupernode1" : "...",
  "baseTunnelingSupernode0" : "...",
  "baseTunnelingSupernode1" : "...",
  "s6IP" : "...",
  "clusterIndex" : 12345,
  "managementCIDRSuffix" : "...",
  "lanCIDRSuffix" : "...",
  "wanCIDRSuffix" : "...",
}
```

MMEDay1Conf

MME Day 1 configurations.

Properties

Name	Data Type	Description
mmeRealm	string	Gets the mme realm.
diameterPeer	string	Gets the diameter peer.
diameterPeerIpAddress	string	Gets the diameter peer ip address.
realmPeerMap	string	Gets the realm peer map.
diameterClientInstanceIpAddress	string	Gets the diameter client instance ip address.
gtpProfileIpAddress	string	Gets the gtp profile ip address.
s1IflIpAddress	string	Gets the s 1 if ip address.
domainName1	string	Gets the domain name 1.
domainName2	string	Gets the domain name 2.
ntpServerIp	string	Gets the ntp server ip.
externalNs1	string	Gets the external ns 1.
mmeVmName	string	Gets the mme vm name.
node1DiameterPeer	string	Gets the node 1 diameter peer.
node1DiameterPeerIpAddress	string	Gets the node 1 diameter peer ip address.
node1RealmPeerMap	string	Gets the node 1 realm peer map.
node1ExternalNs1	string	Gets the node 1 external ns 1.
node1MmeVmName	string	Gets the node 1 mme vm name.
node1MrmVmName	string	Gets the node 1 mrm vm name.
nextHopGWIp	string	Gets the next hop Gateway ip.
routingInstancesBfd	string	Gets the routing instances bfd.
tacLowByte	string	TAC low byte value - hexadecimal value
tacHighByte	string	TAC high byte value - hexadecimal value
Properties inherited from Day1Base		
networkName	string	Unique Network Name.
shortNetworkName	string	Shorter Name for the Network.
apnName	string	APN Name.
homePlmnId	string	Home PLMN Identifier.

Name	Data Type	Description
pgwIpAddress	string	PGateway IP Address
sgwS11IpAddress	string	SGateway S11 IP Address

Example

```

{
  "mmeRealm" : "...",
  "diameterPeer" : "...",
  "diameterPeerIpAddress" : "...",
  "realmPeerMap" : "...",
  "diameterClientInstanceIpAddress" : "...",
  "gtpProfileIpAddress" : "...",
  "s11fIpAddress" : "...",
  "domainName1" : "...",
  "domainName2" : "...",
  "ntpServerIp" : "...",
  "externalNs1" : "...",
  "mmeVmName" : "...",
  "node1DiameterPeer" : "...",
  "node1DiameterPeerIpAddress" : "...",
  "node1RealmPeerMap" : "...",
  "node1ExternalNs1" : "...",
  "node1MmeVmName" : "...",
  "node1MrmVmName" : "...",
  "nextHopGWIp" : "...",
  "routingInstancesBfd" : "...",
  "tacLowByte" : "...",
  "tacHighByte" : "...",
  "networkName" : "...",
  "shortNetworkName" : "...",
  "apnName" : "...",
  "homePlmnId" : "...",
  "pgwIpAddress" : "...",
  "sgwS11IpAddress" : "..."
}

```

MmeEnbOperEntity Data Type

Properties

name	data type	constraints	description
plmn-id	string		
enodeb-id	number	required	
enodeb-type	string		

name	data type	constraints	description
ip-address1-role	string		
ip-address2-role	string		
enodeb-name	string		
process-pid	number	required	
overload-start-sent	string		
connection-state	string		
cell-access-mode	string		
local-ip-address1-role	string		
local-ip-address2-role	string		
local-port	number	required	
connection-time	string		
raninfoId	number	required	the raninfoId
plmnId	string		the plmnId
enodebId	number	required	the enodebId
enodebType	string		the enodebType
ipAddress1Role	string		the ipAddress1Role
ipAddress2Role	string		the ipAddress2Role
port	number	required	the port
enodebName	string		the enodebName
processPid	number	required	the processPid
service	string		the service
overloadStartSent	string		the overloadStartSent
connectionState	string		the connectionState
cellAccessMode	string		the cellAccessMode
localIpAddress1Role	string		the localIpAddress1Role
localIpAddress2Role	string		the localIpAddress2Role
localPort	number	required	the localPort
connectionTime	string		the connectionTime

name	data type	constraints	description
networkId	number	required	the networkId
mmeResourceIP	string		the mmeResourceIP
networkname	string		the networkname

Example

```
{
  "plmn-id" : "...",
  "enodeb-id" : 12345,
  "enodeb-type" : "...",
  "ip-address1-role" : "...",
  "ip-address2-role" : "...",
  "enodeb-name" : "...",
  "process-pid" : 12345,
  "overload-start-sent" : "...",
  "connection-state" : "...",
  "cell-access-mode" : "...",
  "local-ip-address1-role" : "...",
  "local-ip-address2-role" : "...",
  "local-port" : 12345,
  "connection-time" : "...",
  "raninfoId" : 12345,
  "plmnId" : "...",
  "enodebId" : 12345,
  "enodebType" : "...",
  "ipAddress1Role" : "...",
  "ipAddress2Role" : "...",
  "port" : 12345,
  "enodebName" : "...",
  "processPid" : 12345,
  "service" : "...",
  "overloadStartSent" : "...",
  "connectionState" : "...",
  "cellAccessMode" : "...",
  "localIpAddress1Role" : "...",
  "localIpAddress2Role" : "...",
  "localPort" : 12345,
  "connectionTime" : "...",
  "networkId" : 12345,
  "mmeResourceIP" : "...",
  "networkname" : "..."}

```

MMEStatus

MME Day 0 and Day 1 status.

Properties

Name	Data Type	Constraints	Description
day0Status	ActionStatus		Gets the day 0 status. Deployment status
day0message	string		Gets the day 0 message. Deployment status message
day1Status	ActionStatus		Gets the day 1 status. Service Configuration status
day1message	string		Gets the day 1 message. Service Configuration message
optionalNTPServer1Configured	boolean	required, read-only	Internal use - NTP server optional IP 1
optionalNTPServer2Configured	boolean	required, read-only	Internal use - NTP server optional IP 2
requiredConfigurationsCompleted	boolean	required, read-only	Internal use - To find required IP configuration pushed to device or not

Example

```

{
  "day0Status" : "INPROGRESS",
  "day0message" : "...",
  "day1Status" : "INPROGRESS",
  "day1message" : "...",
  "optionalNTPServer1Configured" : true,
  "optionalNTPServer2Configured" : true,
  "requiredConfigurationsCompleted" : true
}

```

NATProfile

NAT Profile configurations. This is and Optional Day 1 configuration.

Properties

Name	Data Type	Description
configStatus	ASAPOperationStatus	Gets the config status.
status	ActionStatus	Gets the status.

Name	Data Type	Description
message	string	Gets the message.
natSubPoolRangeStart	string	Gets the nat sub pool range start.
natSubPoolRangeEnd	string	Gets the nat sub pool range end.
profileName	string	Gets the profile name.

Example

```
{
  "configStatus" : "PROVISIONED",
  "status" : "FAIL",
  "message" : "...",
  "natSubPoolRangeStart" : "...",
  "natSubPoolRangeEnd" : "...",
  "profileName" : "..."}

```

NEBackup

Data input to trigger the Network Element Backup. Backup File Name, File Path, for which Resources the Backup has to be taken and in which FTP Server the Backup to be stored.

Properties

Name	Data Type	Constraints	Description
backupFilePrefix	string	required	Backup file prefix appended to the file name
relativeFilePath	string	required	Relative file path on the file server
ftpServerId	number	required	File server ID
resourceIdList	array of number	required	List of resource IDs to backup

Example

```
{
  "backupFilePrefix" : "...",
  "relativeFilePath" : "...",
  "ftpServerId" : 12345,
  "resourceIdList" : [ 12345, 12345 ]
}

```

NEBackupPurge Data Type

NE Backup Purge Configuration

Properties

name	data type	constraints	description
count	number	required	Purge count
status	boolean	required	Purge status - enable/disable

Example

```
{
  "count" : 12345,
  "status" : true
}
```

NEBackupRestore

Network Element Backup Details. Backup ID, File Name, File Path, Status, from which Resource the Backup was taken and in which FTP Server the Backup is stored

Properties

Name	Data Type	Constraints	Description
nebackuprestoreid	number	required, read-only	Unique NE backup Identifier
resourceid	number	required, read-only	Resource ID of the NE to create the backup
backuptime	number	required, read-only	Epoch timestamp of backup operation
backuptriggtypeid	number	required, read-only	Type of backup triggered - (1) Scheduled or (2) User Requested
statusid	number	required, read-only	Status of backup operation - (1) Scheduled, (2) Running, (3) Success, (4) Failure
ftpprotocolid	number	required, read-only	File server ID to copy the backup file
backupfilename	string	read-only	Backup file name
relativepath	string	read-only	Relative file path on the file server
ftpprotocol	Ftpprotocol	read-only	Details of the file server configured for the backup.
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```
{
  "nebackuprestoreid" : 12345,
  "resourceid" : 12345,
  "backuptime" : 12345,
  "backuptriggtypeid" : 12345,
```

```

    "statusid" : 12345,
    "ftpprotocolid" : 12345,
    "backupfilename" : "...",
    "relativepath" : "...",
    "ftpprotocol" : {
        "secondaryftpipaddress" : "...",
        "protocolparamsid" : 12345,
        "primaryftppassword" : "...",
        "primaryftpipaddress" : "...",
        "secondaryftppassword" : "...",
        "port" : 12345,
        "ftpservername" : "...",
        "protocolType" : "...",
        "default" : true,
        "status" : true,
        "primaryftpusername" : "...",
        "secondaryftpusername" : "...",
        "version" : 12345
    },
    "version" : 12345
}

```

NEBackupSchedule

The Class NEBackupSchedule.

Properties

Name	Data Type	Constraints	Description
backupscheduleid	number	required	Gets the backupscheduleid.
name	string		Gets the name.
operationalstatusid	number	required	Gets the operationalstatusid.
ftpprotocolid	number		Gets the ftpprotocolid.
filenameprefix	string		Gets the filenameprefix.
filerelativepath	string		Gets the filerelativepath.
ftpprotocol	Ftpprotocol		Gets the ftpprotocol.
recurrencePattern	Recurrencepattern		Gets the recurrence pattern.
resourceidList	array of number		Gets the resourceid list.

Example

```

{
    "backupscheduleid" : 12345,
    "name" : "...",
    "operationalstatusid" : 12345,

```

```

"ftpprotocolid" : 12345,
"filenameprefix" : "...",
"filerelativepath" : "...",
"ftpprotocol" : {
  "secondaryftpusername" : "...",
  "port" : 12345,
  "secondaryftppassword" : "...",
  "ftpservername" : "...",
  "primaryftpipaddress" : "...",
  "primaryftppassword" : "...",
  "default" : true,
  "primaryftpusername" : "...",
  "protocolparamsid" : 12345,
  "protocolType" : "...",
  "status" : true,
  "secondaryftpipaddress" : "...",
  "version" : 12345
},
"recurrencePattern" : {
  "minute" : 12345,
  "taskschedulerid" : 12345,
  "day" : 12345,
  "month" : 12345,
  "nextexecutiontime" : 12345,
  "year" : 12345,
  "firstexecutiontime" : 12345,
  "lastexecutedtime" : 12345,
  "week" : 12345,
  "orgrepeatcount" : 12345,
  "hour" : 12345,
  "adminStatusId" : 12345,
  "starttime" : 12345,
  "repeatcount" : 12345,
  "modified" : true,
  "endtime" : 12345,
  "version" : 12345
},
"resourceidList" : [ 12345, 12345 ]
}

```

NetworkDeploymentConfig

Location Subnet and the set of Management IP Addresses reserved/assigned for the ASE Devices

Properties

Name	Data Type	Constraints	Description
networkIndex	number	required	HA Index of the Network

Name	Data Type	Constraints	Description
aseDeviceName	string		Unique ASE Device Name
HADeployment	boolean	required	TRUE if the Network deployment mode is HA. FALSE if the Network deployment mode is Standalone
primaryNode	boolean	required	TRUE if the Network Node is the Primary Node in the HA deployment or if the Network Node is Standalone. FALSE if the Network Node is the Secondary Node in the HA deployment
subnetIndex	string		Subnet Index of the Network
mgmtSubnet	string		Management Subnet
lanSubnet	string		LAN Subnet
wanSubnet	string		WAN Subnet
managementIPAddress	ManagementIPAddress		Management IP Addresses reserved/assigned for the Network
lanIPAddress	LANIPAddress		LAN IP Addresses reserved/assigned for the Network

Example

```
{
  "networkIndex" : 12345,
  "aseDeviceName" : "...",
  "HADeployment" : true,
  "primaryNode" : true,
  "subnetIndex" : "...",
  "mgmtSubnet" : "...",
  "lanSubnet" : "...",
  "wanSubnet" : "...",
  "managementIPAddress" : {
    "mgmtCIDR" : "...",
    "mgmtSubnetPrefix" : "...",
    "mgmtGatewayIPAddress" : "...",
    "mgmtNetMaskIPAddress" : "...",
    "mrmManagementIPAddress" : "...",
    "msmManagementIPAddress" : "...",
    "mmeLogicalManagementIPAddress" : "...",
    "mccManagement0IPAddress" : "...",
    "mccManagement1IPAddress" : "...",
  }
}
```

```

    "mccManagement2IPAddress" : "...",
    "hssManagementNic1IPAddress" : "...",
    "nfManagementIPAddress" : "..." },
  "lanIPAddress" : {
    "lanCIDR" : "...",
    "lanSubnetPrefix" : "...",
    "lanGatewayIPAddress" : "...",
    "lanNetMaskIPAddress" : "...",
    "nfLANIPAddress" : "..." }
}

```

NetworkFunction

Network Function configurations

Properties

Name	Data Type	Description
id	string	
name	string	
type	string	
location	string	
properties	NetworkFunctionPropertiesFormat	

Example

```

{
  "id" : "...",
  "name" : "...",
  "type" : "...",
  "location" : "...",
  "properties" : {
    "provisioningState" : "ACCEPTED",
    "device" : {
      "id" : "..." },
    "skuName" : "...",
    "skuType" : "EVOLVEDPACKETCORE",
    "vendorName" : "...",
    "serviceKey" : "...",
    "vendorProvisioningState" : "DEPROVISIONING",
    "managedApplication" : {
      "id" : "..."
    },
  },
  "managedApplicationParameters" : { },
  "networkFunctionUserConfigurations" : [ {
    "roleName" : "...",
    "userDataParameters" : { },
  }
]
}

```

```

        "networkInterfaces" : [ { }, { } ],
        "osProfile" : { }
    }, {
        "roleName" : "...",
        "userDataParameters" : { },
        "networkInterfaces" : [ { }, { } ],
        "osProfile" : { }
    } ]
}
}

```

NetworkFunctionPropertiesFormat

Network Function properties.

Properties

Name	Data Type	Description
provisioningState	ProvisioningState	
device	SubResource	
skuName	string	
skuType	SkuType	
vendorName	string	
serviceKey	string	
vendorProvisioningState	VendorProvisioningState	
managedApplication	SubResource	
managedApplicationParameters	ManagedApplicationParameters	
networkFunctionContainerConfigurations	object	
networkFunctionUserConfigurations	array of NetworkFunctionUserConfiguration	

Example

```

{
  "provisioningState" : "CANCELED",
  "device" : {
    "id" : "...",
  },
  "skuName" : "...",
  "skuType" : "FIREWALL",
  "vendorName" : "...",
  "serviceKey" : "...",
  "vendorProvisioningState" : "PROVISIONING",
  "managedApplication" : {

```

```

    "id" : "...",
  },
  "managedApplicationParameters" : { },
  "networkFunctionContainerConfigurations" : { },
  "networkFunctionUserConfigurations" : [ {
    "roleName" : "...",
    "userDataParameters" : { },
    "networkInterfaces" : [ {
      "networkInterfaceName" : "...",
      "macAddress" : "...",
      "ipConfigurations" : [ { }, { } ],
      "vmSwitchType" : "WAN"
    }, {
      "networkInterfaceName" : "...",
      "macAddress" : "...",
      "ipConfigurations" : [ { }, { } ],
      "vmSwitchType" : "MANAGEMENT"
    } ],
    "osProfile" : {
      "customData" : "..."
    }
  }, {
    "roleName" : "...",
    "userDataParameters" : { },
    "networkInterfaces" : [ {
      "networkInterfaceName" : "...",
      "macAddress" : "...",
      "ipConfigurations" : [ { }, { } ],
      "vmSwitchType" : "WAN"
    }, {
      "networkInterfaceName" : "...",
      "macAddress" : "...",
      "ipConfigurations" : [ { }, { } ],
      "vmSwitchType" : "WAN"
    } ],
    "osProfile" : {
      "customData" : "..."
    }
  } ]
}

```

NetworkFunctionSKU

Network Function SKU Details.

Properties

Name	Data Type	Description
vhdName	string	SKU VHD Name

Name	Data Type	Description
imageName	string	SKU Image Name
networkfunctionsuuid	number	Unique Network Function SKU Identifier
role	string	SKu Role
name	string	Unique SKU Name
upgradeVersion	string	SKU Upgrade Version
releaseVersion	string	SKU Release Version
upgradeImageLocation	string	Upgrade Image File Path
softwareId	number	Unique Software Identifier
Properties inherited from EJBBaseEntity		
version	number	

Example

```

{
  "vhdName" : "...",
  "imageName" : "...",
  "networkfunctionsuuid" : 12345,
  "role" : "...",
  "name" : "...",
  "upgradeVersion" : "...",
  "releaseVersion" : "...",
  "upgradeImageLocation" : "...",
  "softwareId" : 12345,
  "version" : 12345
}

```

NetworkFunctionUserConfiguration

Network Function User configuration.

Properties

Name	Data Type	Description
roleName	string	
userDataParameters	UserDataParameters	
networkInterfaces	array of NetworkInterface	
osProfile	NetworkFunctionUserConfigurationOsProfile	

Example


```

{
  "roleName" : "...",
  "userDataParameters" : { },
  "networkInterfaces" : [ {
    "networkInterfaceName" : "...",
    "macAddress" : "...",
    "ipConfigurations" : [ {
      "ipAllocationMethod" : "DYNAMIC",
      "ipAddress" : "...",
      "subnet" : "...",
      "gateway" : "...",
      "ipVersion" : "IPv4",
      "dnsServers" : [ "...", "..." ]
    }, {
      "ipAllocationMethod" : "DYNAMIC",
      "ipAddress" : "...",
      "subnet" : "...",
      "gateway" : "...",
      "ipVersion" : "IPv4",
      "dnsServers" : [ "...", "..." ]
    } ],
    "vmSwitchType" : "LAN" }, {
    "networkInterfaceName" : "...",
    "macAddress" : "...",
    "ipConfigurations" : [ {
      "ipAllocationMethod" : "DYNAMIC",
      "ipAddress" : "...",
      "subnet" : "...",
      "gateway" : "...",
      "ipVersion" : "UNKNOWN",
      "dnsServers" : [ "...", "..." ]
    }, {
      "ipAllocationMethod" : "UNKNOWN",
      "ipAddress" : "...",
      "subnet" : "...",
      "gateway" : "...",
      "ipVersion" : "IPv4",
      "dnsServers" : [ "...", "..." ]
    } ],
    "vmSwitchType" : "UNKNOWN" } ],
  "osProfile" : {
    "customData" : "..." }
}

```

NetworkFunctionUserConfigurationOsProfile

Specifies the operating system settings for the role instance.

Properties

Name	Data Type	Description
customData	string	

Example

```
{
  "customData" : "..."}

```

NetworkInterface

Network Interface properties.

Properties

Name	Data Type	Description
networkInterfaceName	string	
macAddress	string	
ipConfigurations	array of NetworkInterfaceIPConfiguration	
vmSwitchType	VmSwitchTypeEnum	

Example

```
{
  "networkInterfaceName" : "...",
  "macAddress" : "...",
  "ipConfigurations" : [ {
    "ipAllocationMethod" : "DYNAMIC",
    "ipAddress" : "...",
    "subnet" : "...",
    "gateway" : "...",
    "ipVersion" : "IPv4",
    "dnsServers" : [ "...", "..." ]
  }, {
    "ipAllocationMethod" : "UNKNOWN",
    "ipAddress" : "...",
    "subnet" : "...",
    "gateway" : "...",
    "ipVersion" : "UNKNOWN",
    "dnsServers" : [ "...", "..." ]
  } ],
  "vmSwitchType" : "UNKNOWN"}

```

NetworkInterfaceIPConfiguration

Network Interface IP configurations.

Properties

Name	Data Type	Description
ipAllocationMethod	IpAllocationMethodEnum	
ipAddress	string	
subnet	string	
gateway	string	
ipVersion	IpVersionEnum	
dnsServers	array of string	

Example

```
{
  "ipAllocationMethod" : "UNKNOWN",
  "ipAddress" : "...",
  "subnet" : "...",
  "gateway" : "...",
  "ipVersion" : "IPV4",
  "dnsServers" : [ "...", "..." ]
}
```

NetworkRetryConfig

Retry Configuration information. Will be used during network retry.

Properties

Name	Data Type	Constraints	Description
retryStatus	ActionStatus		Status of the Network deployment retry INPROGRESS during deletion SUCCESS during resource creation Failed when any of the identified resource failed while removal from portal.
deploymentRetryCount	number	required	Count to hold number of retries
nfToBeDeleted	array of string		List of networks identified for removal from portal

Example

```
{
  "retryStatus" : "FAIL",
  "deploymentRetryCount" : 12345,
  "nfToBeDeleted" : [ "...", "..." ]
}
```

NFBase

Resource configuration base for MME, MCC and HSS resources.

Properties

Name	Data Type	Description
networkFunction	NetworkFunction	Gets the network function. Complete Azure configuration object. Mainly for service Manager internal use
provisioningState	ProvisioningState	Gets the provisioning state. Provisioning status of individual resources like HSS, MME and MCC. for Service manager internal use
detailedStatusURL	string	Gets the detailed status URL. Url to fetch the detailed error message when the deployment failed in the Azure portal.
nfPortalName	string	Gets the nf portal name. Name of the resource in Azure portal
smName	string	Gets the sm name. Name of the resource in Service Manager left side tree view
asapName	string	Gets the asap name. Name of the resource in ASAP server
createCorrelationId	string	Gets the creates the correlation id.

Example

```
{
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : {
      "provisioningState" : "FAILED",
      "device" : { },
      "skuName" : "...",
      "skuType" : "FIREWALL",
      "vendorName" : "...",
      "serviceKey" : "...",
      "vendorProvisioningState" : "NOTPROVISIONED",
      "managedApplication" : { },
      "managedApplicationParameters" : { },
      "networkFunctionUserConfigurations" : [ { }, { } ]
    }
  },
  "provisioningState" : "DELETED",
  "detailedStatusURL" : "...",
```

```

"nfPortalName" : "...",
"smName" : "...",
"asapName" : "...",
"createCorrelationId" : "..."}

```

NFProvisioningState

Network Function Provisioning State.

Values: **INPROGRESS**, **SUCCEEDED**, **PROVISIONED**, **FAILED**, **DELETING**

Values

Value	Description
INPROGRESS	Network Function Provisioning is Running.
SUCCEEDED	Network Function Provisioning completed successfully.
PROVISIONED	Network Function Provisioning completed successfully.
FAILED	Network Function Provisioning Failed.
DELETING	Network Function is in Deleting state.

NFServiceInstance

Network Instance configurations

Properties

Name	Data Type	Constraints	Description
nfServiceInstanceId	number	required	Unique Network Function Identifier
networkId	number	required	Unique Network Identifier
serviceManagerSkuId	number	required	Unique Service Manager SKU Identifier
networkName	string		Unique Network Name
resourceGroupName	string		Resource Group Name
regionId	number	required	Unique Region Identifier in which the Networks are deployed
serviceConfig	string		Network Instance Configurations
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```

{
  "nfServiceInstanceId" : 12345,
  "networkId" : 12345,

```

```

    "serviceManagerSkuId" : 12345,
    "networkName" : "...",
    "resourceGroupName" : "...",
    "regionId" : 12345,
    "serviceConfig" : "...",
    "version" : 12345
  }

```

NUDetails

Network Element details that runs in the older versions of Software and the latest version to Upgrade

Properties

Name	Data Type	Constraints	Description
networkId	number	required, read-only	Unique Network Identifier
networkName	string	read-only	Gets the network name. Unique Network Name
networkLocation	string	read-only	Gets the network location. Unique Location Name in which the Network has been deployed
currentVersion	string	read-only	Current Software Version of the Network
availableVersion	string	read-only	Gets the available version. Latest available Software Version to which the Network can be Upgraded

Example

```

{
  "networkId" : 12345,
  "networkName" : "...",
  "networkLocation" : "...",
  "currentVersion" : "...",
  "availableVersion" : "..."}

```

ProvisioningState

Current Provisioning State.

Values: **UNKNOWN, SUCCEEDED, ACCEPTED, DELETING, FAILED, CANCELED, DELETED**

Values

Value	Description
UNKNOWN	Network Function is in Unknown state

Value	Description
SUCCEEDED	Provisioning completed successfully
ACCEPTED	Provisioning Accepted
DELETING	Network Function is in Deleting state
FAILED	Provisioning Failed
CANCELED	Provisioning Canceled
DELETED	Network Function Deleted

ProvisioningState

Current Provisioning State.

Values: **UNKNOWN, SUCCEEDED, ACCEPTED, DELETING, FAILED, CANCELED, DELETED, PROVISIONING**

Values

Value	Description
UNKNOWN	Network Function is in Unknown state
SUCCEEDED	Provisioning completed successfully
ACCEPTED	Provisioning Accepted
DELETING	Network Function is in Deleting state
FAILED	Provisioning Failed
CANCELED	Provisioning Canceled
DELETED	Network Function Deleted
PROVISIONING	Network Function is in Provisioning state

Recurrencepattern

Properties

Name	Data Type	Constraints	Description
minute	number	required	
taskschedulerid	number	required	
day	number	required	
month	number	required	
nextexecutiontime	number	required	

Name	Data Type	Constraints	Description
year	number	required	
firstexecutiontime	number	required	
lastexecutedtime	number	required	
week	number	required	
orgrepeatcount	number	required	
hour	number	required	
adminStatusId	number	required	
starttime	number	required	
repeatcount	number	required	
modified	boolean	required	
endtime	number	required	
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```
{
  "minute" : 12345,
  "taskschedulerid" : 12345,
  "day" : 12345,
  "month" : 12345,
  "nextexecutiontime" : 12345,
  "year" : 12345,
  "firstexecutiontime" : 12345,
  "lastexecutedtime" : 12345,
  "week" : 12345,
  "orgrepeatcount" : 12345,
  "hour" : 12345,
  "adminStatusId" : 12345,
  "starttime" : 12345,
  "repeatcount" : 12345,
  "modified" : true,
  "endtime" : 12345,
  "version" : 12345
}
```

Region

Network deployment Region details

Properties

Name	Data Type	Descriptoin
regionid	number	Unique Region Identifier
regionName	string	Unique Region Name
Properties inherited from EJBBaseEntity		
version	number	

Example

```
{
  "regionid" : 12345,
  "regionName" : "...",
  "version" : 12345
}
```

Resourcecomponent

Properties

Name	Data Type	Constraint	Description
parentcomponentid	number	required	
componentid	number	required	
componentIndex	string		
resourceid	number	required	

Example

```
{
  "parentcomponentid" : 12345,
  "componentid" : 12345,
  "componentIndex" : "...",
  "resourceid" : 12345
}
```

Resourcedetails

Properties

Name	Data Type	Constraints	Descriptions
subnetMask	string		
resourceVersion	string		
resourceName	string		

Name	Data Type	Constraints	Descriptions
resourceStatusID	number	required	
resourceid	number	required	
preProvisioned	boolean	required	
displayName	string		
resourceType	Resource type		
resourceGroupID	number	required	
IPAddress	string		
resourceTypeID	number	required	
snmpCommunity	string		
resourceComponent	Resource component		
componentid	array of number		
resyncInterval	number	required	
identifier	number	required	
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```
{
  "subnetMask" : "...",
  "resourceVersion" : "...",
  "resourceName" : "...",
  "resourceStatusID" : 12345,
  "resourceid" : 12345,
  "preProvisioned" : true,
  "displayName" : "...",
  "resourceType" : {
    "type" : "...",
    "description" : "...",
    "resyncInterval" : 12345,
    "primaryKey" : "...",
    "resourcetypeid" : 12345,
    "isdefault" : true,
    "iconname" : "...",
    "version" : 12345
  },
  "resourceGroupID" : 12345,
  "IPAddress" : "...",
  "resourceTypeID" : 12345,
```

```

    "snmpCommunity" : "...",
    "resourceComponent" : {
        "parentcomponentid" : 12345,
        "componentid" : 12345,
        "componentIndex" : "...",
        "resourceid" : 12345
    },
    "componentid" : [ 12345, 12345 ],
    "resyncInterval" : 12345,
    "identifier" : 12345,
    "version" : 12345
}

```

ResourceIMSIStatus

SIM Instance Status

Properties

Name	Data Type	Description
deletetime	string	SIM Instance Delete Time
imsino	string	Unique SIM IMSI Number
status	string	SIM Instance Status
rstatus	string	SIM Deployed Resource Status
comment	string	SIM Instance Comment

Example

```

{
    "deletetime" : "...",
    "imsino" : "...",
    "status" : "...",
    "rstatus" : "...",
    "comment" : "..."
}

```

Resourceparams

Properties

Name	Data Type	Constraints	Description
port	number	required	
resourceid	number	required	
timeout	number	required	

Name	Data Type	Constraints	Description
resourceparamsid	number	required	
retries	number	required	
protocolid	number	required	
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```
{
  "port" : 12345,
  "resourceid" : 12345,
  "timeout" : 12345,
  "resourceparamsid" : 12345,
  "retries" : 12345,
  "protocolid" : 12345,
  "version" : 12345
}
```

ResourceSimManagementStatus

SIM Instance Status in the Activated Resource Instances

Properties

Name	Data Type	Description
resourcesimid	number	Unique Resource-SIM Identifier
simInstanceid	number	Unique SIM Instance Identifier
resourceid	number	Unique Resource Identifier
status	string	Resource-Sim Status
simInstance	SimInstance	SIM Instances
resourceDetails	Resourcedetails	Resource Details
comment	string	Resource-SIM Comment
state	string	Resource-SIM State
Properties inherited from EJBBaseEntity		
version	number	

Example

```

{
  "resourcesimid" : 12345,
  "simInstanceId" : 12345,
  "resourceId" : 12345,
  "status" : "...",
  "simInstance" : {
    "deletetime" : 12345,
    "siminstanceid" : 12345,
    "simfileid" : 12345,
    "status" : "...",
    "imsiNo" : "...",
    "simFileMaster" : {
      "simfileid" : 12345,
      "simFile" : "...",
      "fileName" : "...",
      "centralHssEnabled" : true,
      "version" : 12345
    },
    "userId" : "...",
    "phoneNumber" : "...",
    "opcValue" : "...",
    "kiValue" : "...",
    "version" : 12345
  },
  "resourceDetails" : {
    "subnetMask" : "...",
    "resourceVersion" : "...",
    "resourceName" : "...",
    "resourceStatusID" : 12345,
    "resourceid" : 12345,
    "preProvisioned" : true,
    "displayName" : "...",
    "resourceType" : {
      "type" : "...",
      "description" : "...",
      "resyncInterval" : 12345,
      "primaryKey" : "...",
      "resourcetypeid" : 12345,
      "isdefault" : true,
      "iconname" : "...",
      "version" : 12345
    },
    "resourceGroupID" : 12345,
    "IPAddress" : "...",
    "resourceTypeID" : 12345,
    "snmpCommunity" : "...",
    "resourceComponent" : {
      "parentcomponentid" : 12345,

```

```

        "componentid" : 12345,
        "componentIndex" : "...",
        "resourceid" : 12345
    },
    "componentid" : [ 12345, 12345 ],
    "resyncInterval" : 12345,
    "identifier" : 12345,
    "version" : 12345
},
"comment" : "...",
"state" : "...",
"version" : 12345
}

```

Resourcetype

Properties

Name	Data Type	Constraints	Description
type	string		
description	string		
resyncInterval	number	required	
primaryKey	string		
resourcetypeid	number	required	
isdefault	boolean	required	
iconname	string		
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```

{
  "type" : "...",
  "description" : "...",
  "resyncInterval" : 12345,
  "primaryKey" : "...",
  "resourcetypeid" : 12345,
  "isdefault" : true,
  "iconname" : "...",
  "version" : 12345
}

```

ServiceInstanceConfig

Configurations for all the [DeploymentMode](#)

Properties

Name	Data Type	Constraints	Description
serviceInstanceId	number	required	Gets the service instance id.
parentResourceGroupId	number	required	Gets the parent resource group id. Internally generated and used to link the Network in Left side tree view of the UI.
nfsProvisioningState	ProvisioningState		Gets the Network provisioning state.
provisioningState	NFProvisioningState		Gets the provisioning state.
deploymentMode	DeploymentMode		Gets the deployment mode. Mode of the deployment.
locations	array of Location		Gets the locations. ASE device configuration when Deployment mode is set to ONPREM or CUPS .
cloudContainers	array of CloudContainer		Gets the cloud containers. Cloud configuration when Deployment mode is set to CUPS or CLOUD.
networkName	string		Gets the network name.
networkId	number	required	Gets the network id.
regionId	number	required	Gets the region id. Internal use.
regionName	string		Gets the region name. Under this region resources are deployed in the azure portal.
resourceGroupName	string		Gets the resource group name. Networks are deployed under this resource group.
mmePrivateKey	string		Gets the mme private key. Internal use. New private key is generated for MME operations.
skuName	string		Gets the sku name. Name of the SKU selected.
serviceManagerSkuId	number	required	Gets the service manager sku id. Selected SKU, APNS 1.0 like that.
createdTime	number	required	Gets the created time. Network created time.

Name	Data Type	Constraints	Description
resourcesAvailableForDay1	boolean	required	Checks if is resources available for day 1. Flag to find the Resources are added to ASAP server. Once all the resources are added to the ASAP server, this flag sets to true. Then only User can perform for the Service Configuration.
useSystemGeneratedIPs	boolean	required	Checks if is use system generated Ips. For Service Manager use. Refer whether IPs are user given or System generated IPs.
serviceConfigEdited	boolean	required	Checks if is use system generated Ips. For Service Manager use. Refer whether IPs and user given or System generated IPs.
mccPart	string		Gets the mcc part. Mobile Country Code
mncPart	string		Gets the mnc part. Mobile Network Code
ntpIP	string		Gets the ntp IP. Network Time Protocol IP
optionalNtpP1	string		Optional NTP IP configuration
optionalNtpP2	string		Optional second NTP IP configuration
timeZone	string		Time zone configuration for resources.
networkRetryConfig	NetworkRetryConfig	read-only	Get the network Retry related configuration
lastNetworkAction	string		Last Action performed on the Network.

Example

```

{
  "serviceInstanceId" : 12345,
  "parentResourceGroupId" : 12345,
  "nfsProvisioningState" : "ACCEPTED",
  "provisioningState" : "INPROGRESS",
  "deploymentMode" : "ONPREMHA",
  "locations" : [ {
    "name" : "...",
    "displayName" : "...",

```



```

"HADeploymentMode" : true,
"mgmtsubnet" : "...",
"lansubnet" : "...",
"wansubnet" : "...",
"locationNetworkConfig" : {
  "nwIDSubnetMap" : {
    "property1" : { },
    "property2" : { }
  },
  "mgmtSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "lanSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "wanSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "networkList" : [ {
    "property1" : "...",
    "property2" : "..."
  }, {
    "property1" : "...",
    "property2" : "..."
  } ],
"aseNetworkDeploymentConfig" : {
  "property1" : { },
  "property2" : { }
}
},
"asedeviceContainers" : [ {
  "node0" : { },
  "regionName" : "...",
  "node1" : { },
  "regionId" : 12345
}, {
  "node0" : { },
  "regionName" : "...",
  "node1" : { },
  "regionId" : 12345
} ],
"useSystemGeneratedIPs" : true,
"mgmtgw" : "...",
"langw" : "...",
"wangw" : "..."
}, {
  "name" : "...",

```

```

"displayName" : "...",
"HADeploymentMode" : true,
"mgmtsubnet" : "...",
"lansubnet" : "...",
"wansubnet" : "...",
"locationNetworkConfig" : {
  "nwIDSubnetMap" : {
    "property1" : { },
    "property2" : { }
  },
  "mgmtSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "lanSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "wanSubnet" : {
    "property1" : { },
    "property2" : { }
  },
  "networkList" : [ {
    "property1" : "...",
    "property2" : "...",
  } ],
  "aseNetworkDeploymentConfig" : {
    "property1" : { },
    "property2" : { }
  }
},
"asedeviceContainers" : [ {
  "node0" : { },
  "regionName" : "...",
  "node1" : { },
  "regionId" : 12345
}, {
  "node0" : { },
  "regionName" : "...",
  "node1" : { },
  "regionId" : 12345
} ],
"useSystemGeneratedIPs" : true,
"mgmtgw" : "...",
"langw" : "...",
"wangw" : "..."
} ],
"cloudContainers" : [ {
  "regionName" : "...",
  "storageAccountName" : "....",

```

```

    "regionId" : 12345
    "storageAccountContainerName" : "...",
    "resourceGroupName" : "...",
    "vnetName" : "...",
    "node0" : {
        "commonResources" : { },
        "name" : "...",
        "mme" : { },
        "mcc" : { },
        "hss" : { },
        "clusterIndex" : 12345
    },

    "node1" : {
        "commonResources" : { },
        "name" : "...",
        "mme" : { },
        "mcc" : { },
        "hss" : { },
        "clusterIndex" : 12345
    },
    "regionName" : "...",
    "storageAccountName" : "...",
    "regionId" : 12345,
    "storageAccountContainerName" : "...",
    "resourceGroupName" : "...",
    "vnetName" : "...",
    "node0" : {
        "commonResources" : { },
        "name" : "...",
        "mme" : { },
        "mcc" : { },
        "hss" : { },
        "clusterIndex" : 12345
    },
    "node1" : {
        "name" : "...",
        "mme" : { },
        "mcc" : { },
        "hss" : { },
        "clusterIndex" : 12345
    },
} ],
"networkName" : "...",
"networkId" : 12345,
"regionId" : 12345,
"regionName" : "...",
"resourceGroupName" : "...",
"mmePrivateKey" : "...",

```

```

    "skuName" : "...",
    "serviceManagerSkuId" : 12345,
    "createdTime" : 12345,
    "resourcesAvailableForDay1" : true,
    "useSystemGeneratedIPs" : true,
    "mccPart" : "...",
    "mncPart" : "...",
    "ntpIP" : "...",
    "optionalNtpIp1" : "...",
    "optionalNtpIp2" : "...",
    "timeZone" : "...",
    "networkRetryConfig" : {
        "retryStatus" : "INPROGRESS",
        "deploymentRetryCount" : 12345,
        "nfToBeDeleted" : [ "...", "..." ]
    },
    "lastNetworkAction" : "..."
}

```

ServiceInstanceSoftware

Software Package Details. Software Package Name, Version and SKU Details of each Node Type

Properties

Name	Data Type	Constraints	Description
name	string	required	Unique Software Package Name
version	string	required	Unique Software Package Version
rollbackVersion	string	required	Unique Software Rollback Package Version
skuDetails	array of SkuDetails	required	List of SKU Details for MME, MCC and HSS Nodes

Example

```

{
    "name" : "...",
    "version" : "...",
    "rollbackVersion" : "...",
    "skuDetails" : [ {
        "skuType" : "...",
        "skuName" : "...",
        "relVersion" : "...",
        "imageFilePath" : "...",
        "signatureFilePath" : "...",
        "armTempFilePath" : "...",
        "skuVhdName" : "...",
        "skuImageName" : "...",
        "rollbackSkuName" : "..."
    }
]

```

```

    "rollbackRelVersion" : "...",
    "rollbackImageFilePath" : "...",
    "rollbackSignatureFilePath" : "...",
    "rollbackArmTempFilePath" : "...",
    "rollbackSkuVhdName" : "...",
    "rollbackSkuImageName" : "..."
  }, {
    "skuType" : "...",
    "skuName" : "...",
    "relVersion" : "...",
    "imageFilePath" : "...",
    "signatureFilePath" : "...",
    "armTempFilePath" : "...",
    "skuVhdName" : "...",
    "skuImageName" : "...",
    "rollbackSkuName" : "...",
    "rollbackRelVersion" : "...",
    "rollbackImageFilePath" : "...",
    "rollbackSignatureFilePath" : "...",
    "rollbackArmTempFilePath" : "...",
    "rollbackSkuVhdName" : "...",
    "rollbackSkuImageName" : "..."
  } ]
}

```

ServiceManagerSKU

Service Manager SKU Details.

Properties

Name	Data Type	Constraints	Description
servicemanagerskuid	number		Unique Service Manager SKU Identifier
name	string		Unique SKU Name
type	string		Unique SKU Type
skuVersion	string		Unique SKU Version
active	boolean	required	Service Manager SKU Status

Properties inherited from [EJBBaseEntity](#)

version	number	required	
---------	--------	----------	--

Example

```

{
  "servicemanagerskuid" : 12345,
  "name" : "...",

```

```

    "type" : "...",
    "skuVersion" : "...",
    "active" : true,
    "version" : 12345
  }

```

SimFileInstance

SIM File Details and the List of SIM Instances

Properties

Name	Data Type	Constraints	Description
simFileFileName	string	required	SIM File name
instanceList	SimInstance	required	SIM Instance List
addToCentralHSS	boolean	required	Refers to add the IMSI numbers added as part of this particular file to Roaming HSS or not

Example

```

{
  "simFileFileName" : "...",
  "instanceList" : {
    "deletetime" : 12345,
    "siminstanceid" : 12345,
    "simfileid" : 12345,
    "status" : "...",
    "imsiNo" : "...",
    "simFileMaster" : {
      "simfileid" : 12345,
      "simFile" : "...",
      "fileName" : "...",
      "centralHssEnabled" : true,
      "version" : 12345
    },
    "userId" : "...",
    "phoneNumber" : "...",
    "opcValue" : "...",
    "kiValue" : "...",
    "version" : 12345
  },
  "addToCentralHSS" : true
}

```

SimFileMaster

SIM File Entity which holds the details of SIM file and Instances added as part of it.

Properties

Name	Data Type	Constraints	Description
simfileid	number	read-only	SIM File Unique identifier
simFile	string		
fileName	string	read-only	Name of the JSON file which holds the IMSI number list
centralHssEnabled	boolean	required, read-only	Refers to add the IMSI numbers added as part of this particular file to Raoming HSS or not

Properties inherited from [EJBBaseEntity](#)

version	number	required	
---------	--------	----------	--

Example

```
{
  "simfileid" : 12345,
  "simFile" : "...",
  "fileName" : "...",
  "centralHssEnabled" : true,
  "version" : 12345
}
```

SimInstance

SIM Instance Entity which holds the details of SIM (IMSI Number, User Id, Phone Number, Profile) and its status.

Properties

Name	Data Type	Constraints	Description
deletetime	number	read-only	Time which the instance marked for delete
siminstanceid	number	read-only	SIM Instance Unique identifier
simfileid	number	read-only	File identifier in which the SIM added as part.
status	string	read-only	Status of the SIM like ACTIVATED/DEACTIVATED/DELETED.
imsiNo	string	read-only	IMSI Number of the SIM.
simFileMaster	SimFileMaster	read-only	JSON File details in the SIM added as part.
userId	string	read-only	User name of the SIM user
phoneNumber	string	read-only	Assigned phone number of the SIM.

Name	Data Type	Constraints	Description
opcValue	string	read-only	OPC value of the SIM.
kiValue	string	read-only	KI value of the SIM.
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```
{
  "deletetime" : 12345,
  "siminstanceid" : 12345,
  "simfileid" : 12345,
  "status" : "...",
  "imsiNo" : "...",
  "simFileMaster" : {
    "simfileid" : 12345,
    "simFile" : "...",
    "fileName" : "...",
    "centralHssEnabled" : true,
    "version" : 12345
  },
  "userId" : "...",
  "phoneNumber" : "...",
  "opcValue" : "...",
  "kiValue" : "...",
  "version" : 12345
}
```

SKUDetails

SKU Details for MME, MCC and HSS Nodes. SKU Name, Version, Image File Path, Signature File Path and ARM Template File Path

Properties

Name	Data Type	Constraints	Description
skuType	string	required	SKU Type - mme/mcc/hss
skuName	string	required	Unique SKU Name
relVersion	string	required	Unique SKU Version
imageFilePath	string	required	SKU Image Absolute File Path
signatureFilePath	string	required	SKU Image Signature Absolute File Path
armTempFilePath	string	required	SKU ARM Template Absolute File Path

Name	Data Type	Constraints	Description
skuVhdName	string		SKU VHD Name
skuImageName	string		SKU Image Name
rollbackSkuName	string		Unique Rollback SKU Name
rollbackRelVersion	string		Unique Rollback SKU Version
rollbackImageFilePath	string		Rollback SKU Image Absolute File Path
rollbackSignatureFilePath	string		Rollback SKU Image Signature Absolute File Path
rollbackArmTempFilePath	string		Rollback SKU ARM Template Absolute File Path
rollbackSkuVhdName	string		Rollback SKU VHD Name
rollbackSkuImageName	string		Rollback SKU Image Name

Example

```
{
  "skuType" : "...",
  "skuName" : "...",
  "relVersion" : "...",
  "imageFilePath" : "...",
  "signatureFilePath" : "...",
  "armTempFilePath" : "...",
  "skuVhdName" : "...",
  "skuImageName" : "...",
  "rollbackSkuName" : "...",
  "rollbackRelVersion" : "...",
  "rollbackImageFilePath" : "...",
  "rollbackSignatureFilePath" : "...",
  "rollbackArmTempFilePath" : "...",
  "rollbackSkuVhdName" : "...",
  "rollbackSkuImageName" : "..."
}
```

SkuMapping

Mapping between Network Function SKU and Service Manager SKU

Properties

Name	Data Type	Description
skumappingid	number	Unique SKU Mapping Identifier
networkfunctionsuuid	number	Unique Network Function SKU Identifier

Name	Data Type	Description
servicemanagerskuid	number	Unique Service Manager SKU Identifier
Properties inherited from EJBBaseEntity		
version	number	

Example

```
{
  "skumappingid" : 12345,
  "networkfunctionskuid" : 12345,
  "servicemanagerskuid" : 12345,
  "version" : 12345
}
```

SkuType

SKU Type

Values

Value	Description
UNKNOWN	SKU Type Unknown
EVOLVEDPACKETCORE	SKU Type EvolvedPacketCore
SDWAN	SKU Type SDWAN
FIREWALL	SKU Type Firewall

SMDevice

Node Resources configuration

Properties

Name	Data Type	Constraints	Description
name	string		Unique Node Name.
mme	MME		MME resource configuration.
mcc	MCC		MCC resource configuration.
hss	HSS		HSS resource configuration.
clusterIndex	number	required	Unique Cluster Index. Internal Service Manager usage.

Example

```

{
  "name" : "...",
  "mme" : {
    "day0Conf" : {
      "msmVMServiceName" : "...",
      "mrmVMServiceName" : "...",
      "mrmManagement" : "...",
      "msmManagement" : "...",
      "logicalManagement" : "...",
      "managementNetworkAddress" : "...",
      "managementGatewayAddress" : "...",
      "mrmBase" : "...",
      "msmBase" : "...",
      "baseNetworkAddress" : "...",
      "baseGatewayAddress" : "...",
      "mrmData" : "...",
      "msmData" : "...",
      "dataNetworkAddress" : "...",
      "dataGatewayAddress" : "...",
      "ns1" : "...",
      "ns1NetworkAddress" : "...",
      "ns1GatewayAddress" : "...",
      "cloudMMEManagementNetMask" : "...",
      "cloudMMEBaseNetMask" : "...",
      "cloudMMEDataNetMask" : "...",
      "dataTunnelingSupernode0" : "...",
      "dataTunnelingSupernode1" : "...",
      "baseTunnelingSupernode0" : "...",
      "baseTunnelingSupernode1" : "...",
      "s6IP" : "...",
      "clusterIndex" : 12345,
      "managementCIDRSuffix" : "...",
      "lanCIDRSuffix" : "...",
      "wanCIDRSuffix" : "...",
    },
    "day1Conf" : {
      "mmeRealm" : "...",
      "diameterPeer" : "...",
      "diameterPeerIpAddress" : "...",
      "realmPeerMap" : "...",
      "diameterClientInstanceIpAddress" : "...",
      "gtpProfileIpAddress" : "...",
      "s1IfIpAddress" : "...",
      "domainName1" : "...",
      "domainName2" : "...",
      "ntpServerIp" : "...",
      "externalNs1" : "...",
      "mmeVmName" : "...",
      "node1DiameterPeer" : "...",
      "node1DiameterPeerIpAddress" : "...",
    }
  }
}

```

```

    "node1RealmPeerMap" : "...",
    "node1ExternalNs1" : "...",
    "node1MmeVmName" : "...",
    "node1MrmVmName" : "...",
    "nextHopGWIp" : "...",
    "routingInstancesBfd" : "...",
    "tacLowByte" : "...",
    "tacHighByte" : "...",
    "networkName" : "...",
    "shortNetworkName" : "...",
    "apnName" : "...",
    "homePlmnId" : "...",
    "pgwIpAddress" : "...",
    "sgwS11IpAddress" : "...",
  },
  "mmeStatus" : {
    "day0Status" : "INPROGRESS",
    "day0message" : "...",
    "day1Status" : "SUCCESS",
    "day1message" : "...",
    "optionalNTPServer1Configured" : true,
    "optionalNTPServer2Configured" : true,
    "requiredConfigurationsCompleted" : true
  },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : { }
  },
  "provisioningState" : "PROVISIONING",
  "detailedStatusURL" : "...",
  "nfPortalName" : "...",
  "smName" : "...",
  "asapName" : "...",
  "createCorrelationId" : "...",
},
"mcc" : {
  "day0Conf" : {
    "management0" : "...",
    "management1" : "...",
    "management2" : "...",
    "managementNetworkAddress" : "...",
    "managementGatewayAddress" : "...",
    "base0" : "...",
    "base1" : "...",
    "base2" : "...",
    "baseNetworkAddress" : "...",
    "baseGatewayAddress" : "...",
  }
}

```

```

    "ew" : "...",
    "ewNetworkAddress" : "...",
    "ewGatewayAddress" : "...",
    "ns1" : "...",
    "ns1NetworkAddress" : "...",
    "ns1GatewayAddress" : "...",
    "ns2" : "...",
    "ns2NetworkAddress" : "...",
    "ns2GatewayAddress" : "...",
    "cloudBaseInternalCIDR" : "...",
    "clusterIndex" : 12345,
    "managementCIDRSuffix" : "...",
    "lanCIDRSuffix" : "...",
    "wanCIDRSuffix" : "...",
  },
  "day1Conf" : {
    "scRoamingPlmnId" : "...",
    "zoneDefaultQosProfile" : "...",
    "gnInterfaceIpRouteNextHop" : "...",
    "gnInterfaceIpRoutePrefix" : "...",
    "gnInterfaceIpRouteDest" : "...",
    "gnSgwS1" : "...",
    "gnSgwLoopbackIp" : "...",
    "giInterfaceIp" : "...",
    "giInterfaceIpRoutePrefix" : "...",
    "giInterfaceIpRouteDest" : "...",
    "giInterfaceIpRouteNextHop" : "...",
    "giIpSubPoolRangeStart" : "...",
    "giIpSubPoolRangeEnd" : "...",
    "giLoopbackIp" : "...",
    "bgpRouterGN" : { },
    "bgpRouterGI" : { },
    "giV4IpSubPoolRangeStart" : "...",
    "giV4IpSubPoolRangeEnd" : "...",
    "giRemoteIpSubPoolRangeStart" : "...",
    "giRemoteIpSubPoolRangeEnd" : "...",
    "dataLoopIP" : "...",
    "gnInterfaceIp" : "...",
    "dataFabricIpAddress" : "...",
    "dnsPrimaryIp" : "...",
    "dnsSecondaryIp" : "...",
    "pfcPPeerIp" : "...",
    "networkName" : "...",
    "shortNetworkName" : "...",
    "apnName" : "...",
    "homePlmnId" : "...",
    "pgwIpAddress" : "...",
    "sgwS11IpAddress" : "...",
  },
  "mccStatus" : {

```

```

    "day0Status" : "FAIL",
    "day0message" : "...",
    "day1GISStatus" : "FAIL",
    "day1GImessage" : "...",
    "day1GNStatus" : "FAIL",
    "day1GNmessage" : "...",
    "day1SCStatus" : "SUCCESS",
    "day1SCmessage" : "...",
    "day1ServiceStatus" : "FAIL",
    "day1Servicemessage" : "...",
    "day1Status" : "SUCCESS",
    "day1Message" : "...",
    "dataFabricConfigured" : true,
    "optionalNTPServer1Configured" : true,
    "optionalNTPServer2Configured" : true,
    "requiredConfigurationsCompleted" : true,
    "pfcpcConfigurationsCompleted" : true,
    "dnsPrimaryConfigured" : true,
    "dnsSecondaryConfigured" : true
  },
  "natProfile" : {
    "configStatus" : "PROVISIONED",
    "status" : "FAIL",
    "message" : "...",
    "natSubPoolRangeStart" : "...",
    "natSubPoolRangeEnd" : "...",
    "profileName" : "..."
  },
  "edrProfile" : {
    "configStatus" : "UNPROVISIONED",
    "status" : "FAIL",
    "message" : "...",
    "ipAddress" : "...",
    "profileName" : "..."
  },
  "networkFunction" : {
    "id" : "...",
    "name" : "...",
    "type" : "...",
    "location" : "...",
    "properties" : { }
  },
  "provisioningState" : "DELETING",
  "detailedStatusURL" : "...",
  "nfPortalName" : "...",
  "smName" : "...",
  "asapName" : "...",
  "createCorrelationId" : "..."
},
"hss" : {

```

```

"day0Conf" : {
  "managementNic1" : "...",
  "managementGateway" : "...",
  "managementNetworkAddress" : "...",
  "lanNic1" : "...",
  "lanGateway" : "...",
  "lanNetworkAddress" : "...",
  "s6MMEIp" : "...",
  "s6MMEGateway" : "...",
  "clusterIndex" : 12345,
  "managementCIDRSuffix" : "...",
  "lanCIDRSuffix" : "...",
  "wanCIDRSuffix" : "...",
},
"hssStatus" : {
  "day0Status" : "SUCCESS",
  "day0message" : "...",
  "auditStatus" : "FAIL",
  "auditTime" : 12345
},
"networkFunction" : {
  "id" : "...",
  "name" : "...",
  "type" : "...",
  "location" : "...",
  "properties" : { }
},
"provisioningState" : "UNKNOWN",
"detailedStatusURL" : "...",
"nfPortalName" : "...",
"smName" : "...",
"asapName" : "...",
"createCorrelationId" : "..."
},
"clusterIndex" : 12345
}

```

SoftwareScheduleDetails

Software Upgrade Schedule Details

Properties

Name	Data Type	Constraints	Description
scheduleID	number	required	Unique Software Upgrade Schedule Identifier
scheduleName	string	read-only	Software Upgrade Name

Name	Data Type	Constraints	Description
networkID	number	required	the networkID
resourceID	number	required	Unique Resource Identifier
scheduleDetailsID	number	required	Unique Software Upgrade Schedule Details Identifier
scheduleStatus	string	read-only	Software Upgrade Status

Example

```
{
  "scheduleID" : 12345,
  "scheduleName" : "...",
  "networkID" : 12345,
  "resourceID" : 12345,
  "scheduleDetailsID" : 12345
  "scheduleStatus" : "..."}

```

SoftwareUpgradeDetails

Software Upgrade Schedule Details

Properties

Name	Data Type	Description
scheduleId	string	Unique Software Upgrade Schedule Identifier
scheduleDetailsId	string	Unique Software Upgrade Details Identifier
upgradeName	string	Unique Software Upgrade Name
upgradeVersion	string	Unique Software Upgrade Version
softwareRelease	string	Unique Software Upgrade Release Version
elementName	string	Resource Element Name
executedTime	string	Software Upgrade Executed Time
nextScheduleTime	string	Software Upgrade next Schedule Time
operationalStatus	string	Operational Status
status	string	Software Upgrade Status

Example

```
{
  "scheduleId" : "...",
  "scheduleDetailsId" : "...",
  "upgradeName" : "...",

```



```

    "upgradeVersion" : "...",
    "softwareRelease" : "...",
    "elementName" : "...",
    "executedTime" : "...",
    "nextScheduleTime" : "...",
    "operationalStatus" : "...",
    "status" : "..."
  }

```

StatusEnum

Device Status.

Values: **UNKNOWN**, **NOTREGISTERED**, **REGISTERED**, **DELETED**

Values

Value	Description
UNKNOWN	Device status is Unknown
NOTREGISTERED	Device is not Registered
REGISTERED	Device is Registered successfully
DELETED	Device has been Deleted

SubResource

Reference to another Sub Resource.

Properties

Name	Data Type	Description
id	string	

Example

```

{
  "id" : "..."
}

```

SubResource

Reference to another Sub Resource.

Properties

Name	Data Type	Description
id	string	

Example

```
{
  "id" : "...
}
```

SubscriberTracingEntity

Network Subscriber Tracing Details

Properties

Name	Data Type	Constraints	Description
traceID	number	required	Subscriber Tracing ID
traceName	string		Subscriber Tracing Name
sessionTraceKey	string		Unique MME Session Trace Key
networkID	number	required	Network ID
state	string		Subscriber Tracing State
startTime	number		Subscriber Tracing Start Time
endTime	number		Subscriber Tracing End Time
status	string		Subscriber Tracing Status
downloadPath	string		Subscriber Tracing Logs Download Path
traceData	string		Subscriber Tracing Data
traceTemplateID	number	required	Subscriber Tracing Template ID
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```
{
  "traceID" : 12345,
  "traceName" : "...",
  "sessionTraceKey" : "...",
  "networkID" : 12345,
  "state" : "...",
  "startTime" : 12345,
  "endTime" : 12345,
  "status" : "...",
  "downloadPath" : "...",
  "traceData" : "...",
  "traceTemplateID" : 12345,
  "version" : 12345
}
```

SubscriberTracingTemplateEntity Data Type

Network Subscriber Tracing Template Details

Properties

Name	Data Type	Constraints	Description
traceTemplateID	number	required	Subscriber Tracing Template ID
traceTemplateName	string		Subscriber Tracing Template Name
traceTemplateData	string		Subscriber Tracing Template Data
Properties inherited from EJBBaseEntity			
version	number	required	

Example

```
{
  "traceTemplateID" : 12345,
  "traceTemplateName" : "...",
  "traceTemplateData" : "...",
  "version" : 12345
}
```

TroubleshootBundleDetails

Troubleshoot Bundle Details provides detail of its creation , location and name.

Properties

name	data type	constraints	description
creationTime	number	required, read-only	Gets the creation time of the bundle.
filename	string	read-only	Gets the filename of the bundle.
filePath	string	read-only	Gets the filePath of the bundle.

Example

```
{
  "creationTime" : 12345,
  "filename" : "...",
  "filePath" : "..."}
}
```

TroubleshootScriptResponse

Troubleshoot Script Response is representation of the output provided by the troubleshootInformationCollector script. It has the collection status and relevant details based on the status.

Properties

Name	Data Type	Constraints	Description
collectionStatus	boolean		the collectionStatus
message	string		the message
creationTime	number	required	the creationTime
filename	string		the filename
path	string		the path
storageMode	string		the storageMode

Example

```
{
  "collectionStatus" : true,
  "message" : "...",
  "creationTime" : 12345,
  "filename" : "...",
  "path" : "...",
  "storageMode" : "..."}

```

TroubleshootStorageDetails Data Type

This class represent the request body to create the troubleshoot bundle.

Properties

Name	Data Type	Constraints	Description
storageMode	string		Storage Mode
fileserverId	number	required	Fileserver Id
filePath	string		File Path

Example

```
{
  "storageMode" : "...",
  "fileserverId" : 12345,
  "filePath" : "..."}

```

UserDataParameters

User Data Parameters

Properties

Name	Data Type	Description
------	-----------	-------------

Example

```
{ }
```

VendorProvisioningState

Vendor Provisioning State.

Values: **UNKNOWN, NOTPROVISIONED, PROVISIONING, PROVISIONED, DEPROVISIONED, USERDATAVALIDATIONFAILED**

Values

Value	Description
UNKNOWN	Vendor Provisioning State Unknown
NOTPROVISIONED	Vendor Provisioning State NotProvisioned
PROVISIONING	Vendor Provisioning State Provisioning
PROVISIONED	Vendor Provisioning State Provisioned
DEPROVISIONED	Vendor Provisioning State Deprovisioned
USERDATAVALIDATIONFAILED	Vendor Provisioning State UserDataValidationFailed

VmSwitchTypeEnum

VM Switch Type.

Values: **UNKNOWN, MANAGEMENT, WAN, LAN**

Values

Value	Description
UNKNOWN	VM Switch Type Unknown
MANAGEMENT	Management VM Switch
WAN	WAN VM Switch
LAN	LAN VM Switch